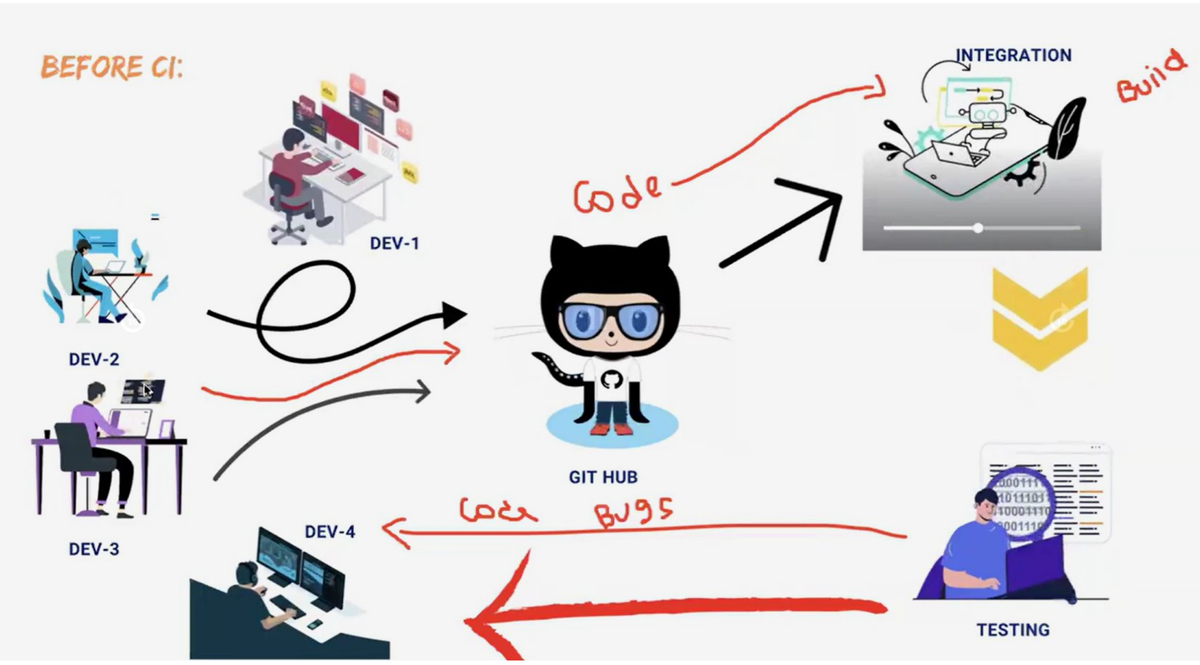
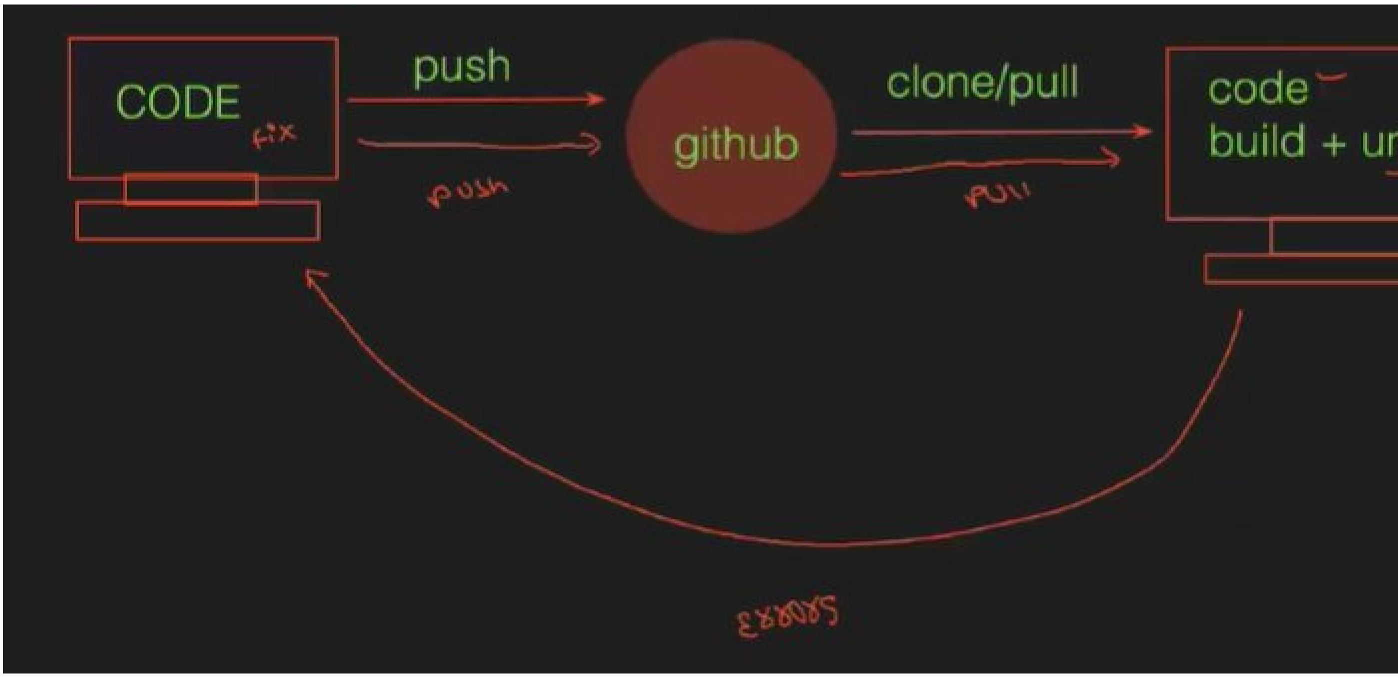
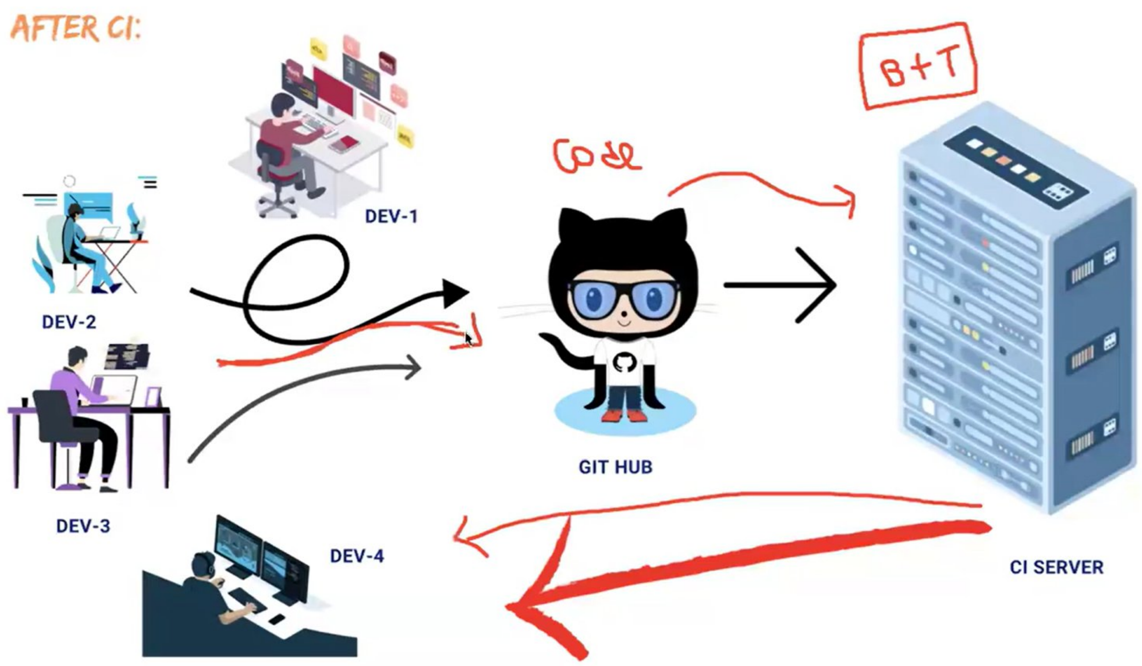
CLASS-1 JENKINS INTRO N INSTALLATION
CLASS-2 JENKINS INTEGRATION VTH GIT,WEBHOOKS,POLL SCM n BUILD PERIODICALLY
CLASS-3 HOW TO BUILD JAVA PROJECTS USING MAVEN
CLASS-4 DEPLOY JAVA APPLICATION USING JENKINS
CLASS-5 MASTER-SLAVE
CLASS-6 ee class lo previous 3 classes-getting code,build+test n deploy ey revise chesaru
CLASS-7 MASTER-SLAVE integrating artifact
CLASS-8 FREE-STYLE OPTIONS DISCARD OLD BUILDS,THROTTLE BULD, BUILD AFTER PROJECTS ARE BUILT,DELETE WORKSPACE BEFORE BUILD STARTS,EXECUTE SHELLss
CLASS-9 RBAC, PARAMETERS & THIN BACKUP
CLASS-10 PIPELINE TYPE JOB PART-1

DAY – 1 JENKINS INTRO N JENKINS SETUP

->git,github n bitbucket developer tools where Jenkins nunchi devops core tools start avutai




CODE
fix
push
push
github
clone/pull
pull
code
build + ur
errors


AFTER CI:
DEV-1
DEV-2
DEV-3
DEV-4
code
GIT HUB
B+T
CI SERVER

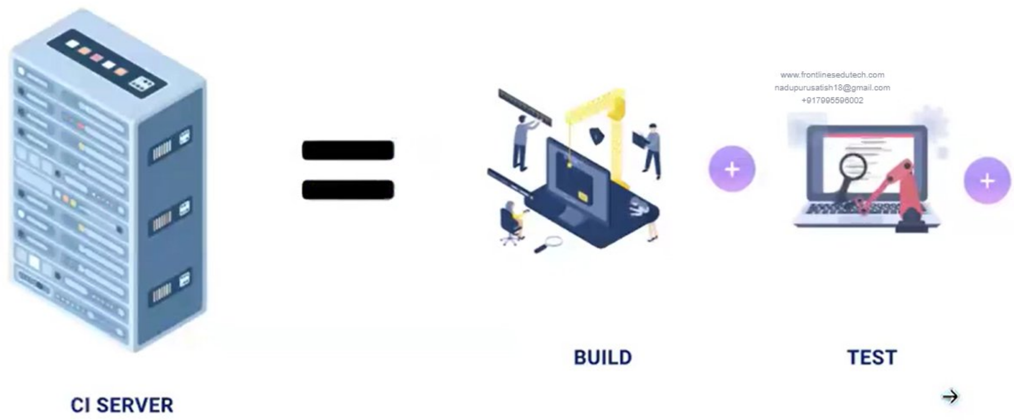
CONTINUOUS INTEGRATION:

It is the combination of continuous build and continuous test

CONTINUOUS INTEGRATION = CONTINUOUS BUILD + CONTINUOUS TEST

CI SERVER:

Here Build, Test & Deploy all these activities are performed in a single CI server



CONTINUOUS INTEGRATION:

Whenever a developer commits the code using source code management like GIT, then the CI pipeline gets the changes of the code runs automatically build and unit test.

- Due to integrating the new code with old code, we can easily get to know the code is a success or failure.
- It finds the errors more quickly
- Delivery the products to client more frequently
- Developers don't need to manual tasks
- Reduces the developers time 20% to 30%

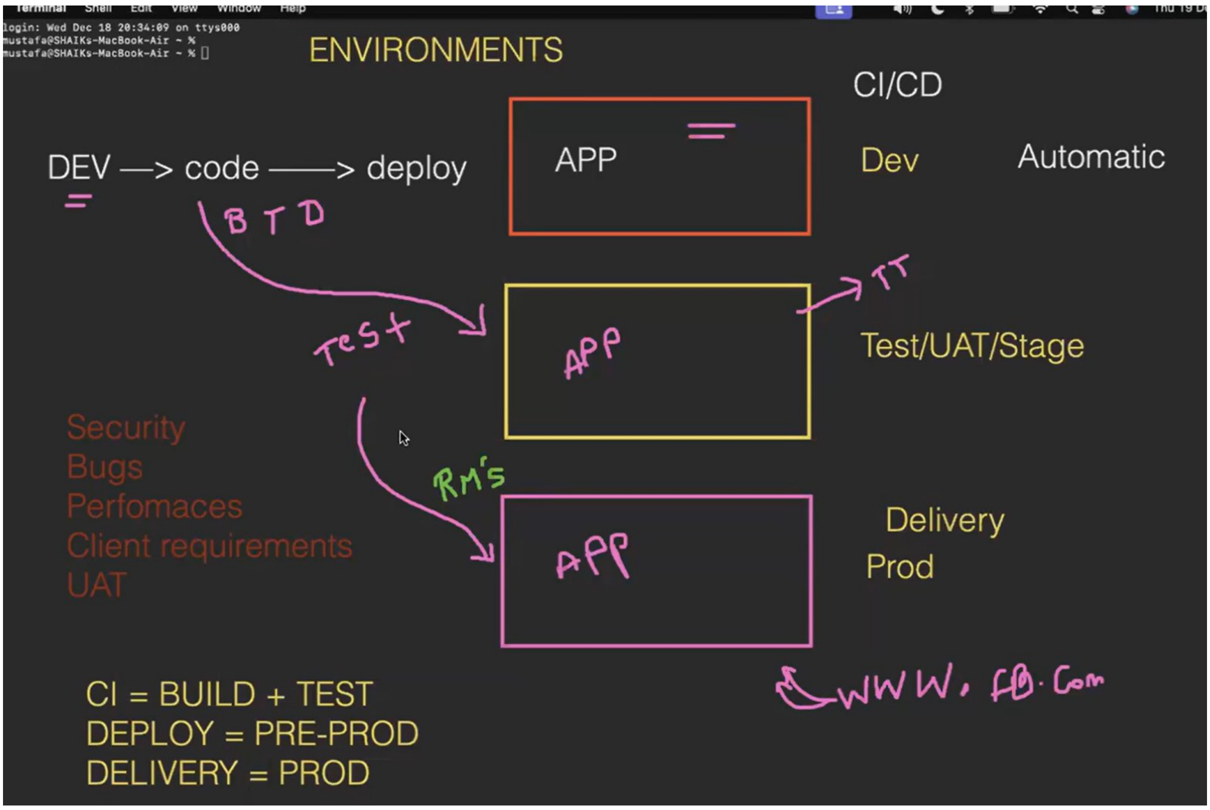
CD: CONTINUOUS DELIVERY/DEPLOYMENT

Continuous Delivery is making it available for deployment. Anytime a new build artifact is available, the artifact is automatically placed in the desired environment and deployed

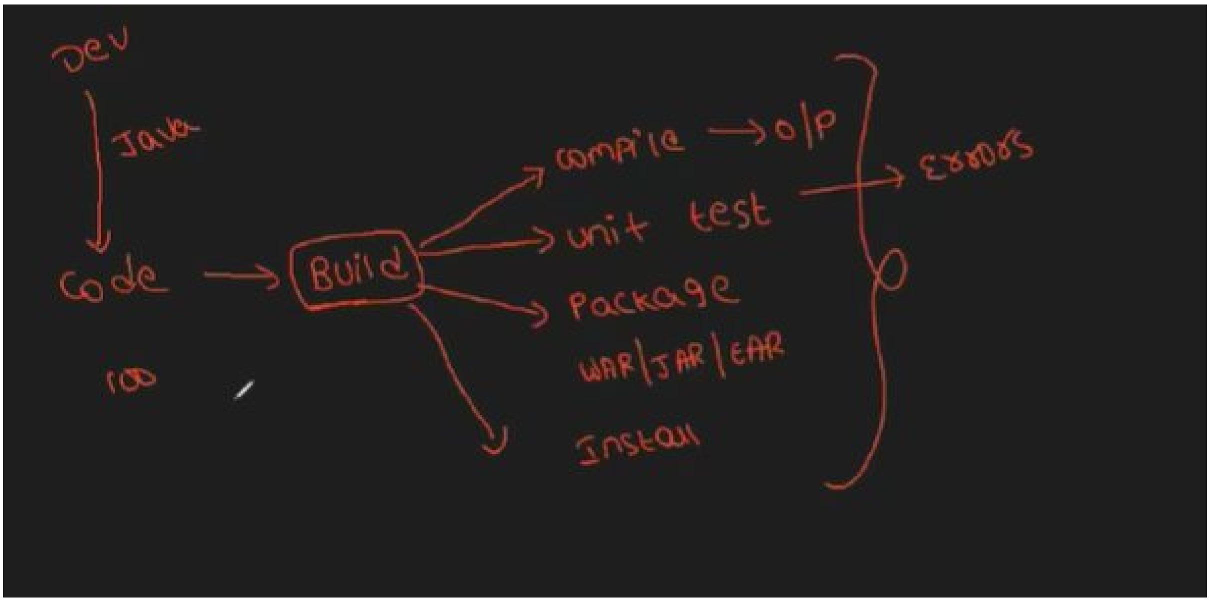
Continuous Deployment is when you commit your code then it gets automatically tested, build and deploy on the production server.

EX:

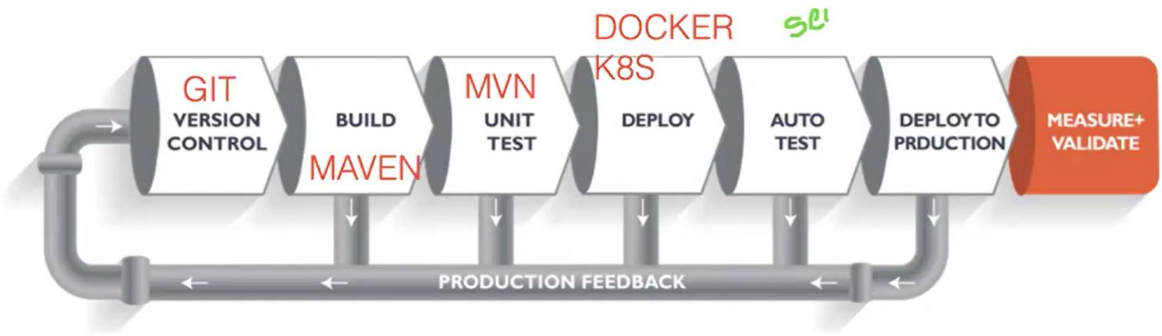
Diff environments



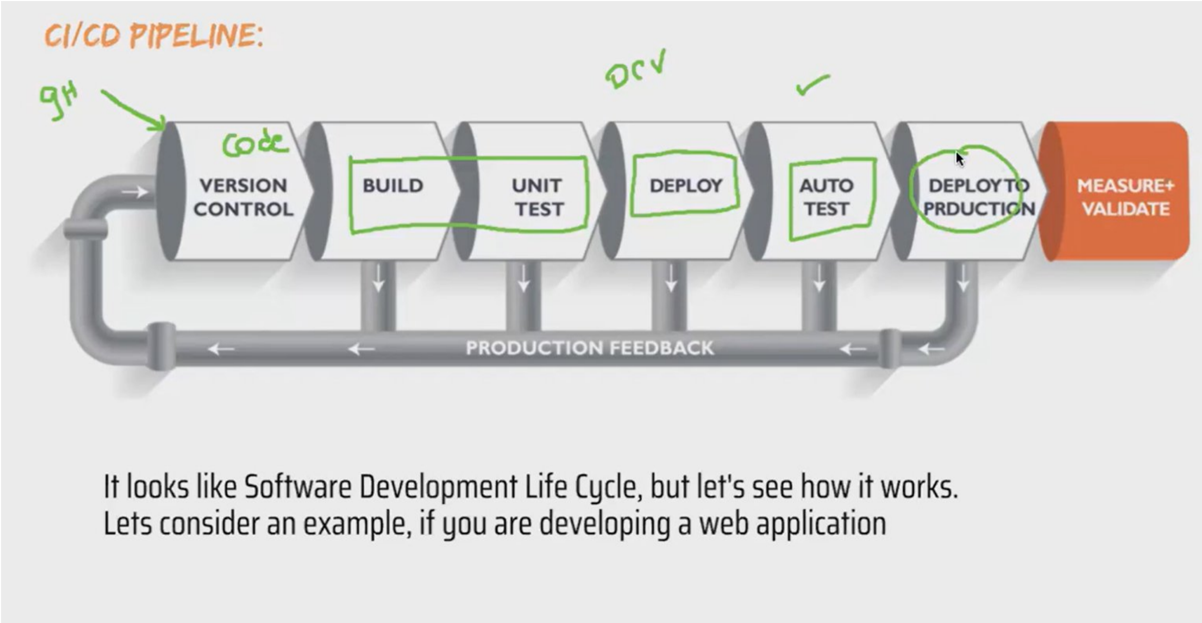
BUILD PROCESSS



CI/CD PIPELINE:



It looks like Software Development Life Cycle, but let's see how it works.
Lets consider an example, if you are developing a web application



Version Control: Here developers need to write code for web applications. So it needs to be committed using version control system like GIT or SVN.

Build: Lets consider your code is written in java, it needs to be compiled before execution. In this build step code gets compiled.

Unit Test: If the build step is completed, then move to testing phase in this step unit step will be done.

Deploy: If the test step is completed, then move to Deploy phase in this step you can deploy your code in dev, testing environment. Here you can see your application output.

FOR CREATING PIPILINES WE CAN USE ANYONE BELOW

CI / CD → pipelines → Tool ?

Jenkins

github actions

gitlab

Az pipelines

aws pipelines

Travis CI

circle CI

Harness

JENKINS:

CI/CD ——> Jenkins
devops ——> deploy

Jenkins is an **open source** project written in **java** by Kohsuke Kawaguchi it runs on the **Window, Linux and Mac OS**

Travis CI
Circle CI
semaphore
buddy
GitLab

It is community-supported, Free to use, and the First choice for Continuous Integration.

AWS —> aws code pipeline
Azure —> Azure pipeline

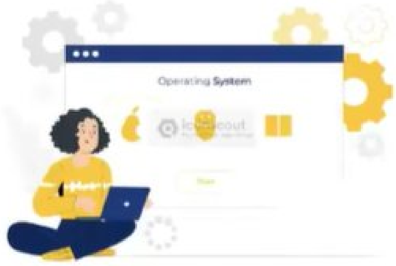
- Consist of Plugins
- Automates the Entire Software Development Life Cycle (SDLC).

- It was originally developed by Sun Microsystem in 2004 as HUDSON.
- Hudson was an enterprise Edition we need to pay for it.

1. The project was renamed Jenkins when Oracle brought the Microsystems.

Plugins-a small software to add some extra feature

- It can run on any major platform without Compatibility issues.



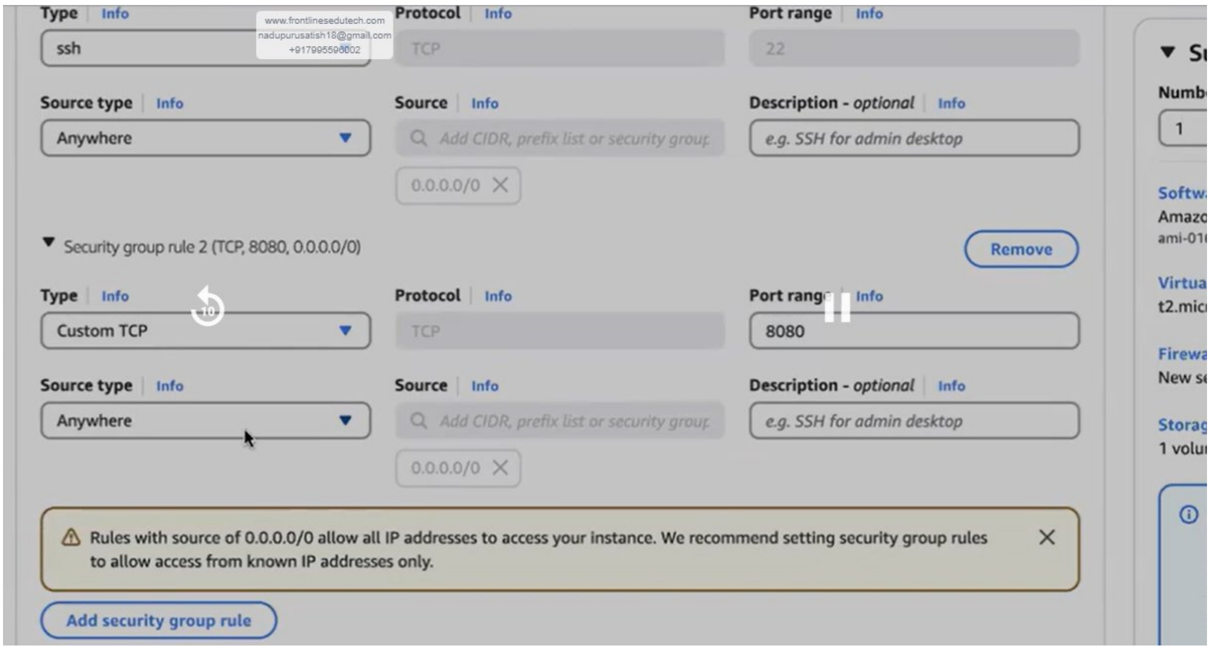
- Whenever developers write code, we integrate all the code of all developers at any point in time and we build, test, and deliver/deploy it to the client. This is called CI/CD.

JENKINS INSTALLATION

->EC2 instance create chestapudu, security grp oka new grp create cheyyandi Jenkins(Jenkins-sec-grp).

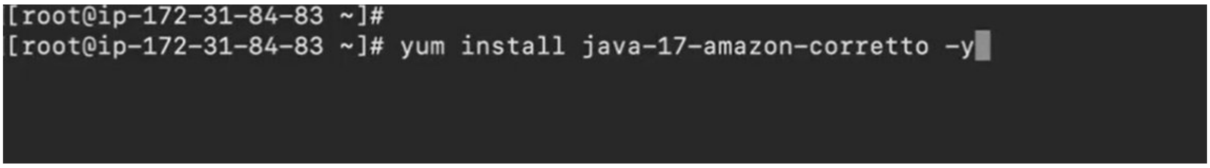
->take c7-flex-large as instance type

->linux server cnt cheyyadaniki type ssh use chestam alaney Jenkins cnt avvadaniki oka port no 8080 kavali

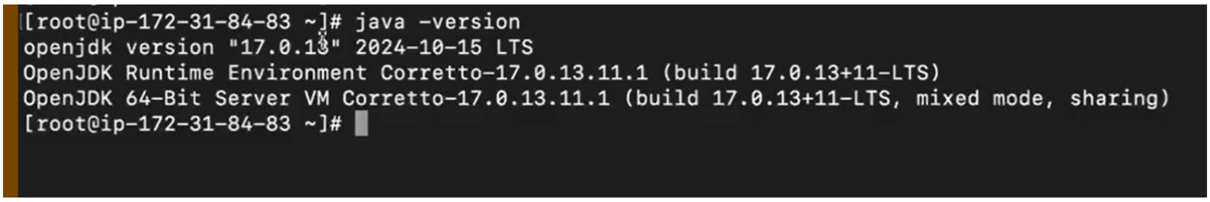


->steps to install Jenkins

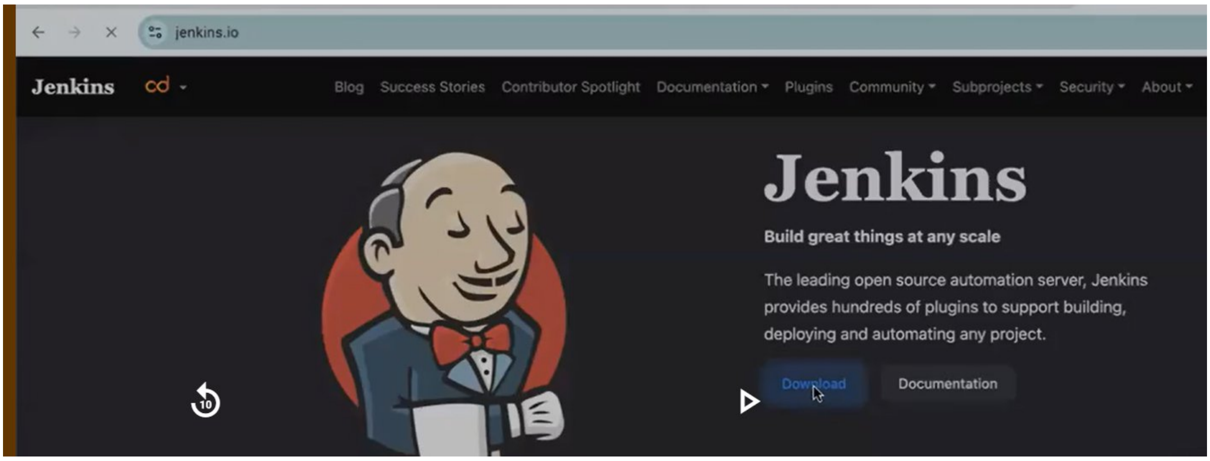
->to install java

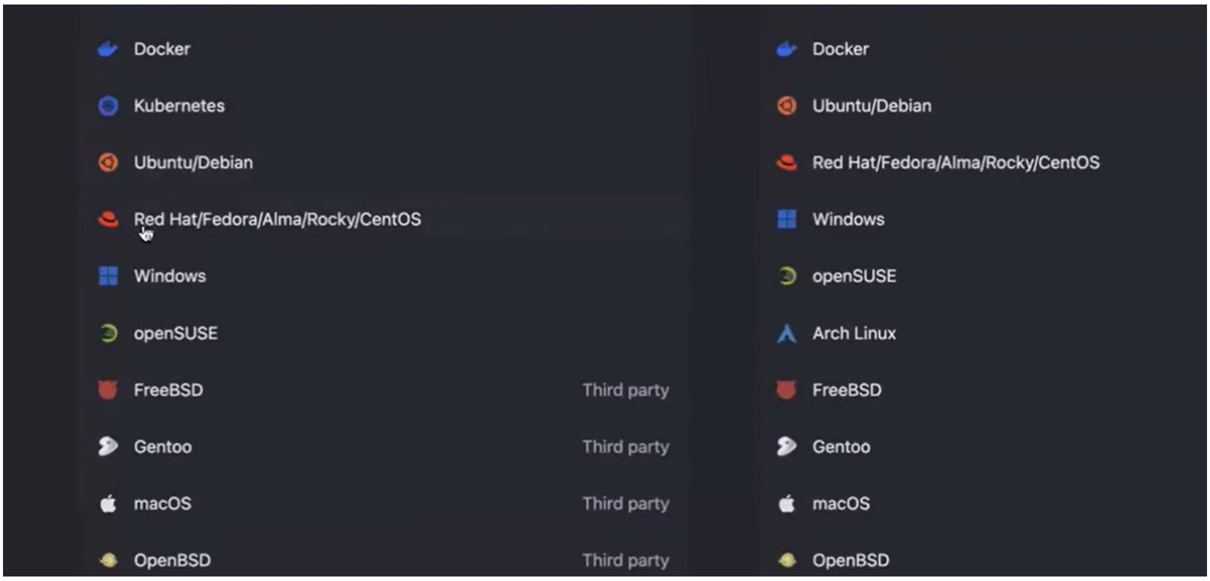


->to chk java installed or not



->to install repo's, goto Jenkins.io> downloads>redhat>copy those cmds one by one in our ec2 instance>

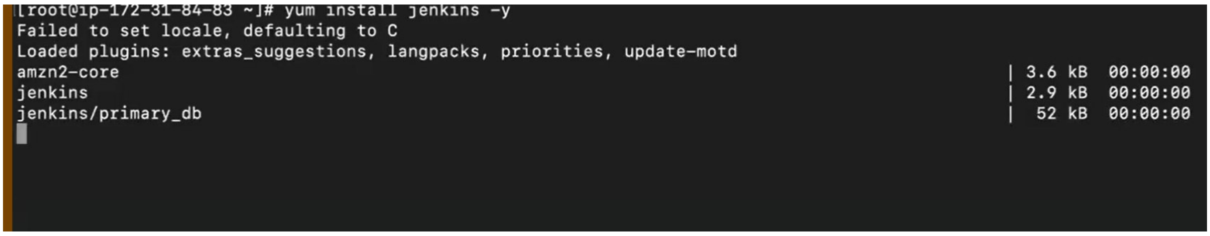




1st cmd – to use Jenkins in ur sys that Jenkins.repo file shd be in ur sys

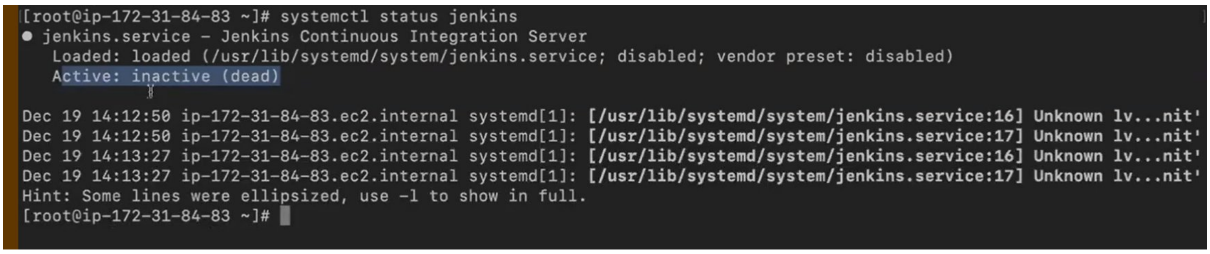
2nd cmd – Jenkins login ayyetapudu oka pword adugutadi, a key mi sys loki ravadaniki a cmd use chestamu

->to install Jenkins

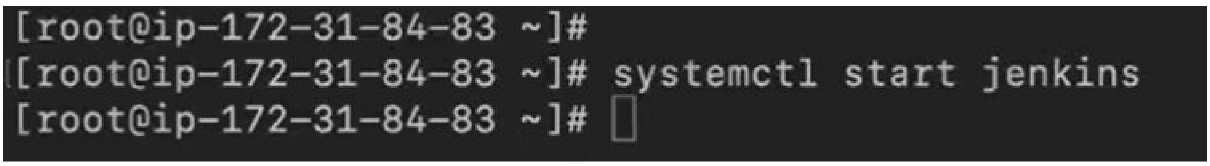

```
[root@ip-172-31-84-83 ~]# yum install jenkins -y
Failed to set locale, defaulting to C
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
amzn2-core                                                | 3.6 kB  00:00:00
jenkins                                                   | 2.9 kB  00:00:00
jenkins/primary_db                                        |  52 kB  00:00:00
```

->start Jenkins(install chesaka by default running lo undadhu, so we need to start jenkins)

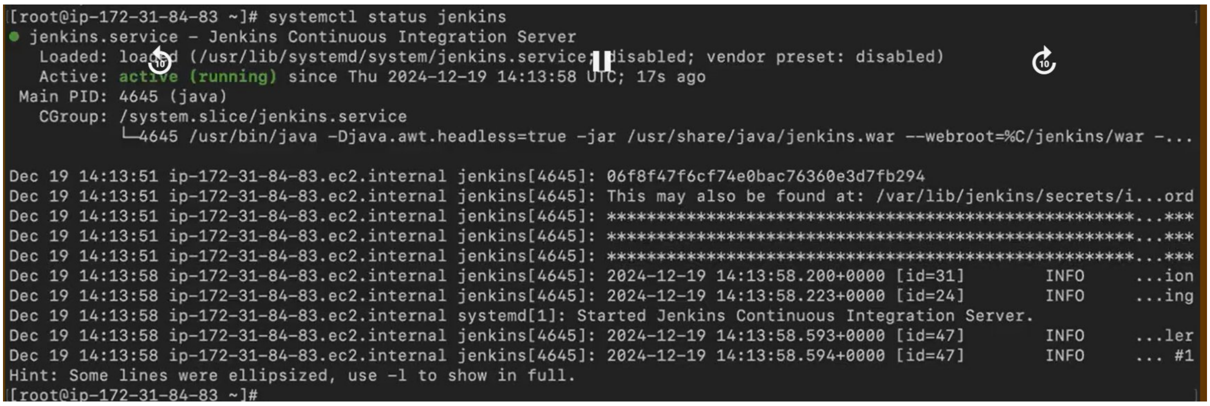
To know Jenkins is running or not – systemctl status jenkins


```
[root@ip-172-31-84-83 ~]# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; disabled; vendor preset: disabled)
   Active: inactive (dead)

Dec 19 14:12:50 ip-172-31-84-83.ec2.internal systemd[1]: [/usr/lib/systemd/system/jenkins.service:16] Unknown lv...nit'
Dec 19 14:12:50 ip-172-31-84-83.ec2.internal systemd[1]: [/usr/lib/systemd/system/jenkins.service:17] Unknown lv...nit'
Dec 19 14:13:27 ip-172-31-84-83.ec2.internal systemd[1]: [/usr/lib/systemd/system/jenkins.service:16] Unknown lv...nit'
Dec 19 14:13:27 ip-172-31-84-83.ec2.internal systemd[1]: [/usr/lib/systemd/system/jenkins.service:17] Unknown lv...nit'
Hint: Some lines were ellipsized, use -l to show in full.
[root@ip-172-31-84-83 ~]#
```

To start, a cmd run chesaka tym pattidi

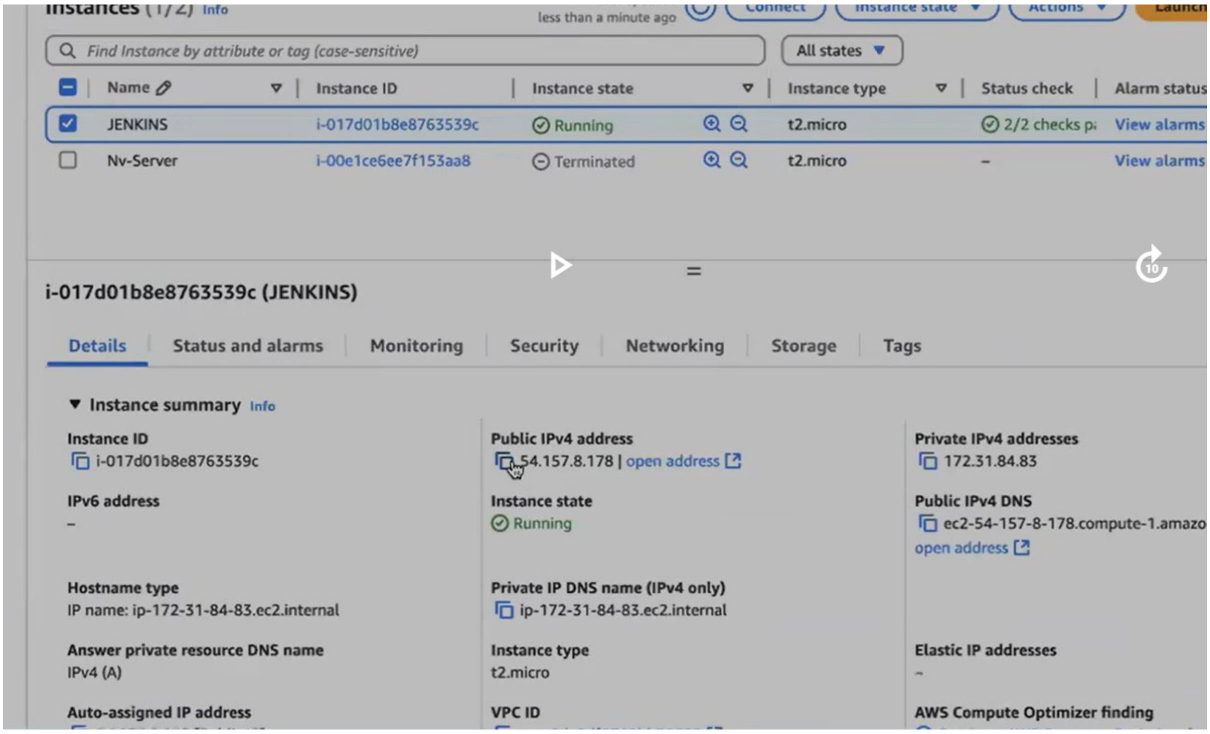

```
[root@ip-172-31-84-83 ~]#
[root@ip-172-31-84-83 ~]# systemctl start jenkins
[root@ip-172-31-84-83 ~]#
```

Now it is active

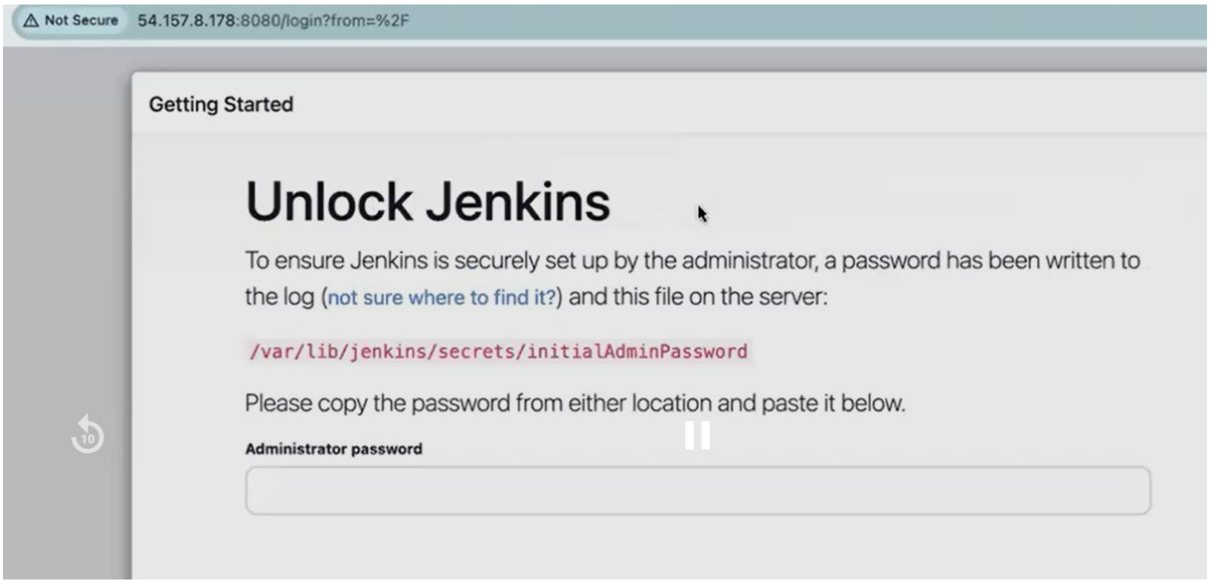


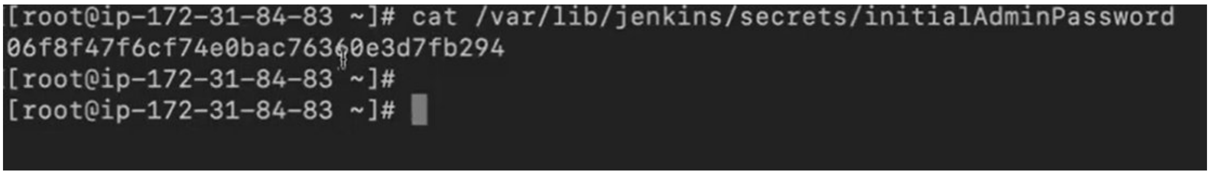
NOTE – 1. Jenkins ni GUI lo use chestamu, so mana server pub-ip tho browser lo cnt aithey GUI ga use chesukovachu

2. pub-ip ki last lo :port_no(which is 8080) ivvali otherwise it won't wrk

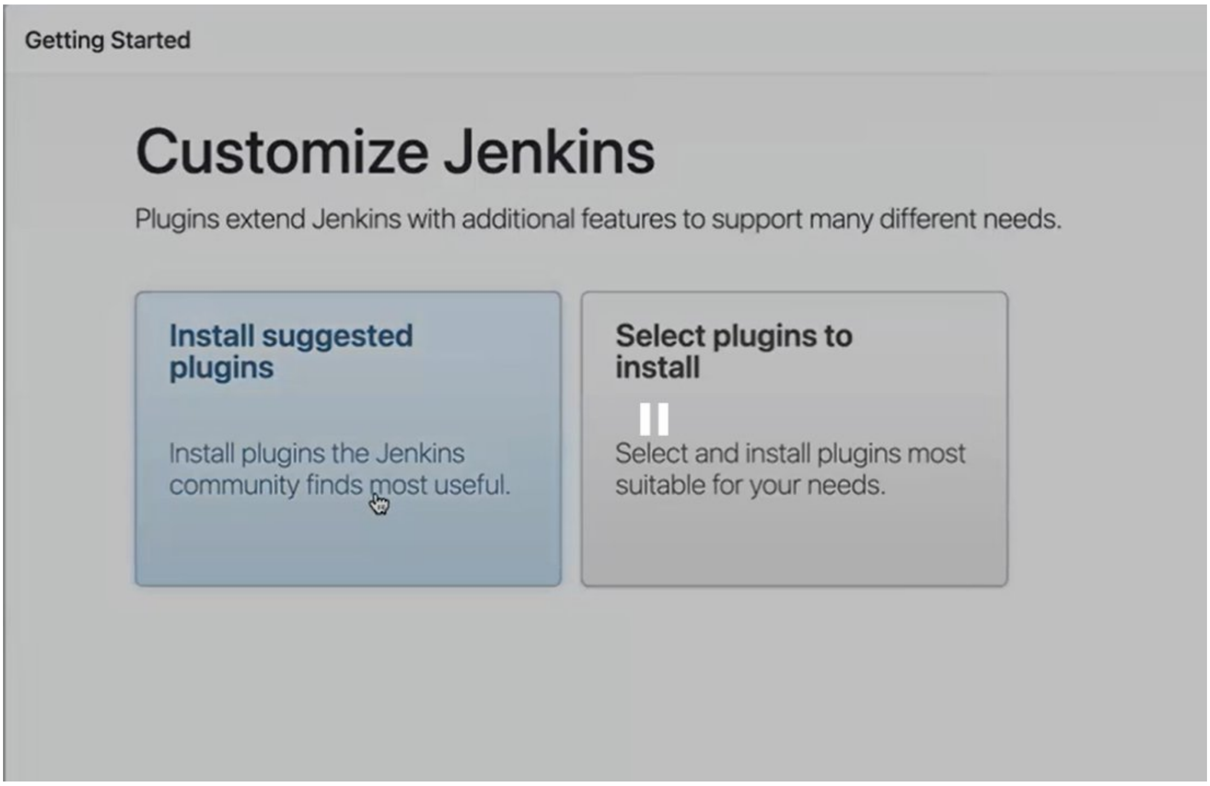


->by default Jenkins lock lo untadi, to un-lock it goto that path

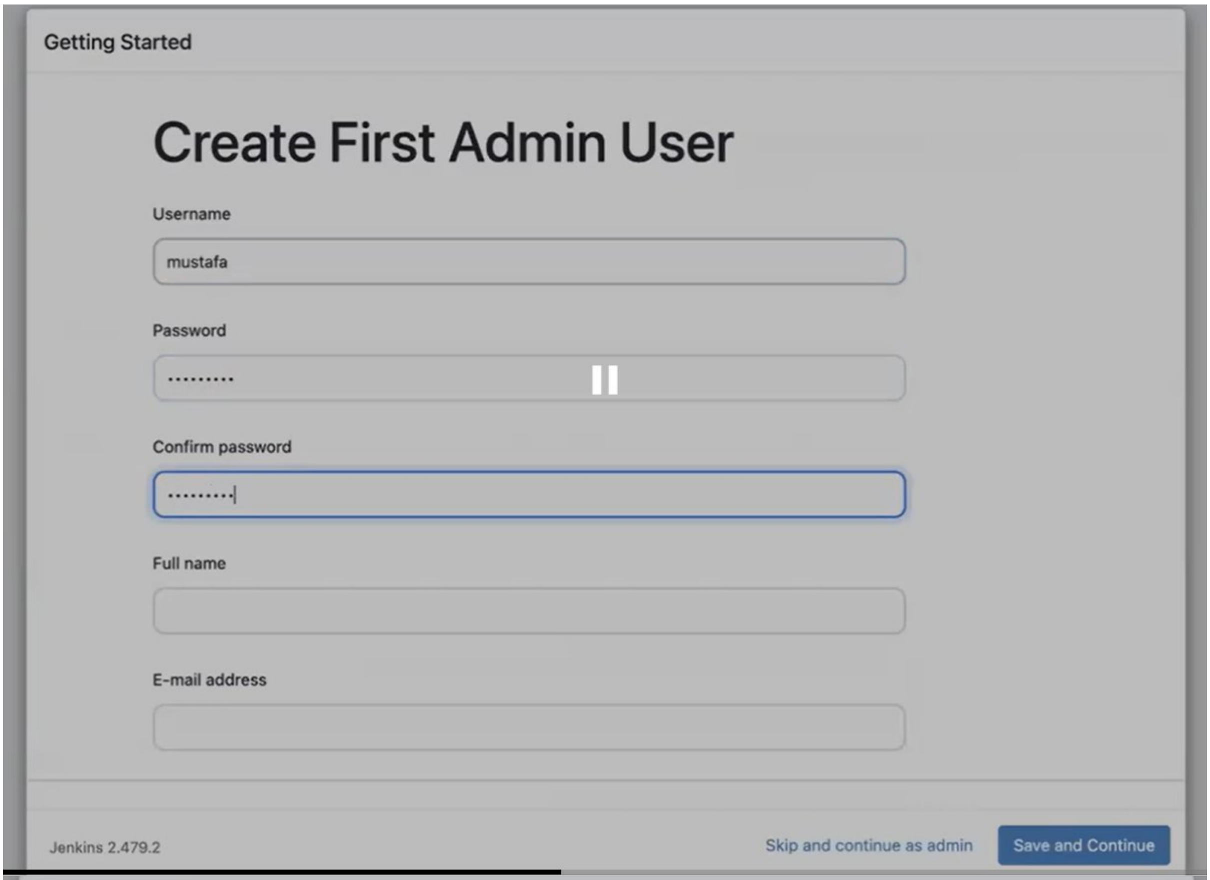


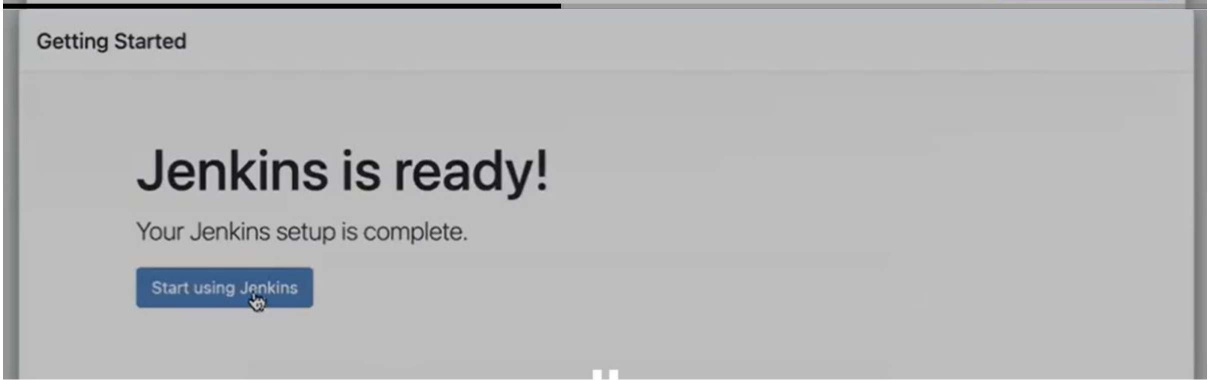


Click on install suggested plugins

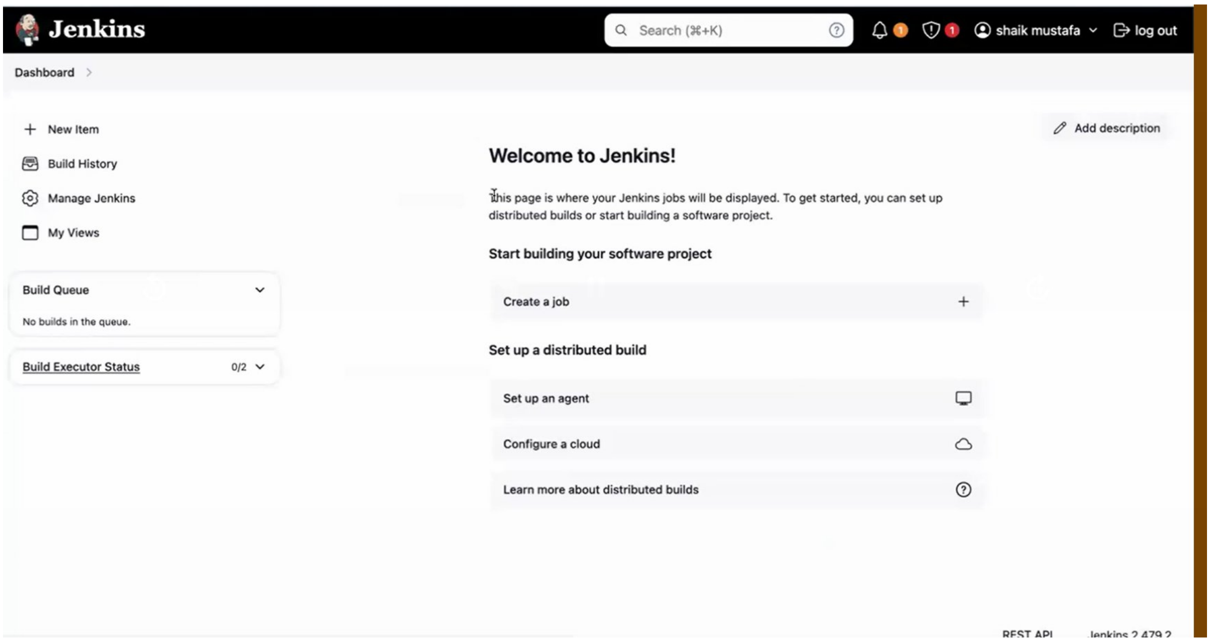


->profile creation





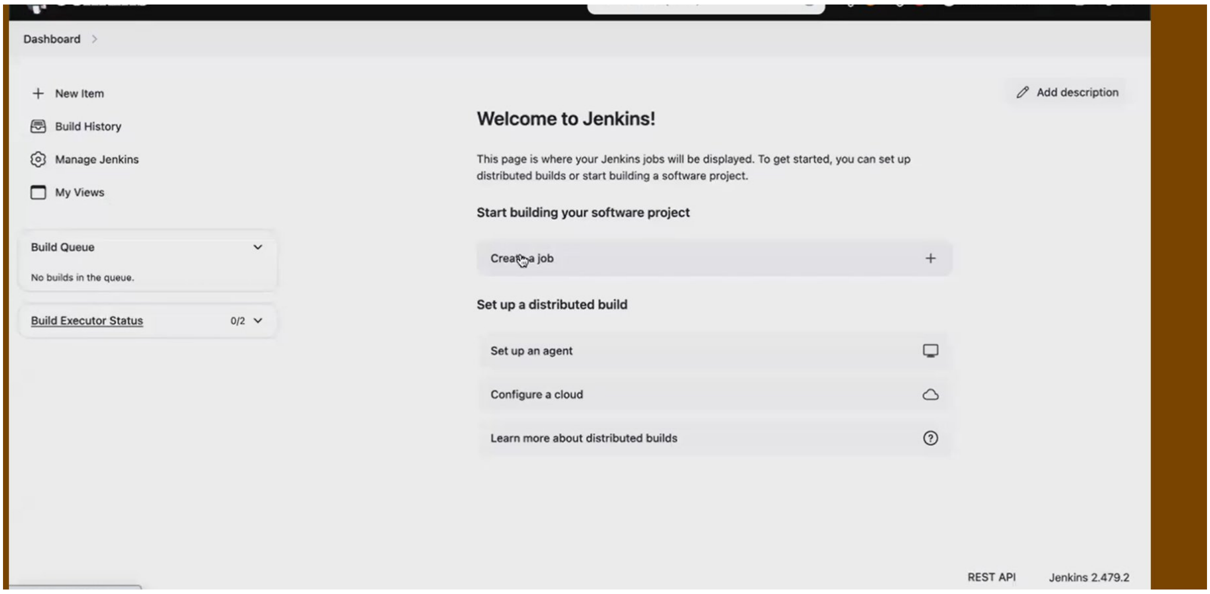
This is user interface



CI PROCESS IN JENKINS FROM GITHUB

->got our github>any repo>code>take only https url

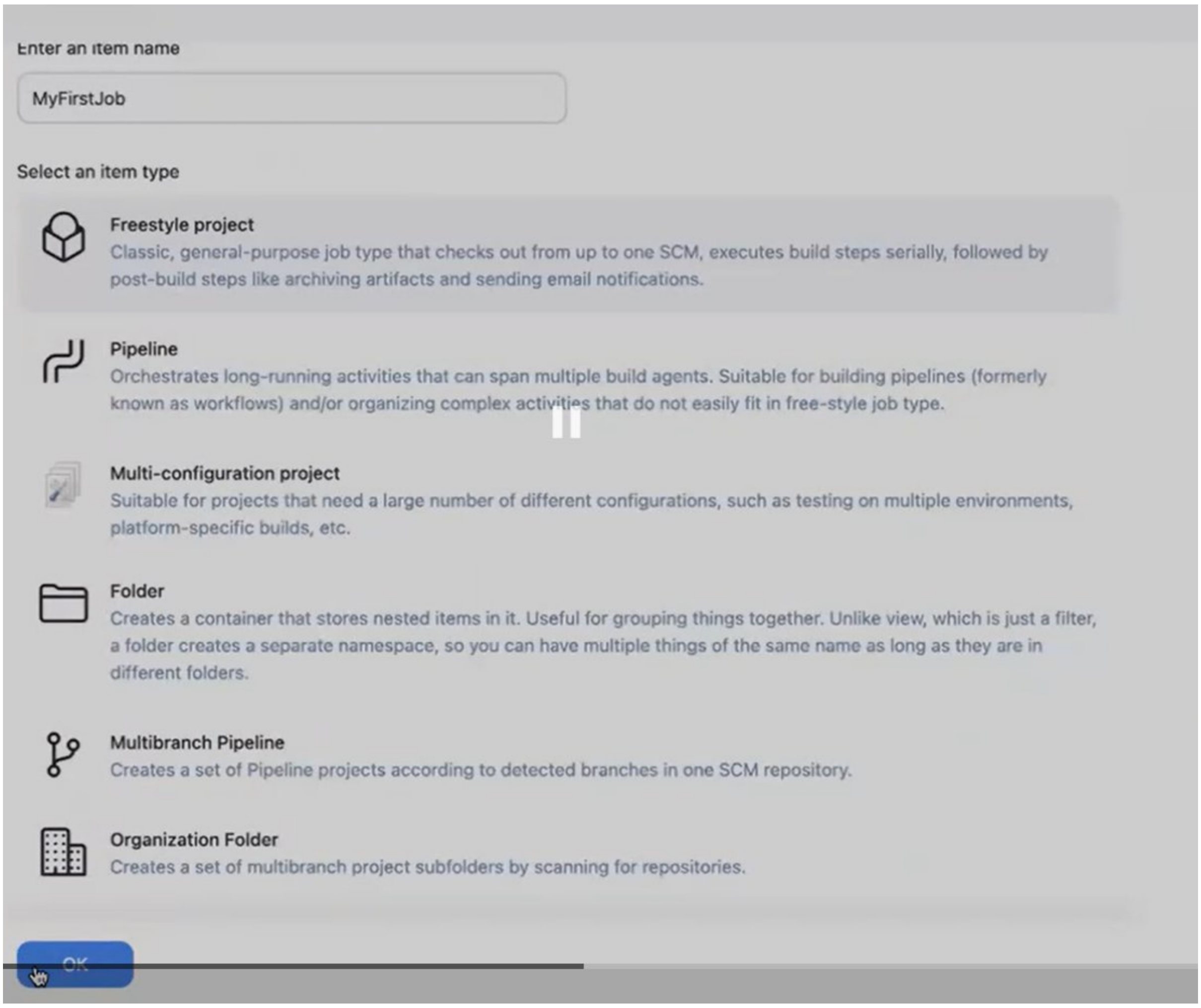
->create a job/new item



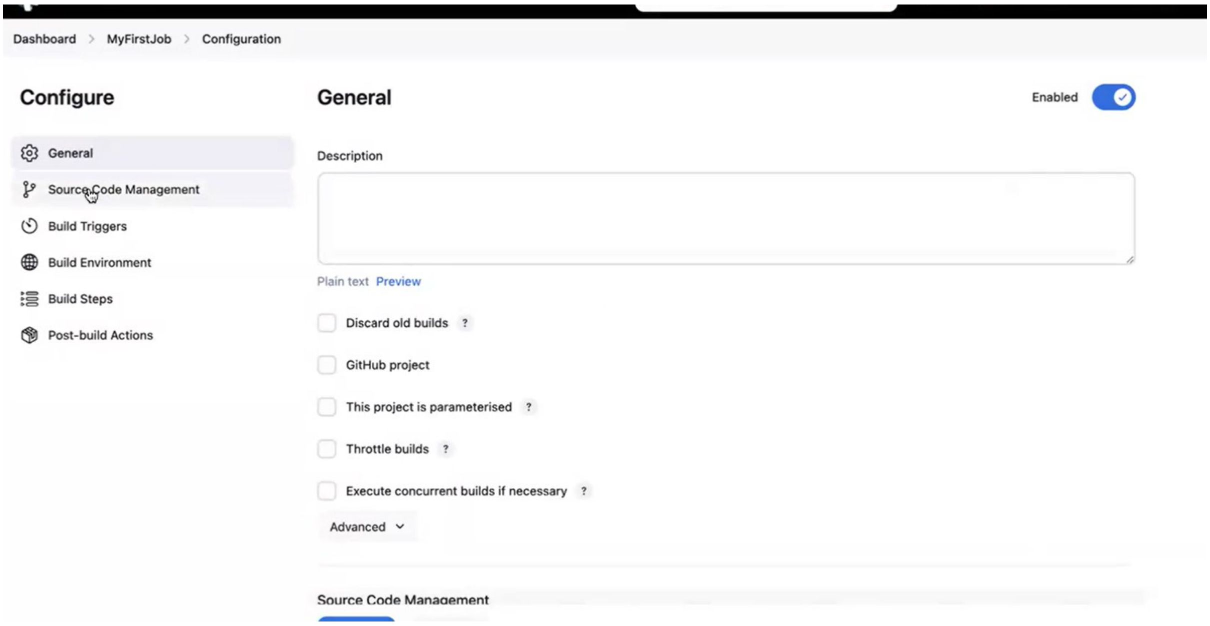
NOTE – 1. Jenkins lo job antey build/deploy/code frm git/testing ….em cheyyali anna job create cheyyali.

2. while creating job don't give spaces

->sel free trial>ok



->click on source code management

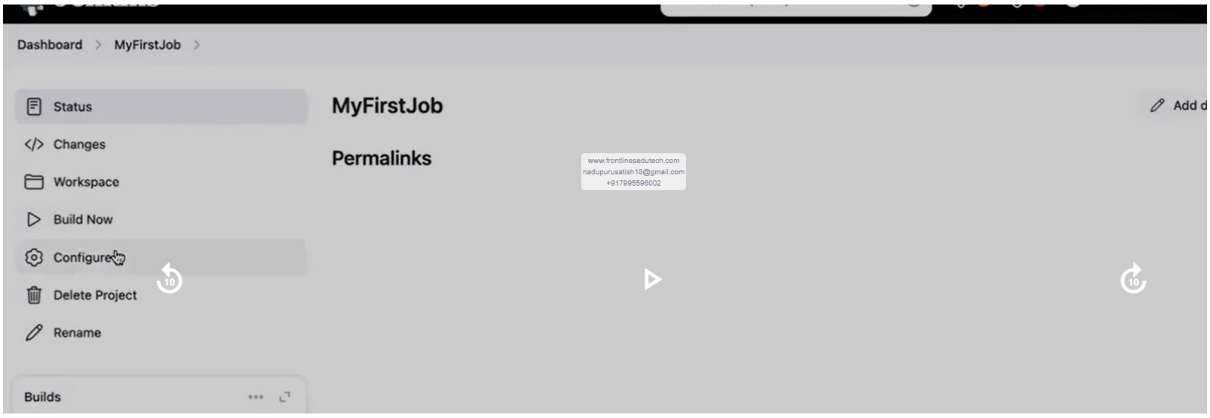


->git install chesi ledu kabatti a err vastundi, edaina git actions cheyyali antey git kavali, so git ni install chedam

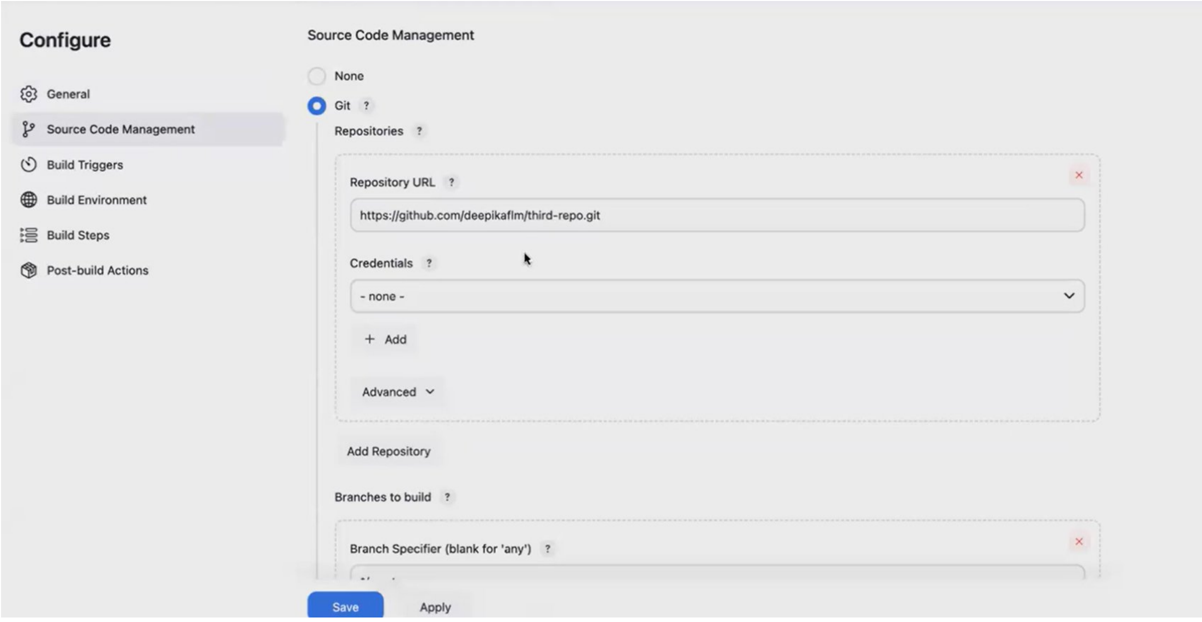
->paste ur github https url(if it a priv repo give credentials otherwise no need)

->install git

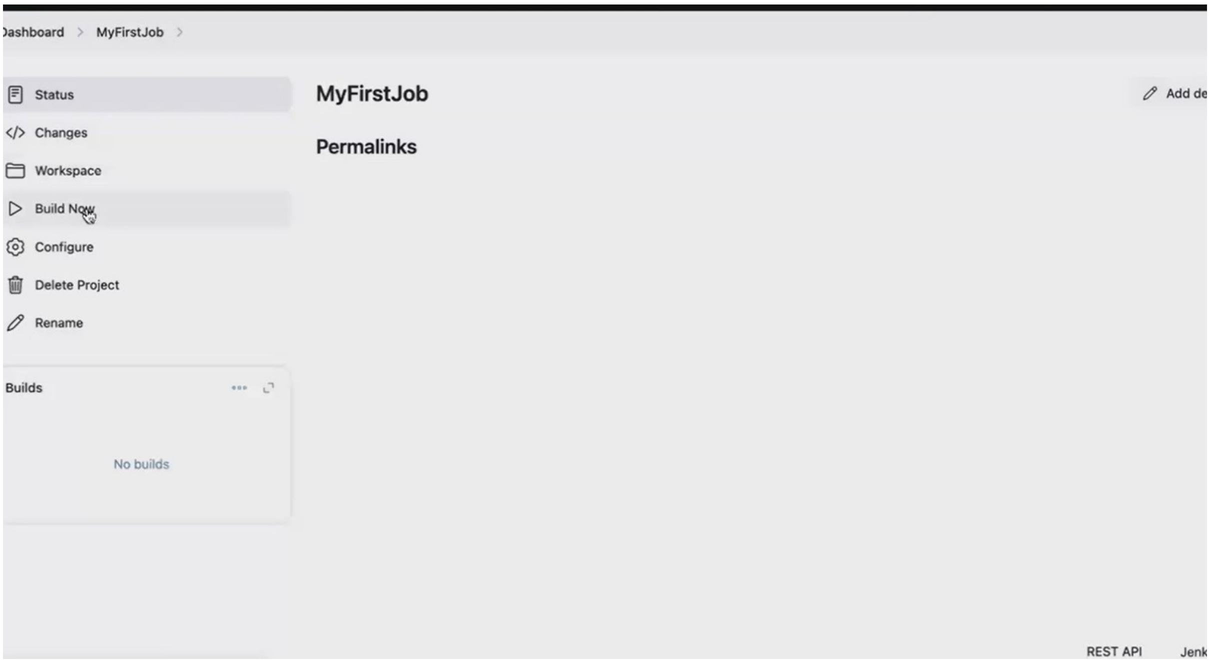
->click on that configure(to open job we need to click on it)

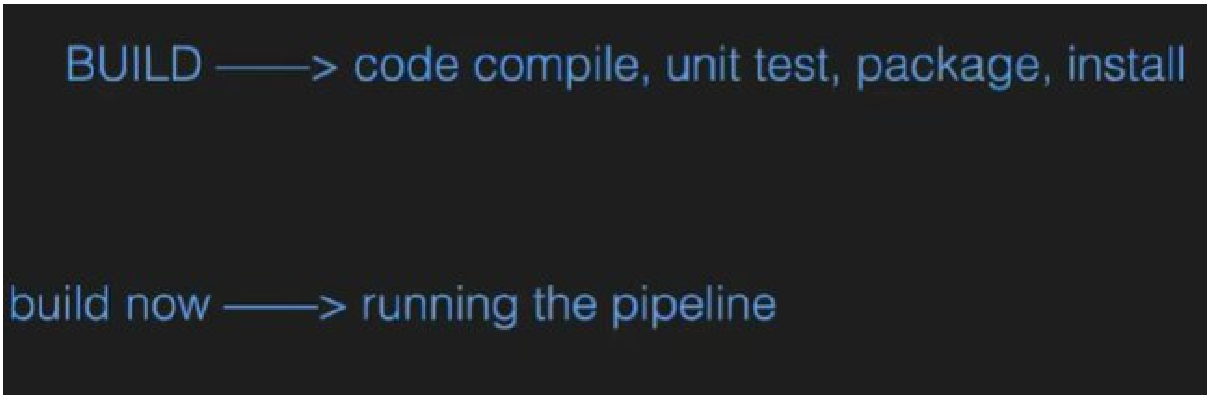


->now no err

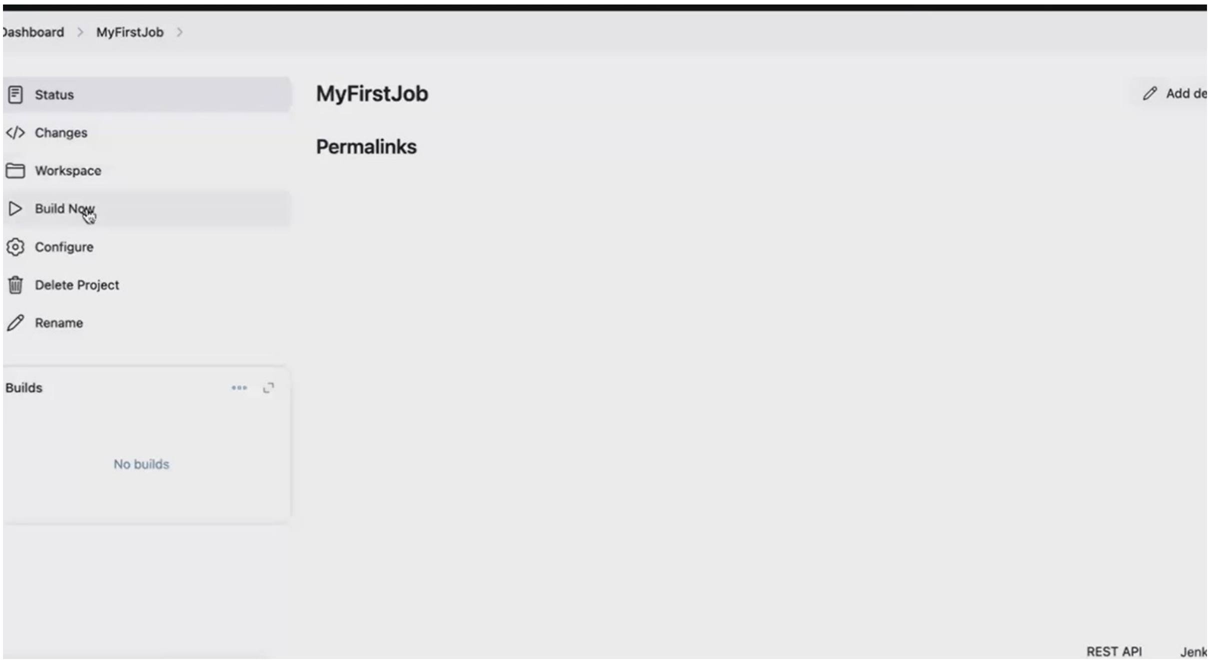


->click on build now(means we are executing that source code)

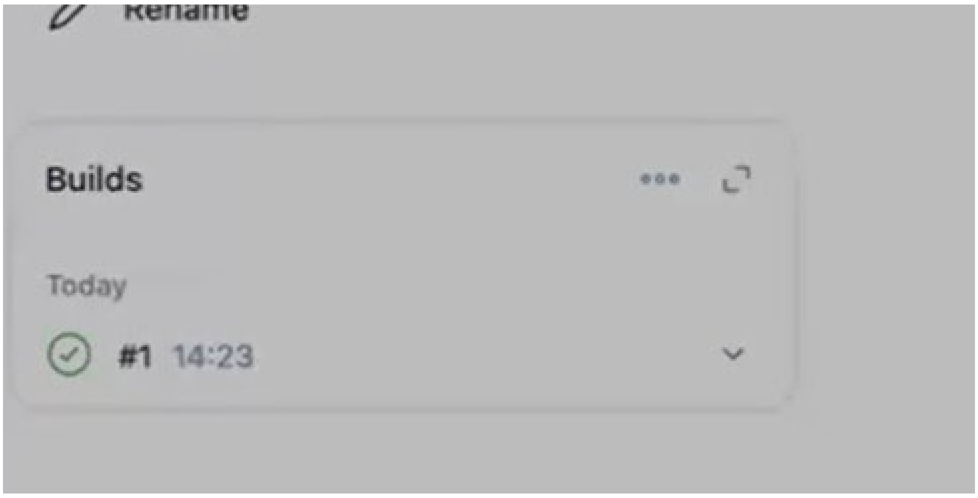




->once we click on build, it starts building

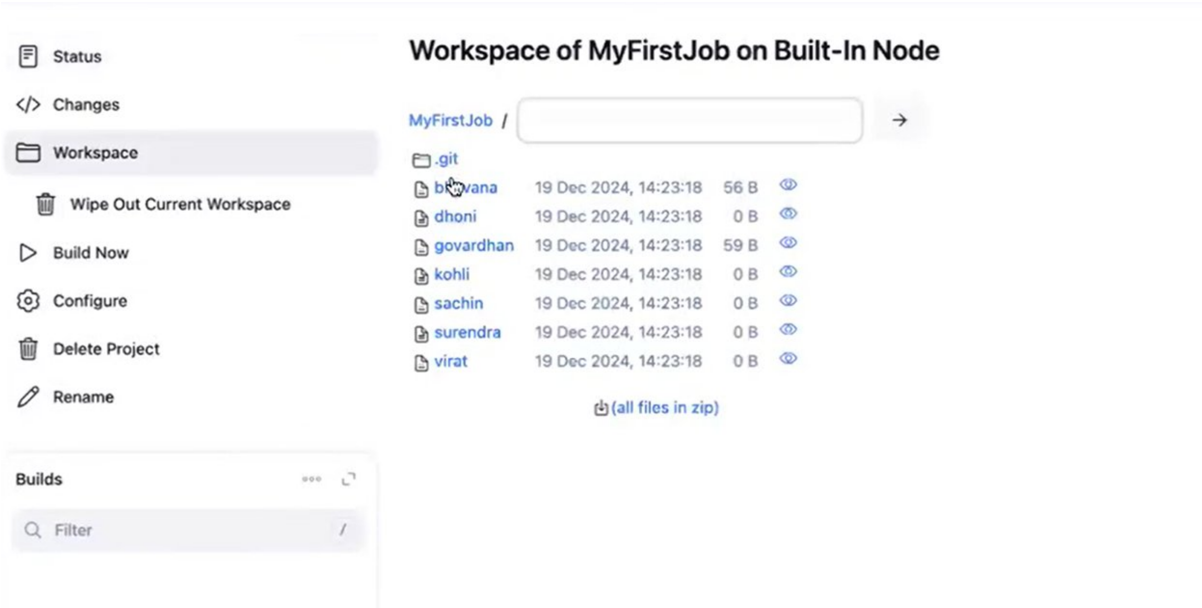


It got success

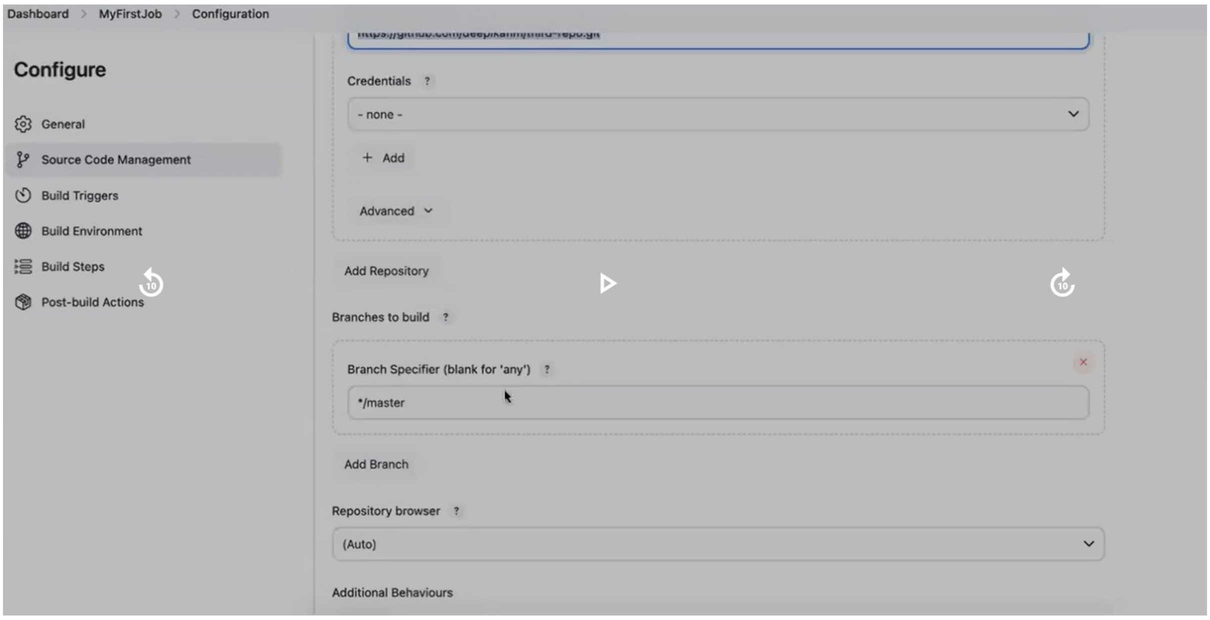


NOTE – IQ -Jenkins job fail aithey em chestavu, I'll chk the logs(a logs console o/p lo untai)

->to chk code is available in Jenkins or not



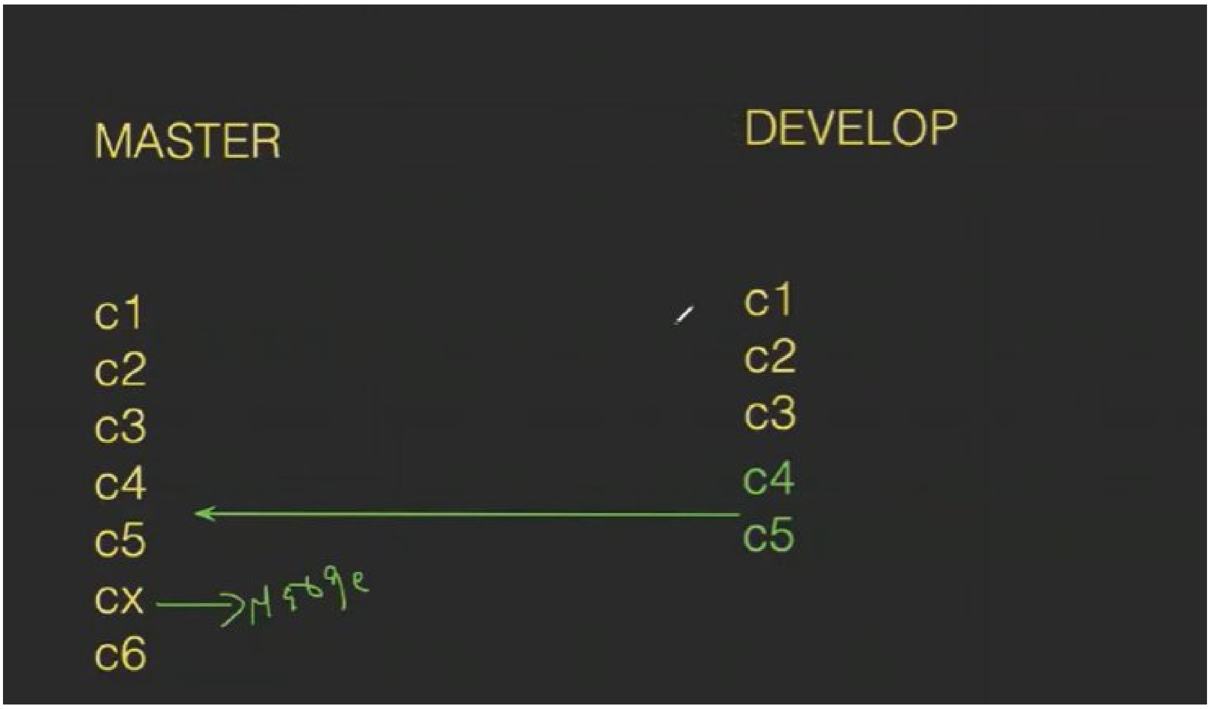
->branch lo master undi kabatti only master ey build ayyindi



NOTE –1. only oka branch ey buld cheyyacha, anni branches cheyyalema cheyyachu with multi-pipiline

2. github lo changes chesthey developer, a changes Jenkins loki vastadi ah????- job ni new ga build chesthey a changes vasthai, but alternate undi automatic ga CI server ki vastadi

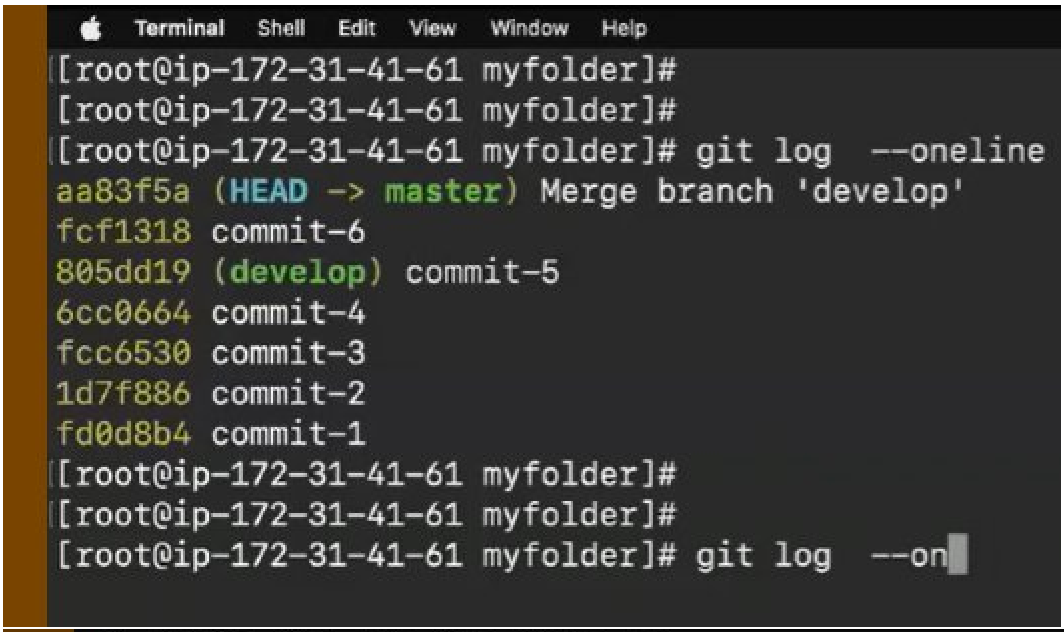
3 diff b/w merege n rebase? – merge aithey new commit vastadi, where rebase lo no commit

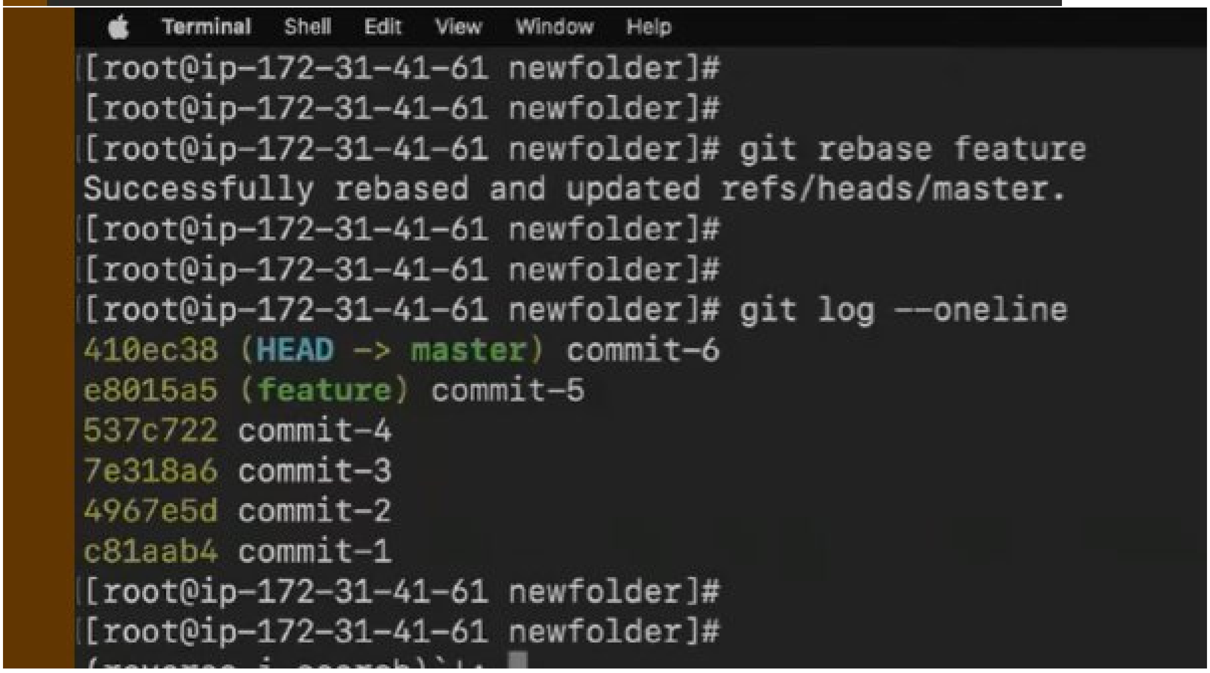


Master branch lo c1 c2 c3 c4 committs taravta c6 chesi,devlop lo c4 c5 merge chesthey cx aney new branch lo cheptadi dev nunchi merge chesaru ani

Ade master branch lo c1 c2 c3 ayyaka, dev lo c4 c5 ni merge chesthey new branch em radhu

C6 taravta dev nunchi rebase chesthey extra commit radhu(which doesn't show any merge data even if we disorder the commits)





AUTOMATIC CHNAGES FRM GITHUB TO CI SERVER

1.webhooks

->in jenkins goto configure> triggers>under build trigger enable github hook trigger for gitscm polling>save

->in github goto repo settings>under code n automation goto webhooks>add webhook >give Jenkins url(http://pub-ip:8080/github-webhook/)>in content give application/json>add webhook(sometimes we'll get dot, jst refresh it, we'll see tick then github is added vth Jenkins)

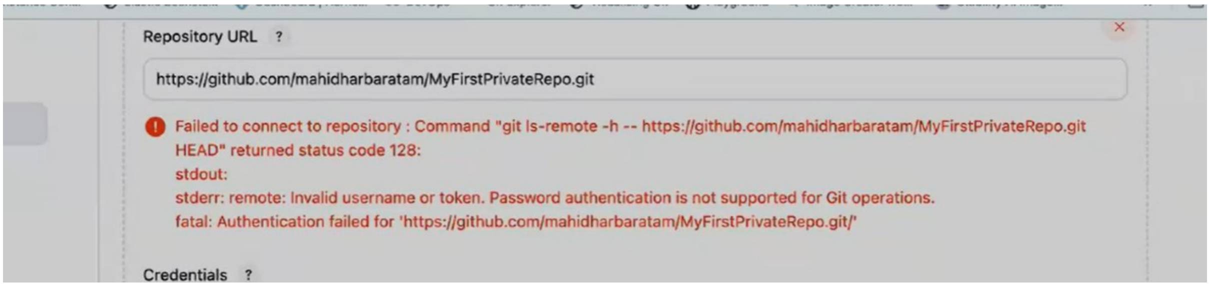
->ippudu manual job ni build cheyyakarledu, eppudu aithey developer github loki push chestado, automatic ga build avutadi

->now change some changes in any file in girhub n commit.

->chk in Jenkins n chk that build is getting created automatically.

TASK – do above vth private repository

->copy priv repo url n create a job vth that



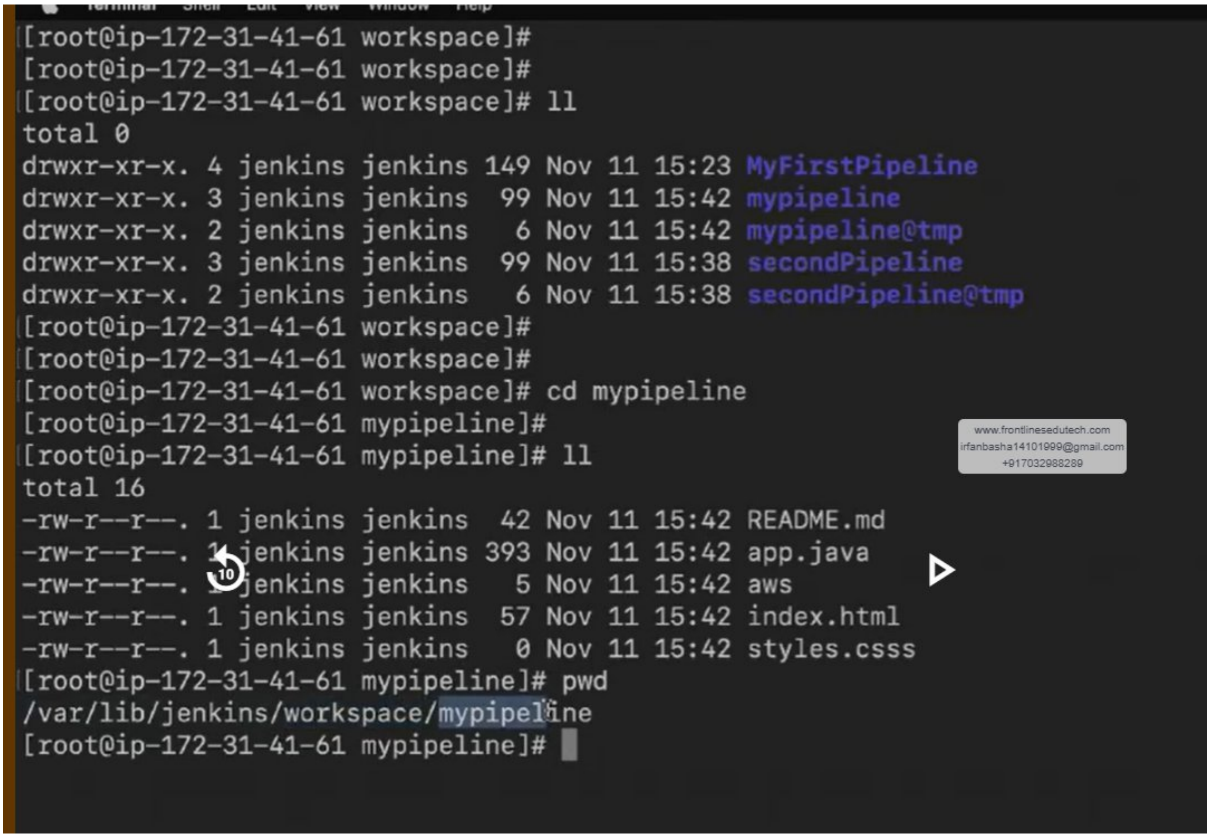
->credentials enter cheyyaledu kabatti err vastundi

->in credentials click on add>sel Jenkins>give github username in username field>in pword give token(profile settings>developer setting>tokens classic>create a token(if exists u can regenerate it))>under id give ur own id>add>now sel that credentials>which branch u want to build>save it

->now build the pipeline

->if we wanna edit that pipeline goto configure

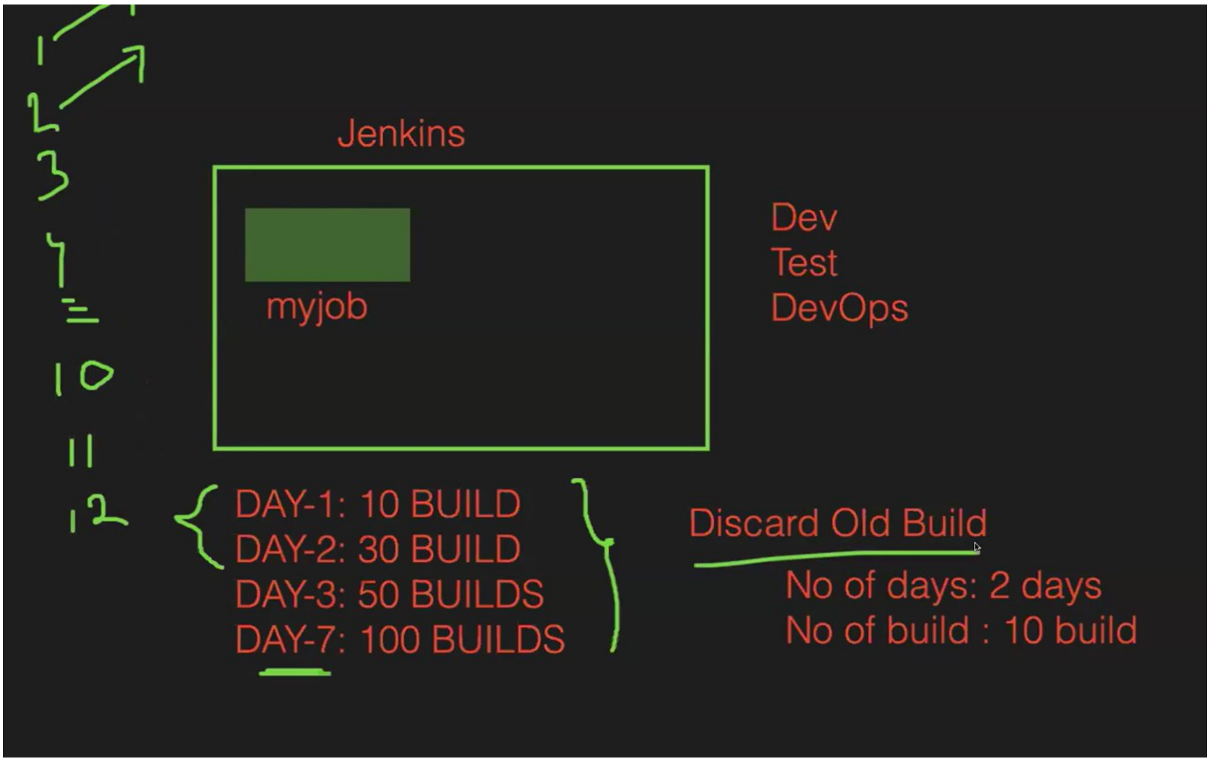
NOTE – IQ: in Jenkins where all pipelines data is stored(it will store in Jenkins def path - /var/lib/Jenkins- ee Jenkins folder lo workspace folder lo mana pipelines untai)



DAY – 2 JENKINS: POLL SCM, BUILD PERIODICALLY, DISCARD OLD BUILDS &THIN BACKUP

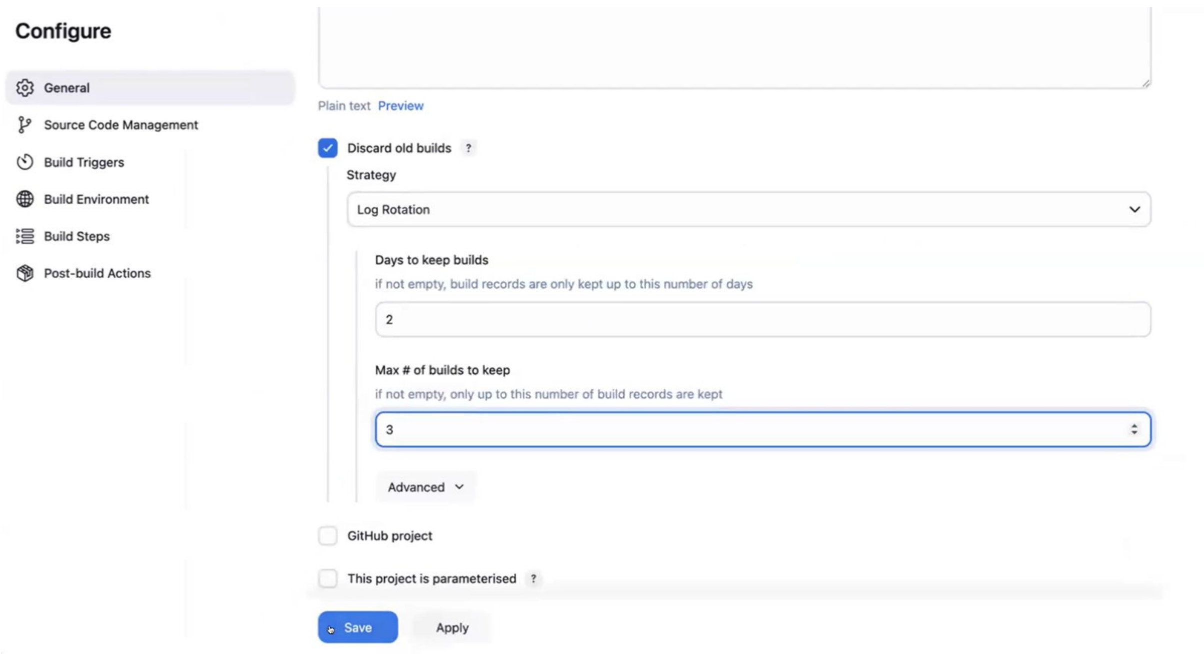
FREE STYLE JOB OPTIONS n HOW TO USE THEM

1.Discard old builds

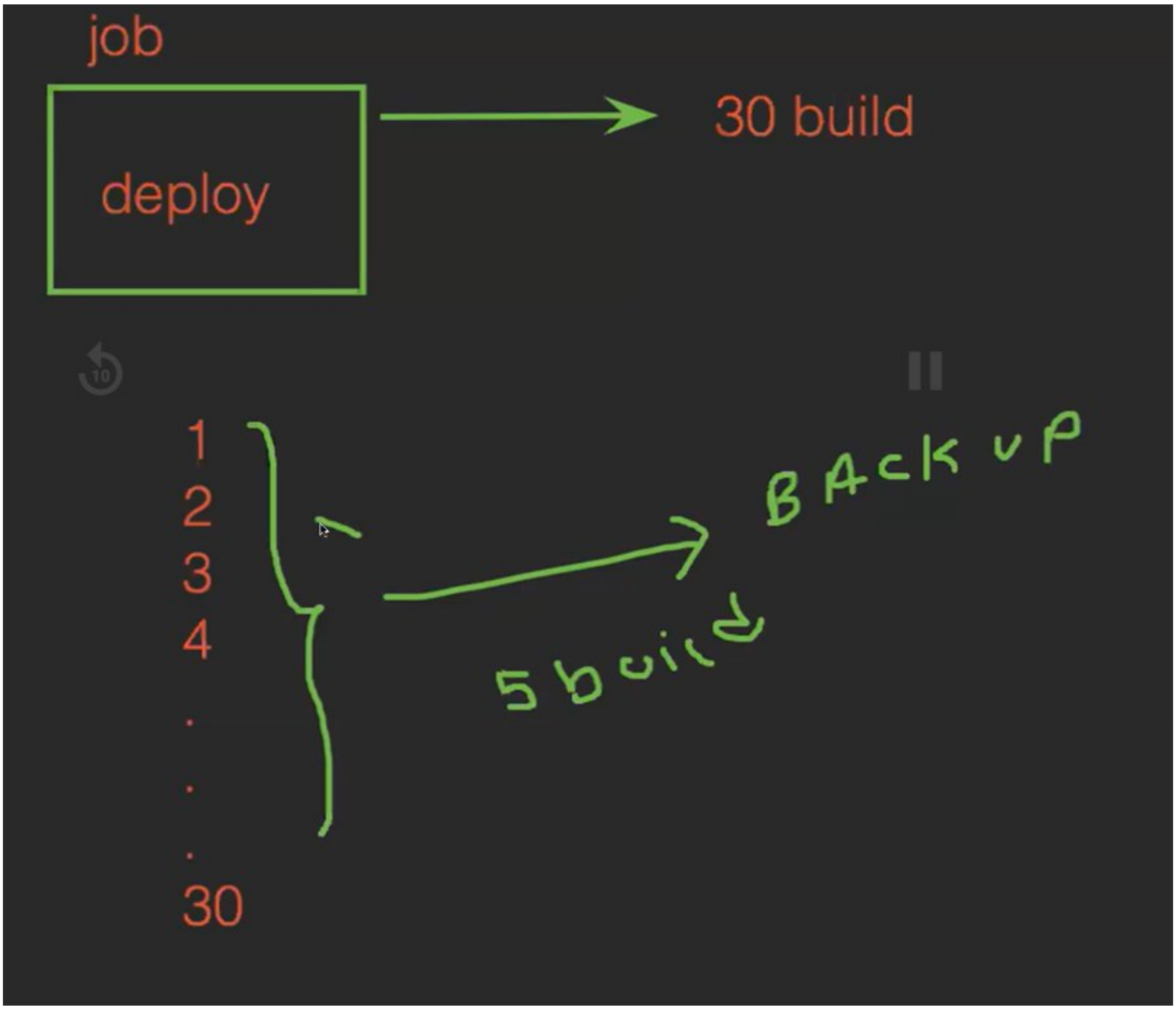


oka job undi a job lo daily build chesthu untam, so ala n no of bulds chestam- that doesn't make any sense. So we use this option discard old builds option.

->goto Jenkins>job>configure>enable discard old build>give days n no of builds>save it

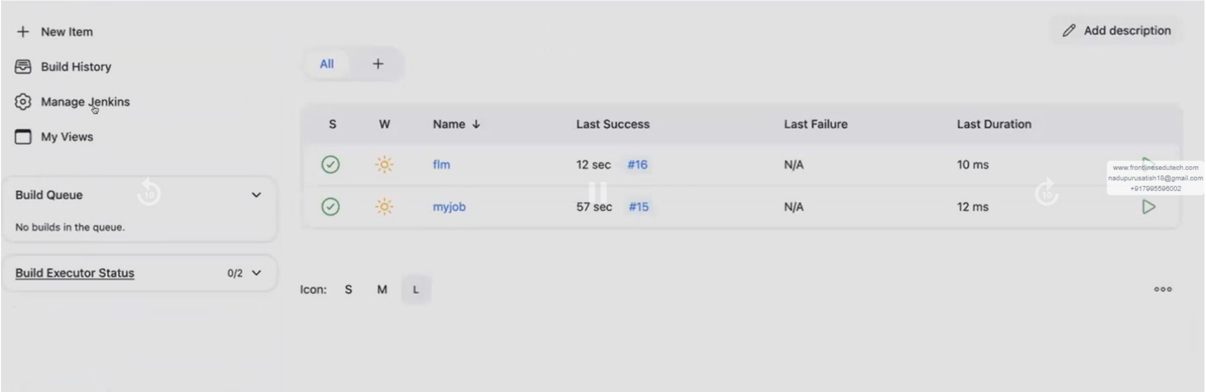


2.BACKUPS



Backup mana server loki ey teesukuntamu, edaina back-up kavali antey techukuntamu

->flm job lo oka 16builds create chesamu, ippudu backup teesundam(goto manage Jenkins>plugins>available plugins>search fr thinbackup>sel>install)



Once installed plugin>manage Jenkins>sys>search fr thinbackup>in backup_directory give ur path to store the back-up(/opt/mybackup-opt anedi default folder untadi, danlo mybachup folder create chesi full permissions ichanu, so that backup store avutadi)>sel move old backups to zip files>save

->goto manage Jenkins>thin backups>backup now(konni sarlu err vastundi so Jenkins ni restart cheyyandi)

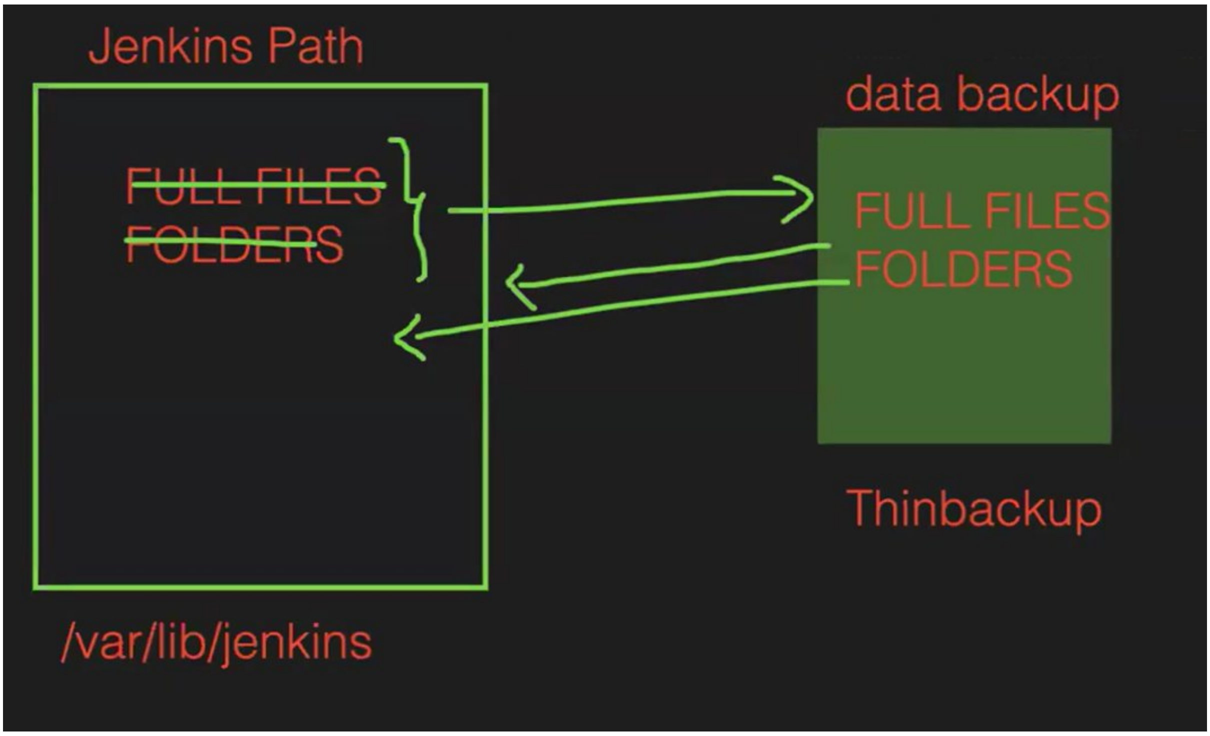
```
[root@ip-172-31-27-147 ~]#
[root@ip-172-31-27-147 ~]# systemctl restart jenkins
```

->now goto that path n chk backup available or not

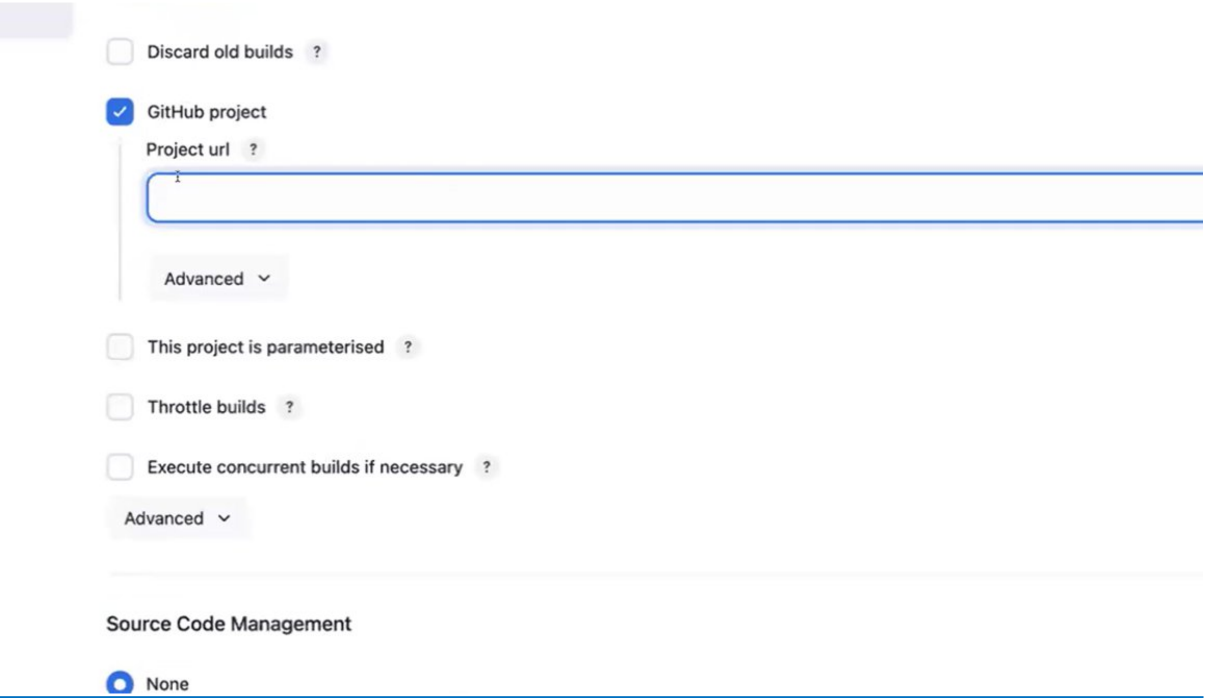
```
[root@ip-172-31-27-147 ~]# cd /opt/mybackup/
[root@ip-172-31-27-147 mybackup]#
[root@ip-172-31-27-147 mybackup]# ll
total 4
drwxr-xr-x 4 jenkins jenkins 4096 Dec 20 14:04 FULL-2024-12-20_14-04
[root@ip-172-31-27-147 mybackup]#
```

Jenkins Def Port: 8080
Jenkins Def Path: /var/lib/jenkins/

HOW WE BACKUP THE FILES IF DELETED



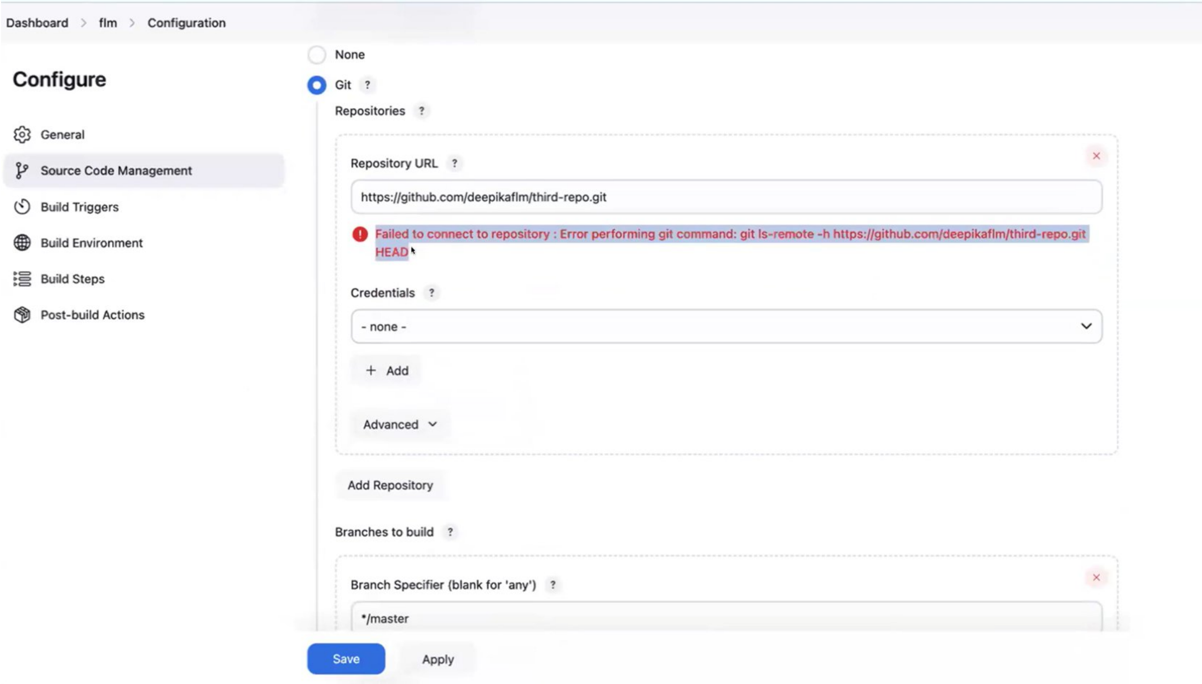
GITHUB PROJECT option



Ee option lo url ichina no use, jst oka description la untadi

3.SOURCE CODE MANAGEMENT

->git install cheyyakapothey, a err vastundi



->pub repo kabatti credentials akkarledu, if it is pri repo need to give credentials.

->branches to build give which branch u need to integrate vth Jenkins>save

4.BUILD TRIGGERS

(i)build periodically(a particular tym ki commits unna lekapoina build avutadi Jenkins lo)

Particular ga a tym ki job schedule avutadi

USE CRON SYNTAX

CRON: THIS IS USED TO SCHEDULE THE JOBS IN JENKINS.
SYNTAX: THIS SYNTAX CONTAINS 5 STARTS

STAR-1 : Minutes
STAR-2: Hours
STAR-3: Days in a months
STAR-4: Months in a year
STAR-5: Day in week

ex: 10:32 AM Tomorrow ----> build

0 = sunday
1 = monday
2 = tuesday
3 = wed
4 = thursday
5 = friday
6 = saturday

ex: 10:32 AM Tomorrow ----> build

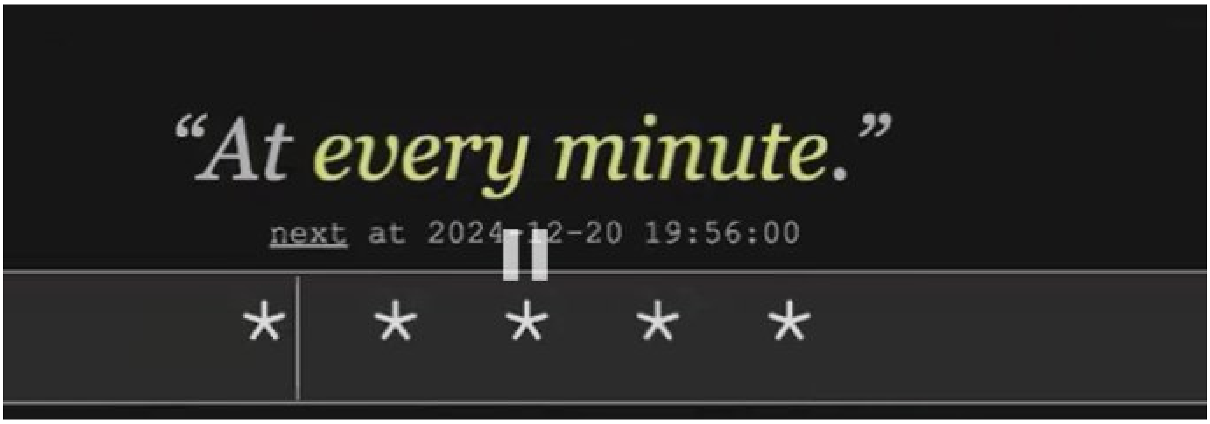
* * * * *
32 10 21 12 6

To chk whetherit is crt/wrong – crontab.guru

ex: 21 apr ---> 5:40 PM ----> 2025

40 17 21 4 1

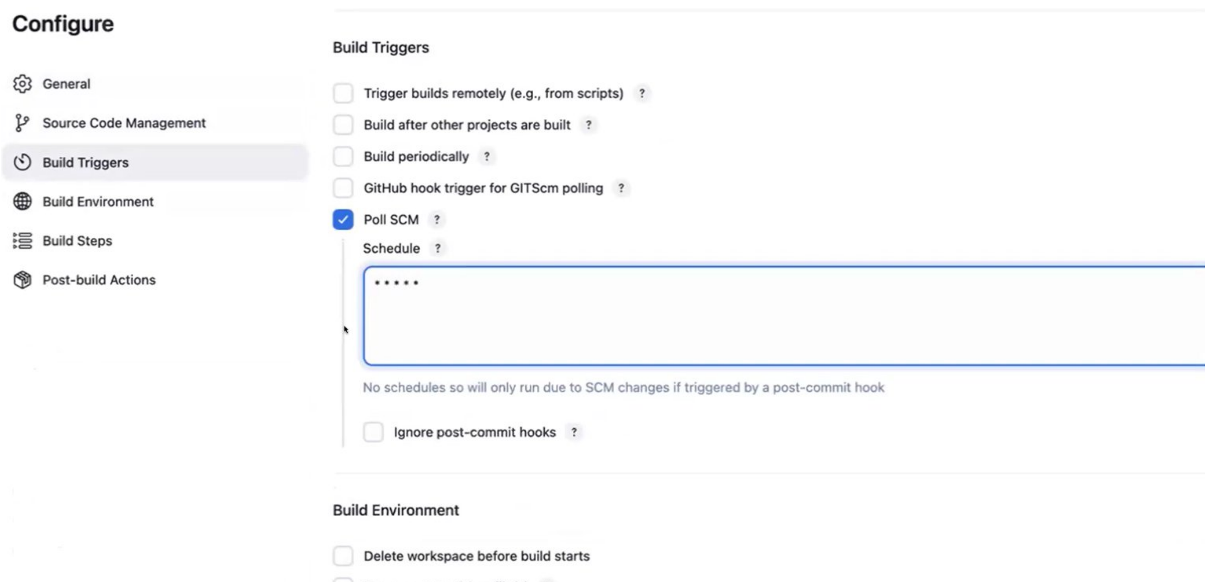
->every minute

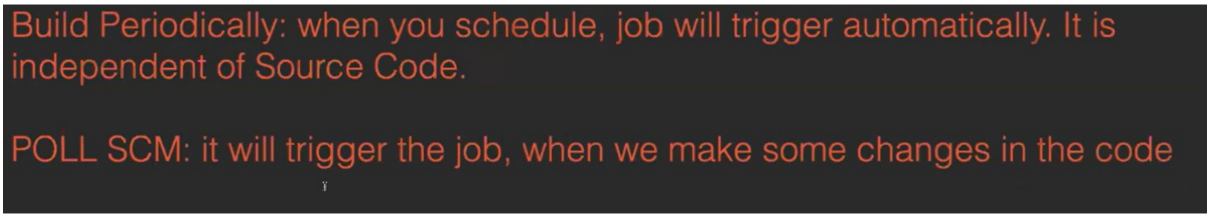


->every 2 mins

->every 3 hrs

(2) poll scm is also used to trigger the jobs automatically(but commits untey ney build avutadi job)





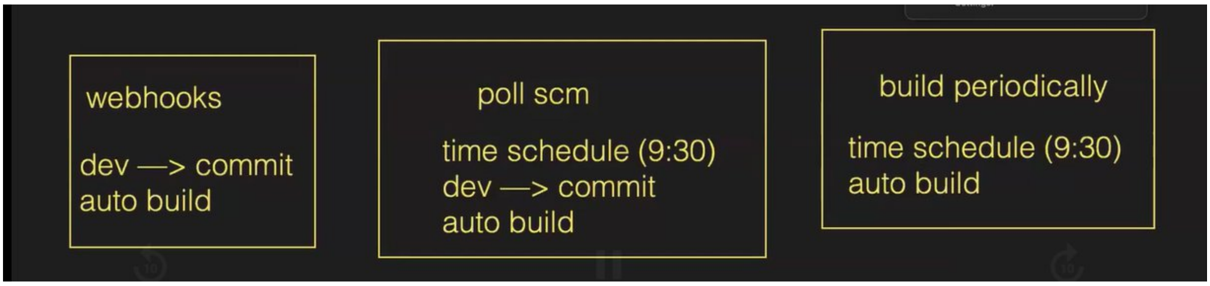
Poll scm lo oka particular tym ki build trigger avvali antey schedule chestamu, a particular tym lopu edaina oka commit aina chesthey ney ey build trigger avutadi otherwise no.

IQ – diff b/w poll scm n webhooks?

IQ– 1. poll scm lo commit aithey ney build avutadi

2. while building we need to use either webhook/poll scm

3. diff among webhooks/periodically/poll scm?



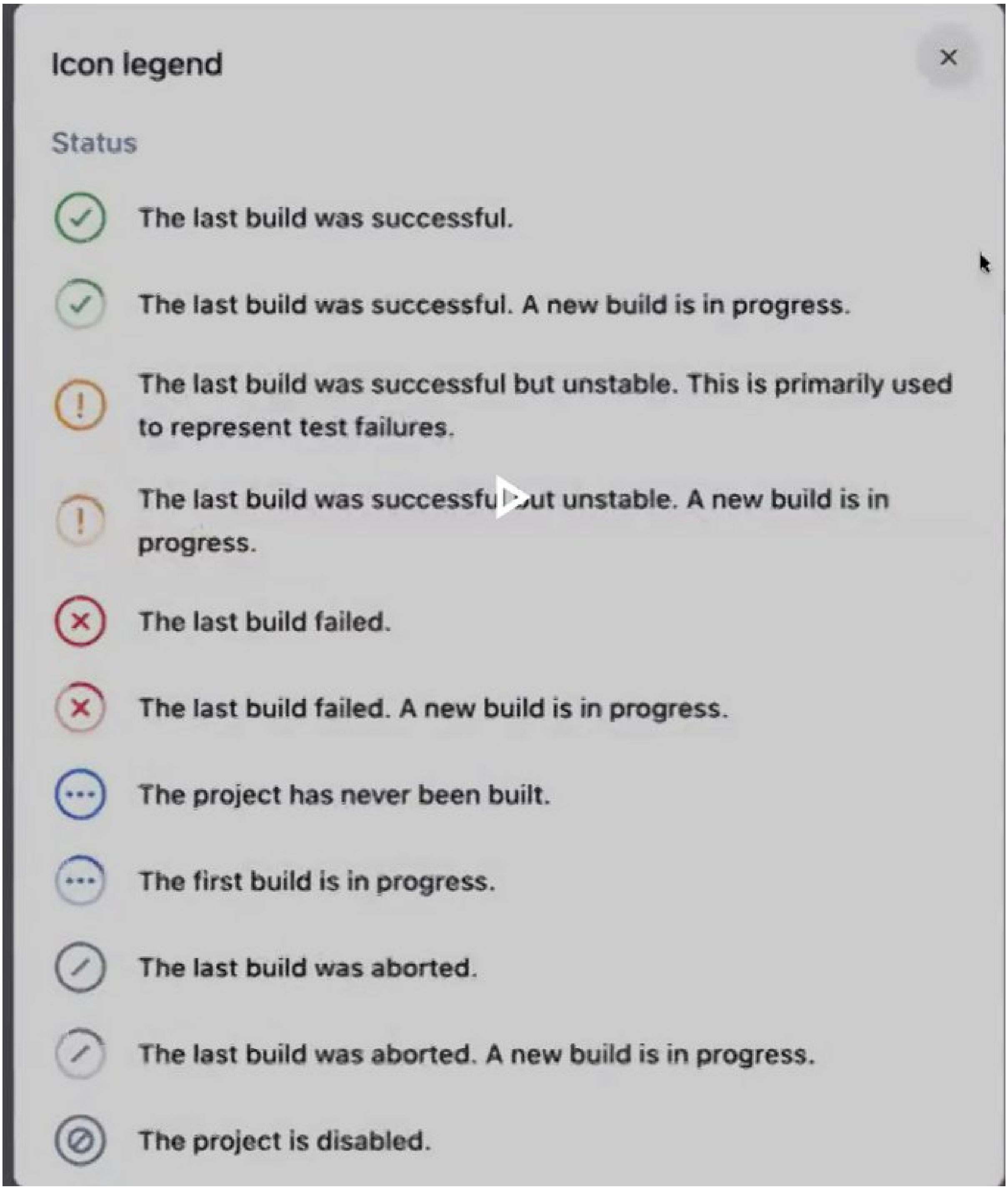
Q/A

DAY – 3 JENKINS FREE STYLE JOB FEATURES

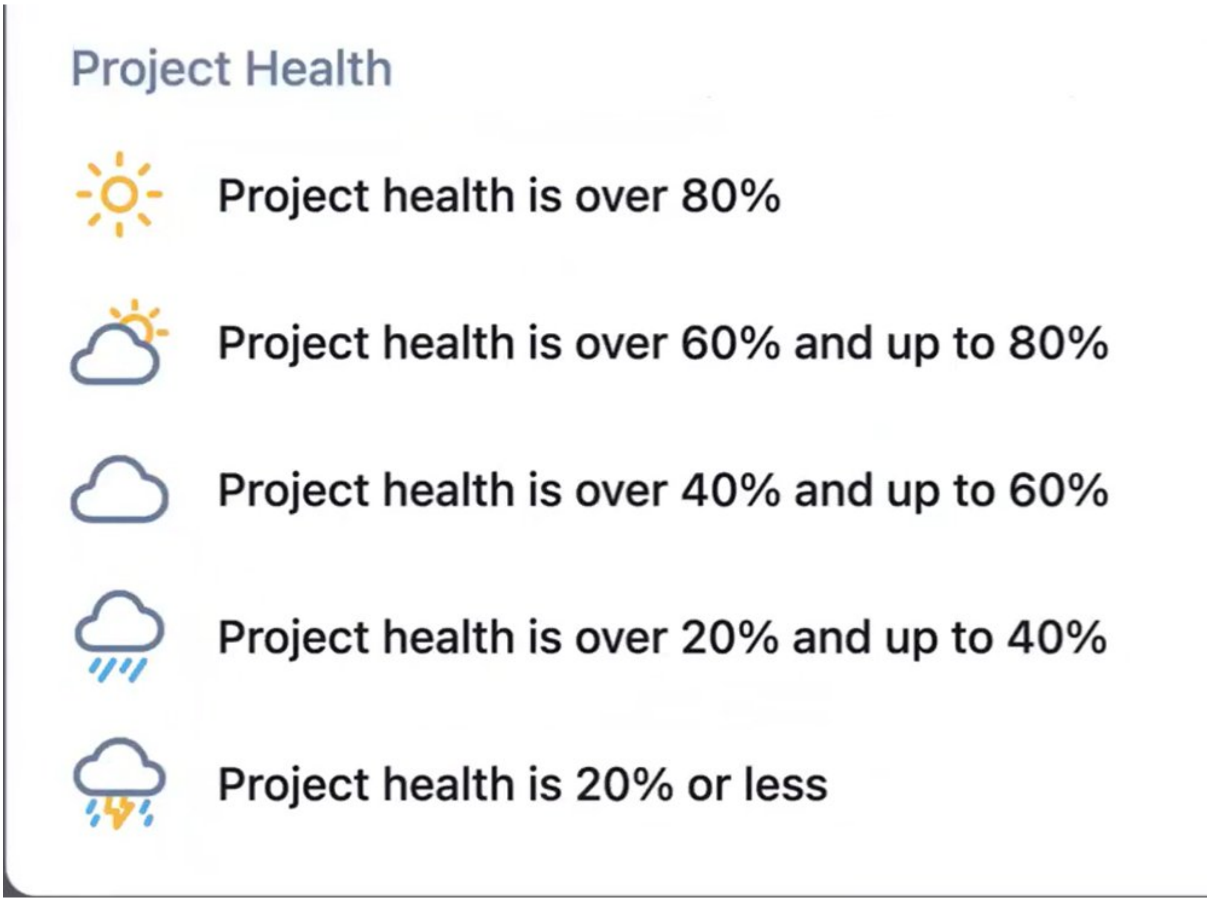
->we can rename the job

->we can del the job(once it is deleted we can't retrieve back)

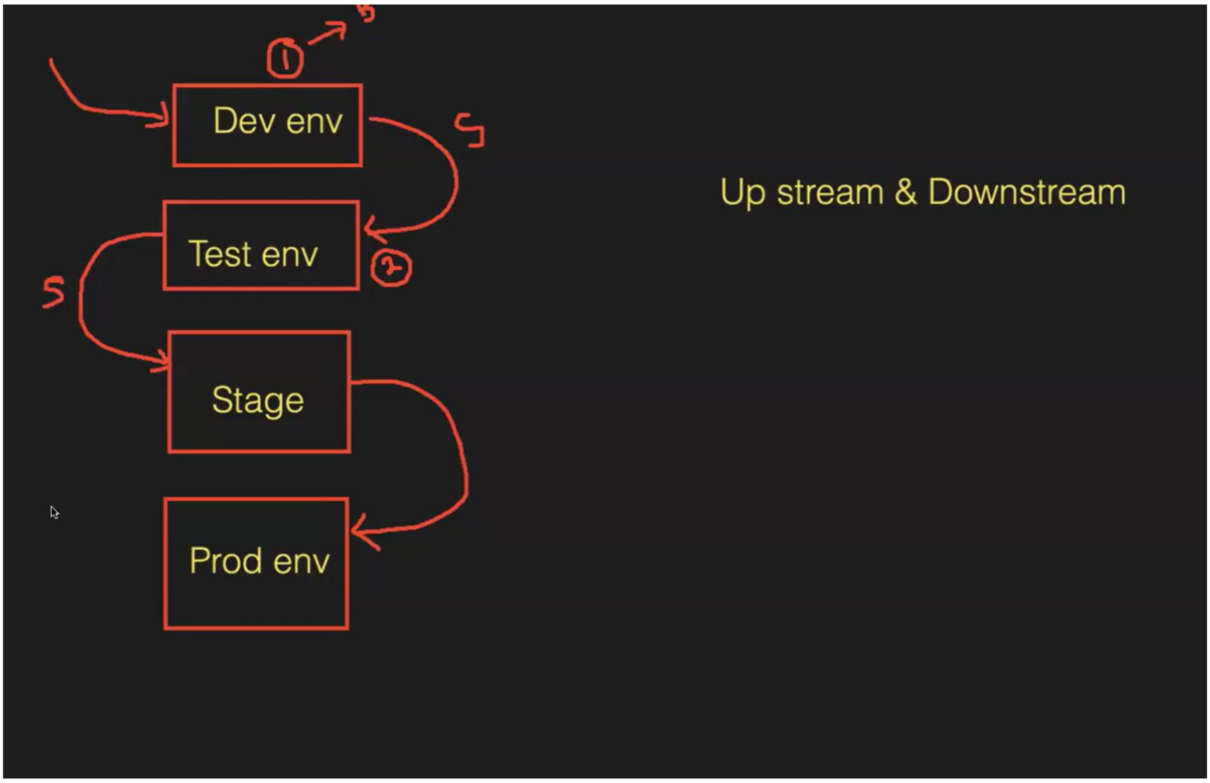
STATUS


Icon legend
Status
The last build was successful.
The last build was successful. A new build is in progress.
The last build was successful but unstable. This is primarily used to represent test failures.
The last build was successful but unstable. A new build is in progress.
The last build failed.
The last build failed. A new build is in progress.
The project has never been built.
The first build is in progress.
The last build was aborted.
The last build was aborted. A new build is in progress.
The project is disabled.

WEATHER

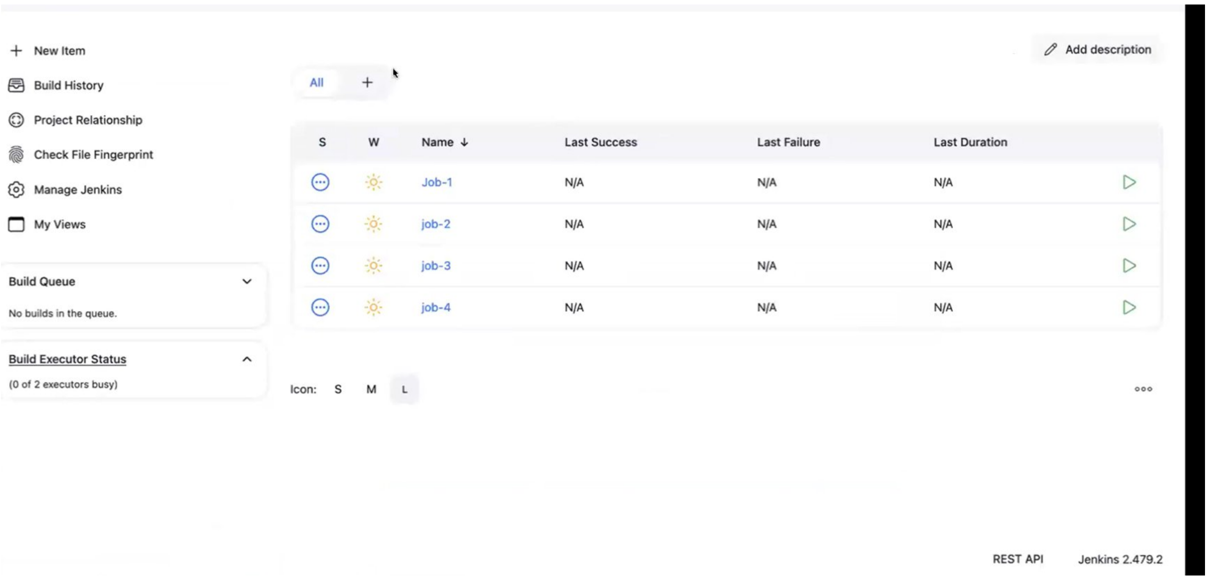


UPSTREAM n DOWNSTREAM

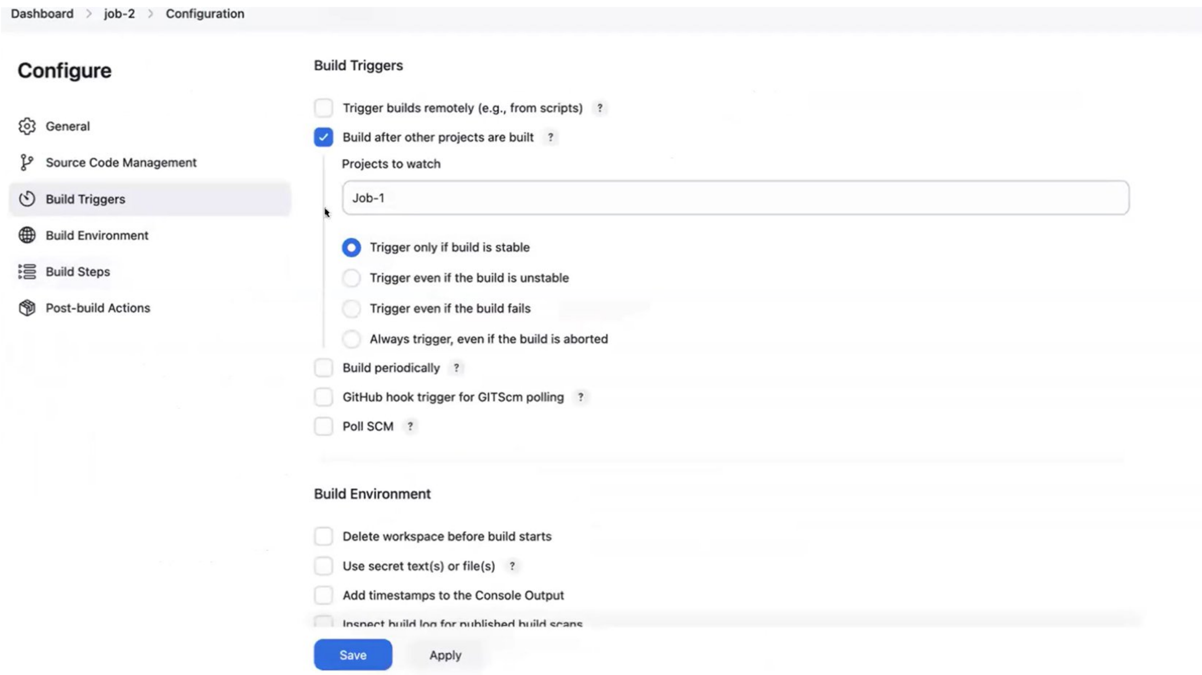


Suppose 4-jobs unnai like dev,test,stage,prod env. 1st dev env lo manual ga build chestanu,adi success aithey automatic ga test env lo build avvali same in other env's as well – up stream & down stram antaru.

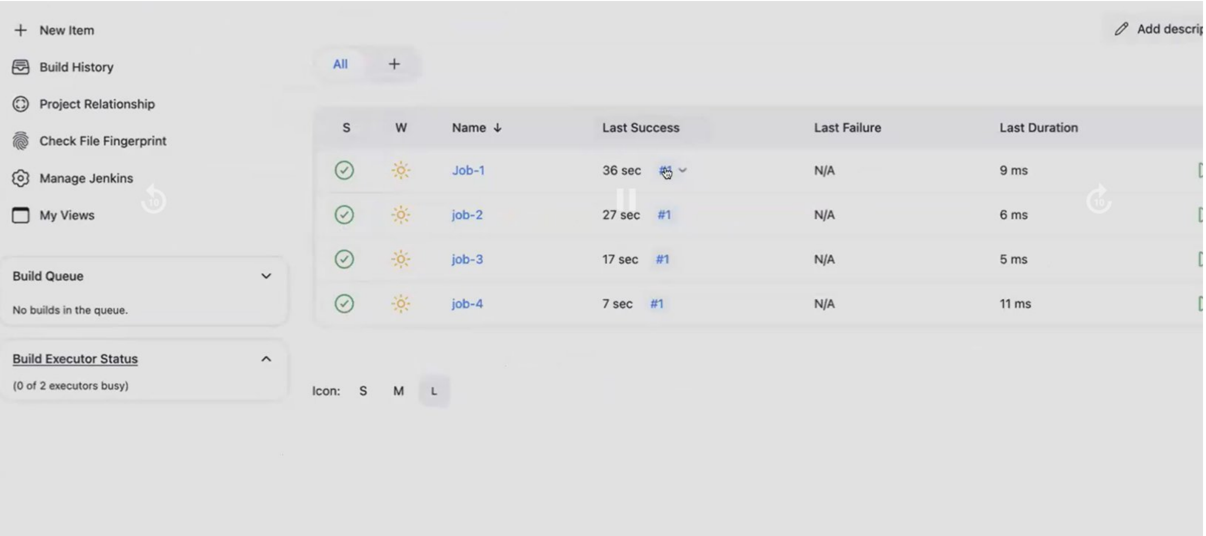
->created 4 empty jobs



->goto job2, give job1 in build after other projects are built same like others jobs also



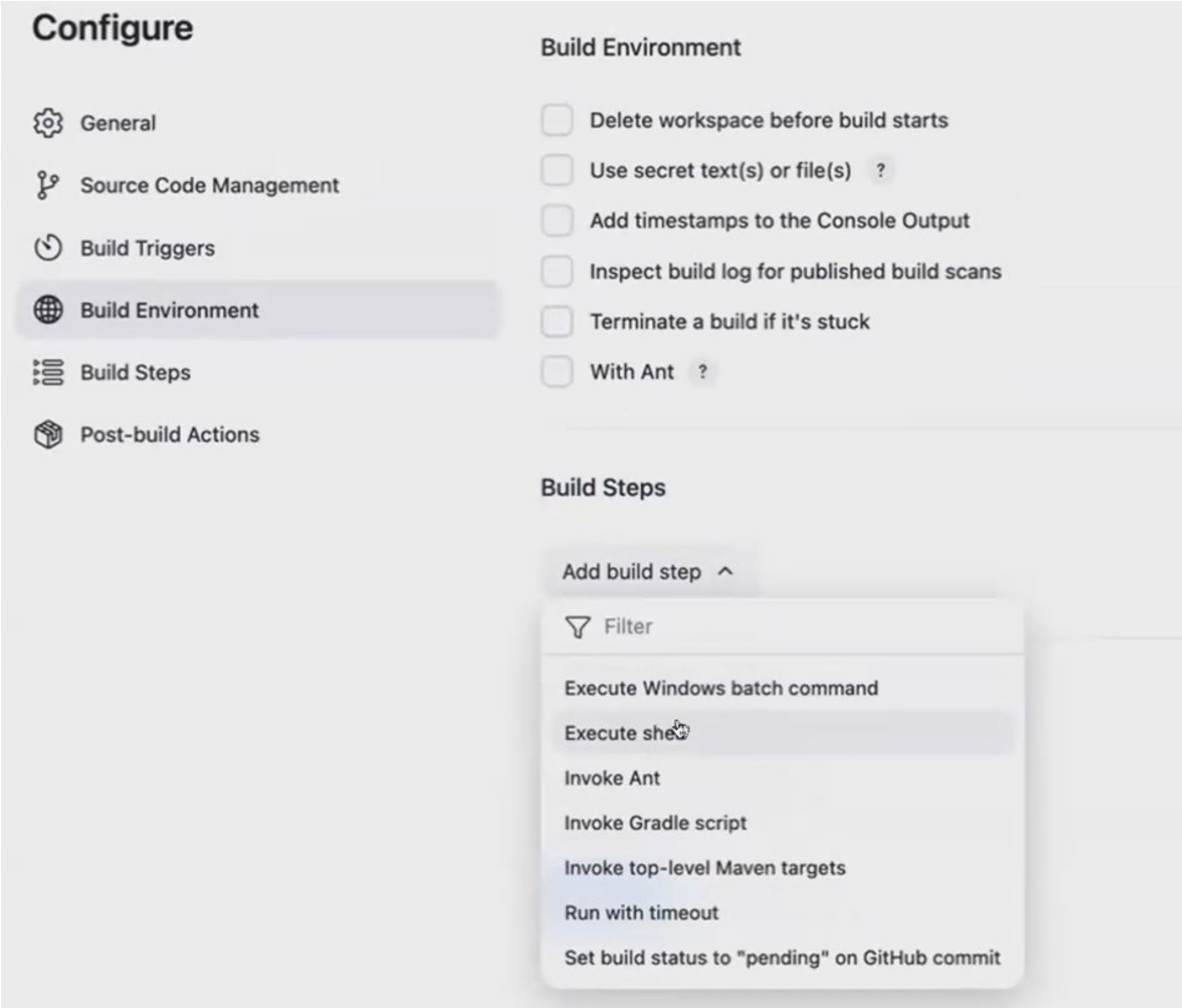
->job1 okkati manual ga build chesamu, others jobs anni automatic ga ave create ayyaye(left side lo queue lo kanipistai nxt job enti ani,once anni jobs build ayyaka no builds in queue ani vastadi)

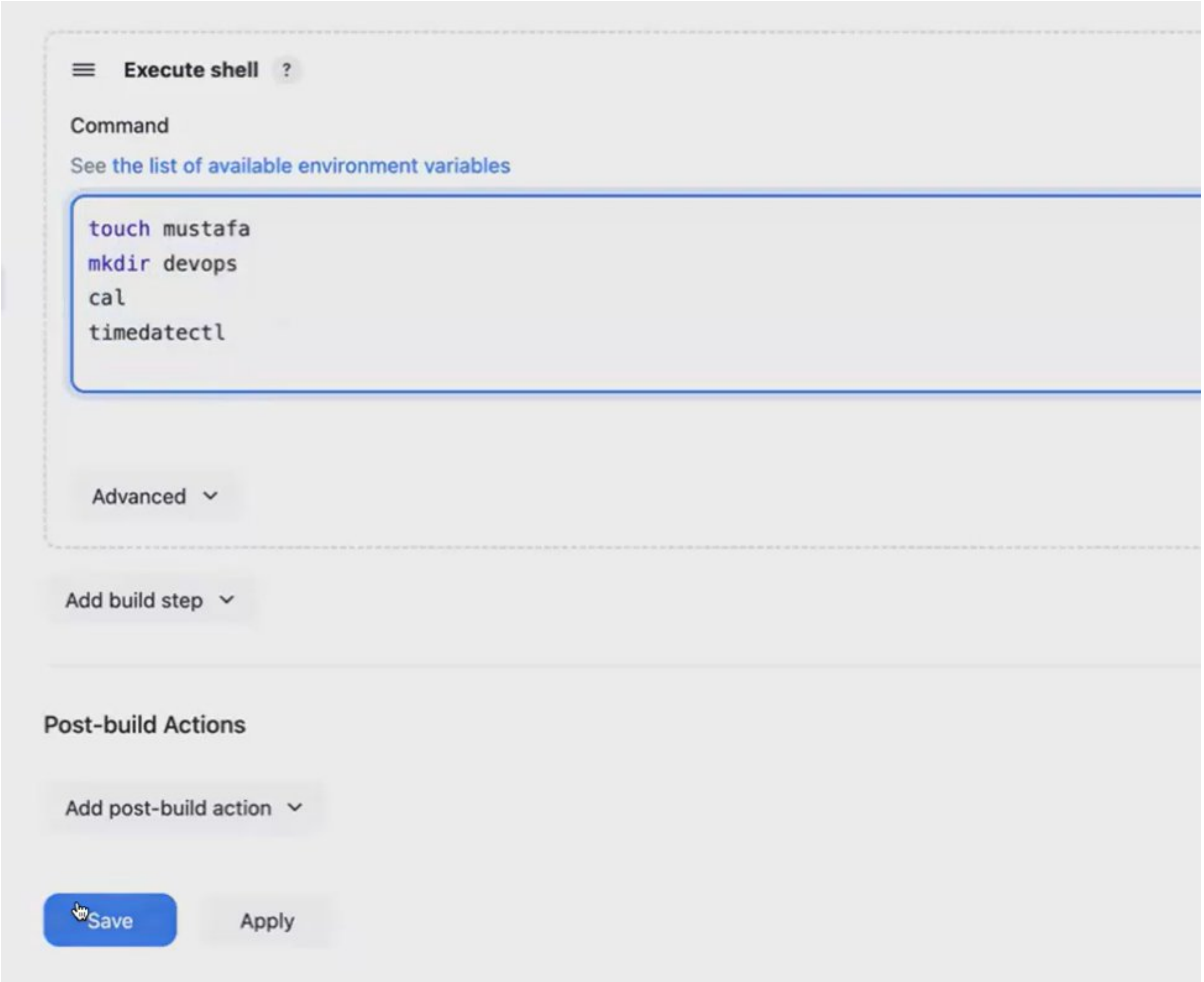


Job3 ki upstream job2 n downstream job4

BUILD STEPS

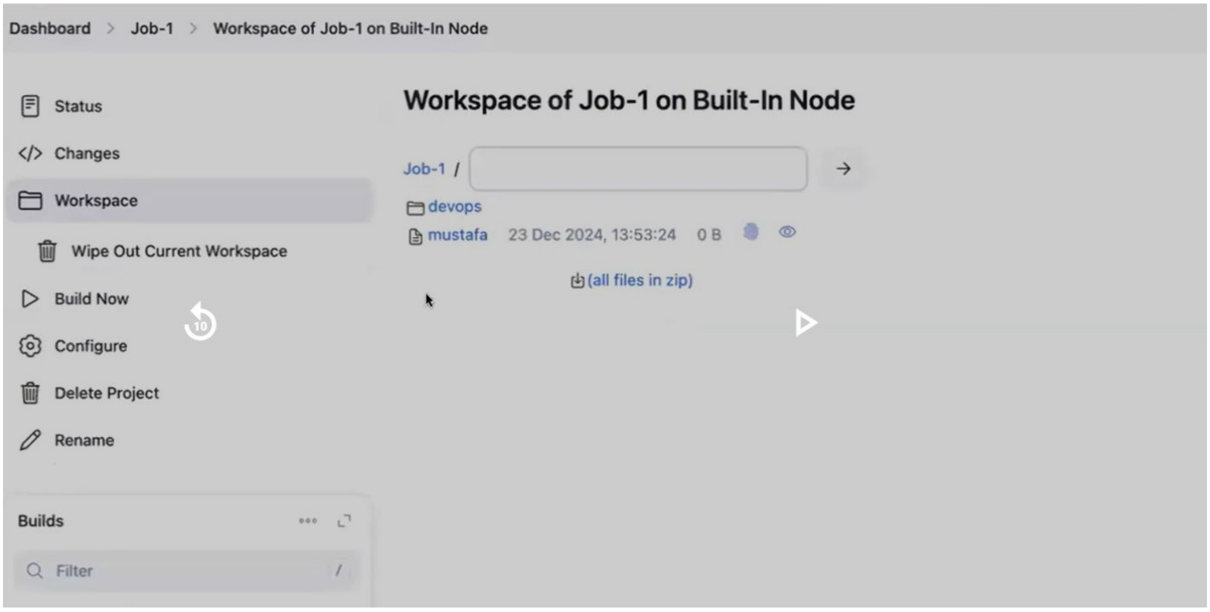
->jenkins job dhwara edaina script run cheyyali antey use avutadi


Configure
General
Source Code Management
Build Triggers
Build Environment
Build Steps
Post-build Actions
Build Environment
Delete workspace before build starts
Use secret text(s) or file(s)
Add timestamps to the Console Output
Inspect build log for published build scans
Terminate a build if it's stuck
With Ant
Build Steps
Add build step
Filter
Execute Windows batch command
Invoke Ant
Invoke Gradle script
Invoke top-level Maven targets
Run with timeout
Set build status to "pending" on GitHub commit

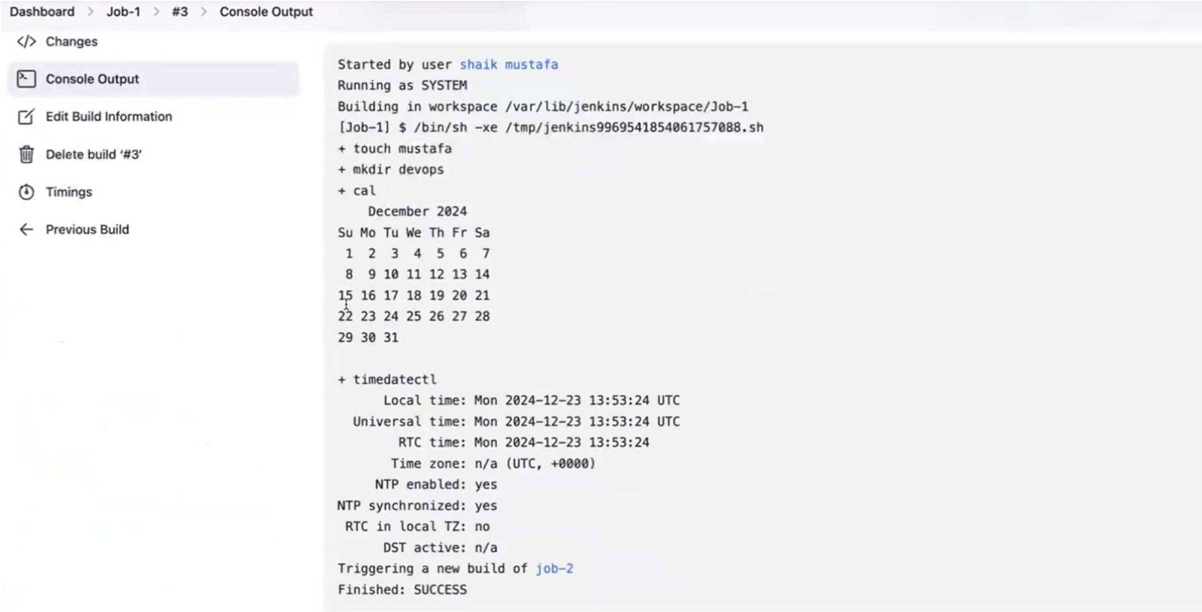


->now click on build now

->we can chk the files created or not in Jenkins-workspace/jenkins_server_defaultpath(/var/lib/jenkins)

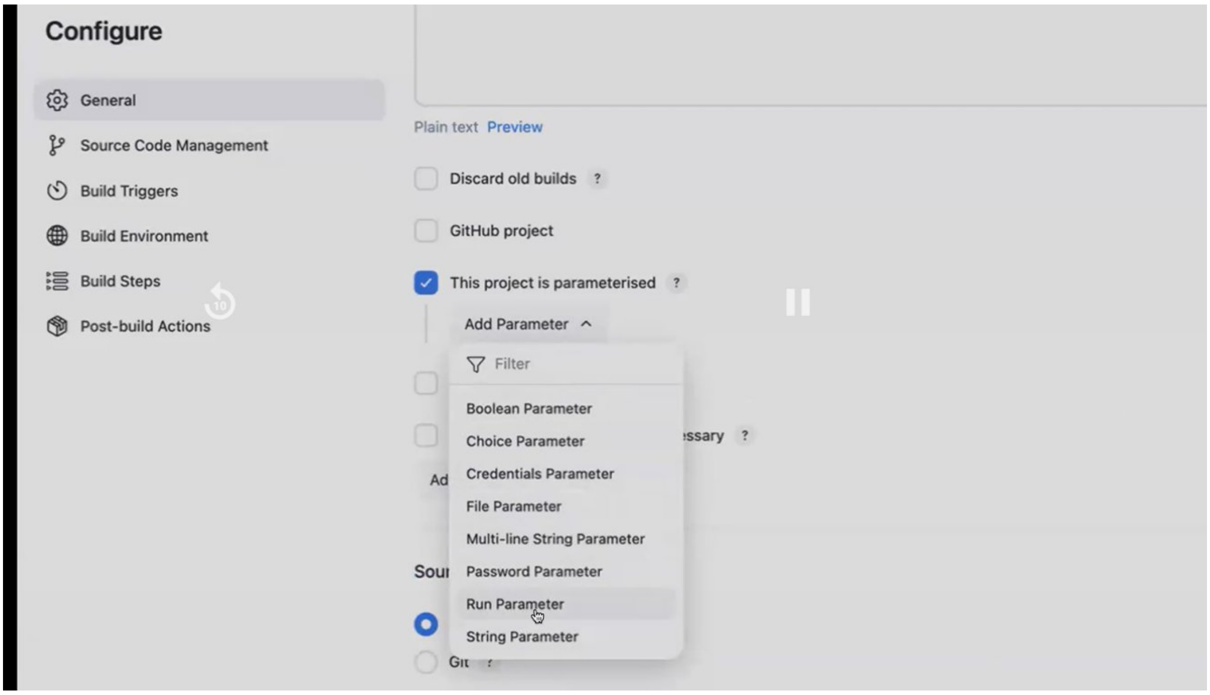


Avi files kabatti chusamu,cmds vi console o/p lo chustamu



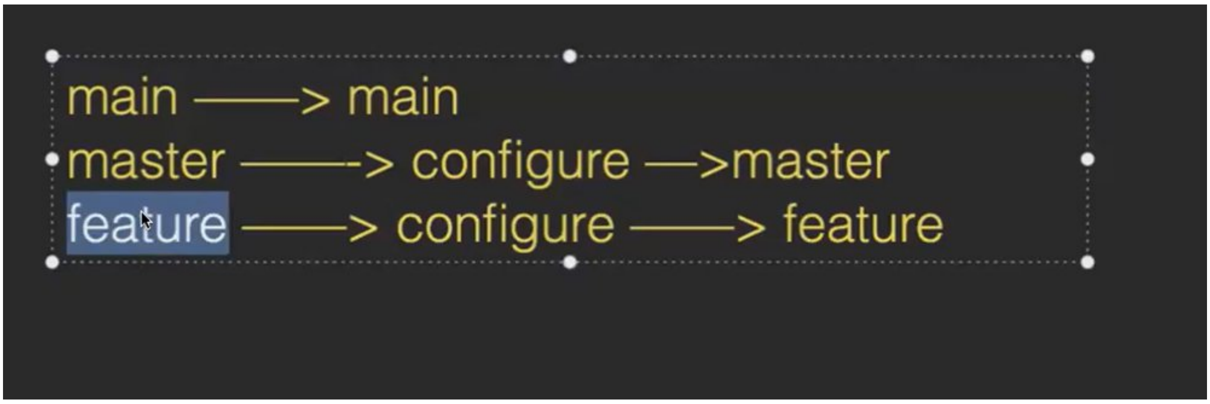
PARAMETER

->diff param's

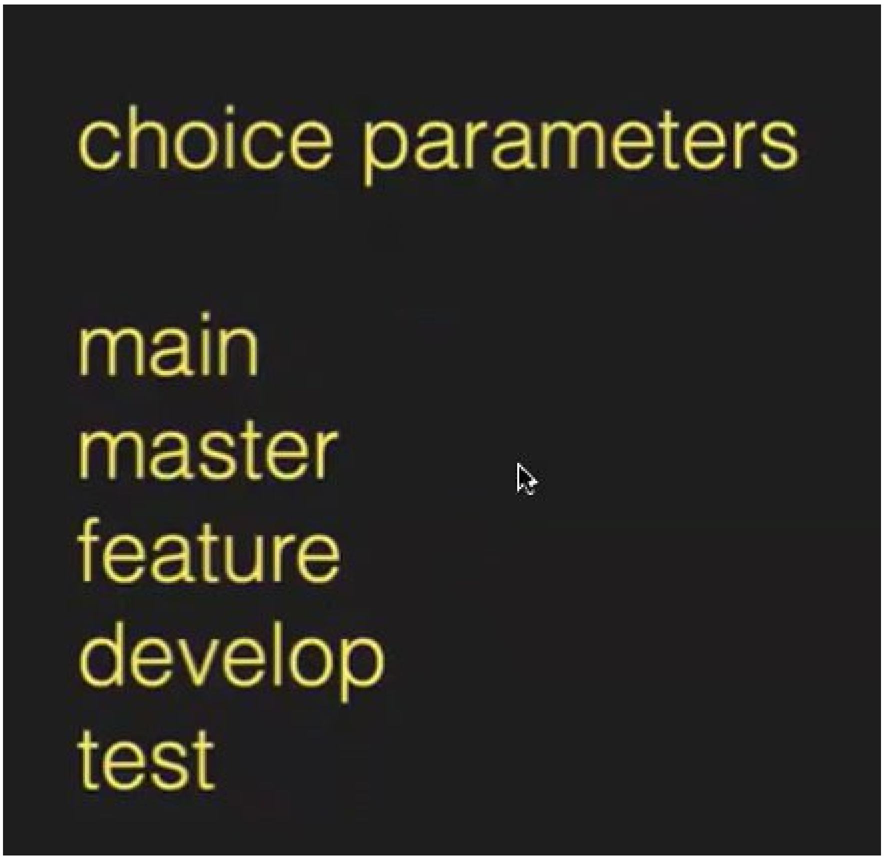


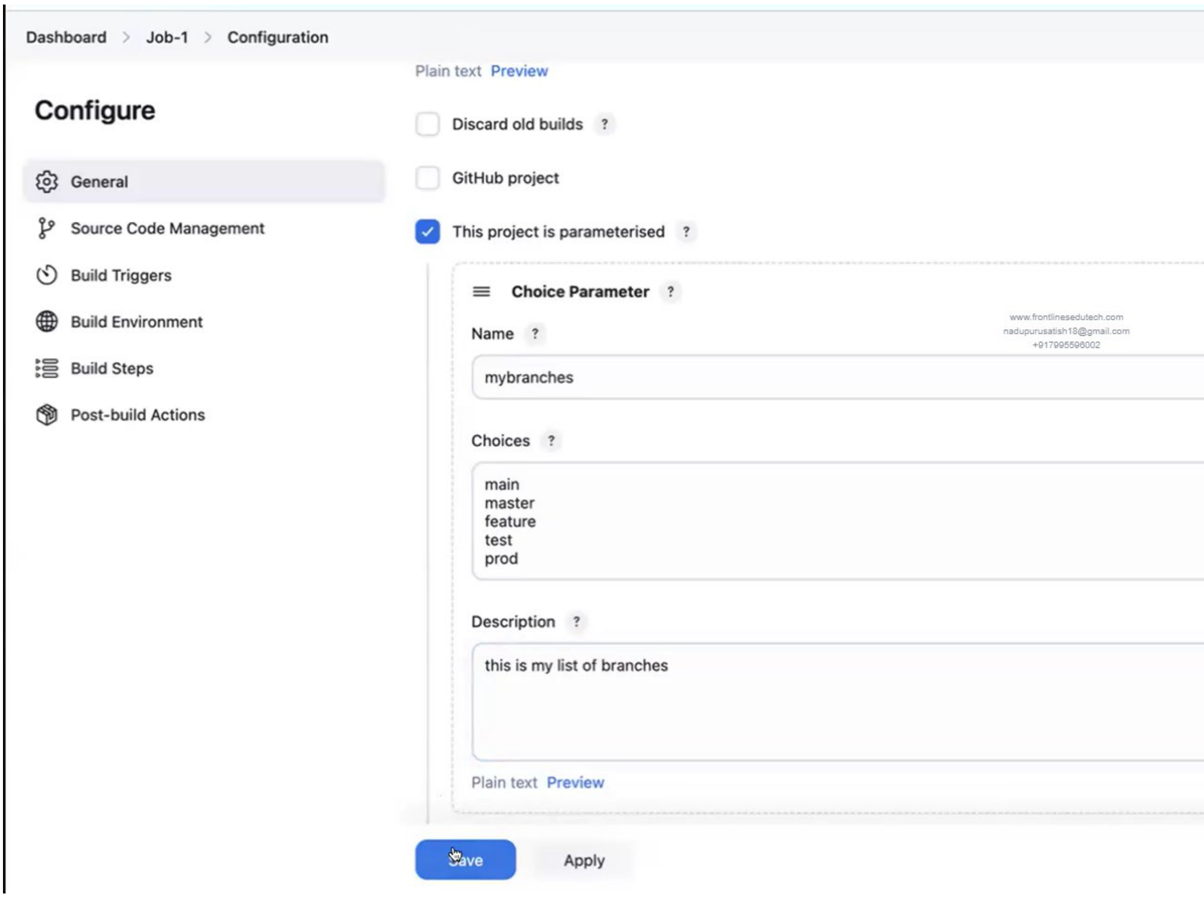
->ekkuva choice param use chestamu

Na manager 1st master branch ki build cheyyamanaru(master ani name istanu), nxt master ki cheyyamanaru(master ni configure chesi master ani name istanu), nxt feature ki cheyyamanaru(feature ni configure chesi feature ani name istanu).But ila cheyyali antey kashtam, so choice param ni use chestanu ee case lo.



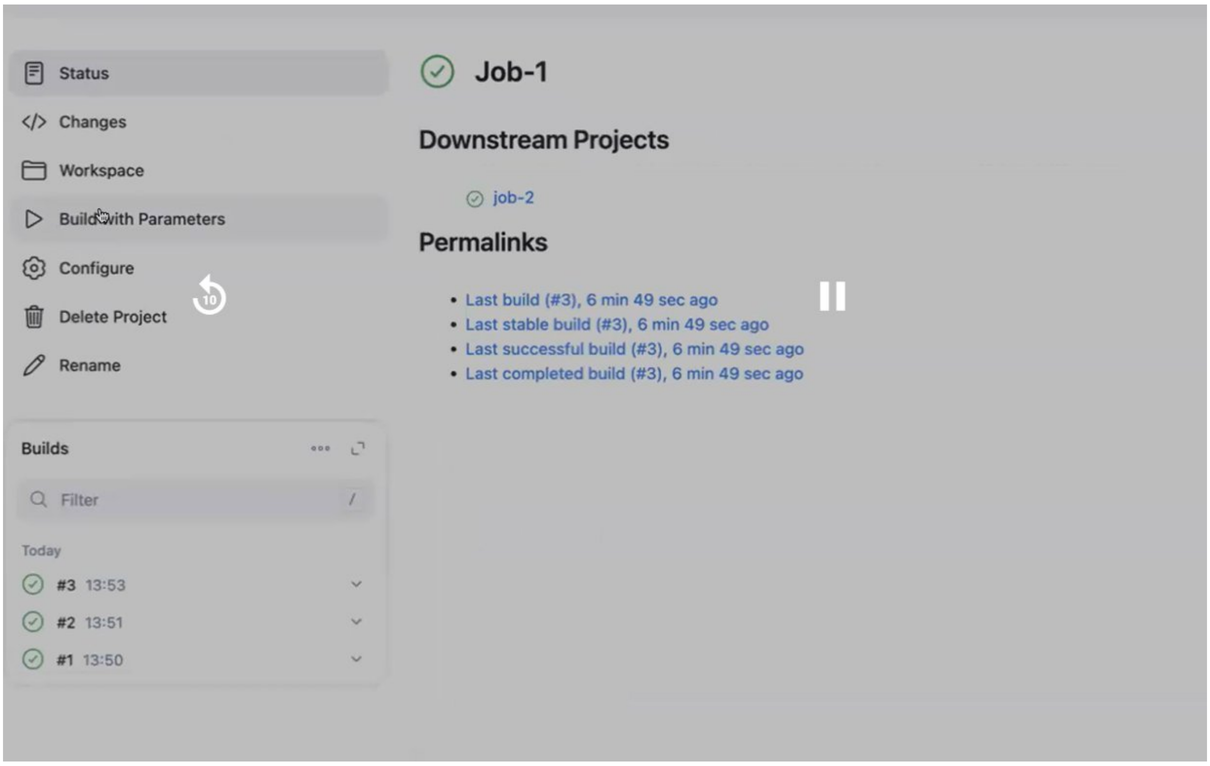
So, prathi sari rename cheyyadam kashtam.Nen ee choice param lo petti edi kavali antey adi pedatanu





Description is optional

->param's add chesamu kabatti, build with param's option vachindi

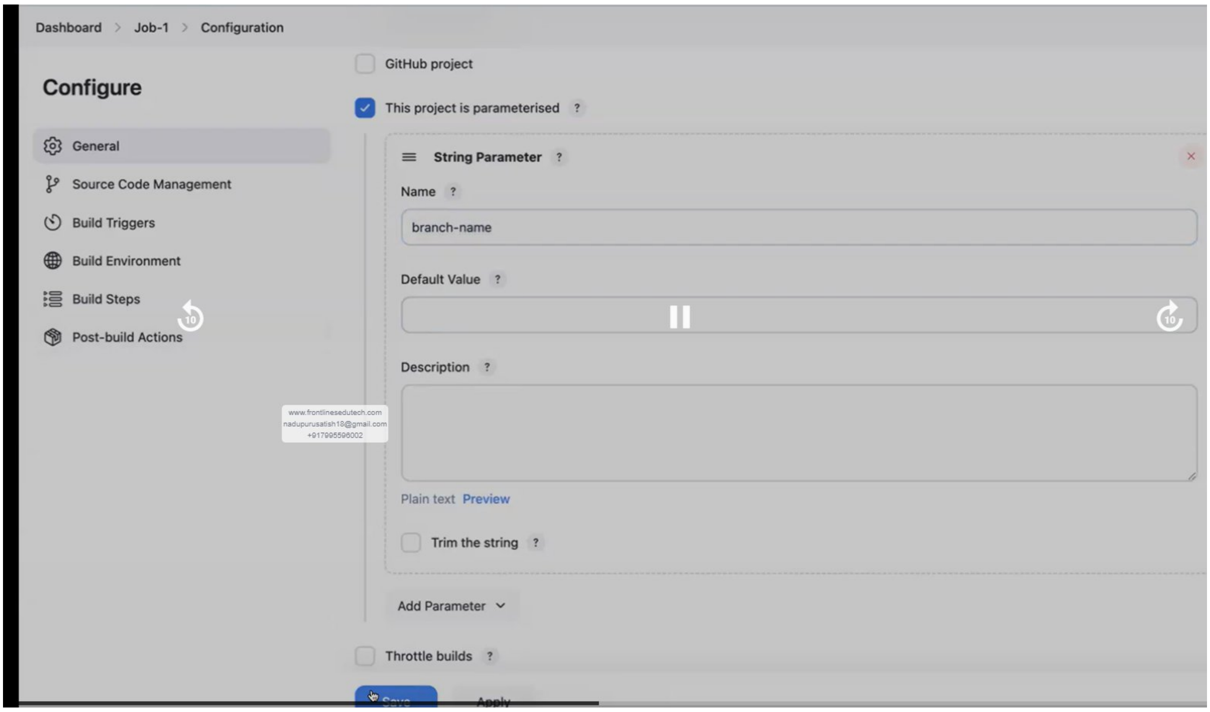


->sel any branch n build

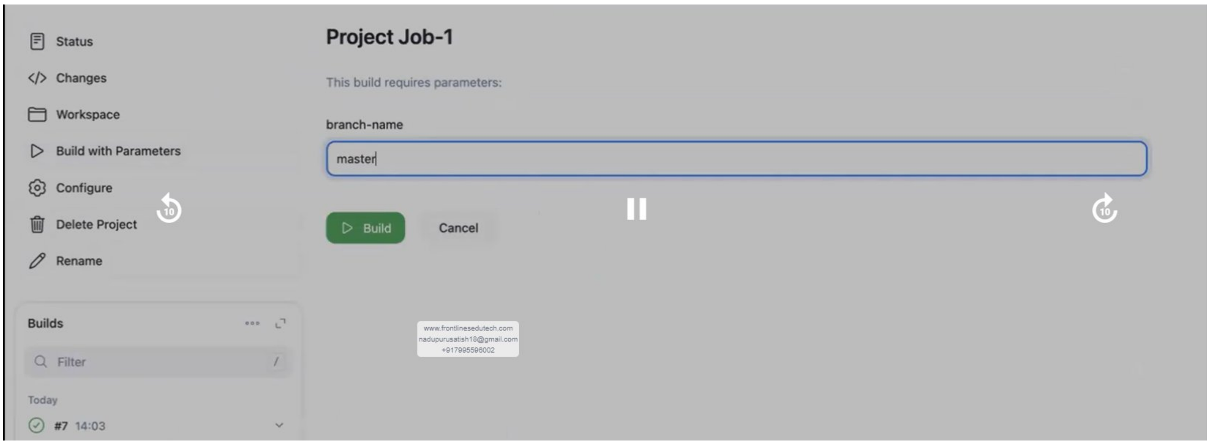
STRING PARAm

->suppose manaki github lo ey branches unnayo telidu+ access ledu, so ee case use avutadi

->name lo edokati ichi, save cheyye



->ippudu build vth param>branch-name lo mi manager cheppina branch name ichi build cheyye



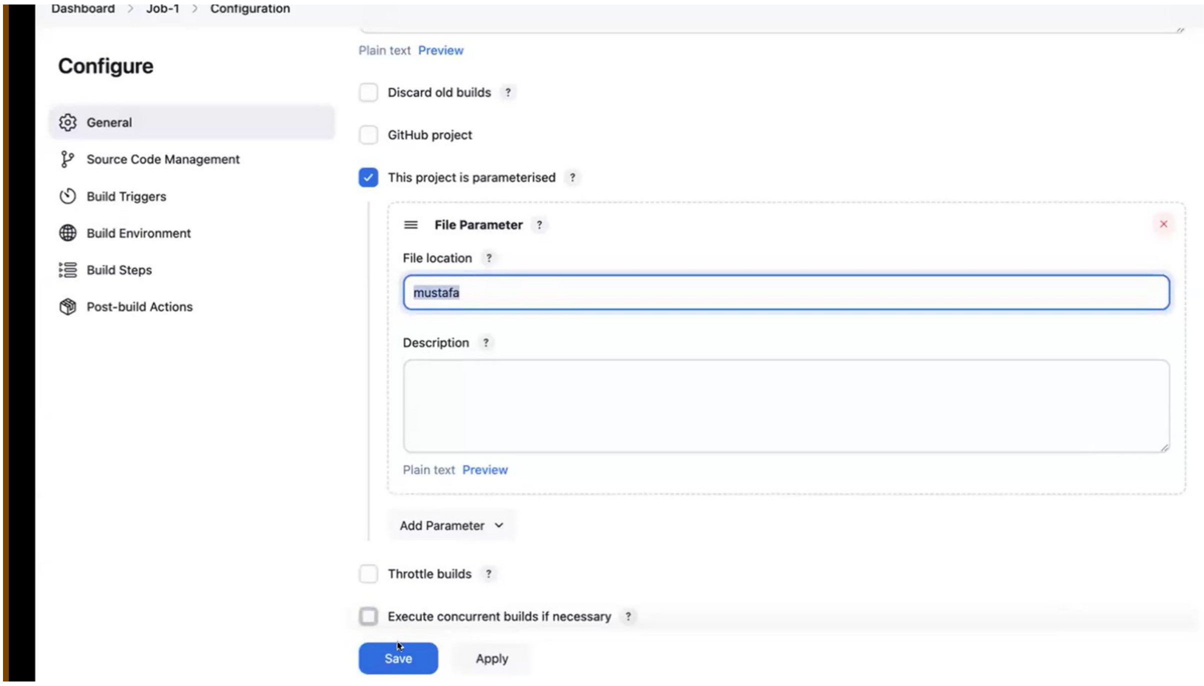
NOTE – 1. If branches telisthey go vth choice param

2. if not go vth string-param

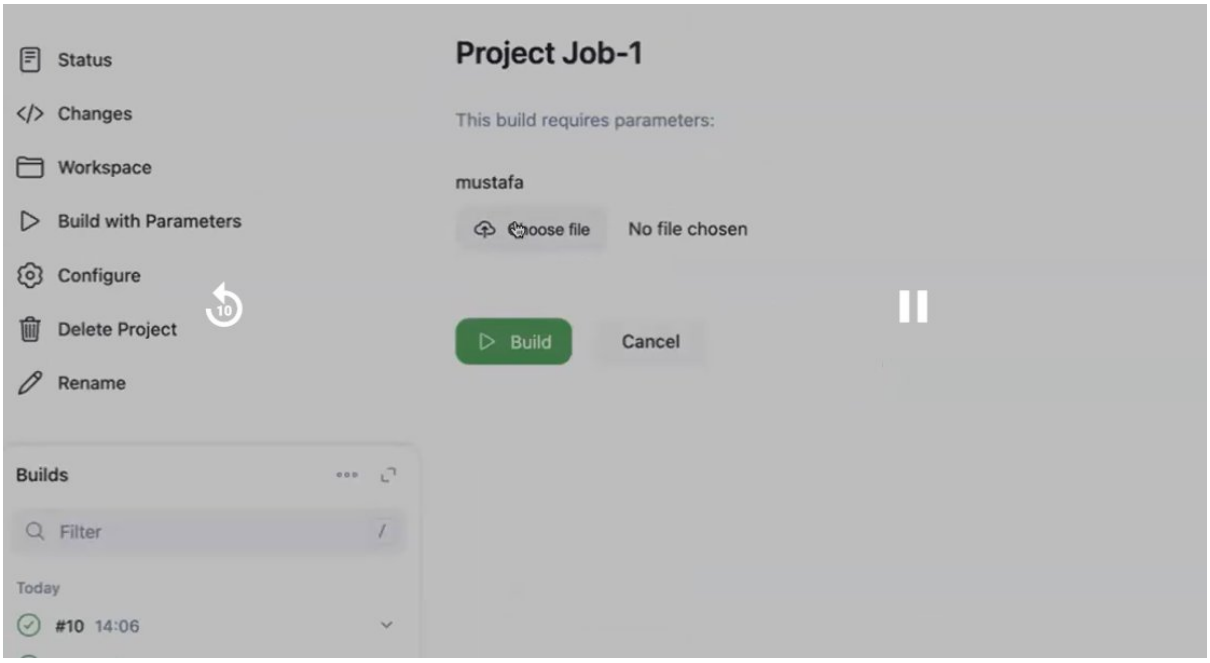
FILE PARAM

->mana local lo undi file, adi Jenkins loki kavali antey use this

->file location lo neeku a file ey name tho store avvali Jenkins lo



->choose file



->Jenkins lo workspace lo chudachu a file vachindho ledho ani

MULTIPLE-LINE STRING PARAM

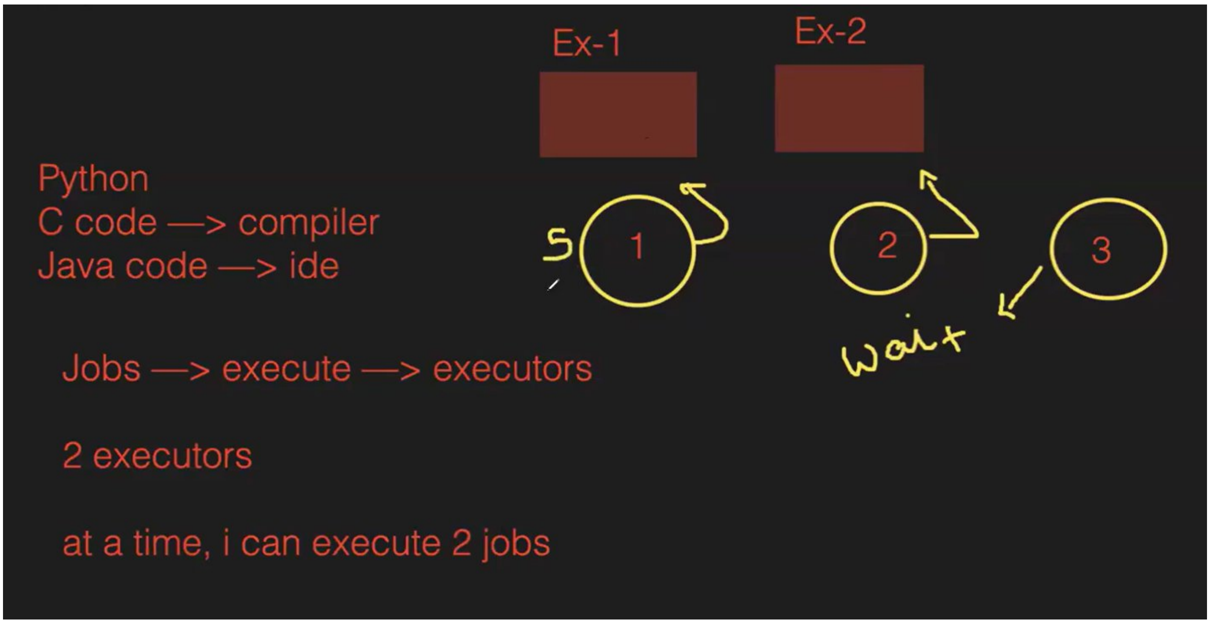
->similar to single-line param but it accepts multiple lines of branch name.

EXECUTOR

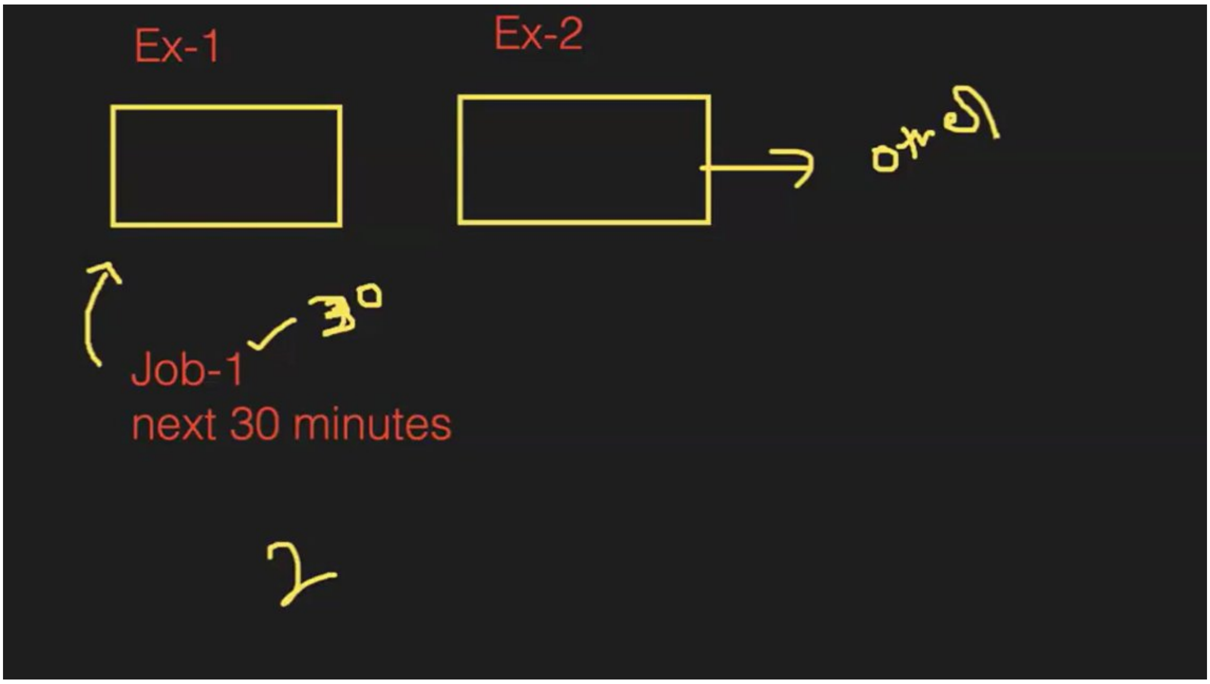
->mana dhagra c code untey adi run cheyyadaniki compiler kavali, python code ki interpreter n java ki ide.....likewise Jenkins lo jobs execute avvadaniki executors untai.

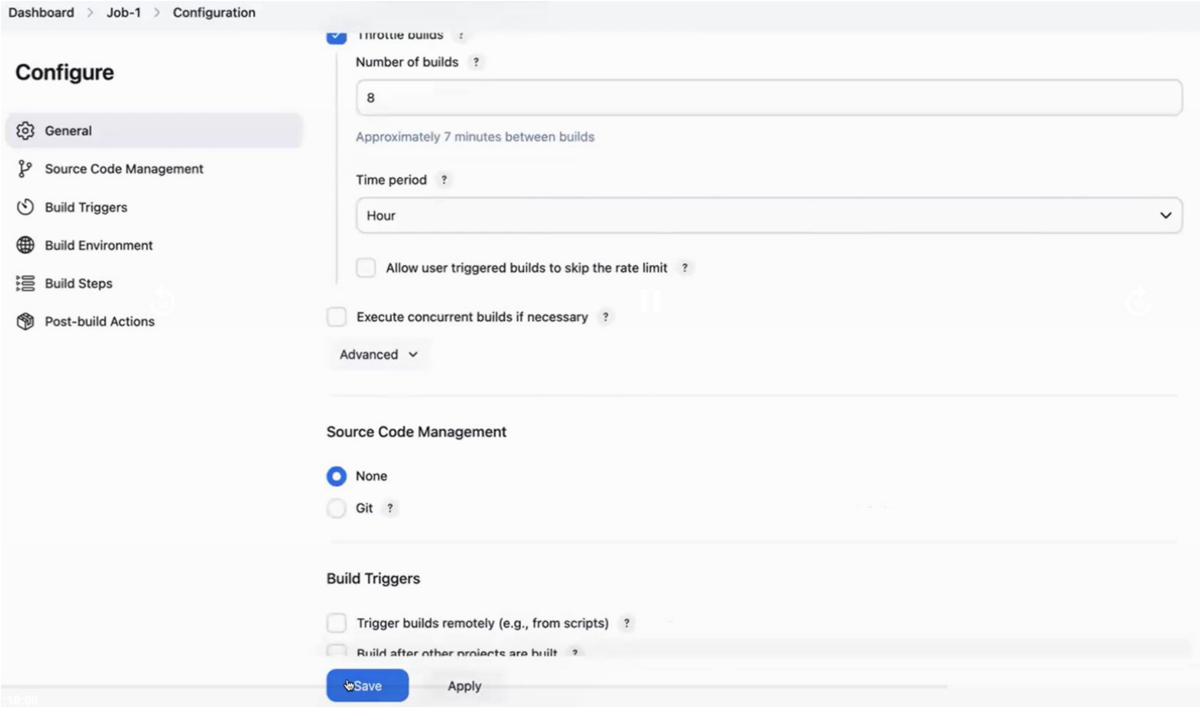
->by def ga manaki Jenkins lo 2 executors untai, suppose mana Jenkins lo oka 10jobs unnai anukundam.

1st job- executor1 ki assign avutadi, 2nd job-executor2 ki….edaina exector free ayyaka other jobs execute avutai.



->job1 ni executor lo run chestunnam adi run avvadaniki oka 30mins tym teesukuntadi anukundam,same fr job2. Job3 ni koda execute cheyyali adi queue lo untadi…..n jobs queue lo unnai anukundam.so, Jenkins meda load padatadi. Jobs ni queue lo lekunda restrict cheyyali using throttle builds



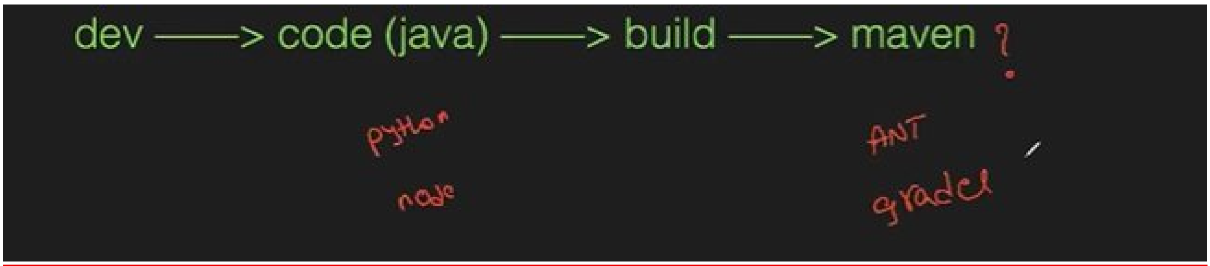


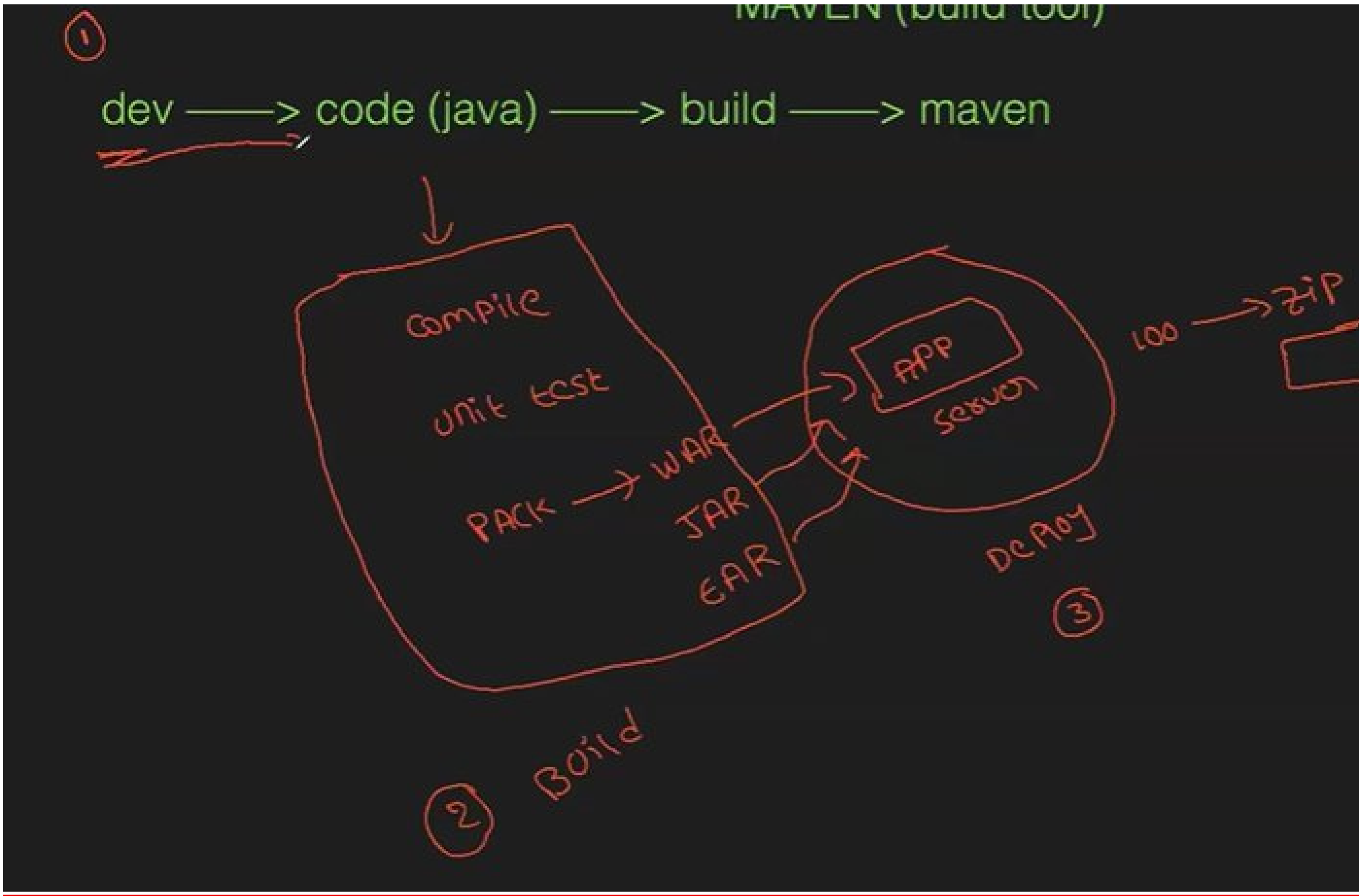
No of builds 8 ichamu, time period lo hour – so prathi build ki approx. 7mins pattdi.

Q/A

->pending at 50min

CLASS-3 HOW TO BUILD JAVA PROJECTS USING MAVEN





->devops raka mundu oka app ni deploy cheyyali antey developcheyyadaniki oka person, build ki person, deploy ki oka person total ga 3 persons kavali, but devops vachaka anta automate cheyyadaniki tools vachai

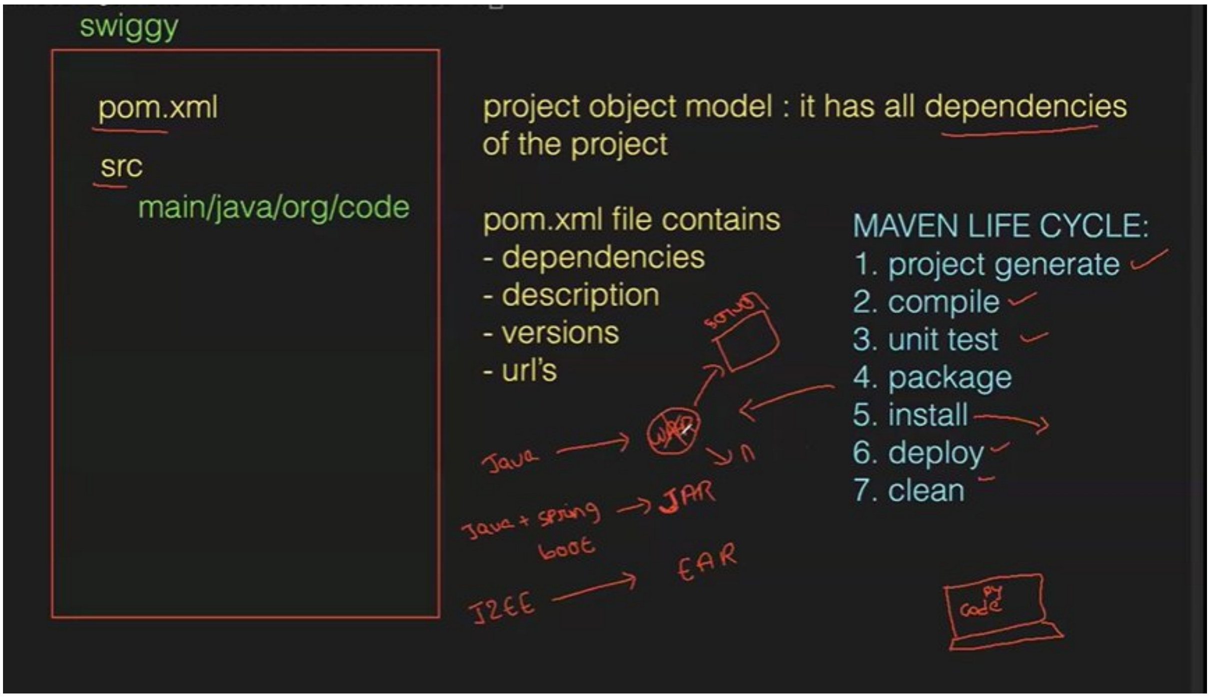
->build, test chesaka dependents tho pack chesaka war/jar/ear files generate avutai

War webarchive= using java

Jar java archive=java+sprongboot

Ear enterprise archive=using j2eee

MAVEN LIFECYCLE



->paina mvn procedure anta manual ga cmds tho cheyyachu but devops vachaka Jenkins tho manam automate chestunnam

->take java project kosam one aney repo fork it from mustafa github acc

->lin otder to build+test we need to have mvn, but no need of installing it, jst need to integrate mvn vth Jenkins

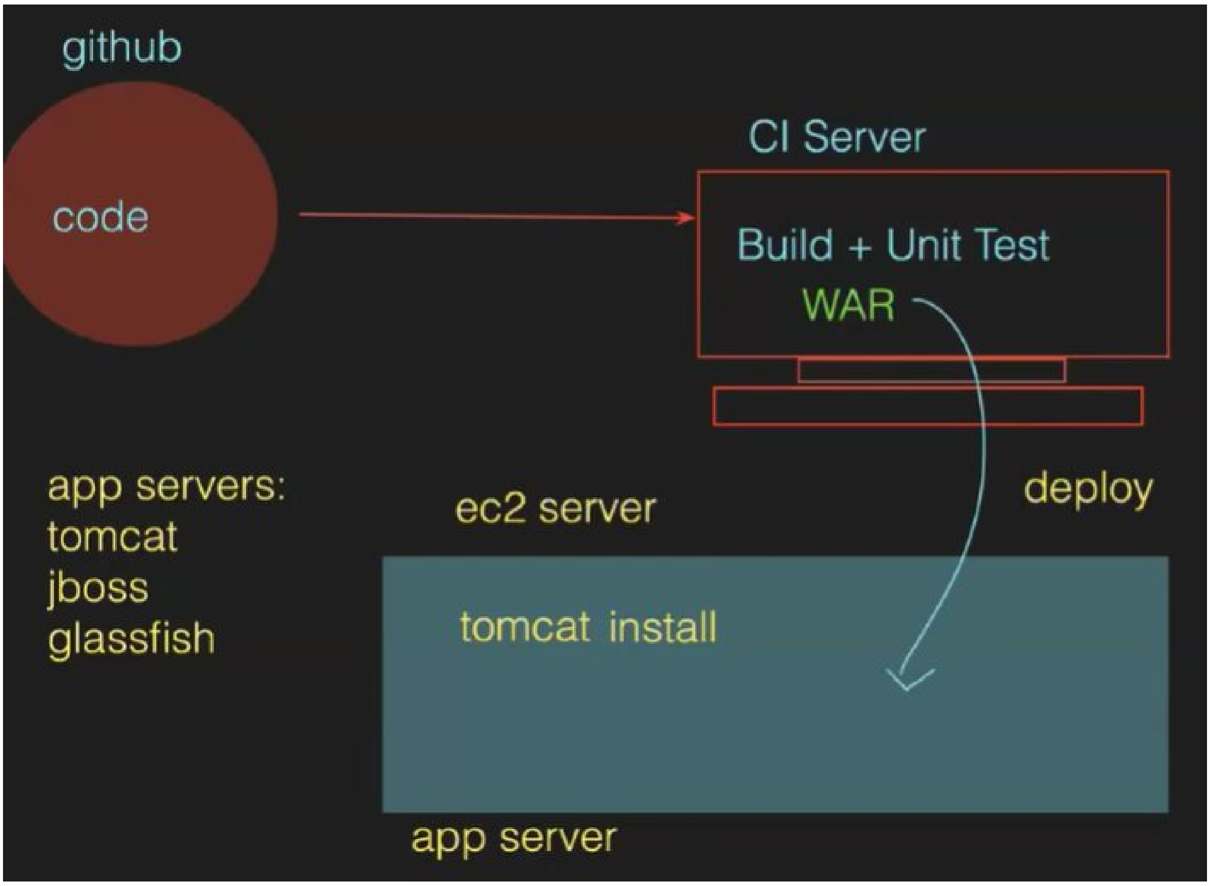
Goto manage Jenkins>tools>scroll-down>we'll see mvn>click on add mvn n name it>save it

->now goto>job>configure>build steps>under build steps sel add build step>sel invoke tp-lvel mvn targets>under mvn version sel whtever name u gave>under goals sel clean package(it'll clean prev build data)>save it>click on build now

->now goto workspace and we see target folder, in which a tar/war/jar file generated

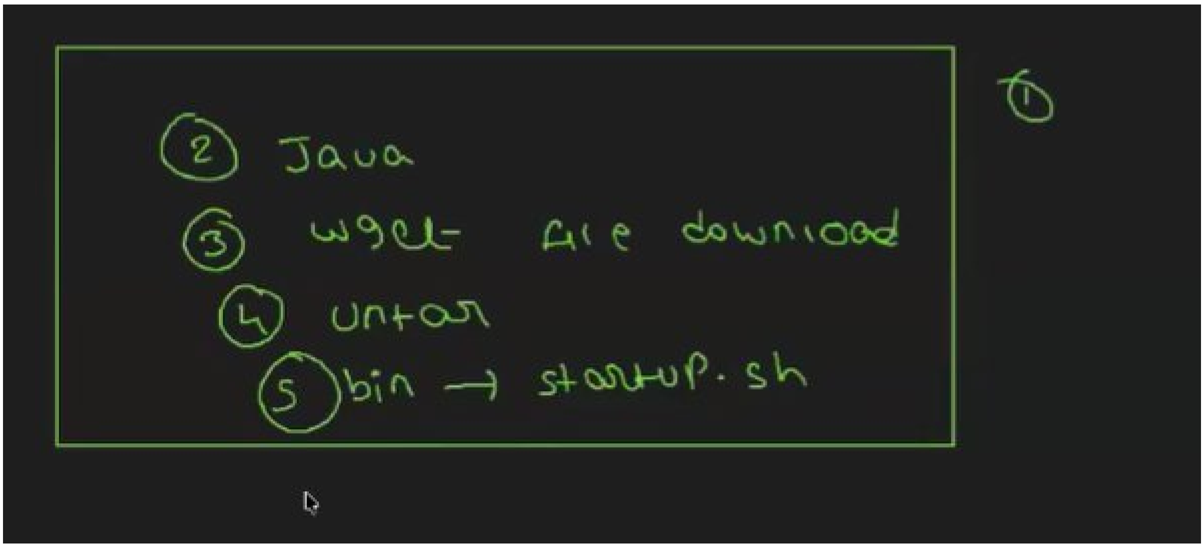
CLASS-4 DEPLOY JAVA APPLICATION USING JENKINS

->last 2 classes lo code github nunchi CI server ki techi build+test chesamu, ee cls lo java app ni build chedam



App server ki painavi use chestamu

Tomcat installation



1.Ec2 instance vth t3 micro,sg is like Jenkins,10gb volume

2.access the server n change hostname vth hostnamectl set-hostname “web-server” ,logout n then login it will change

3.tomcat ki koda java install cheyyali

```
[root@app-server ~]# yum install java-17-amazon-corretto -y
```

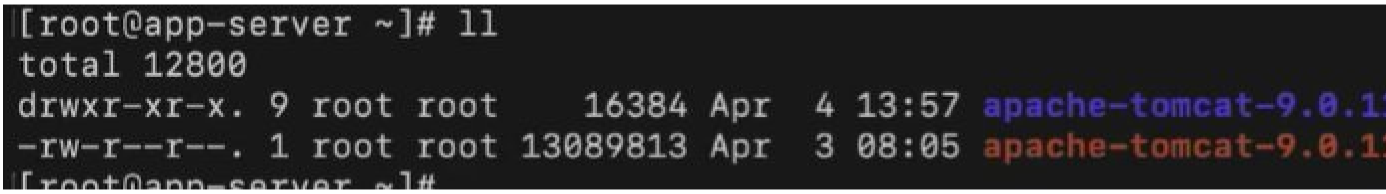
4. installing tomcat binaries

goto dlcdn.apache.org>search fr tomcat>tomcat-9>v9.0.117>bin>---tar.gz>right click n copy link add

now goto server give cmd **wget url**

a tar file to untar

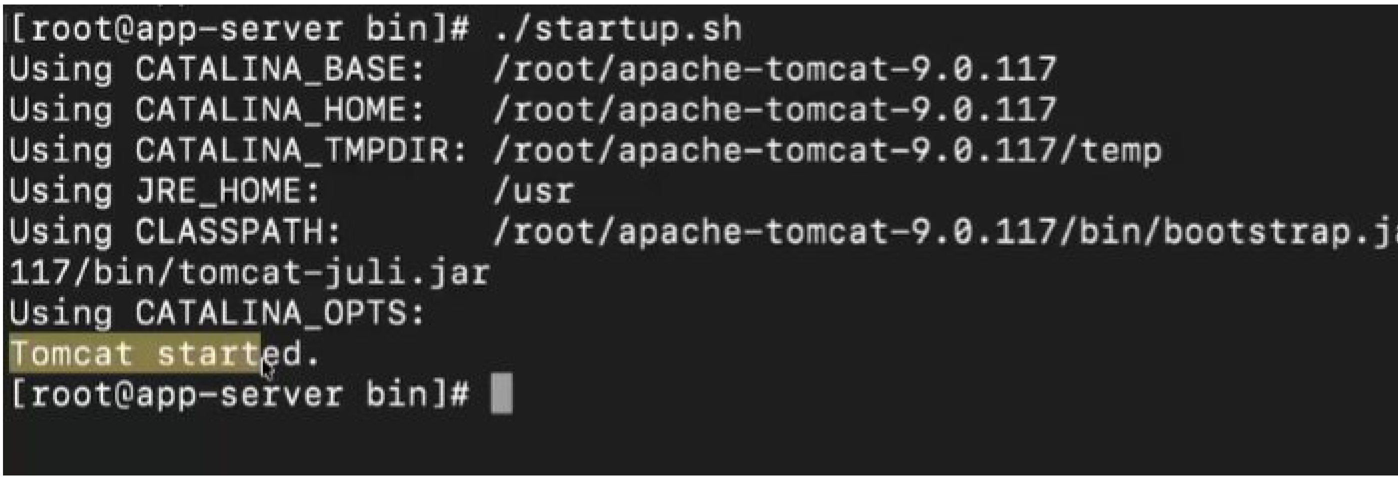
->now file got untarred



->goto that folder>bin>startup.sh

NOTE-to start any service we need to goto bin folder n that file shd have executable permission

->to start ./file_name



It's showing tomcat started, now we can access the tomcat dashboard

->port is same fr tomcat like Jenkins(8080)

Copy pub ip:8080

->now goto manager app, it'll show access denied

403 Access Denied

You are not authorized to view this page.

By default the Manager is only accessible from a browser running on the same machine as Tomcat. If you wish

If you have already configured the Manager application to allow access and you have used your browsers back interface of the Manager application. You will need to reset this protection by returning to the main Manager pa denied message, check that you have the necessary permissions to access this application.

If you have not changed any configuration files, please examine the file `conf/tomcat-users.xml` in your in

For example, to add the `manager-gui` role to a user named `tomcat` with a password of `s3cret`, add the fo

```
<role rolename="manager-gui"/>
<user username="tomcat" password="s3cret" roles="manager-gui"/>
```

Note that for Tomcat 7 onwards, the roles required to use the manager application were changed from the sing

- `manager-gui` - allows access to the HTML GUI and the status pages
- `manager-script` - allows access to the text interface and the status pages
- `manager-jmx` - allows access to the JMX proxy and the status pages
- `manager-status` - allows access to the status pages only

The HTML interface is protected against CSRF but the text and JMX interfaces are not. To maintain the CSRF pr

- Users with the `manager-gui` role should not be granted either the `manager-script` or `manager-j`
- If the text or jmx interfaces are accessed through a browser (e.g. for testing since these interfaces are i

For more information - please see the Manager App How-To.

Adi anta chadivithey access chesukovachu

->context.xml loki velli modify cheyyandi antundi

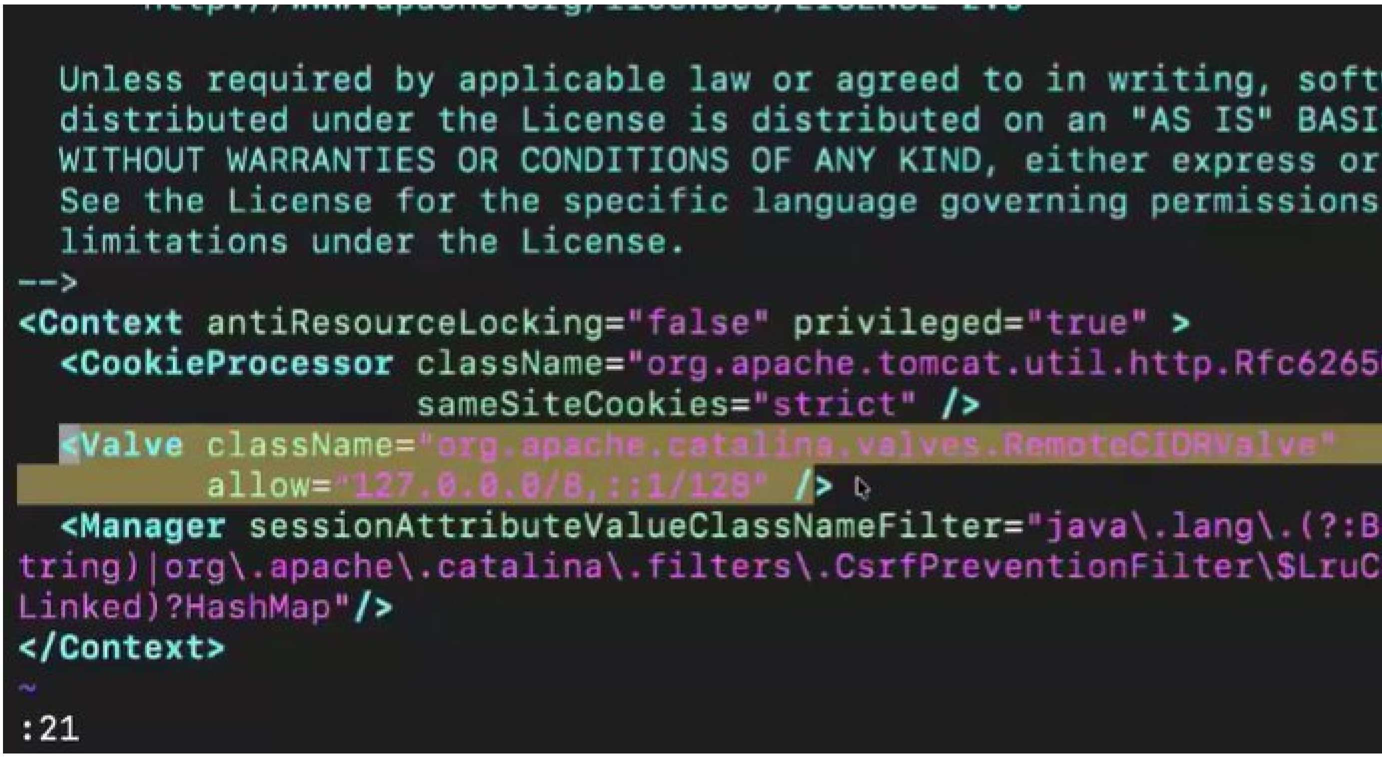
```
[root@app-server apache-tomcat-9.0.117]# ll
total 192
-rw-r-----. 1 root root 26700 Mar 30 18:21 BUILDING.txt
-rw-r-----. 1 root root  6096 Mar 30 18:21 CONTRIBUTING.md
-rw-r-----. 1 root root 57092 Mar 30 18:21 LICENSE
-rw-r-----. 1 root root  2333 Mar 30 18:21 NOTICE
-rw-r-----. 1 root root  3216 Mar 30 18:21 README.md
-rw-r-----. 1 root root  6902 Mar 30 18:21 RELEASE-NOTES
-rw-r-----. 1 root root 16543 Mar 30 18:21 RUNNING.txt
drwxr-x---. 2 root root 16384 Apr  4 13:57 bin
drwx------. 3 root root 16384 Apr  4 13:58 conf
drwxr-x---. 2 root root 16384 Apr  4 13:57 lib
drwxr-x---. 2 root root 16384 Apr  4 13:58 logs
drwxr-x---. 2 root root    30 Apr  4 13:57 temp
drwxr-x---. 7 root root    81 Mar 30 18:21 webapps
drwxr-x---. 3 root root    22 Apr  4 13:58 work
[root@app-server apache-tomcat-9.0.117]#
[root@app-server apache-tomcat-9.0.117]#
[root@app-server apache-tomcat-9.0.117]# find . -name context.xml
./conf/context.xml
./webapps/docs/META-INF/context.xml
./webapps/examples/META-INF/context.xml
./webapps/host-manager/META-INF/context.xml
./webapps/manager/META-INF/context.xml
[root@app-server apache-tomcat-9.0.117]#
```

Manager lo unna context.xml loki vellali

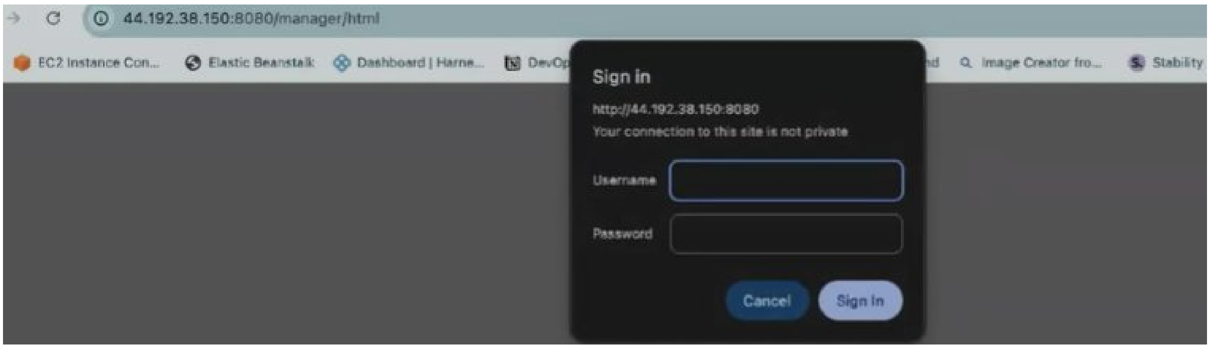
->so open that file

```
[root@app-server META-INF]# vim context.xml
```

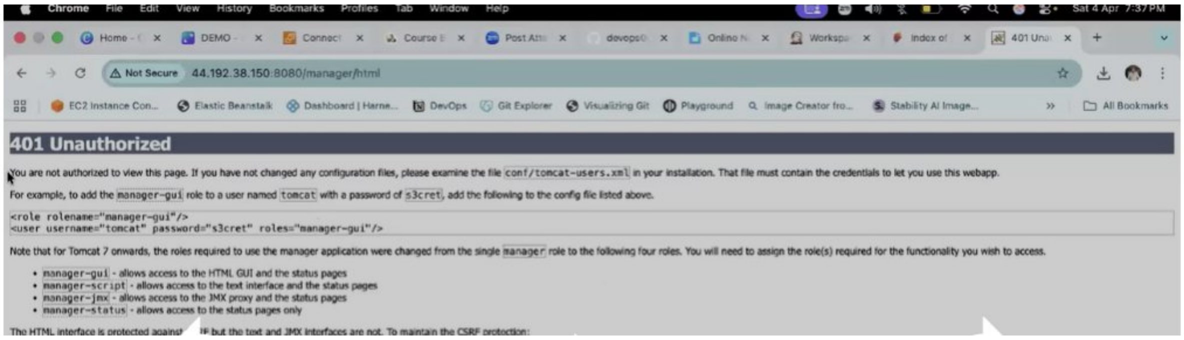
Goto line no21>give 2dd so that it'll del 2 lines>savae n continue



->now access tomcat, it'll show access denied but asking credentials, jst cancel it



Now again access denied and perform whtever it's shows



Goto -apache-tomcat ki velli

```
[root@app-server META-INF]# cd ../
[root@app-server manager]# cd ..
[root@app-server webapps]# cd ..
[root@app-server apache-tomcat-9.0.117]#
```

Here give ll n goto conf folder>open that tomcat-users.xml file

```
[root@app-server conf]# ll
total 228
drwxr-x---. 3 root root     23 Apr  4 13:58 Catalina
-rw-------. 1 root root  12953 Mar 30 18:21 catalina.policy
-rw-------. 1 root root   7654 Mar 30 18:21 catalina.properties
-rw-------. 1 root root   1400 Mar 30 18:21 context.xml
-rw-------. 1 root root   1149 Mar 30 18:21 jaspic-providers.xml
-rw-------. 1 root root   2313 Mar 30 18:21 jaspic-providers.xsd
-rw-------. 1 root root   4003 Mar 30 18:21 logging.properties
-rw-------. 1 root root   8022 Mar 30 18:21 server.xml
-rw-------. 1 root root   2756 Mar 30 18:21 tomcat-users.xml
-rw-------. 1 root root   2558 Mar 30 18:21 tomcat-users.xsd
-rw-------. 1 root root 174047 Mar 30 18:21 web.xml
[root@app-server conf]#
[root@app-server conf]#
```

->now goto lastline(G)>go up n copy those 1st three role lines(3yy)>paste those 3lines after --> and change as below

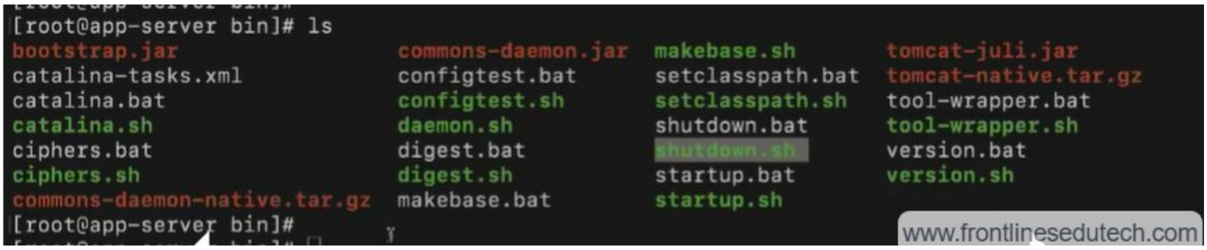
```
<!--
  The sample user and role entries below are intended for use with the
  examples web application. They are wrapped in a comment and thus are ignored
  when reading this file. If you wish to configure these users for use with the
  examples web application, do not forget to remove the <!.. ..> that surrounds
  them. You will also need to set the passwords to something appropriate.
-->
<!--
  <role rolename="tomcat"/>
  <role rolename="role1"/>
  <user username="tomcat" password="<must-be-changed>" roles="tomcat"/>
  <user username="both" password="<must-be-changed>" roles="tomcat,role1"/>
  <user username="role1" password="<must-be-changed>" roles="role1"/>
-->
  <role rolename="manager-gui"/>
  <role rolename="manager-script"/>
  <user username="tomcat" password="admin@123" roles="manager-gui, manager-script"/>
</tomcat-users>
-- INSERT --                                                58,82         Bot
```

NOTE- 1 manaki ela telustadi antey access denied page lo cheptadu

2 to restart/start/stop tomcat we need to got bin folder only

3.configurations change chesthey pakka ga restart cheyyali

->so to restart now we have shutdown.sh file



->now start the service(yum tho install chesthey systemctl tho start/stop/restart chestamu) but manam tomcat ni script tho install chesamu

->now access tomcat by giving credentials

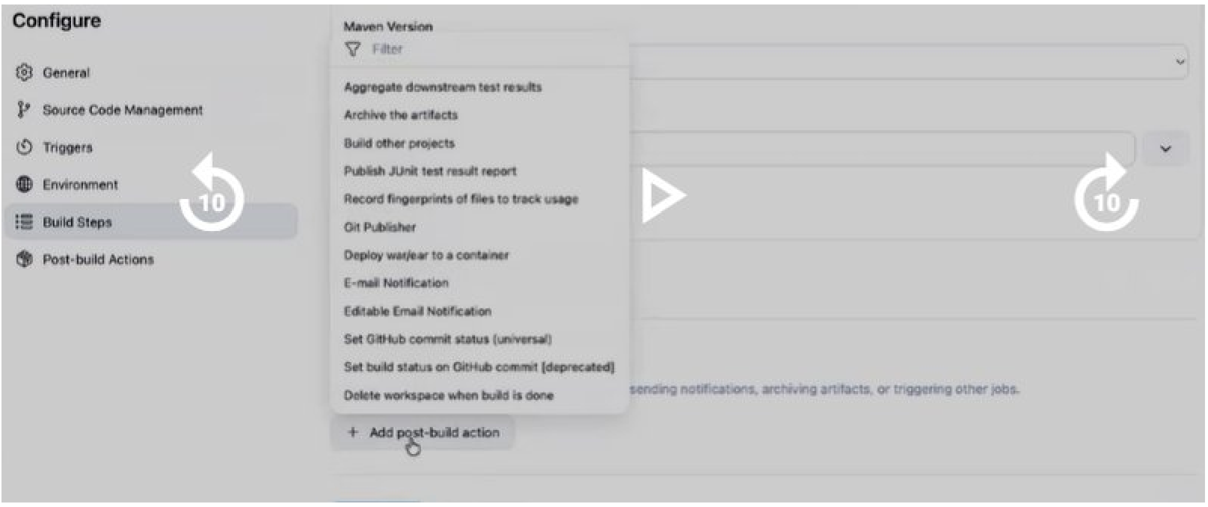
->below are applications but manam manager path lo deploy chestamu

INTEGRATING TOMCAT WITH JENKINS

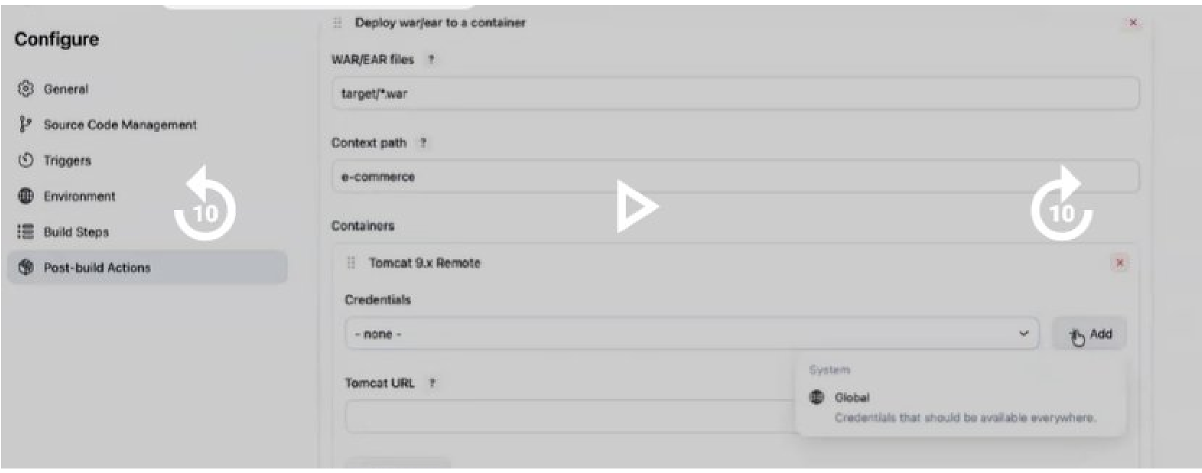
->by def mvn Jenkins tho install avutadi, jst manam configure chestamu but tomcat ni manam manual ga install chesukovali

Goto manage Jenkins>plugins>available plugins>search fr deploy to containers nd sel it>install it

->now goto Jenkins job>configure>build steps>add post-biild steps>sel deploy war/ear to container>give war/ear file path

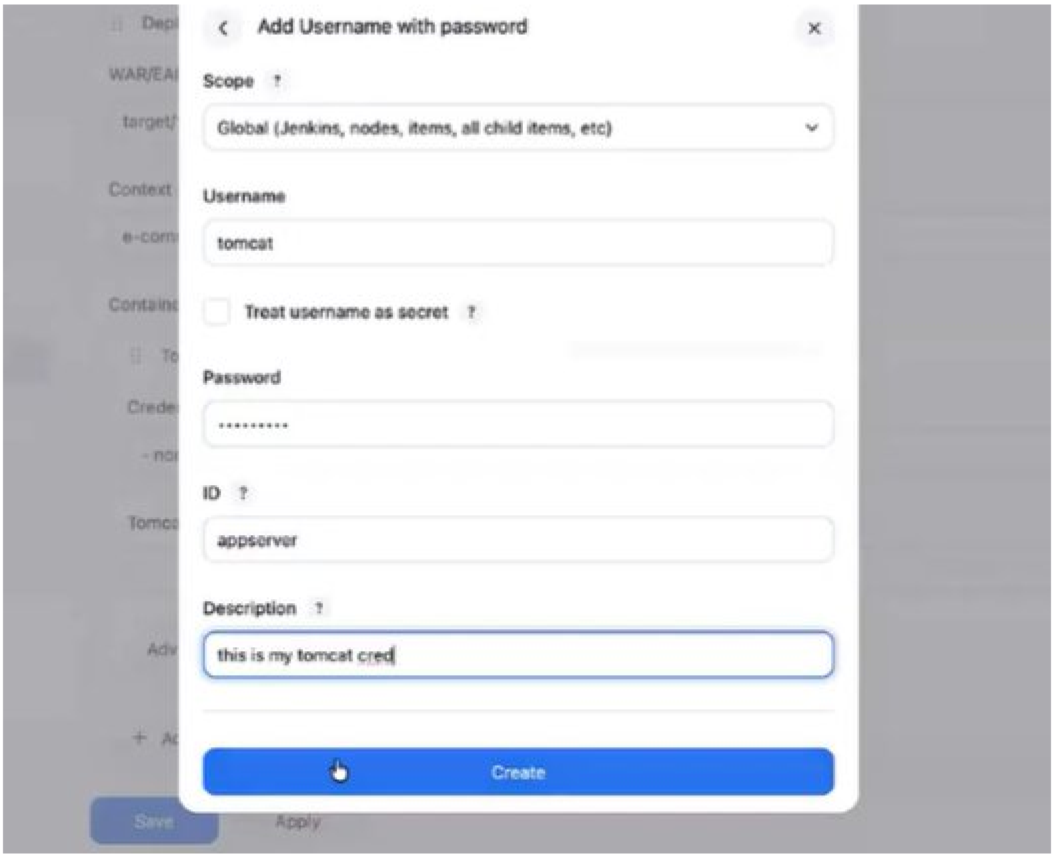


->we can give name to our app in context path>in containers sel tomcat9

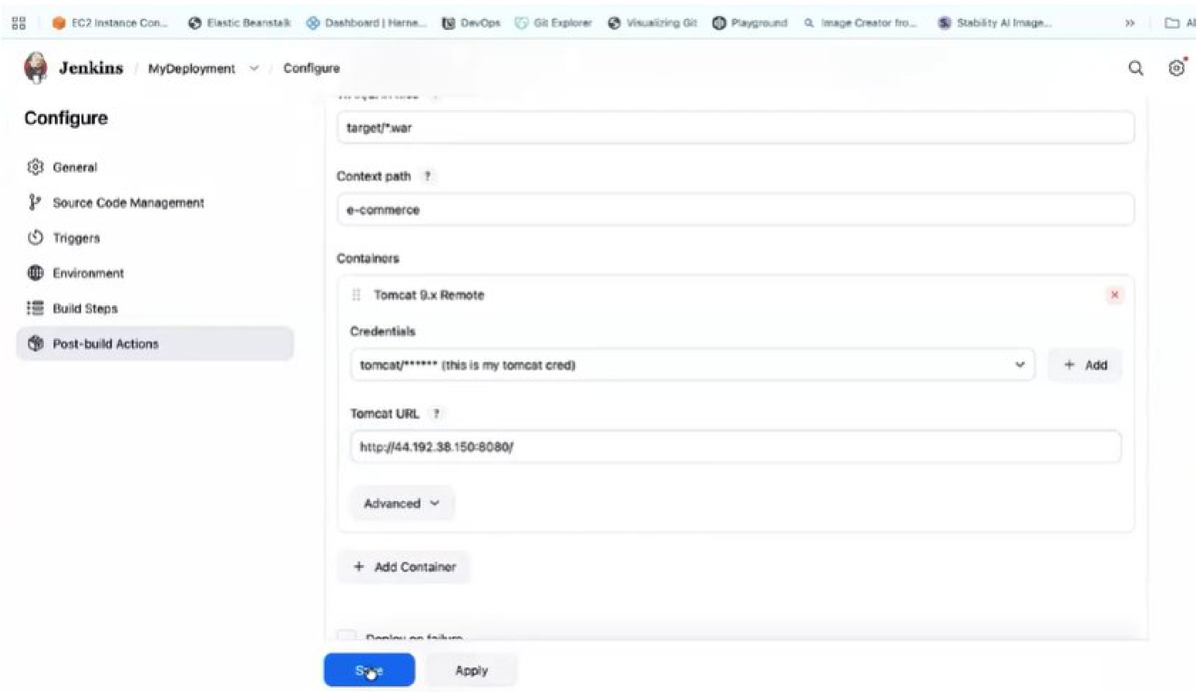


->now we nned to add tomcat credentials

Add>global>username vth password>next>give credentials>give id n description as ur own>create



->select those added credentials>in tomcat url give upto 8080>save



->now we integrated Jenkins vth tomcat, click on build now,goto tomcat dashboard n chk app got deployed or not vth e-commerce name

NOTE-1 java app deploy ki mustafa-github/one-repo/master branch

2 developer changes chesthey application lo reflect avutaya—YES but manual ga build chesthey ney reflect avutadi

3 webhooks/poll scm chesthey automatic ga changes reflect avutai

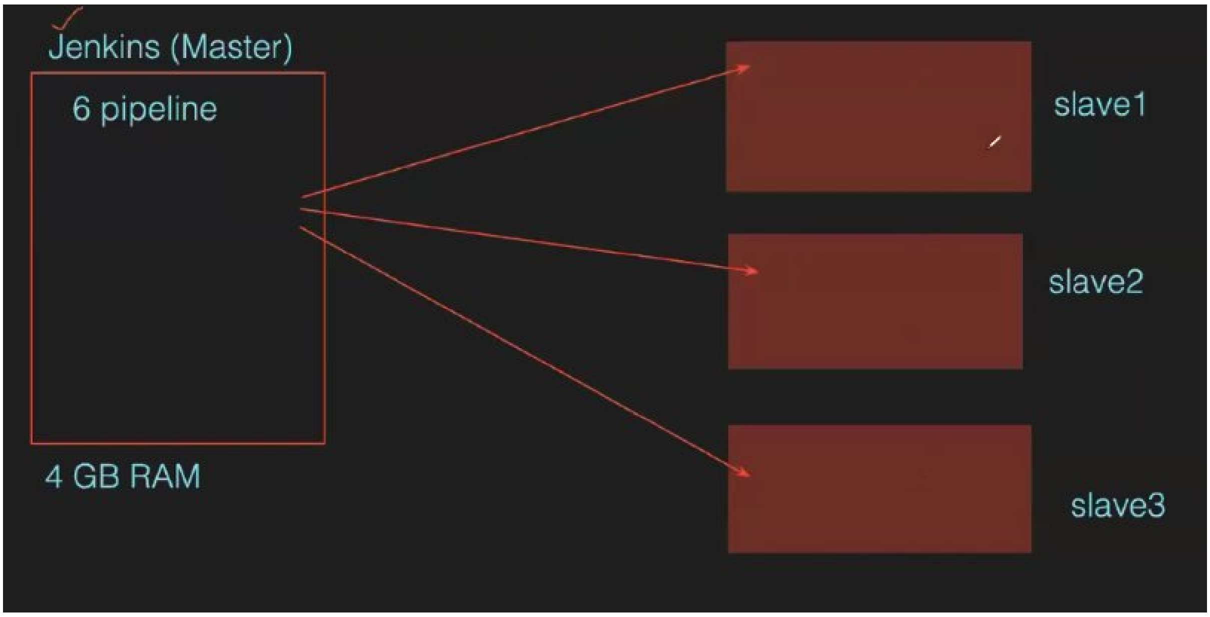
TASK- deploy java app vth webhook/poll scm so that whenever changes occur will reflect automatically

NOTE-1. Realtime lo tomcat ni setup cheyyadaniki scripts ey use chesataru,in github>Mustafa>all-setups>tomcat.sh>edit n change version everywhere >commit

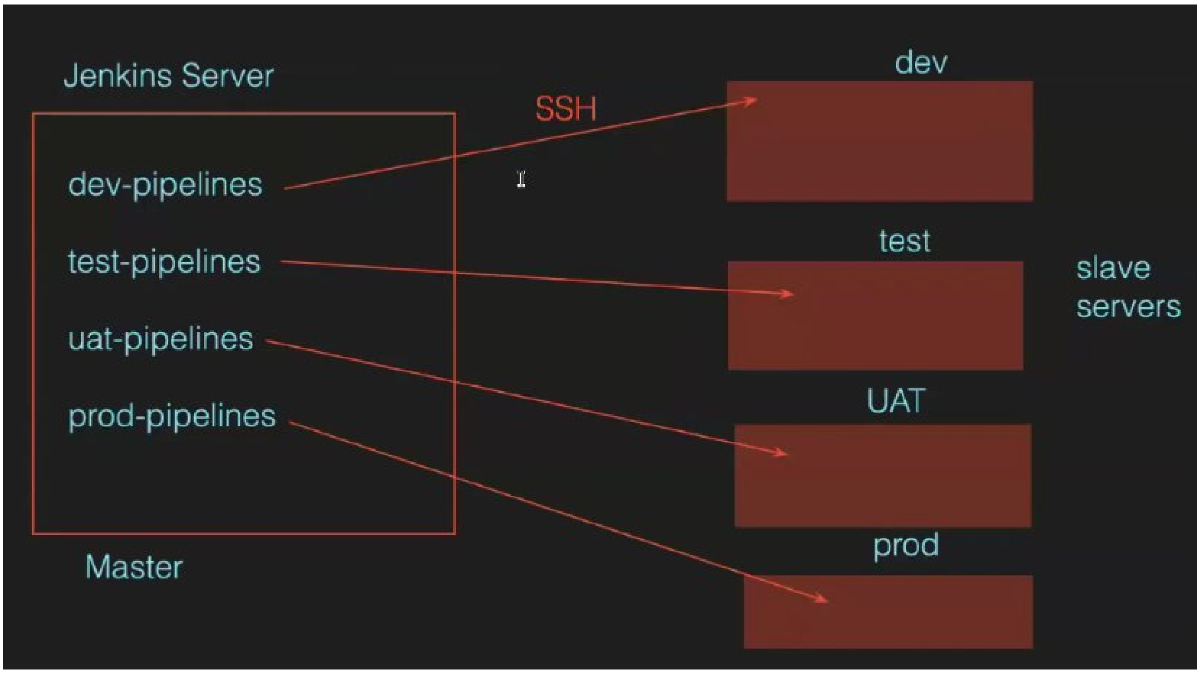
2 now paste those cmds in server

CLASS-5 MASTER-SLAVE

->master n slave concept enduku vachindi antey oka server ekkuva size teesukoni anni app's danloney run cheyyachu but server meda load padi crash/hang so that anni app's impact avutai. So master-slave concept use chestamu



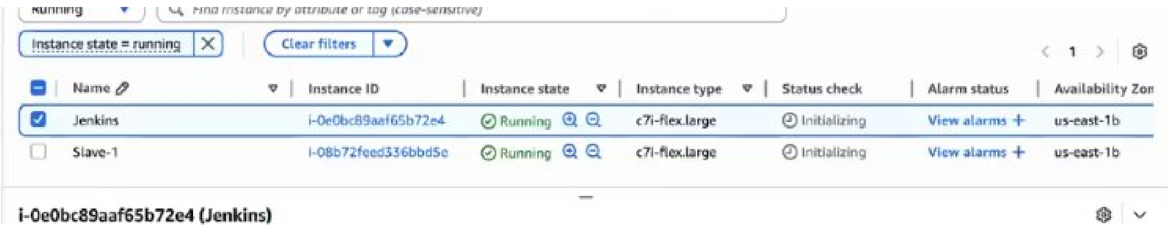
->realtime lo ey case lo use chestamu antey ee master-slave concept manaki diff env's untai line dev,test,pre-prod,prod untai Jenkins server nthg but master ee env salves so master lo pipe-lines create chestamu, slaves lo execute avutai a pipe-lines



Master-slave ki ssh cnt establish chestamu

CREATING 2 SERVERS(1-master,2^{nd}-slave)

->one is Jenkins, 2nd is slave



->install Jenkins in Jenkins server

->git nunchi code techukoni build chesthey, a pipe-line by def a Jenkins server loney exec avutundi

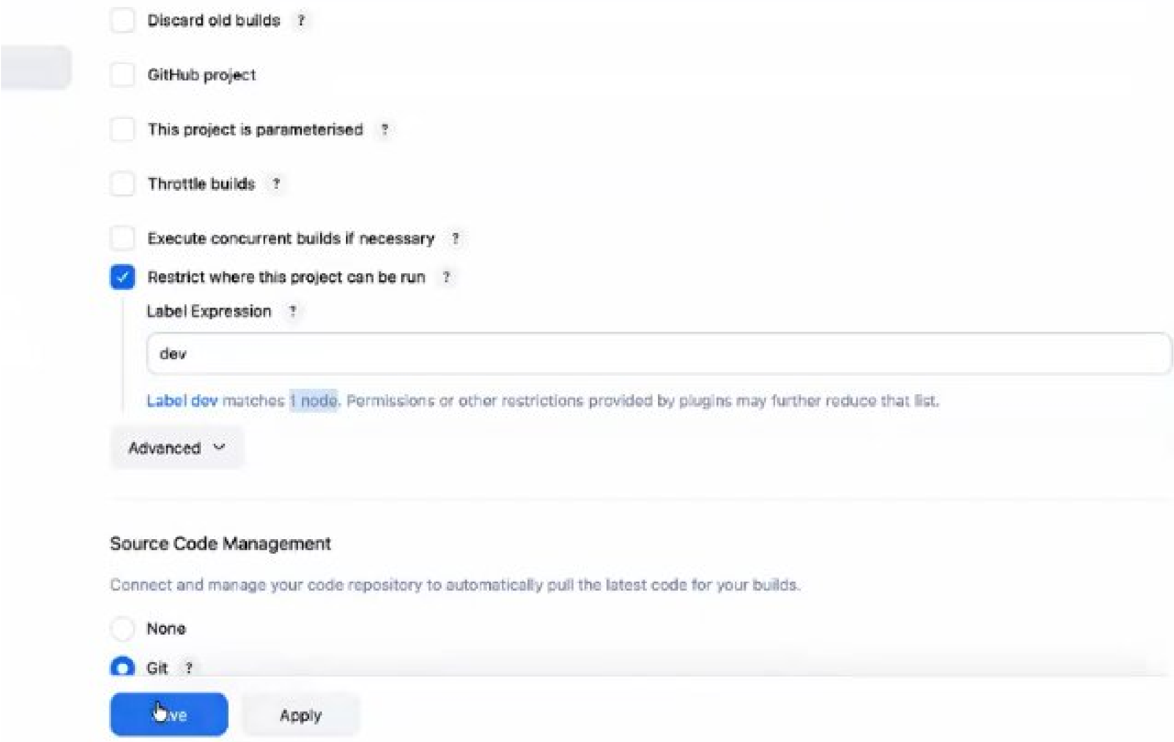
->but nak jenkins lo exec avvakodadhu nak slave server lo avvali

TO EXEC PIPE-LINE IN SLAVE SERVER

->slave server lo java undali, so install it

->now in Jenkins goto>manage Jenkins>nodes>by def ga built-in node untadi, new node>give name>sel permanent agent>give desp>in no of executors give 1or2(enni pipe-lines exec cheyyali ani,we'n give any no but it shd have that capacity)>in remote root dir-ekkada mana pipe-line exec aye store avvali ani, give /home/ec2-user/krishna-ee krishna aney folder automatic ga create aye danlo store avutadi a code>in label give any name like dev/test..>in usage sel only build jobs vth label exp matching this node>in launch method sel launch agents via ssh>in host give slave pub/priv ip>in credentials click add>global>ssh username vth pri key>in id u can give anything desp also>in uname give def user ie ec2-user>in priv key paste that server key-pair>sel that added credentials>in host key verification strategy sel non verifying verification strategy>save it>now that node got added

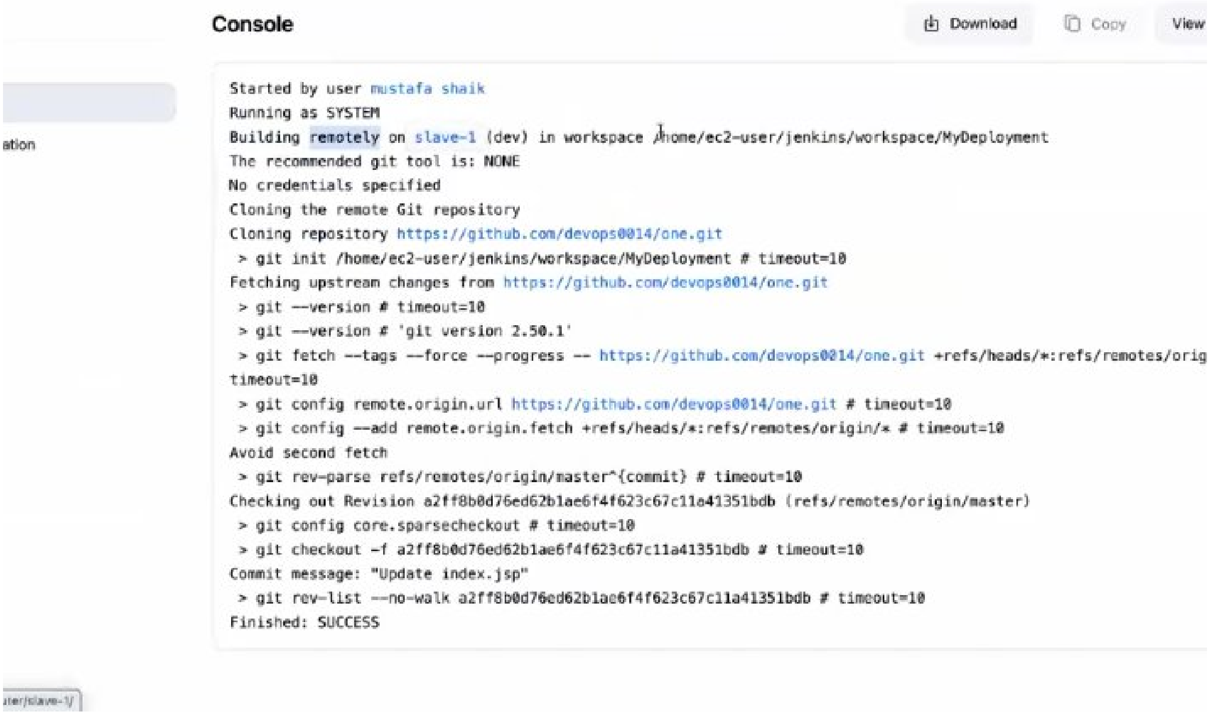
->now open that pipeline>configure>sel restrict where this project can be run>give label name where u mentioned while creating node>save it



->pipeline fail avutadi bcz git install cheyyali, install git n build now

Build gets successfully n chk whether that code is stored in krishna folder or not

Verey path ivvachu /home/ec2-user/folder_nmae but permission pbms vastai avi handle cheyyadam kashtam so go vth server def path



NOTE- 1.master-slave connection ki manam java install chestamu

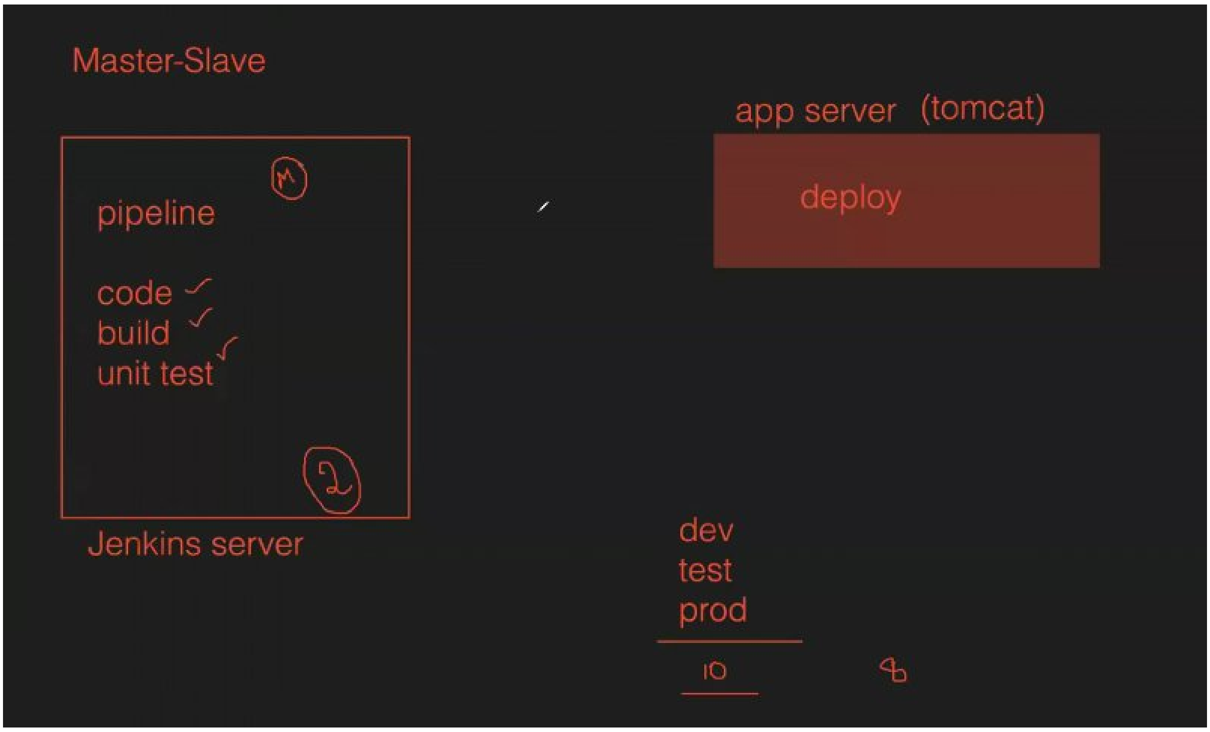
CLASS-6 ee class lo previous 3 classes-getting code,build+test n deploy ey revise chesaru

CLASS-7 MASTER-SLAVE integrating artifact

->kinda chusthey manaki realtime lomanaki diff env's untai, Jenkins server lo anni pipelines run chesthey load padatadi,mem teesukuntadi,server crash avutadi

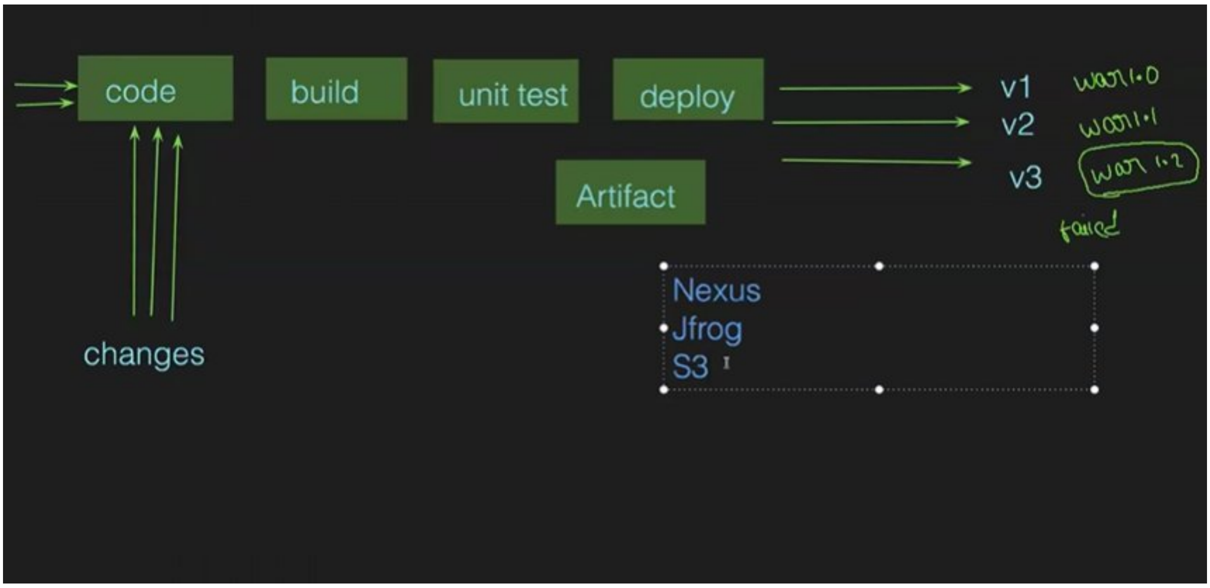
->oka dbt ravali dev server lo ayyaka test nxt pre-pod nxt prod—so one-by-one exec cheyyachu ga

Kinda example chudu oka app lo internation transactions enable chesthey enni chotla modifications cheyyalo, so dev loney 3 pipe-lines vachai ala inka test,prod untai…….so ala pipe-lines ekkuva avutai



->v1 v2 v3 versions release chesamu but v3 anedi crash ayyindi so we need to go n use v2 only manaki backup unetey ney ga v2 version use cheyyagalam

->build+test stage ayyaka artifact stage ni add chestamu(artifact antey create aina war file ni store chestamu)



->launch 3 instances vth c7i flex large, 25gb volume

1.jenkins

2 nexsus

3 tomcat

->1st using master-slave/normally concept deploy app in tomcat server

NEXSUS INTEGRATION VTH JENKINS

->install java

NOTE – for every server we need to install java

->in browser search fr nexus download fr linux, sel install self-hosted repository link>download page>copy link of unix/linux x86-64>now goto /opt run wget url

->unzip it

->goto nuxus-3.91.0-07>bin>nexus file run

->root user ga run avvavu ani official web lo koda mention chesaru

```
.jar
[root@ip-172-31-90-156 bin]#
[root@ip-172-31-90-156 bin]# ./nexus start
WARNING: ************************************************************
WARNING: Detected execution as "root" user.  This is NOT recommended!
WARNING: ************************************************************
Starting nexus
[root@ip-172-31-90-156 bin]#
```

->so create a user nexus n changing owner also for nexus folders

```
[root@ip-172-31-90-156 opt]#
[root@ip-172-31-90-156 opt]# useradd nexus
[root@ip-172-31-90-156 opt]#
[root@ip-172-31-90-156 opt]# ll
total 455680
drwxr-xr-x. 4 root root        33 Apr  4 02:00 aws
drwxr-xr-x. 6 root root        94 Apr 10 14:04 nexus-3.91.0-07
-rw-r--r--. 1 root root 466615526 Apr  7 15:15 nexus-3.91.0-07-linux-x86_64.tar.gz
drwxr-xr-x. 3 root root        20 Apr  6 21:12 sonatype-work
[root@ip-172-31-90-156 opt]#
[root@ip-172-31-90-156 opt]#
[root@ip-172-31-90-156 opt]# chown -R nexus:nexus nexus-3.91.0-07 sonatype-work/
[root@ip-172-31-90-156 opt]#
```

-R is to change files inside a folder also

->we need to swtich user also

```
[root@ip-172-31-90-156 opt]#
[root@ip-172-31-90-156 opt]#
[root@ip-172-31-90-156 opt]# ll
total 455680
drwxr-xr-x. 4 root  root          33 Apr  4 02:00 aws
drwxr-xr-x. 6 nexus nexus         94 Apr 10 14:04 nexus-3.91.0-07
-rw-r--r--. 1 root  root   466615526 Apr  7 15:15 nexus-3.91.0-07-linux-x86_64.tar.
gz
drwxr-xr-x. 3 nexus nexus         20 Apr  6 21:12 sonatype-work
[root@ip-172-31-90-156 opt]#
[root@ip-172-31-90-156 opt]# ll nexus-3.91.0-07
total 16
-rw-r--r--. 1 nexus nexus  1022 Apr  6 21:00 NOTICE.txt
-rw-r--r--. 1 nexus nexus 11249 Apr  6 21:00 OSS-LICENSE.txt
drwxr-xr-x. 2 nexus nexus    89 Apr 10 14:04 bin
drwxr-xr-x. 2 nexus nexus    26 Apr 10 14:04 deploy
drwxr-xr-x. 7 nexus nexus   143 Apr 10 14:04 etc
drwxr-xr-x. 3 nexus nexus    44 Apr  6 21:11 jdk
[root@ip-172-31-90-156 opt]#
[root@ip-172-31-90-156 opt]#
[root@ip-172-31-90-156 opt]# su - nexus
[nexus@ip-172-31-90-156 ~]$
[nexus@ip-172-31-90-156 ~]$
```

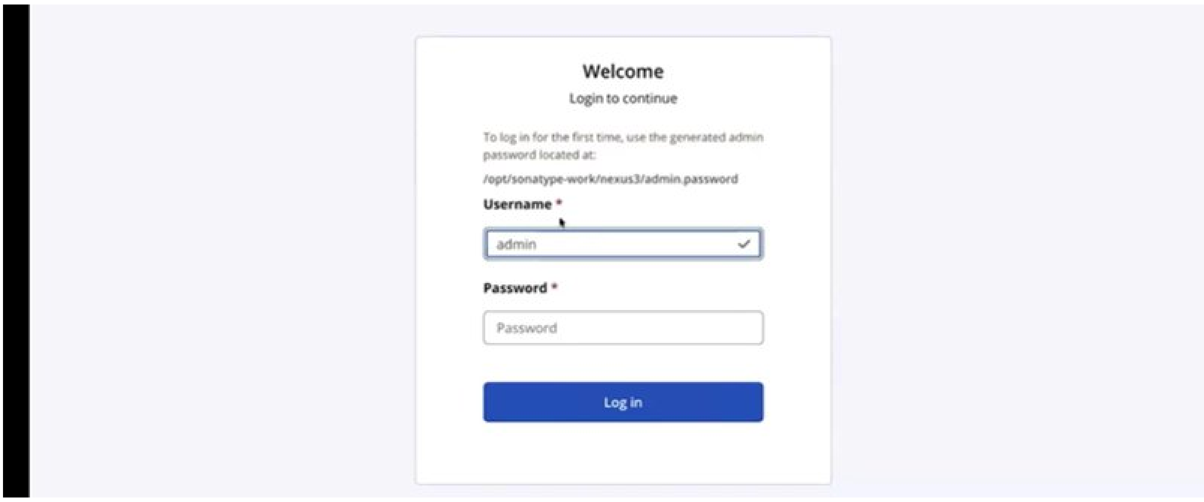
->goto bin and run file n chk status

```
[nexus@ip-172-31-90-156 ~]$ cd /opt/
[nexus@ip-172-31-90-156 opt]$
[nexus@ip-172-31-90-156 opt]$ ll
total 455680
drwxr-xr-x. 4 root  root          33 Apr  4 02:00 aws
drwxr-xr-x. 6 nexus nexus         94 Apr 10 14:04 nexus-3.91.0-07
-rw-r--r--. 1 root  root   466615526 Apr  7 15:15 nexus-3.91.0-07-linux-x86_64.ta
gz
drwxr-xr-x. 3 nexus nexus         20 Apr  6 21:12 sonatype-work
[nexus@ip-172-31-90-156 opt]$
[nexus@ip-172-31-90-156 opt]$ cd nexus-3.91.0-07/
[nexus@ip-172-31-90-156 nexus-3.91.0-07]$
[nexus@ip-172-31-90-156 nexus-3.91.0-07]$ ll
total 16
-rw-r--r--. 1 nexus nexus  1022 Apr  6 21:00 NOTICE.txt
-rw-r--r--. 1 nexus nexus 11249 Apr  6 21:00 OSS-LICENSE.txt
drwxr-xr-x. 2 nexus nexus    89 Apr 10 14:04 bin
drwxr-xr-x. 2 nexus nexus    26 Apr 10 14:04 deploy
drwxr-xr-x. 7 nexus nexus   143 Apr 10 14:04 etc
drwxr-xr-x. 3 nexus nexus    44 Apr  6 21:11 jdk
[nexus@ip-172-31-90-156 nexus-3.91.0-07]$
```

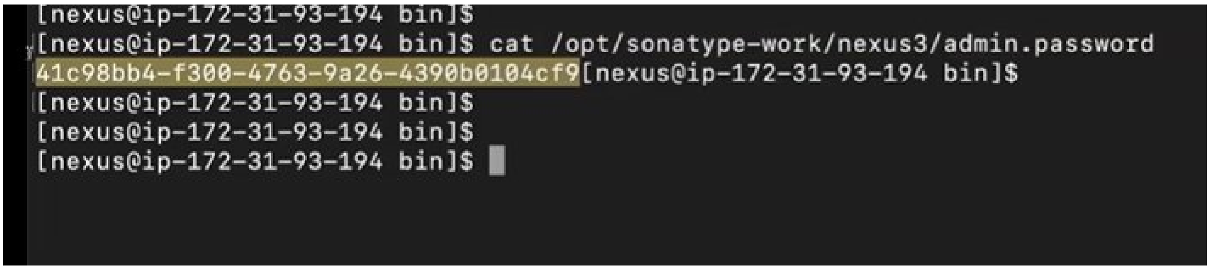
```
[nexus@ip-172-31-93-194 bin]$
[nexus@ip-172-31-93-194 bin]$ ./nexus status
[nexus@ip-172-31-93-194 bin]$
[nexus@ip-172-31-93-194 bin]$
```

```
[nexus@ip-172-31-93-194 bin]$
[nexus@ip-172-31-93-194 bin]$ ./nexus start
Starting nexus
[nexus@ip-172-31-93-194 bin]$
```

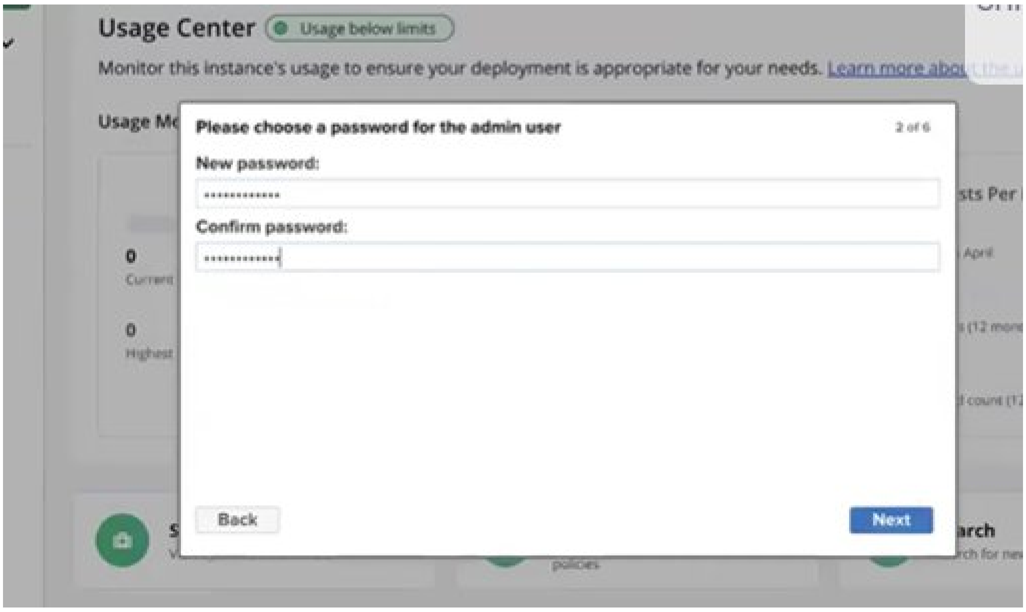
NOTE- nexus ni access cheyadaniki koncham tym pattidi, so wait



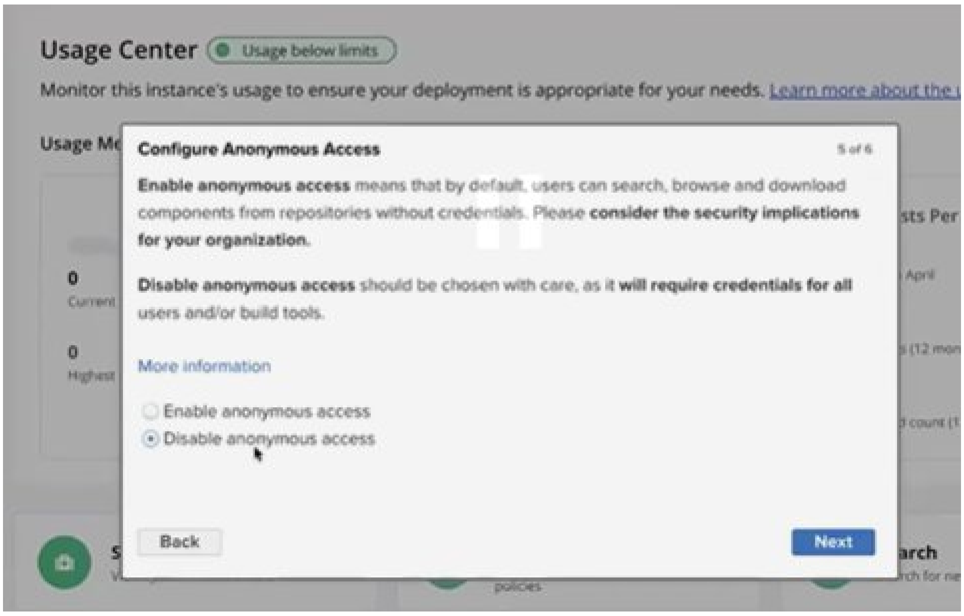
->fr pword goto that path



->set new pword



->next>diable>agree>finish



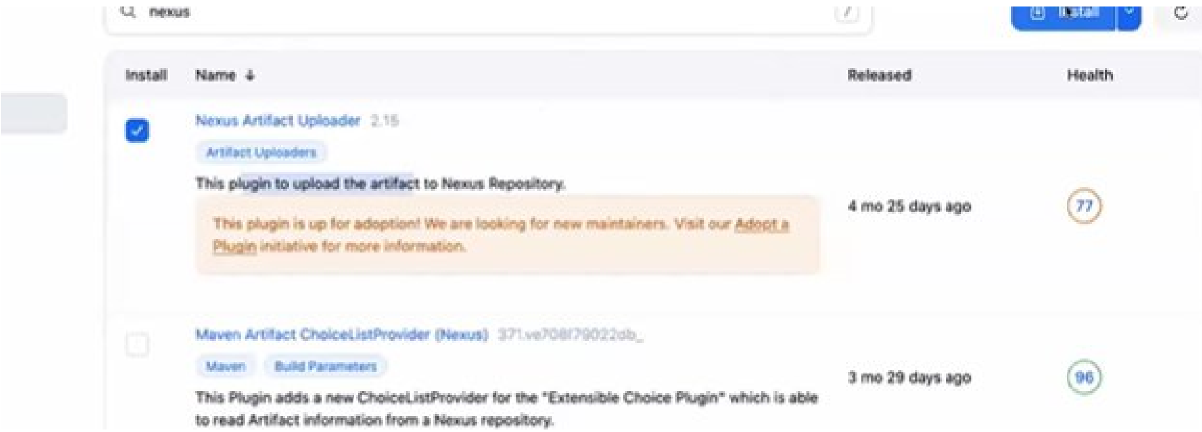
->in nexus create a repo to store our war/ear/jar files

settings>repository>new repo>sel mvn(hosted)>give any repo-name>in deployment policy sel allow redeploy(multiple times war file ni store chesukovachu)>create repo

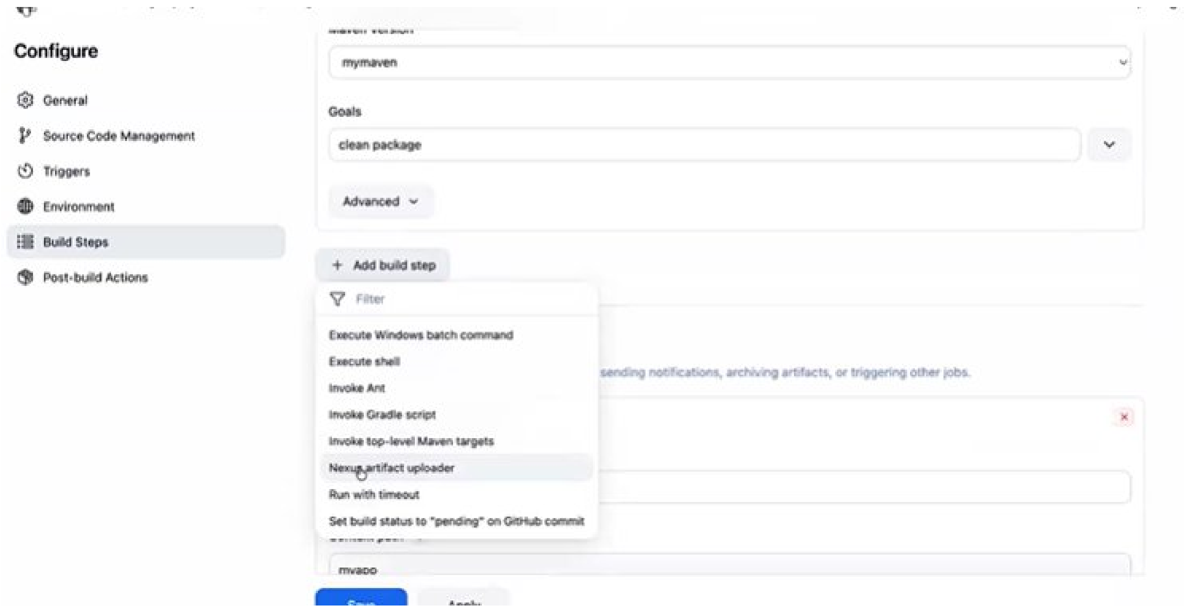
->in LHS database symbol la browse ani untadi sel it>in this we can find our war files

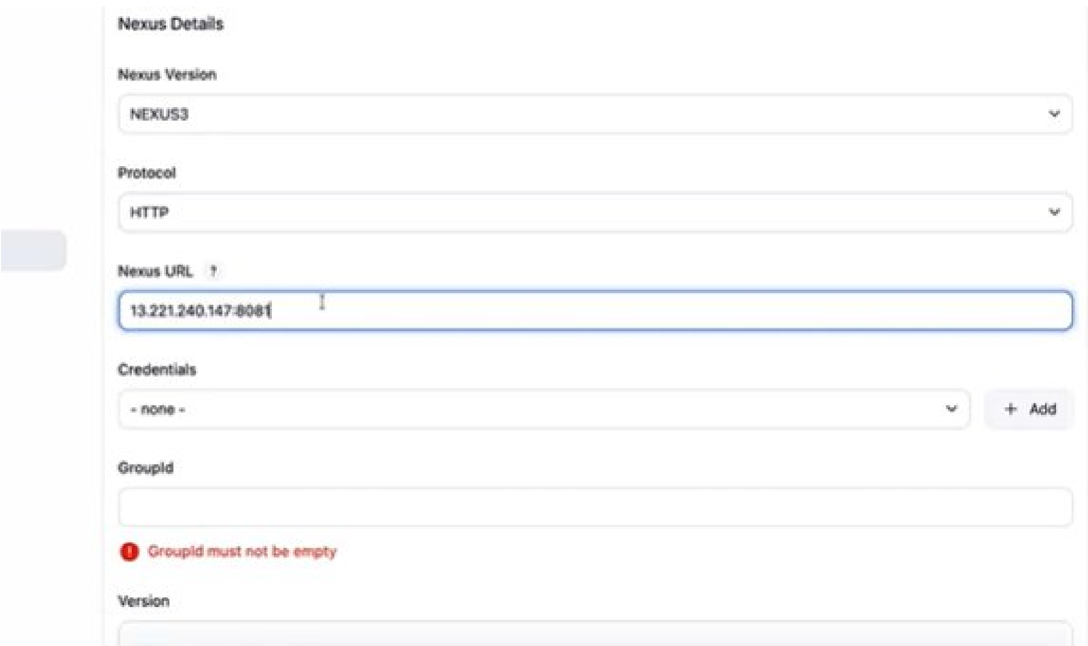
INTEGRATING JENKINS VTH NEXUS

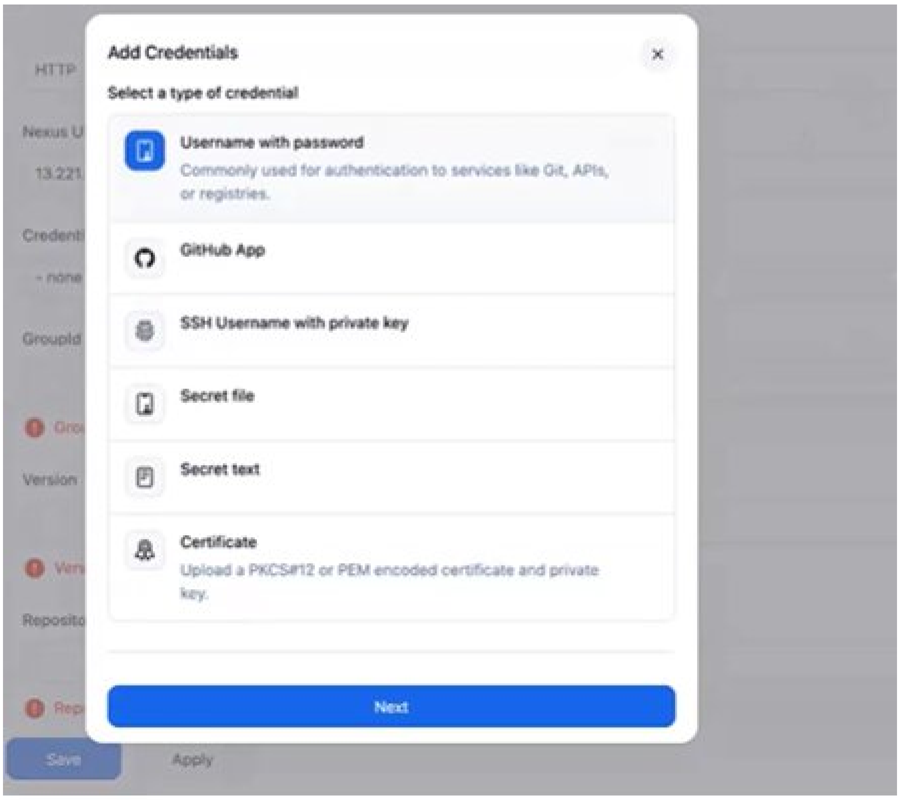
->in Jenkins goto manage Jenkins>plugins>available plugins>search fr nexus > sel nexus artifact uploader>install it

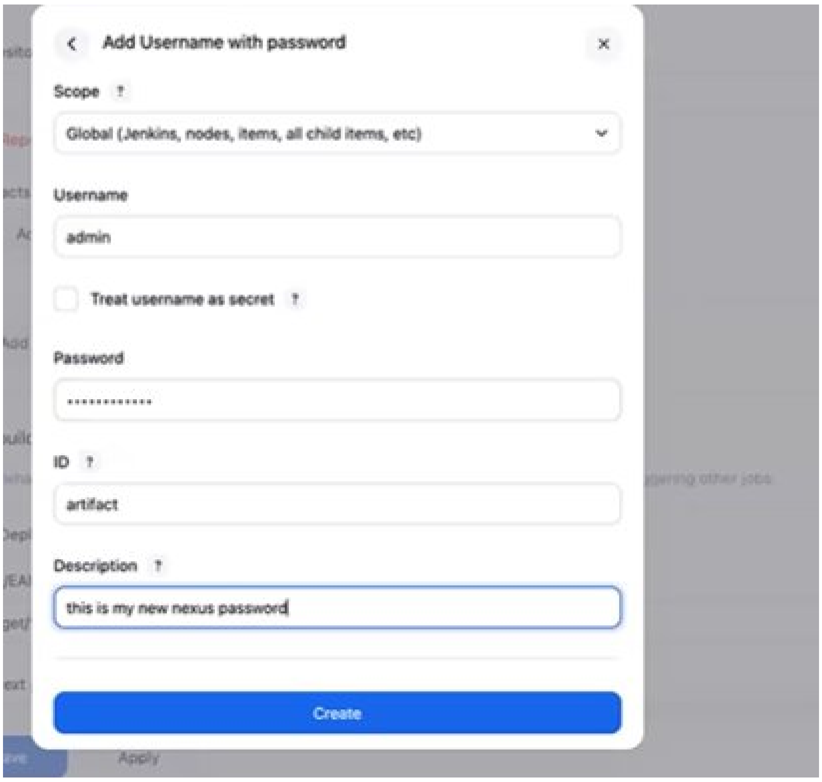


->once nexus is integrated goto pipeline>configure>build steps>after build we shd do artifact>in add build step sel nexus artifact uploader>in nexus version sel whtever version u r using>give nexus ip n port>give credentials>save

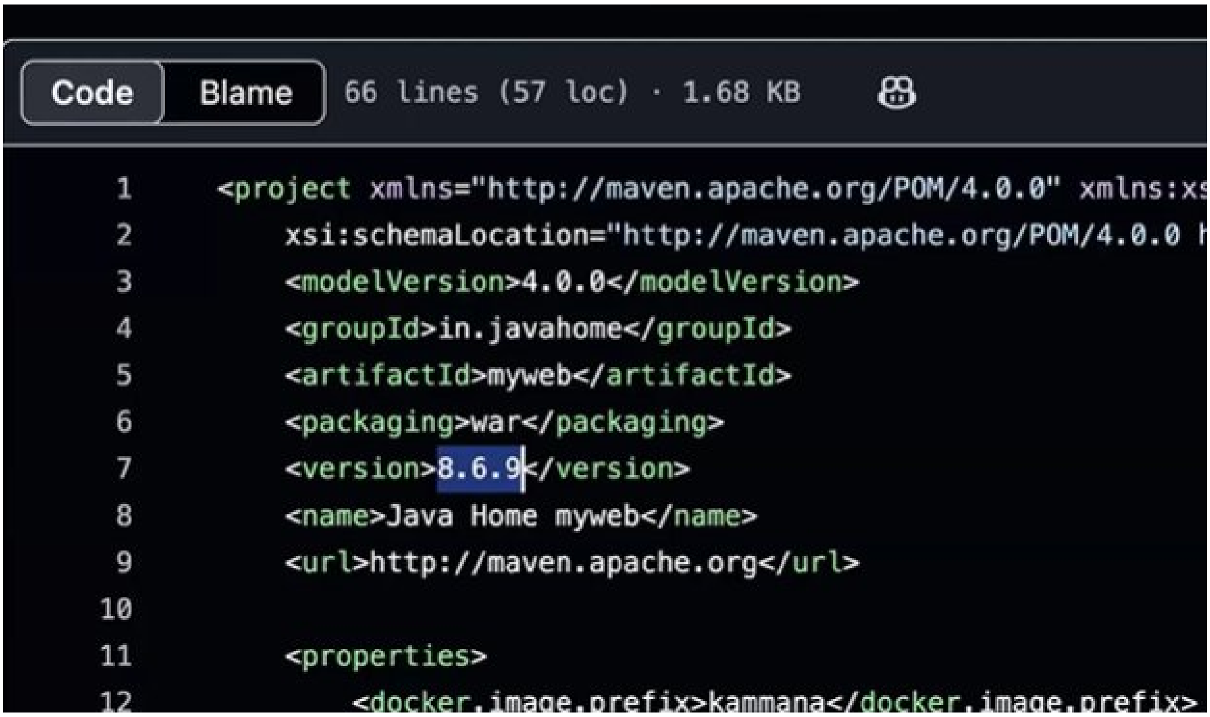

Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
mymaven
Goals
clean package
Advanced
Add build step
Filter
Execute Windows batch command
Execute shell
Invoke Ant
Invoke Gradle script
Invoke top-level Maven targets
Nexus artifact uploader
Run with timeout
Set build status to "pending" on GitHub commit
sending notifications, archiving artifacts, or triggering other jobs.
mvapp

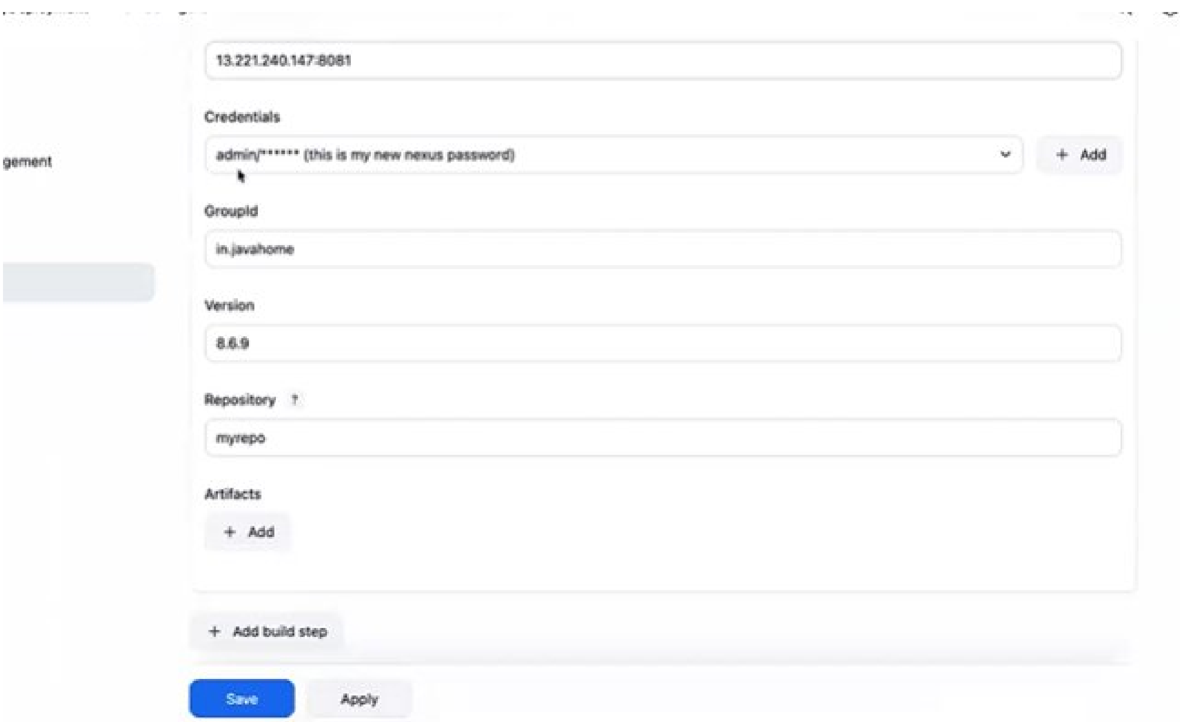

Nexus Details
Nexus Version
NEXUS3
Protocol
HTTP
Nexus URL
13.221.240.147:8081
Credentials
- none -
Add
GroupId
GroupId must not be empty
Version



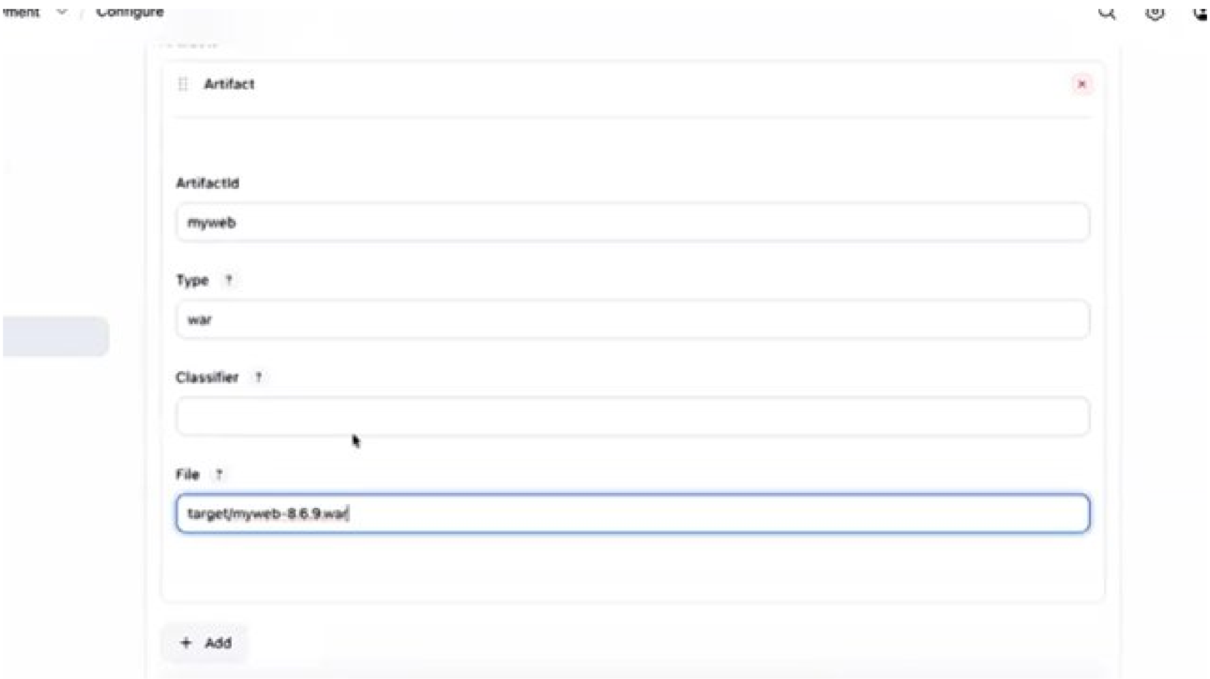


->in pom.xml file we can get the grp d n version which r given by developers



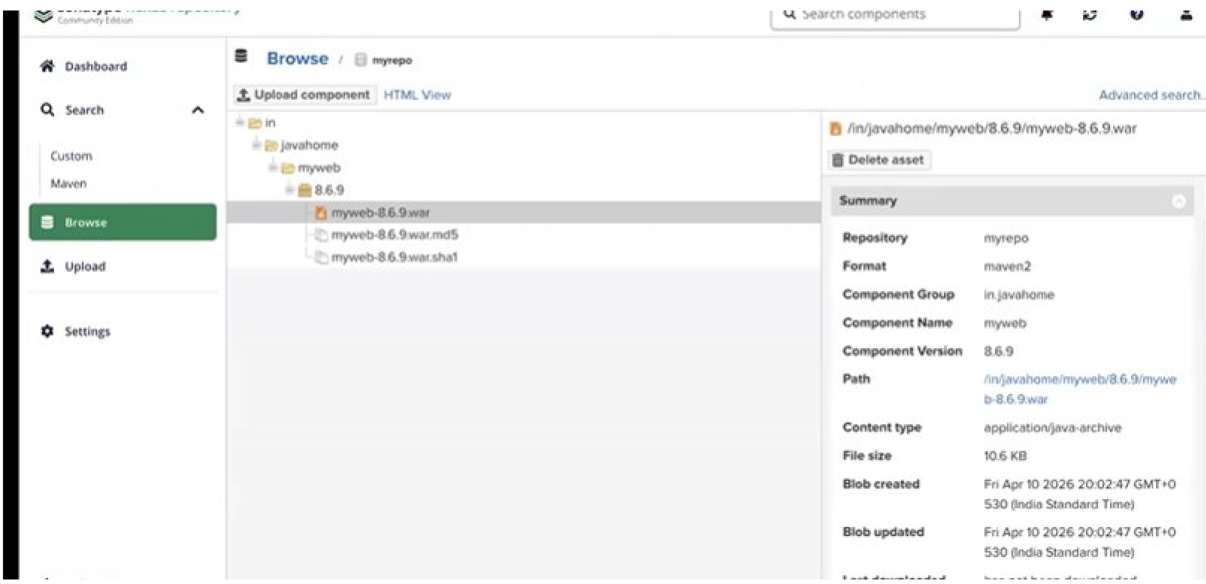


->in artifact give below



NOTE- in file we shd give full ledantey err vastundi

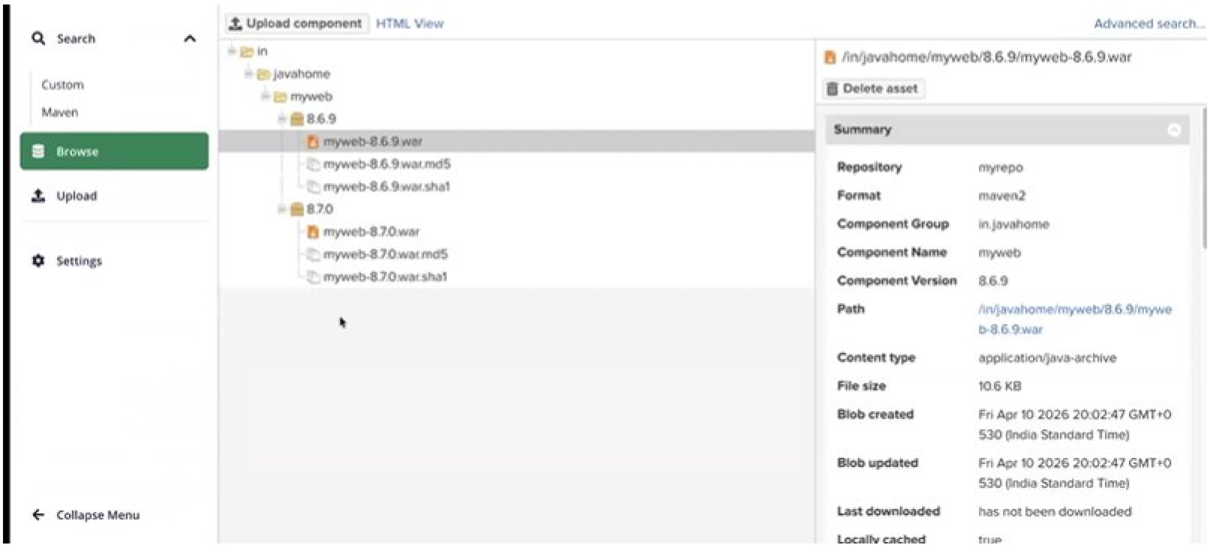
->now build it n refresh the nexus site, we get the wasr file in our repo



NOTE –1. Let’s edit src code ie index.jsp line-no 573 like summer/winter/wedding collection and change version in po.xml file n commit those 2 files

2. before build we need to change version also, now build it in Jenkins(otherwise build gets failed)

3 in nexus refresh n chk whether both versions war files are available



CLASS-8 FREE-STYLE OPTIONS DISCARD OLD BUILDS,THROTTLE BULD, BUILD AFTER PROJECTS ARE BUILT,DELETE WORKSPACE BEFORE BUILD STARTS,EXECUTE SHELL

throttle builds- particular time lo enni builds ey cheyyali annna concepts ki use chestamu

CLASS-9 RBAC, PARAMETERS & THIN BACKUP

RBAC

->role based access control

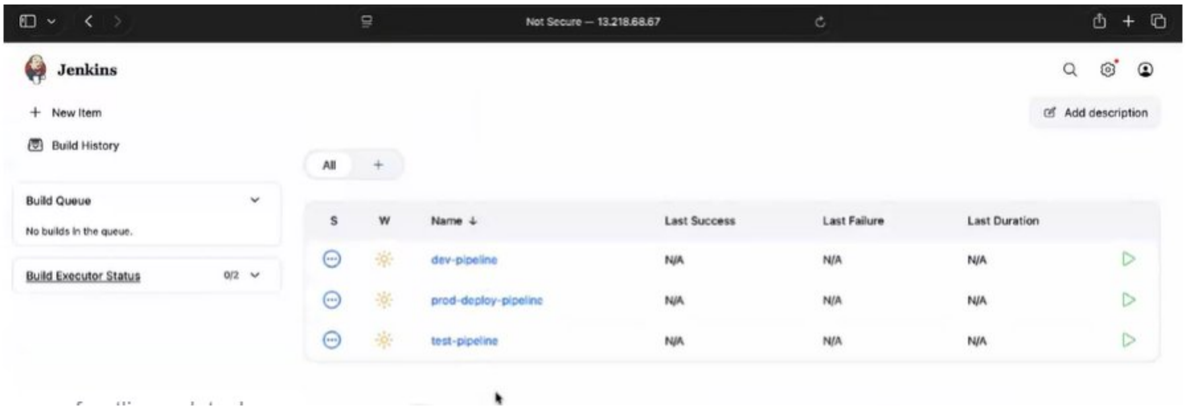
->oka project diff Jenkins dashboard em undadhu

->oka project diff teams like dev,devops,testing—prathi team lo oka new mem vacharu vallaki Jenkins access ivvachu but based on the valla requirement ey ivvali

ADDING AN USER

->manage Jenkins>users>create a user as a tester

->in another browser login created user eith credentials, if we see the dashboard of Mahesh, he has access to every pipeline



->since he has access to everything knowingly/unknowminglly he may do smtg and security issues may come

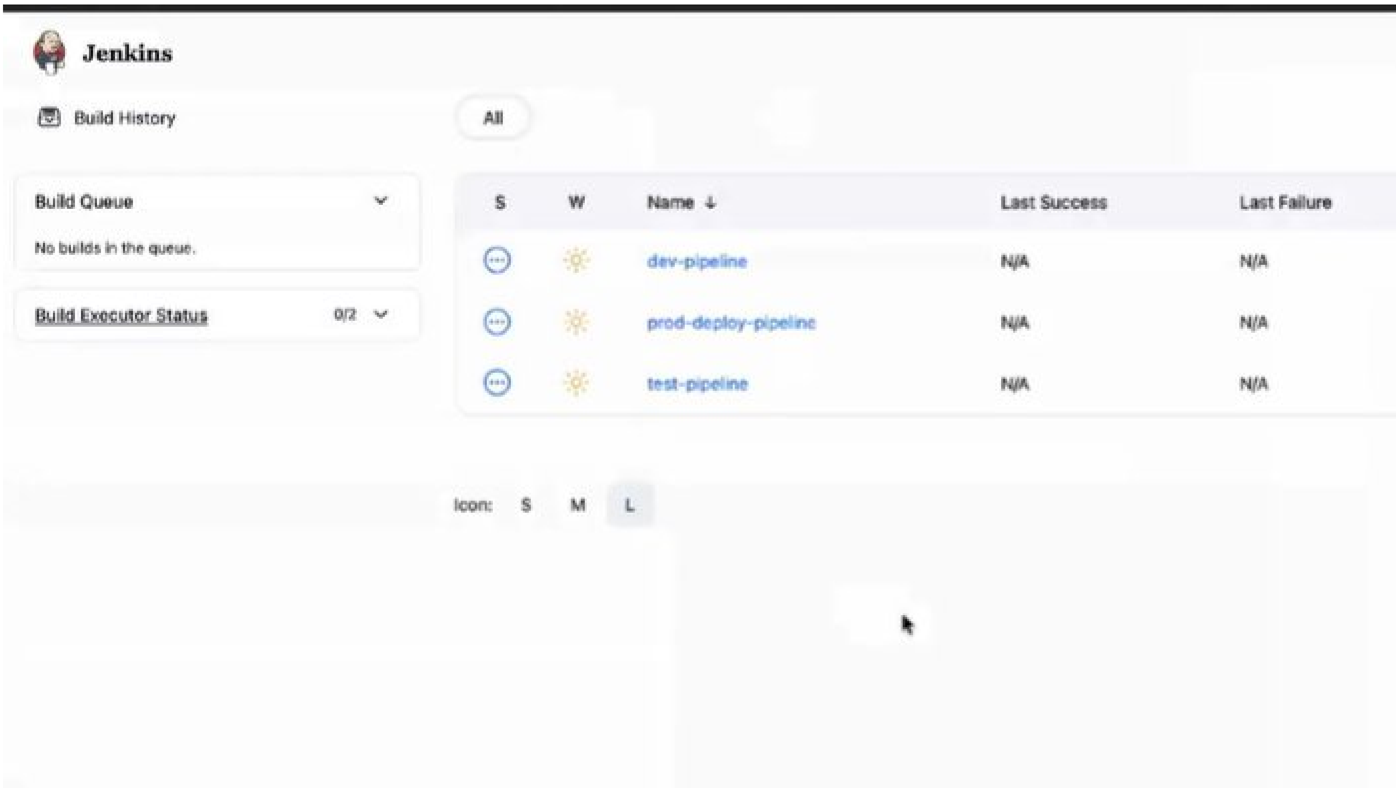
->in order to provide required access to Mahesh

Goto>manage Jenkins>security>under authorization sel project-based matrix authorization strategy add a def user of Jenkins

Now Mahesh dashboard resfresh it, becomes access denied.

->now add Mahesh user also n give only overall read permisiion n in run give read since he is tester n save it

->now refresh it the dashboard of Mahesh, he'll have only read permission

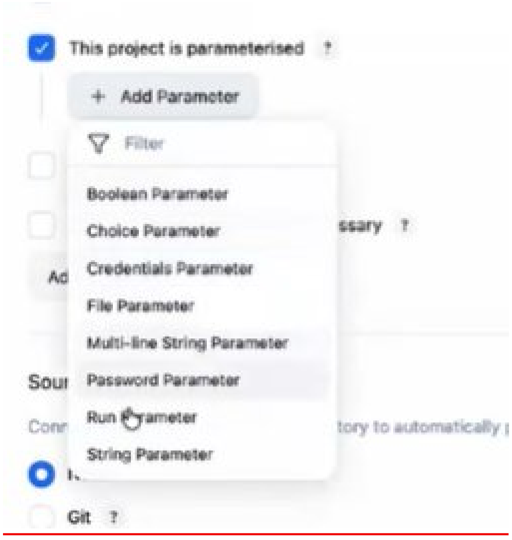


->as a Mahesh is tester he shd have access to only testing pipelines

->as a def user of Jenkins he/she has admin rights, goto testing pipeline>configure>sel enable project-based security,sel Mahesh user give permission to build n read under job>save

->now refresh Mahesh Jenkins, he'll get access to build n run fr test-pipeline

PARAMETERS



->build parameters- passing param's while we're building

->ma manager prathisari okokka branch ni build cheyyamnataru, pratisari build chesetapudu branch name change cheyyali

->so ala kakunda this project is paramterized option ni enable chesi manaki kavalisina branches anni ichesi, manaki ey branch build chesukovali antey adi build chesukuntey chalu

STRING PARAM

->sel this project is param>sel string param>give name>sel git >givr github url>under branches to build give */$string_name>save it

->now we won't see build now option, we get build with parameters option>click on it>give which branch u want to build

CHOICE PARAM

-> sel this project is param>sel choice param>give name>give choices line by line>sel git >givr github url>under branches to build give */$string_name>save it

->now we won't see build now option, we get build with parameters option>click on it>sel which branch u want to build

THINKBACKUP

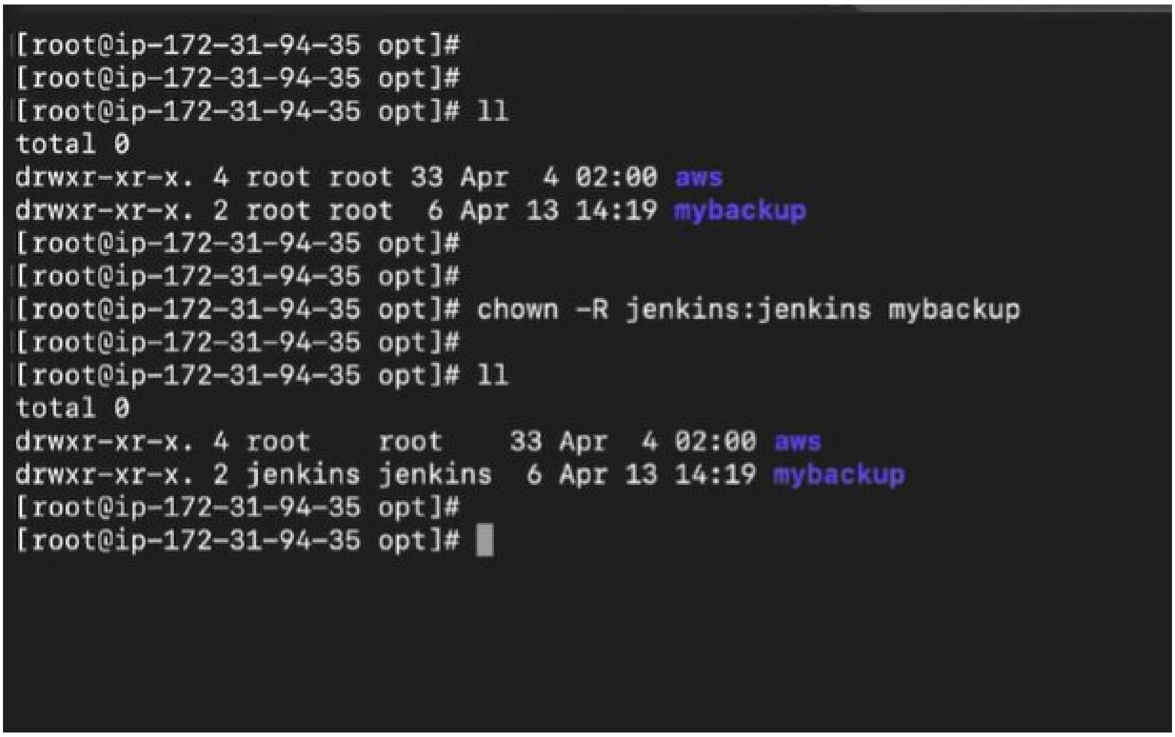
->build information ni back-up teesukovali antey ee thinbackup plugin use chestamu

->manam build info mottam Jenkins def path lo store avutadi

NOTE – console o/p manam server lo jenkin def path>log aney file lo store avutadi

->to install thinbaclup>plugins>available plugins>serch fr thinbackup n install>after installing>at last we see thinbackup option

->create a folder in opt path ie mybackup>change owners to that folder(def Jenkins path lo data store avutundi ga jobs aney folder ki owner Jenkins unnnaru but manam create chestunna user as root)so, need to change owners



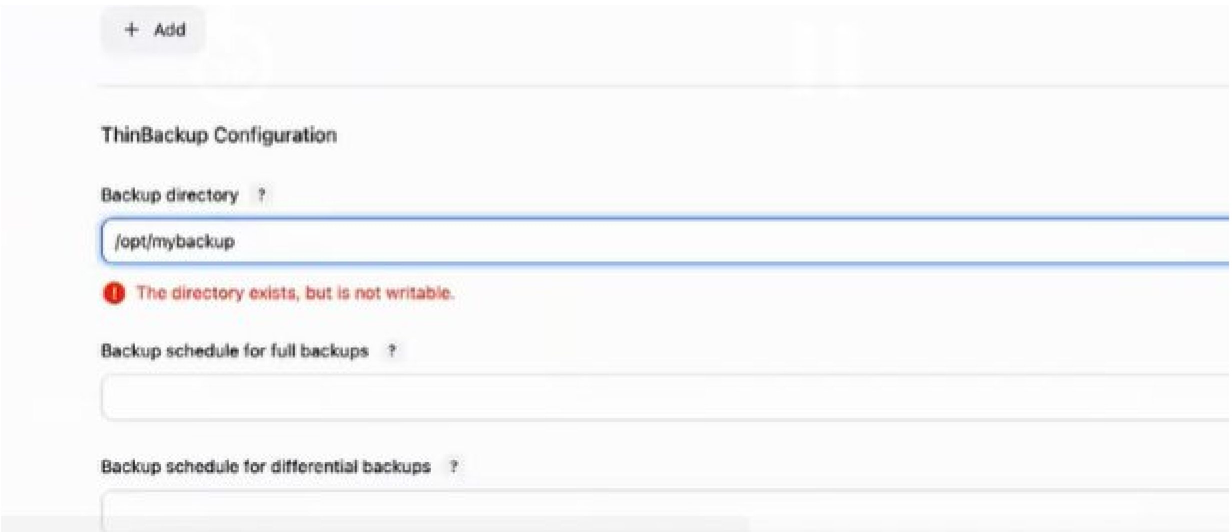
->server lo Jenkins install cheyyaganey def ga Jenkins user create avutadu

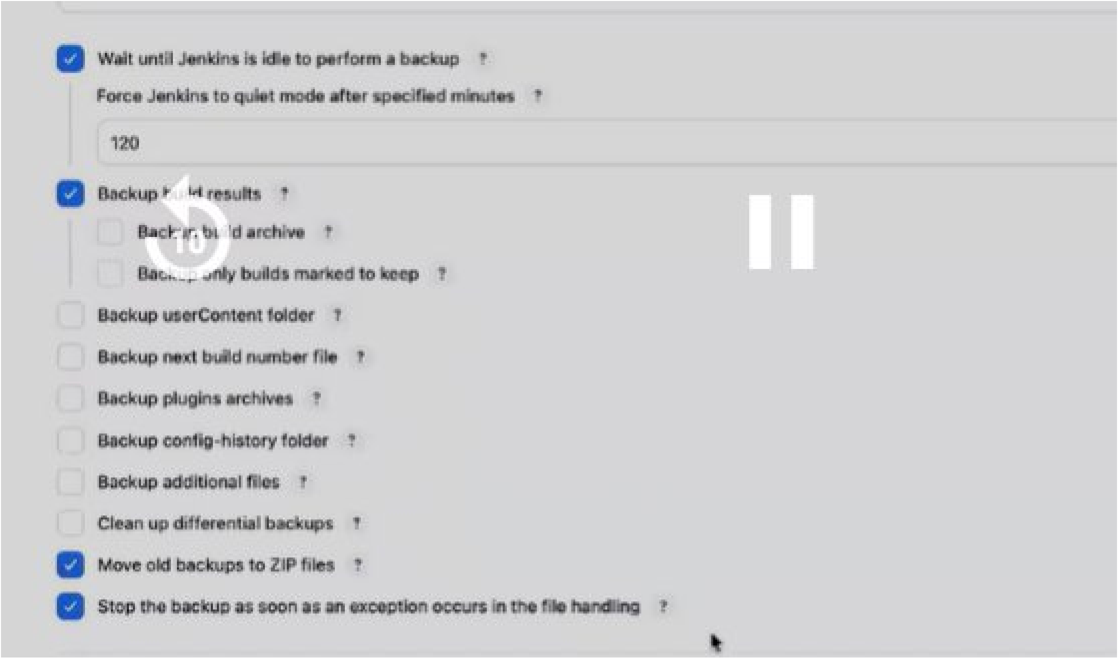
->kotta ga build run cheyye

->now need to configure where our backup shd be stored

in Jenkins goto manage Jenkins>system>thinkbackup config>give path to store underbackup dir >under backup build results sel move old backups to zip files>svae

->now click on backup now





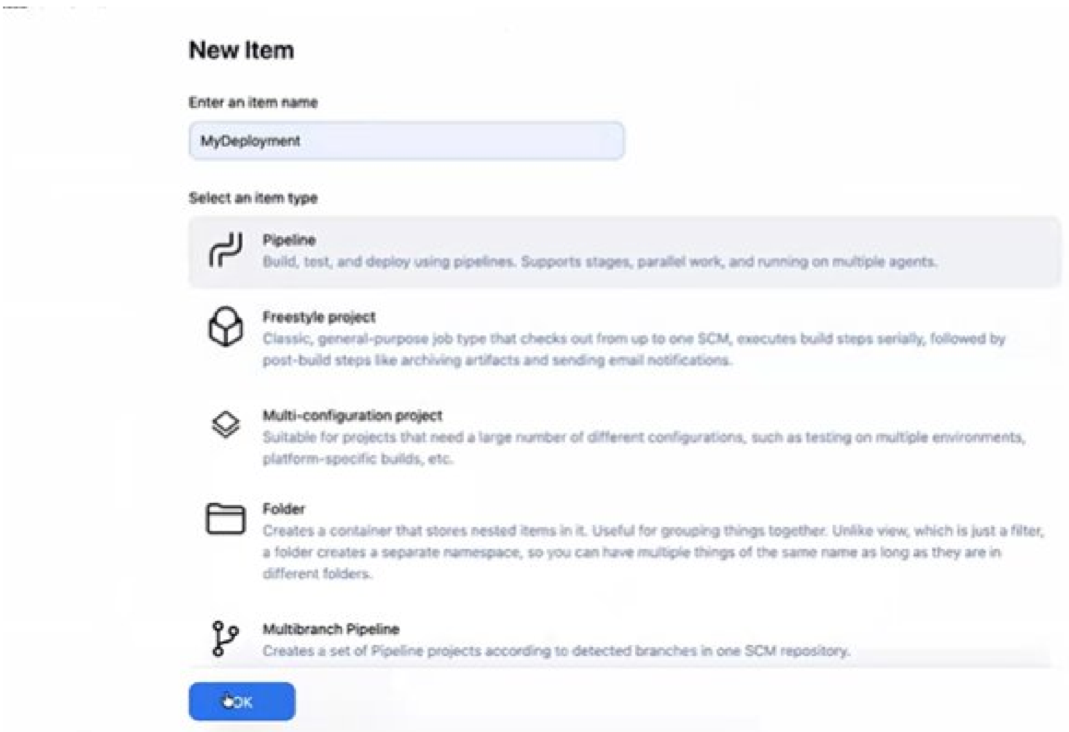
CLASS-10 PIPELINE TYPE JOB PART-1

->jenkins server vth c7i flex large n 10gb

->pipeline are of 2 types

1)declarative 2) scripted

->create Jenkins job by selecting pipeline

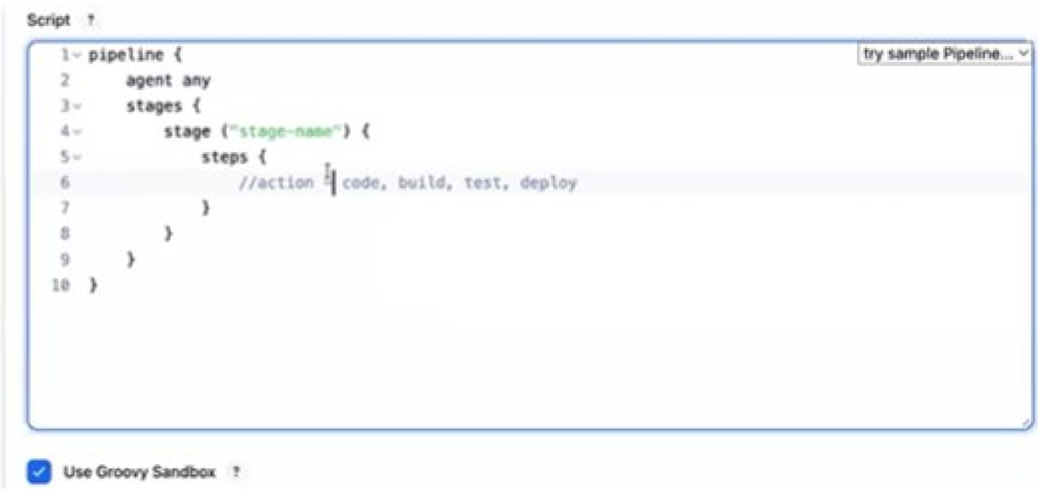


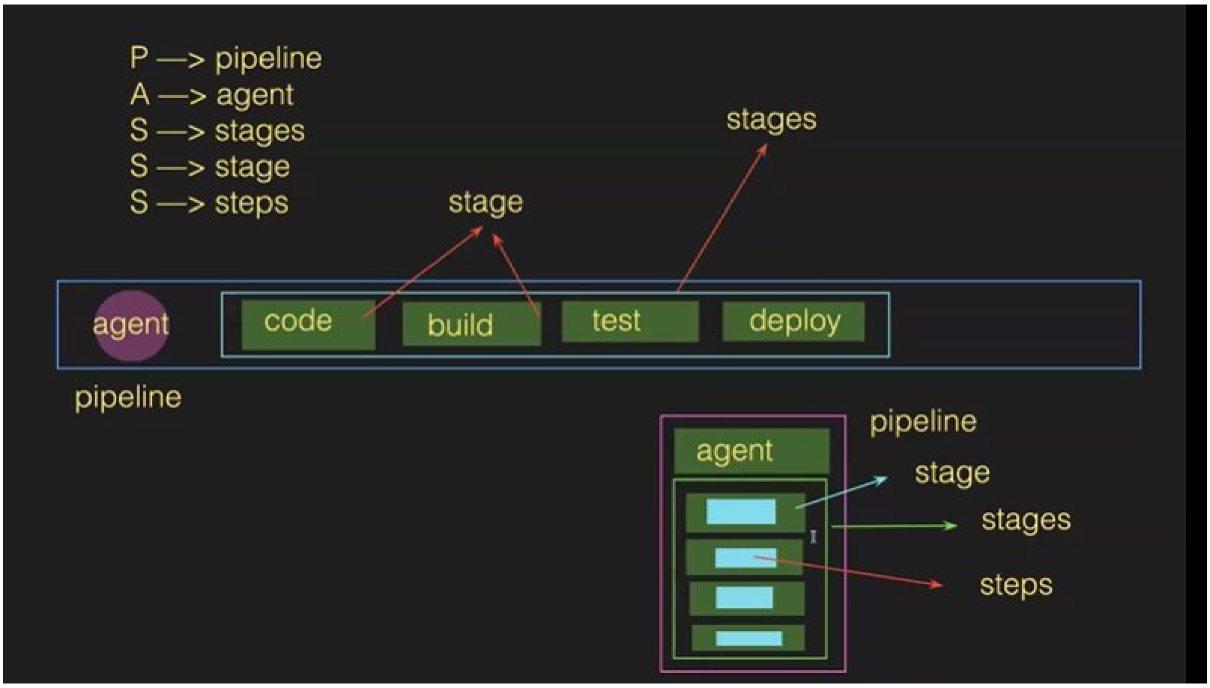
->we'll see all options which we have seen fr freestyle job but here we see script

->2 types of pipeline declarative n scripted

DECLARTIVE PIPELINE

->PASSS





Any=edi available ga untey adi run avvamani

Stage lo ey name aina ivvachu

->save n build now

->pipeline type job ki workspace undadhu

->adding 3 stgaes in pipeline

NOTE- stage lo oka step ey untadi, we can’t use more than 1step

->to run cmds in pipeline

Pipeline script

Script ?

```
                echo "this is stage-1"
            }
        }
        stage ("stage-2") {
            steps {
                sh 'touch mustafa'
            }
        }
        stage ("stage-3") {
            steps {
                sh 'mkdir devops'
            }
        }
    }
}
```

try sample Pipeline...

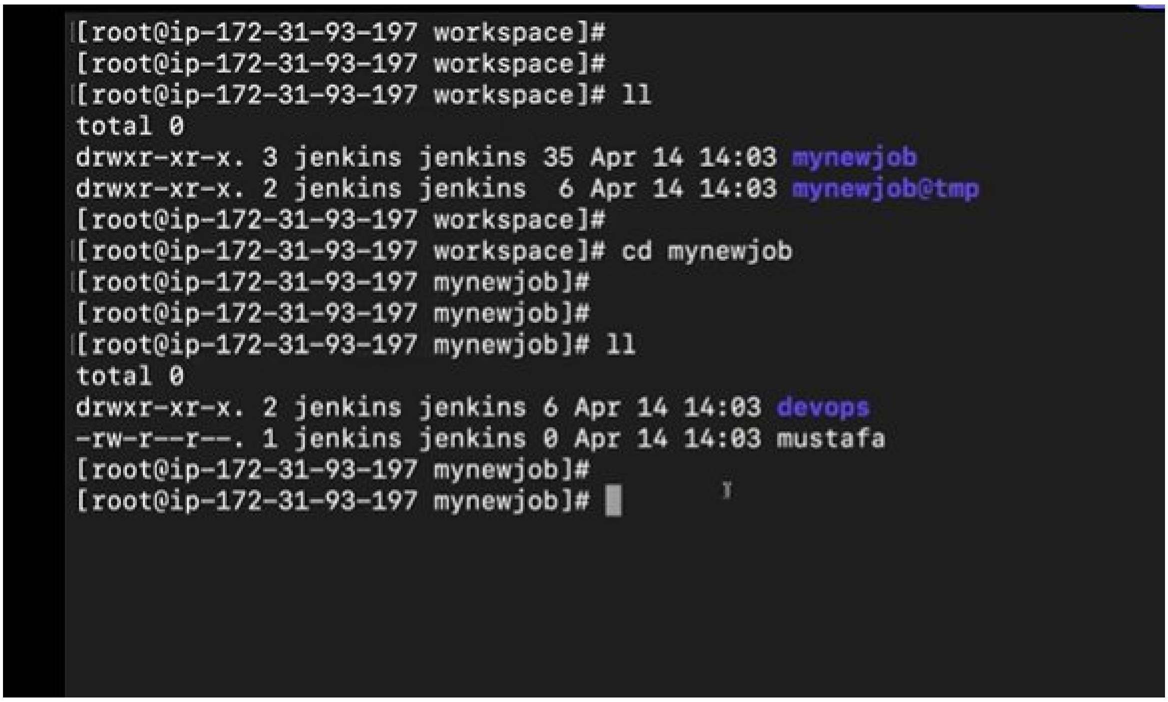
Use Groovy Sandbox ?

Pipeline Syntax

->

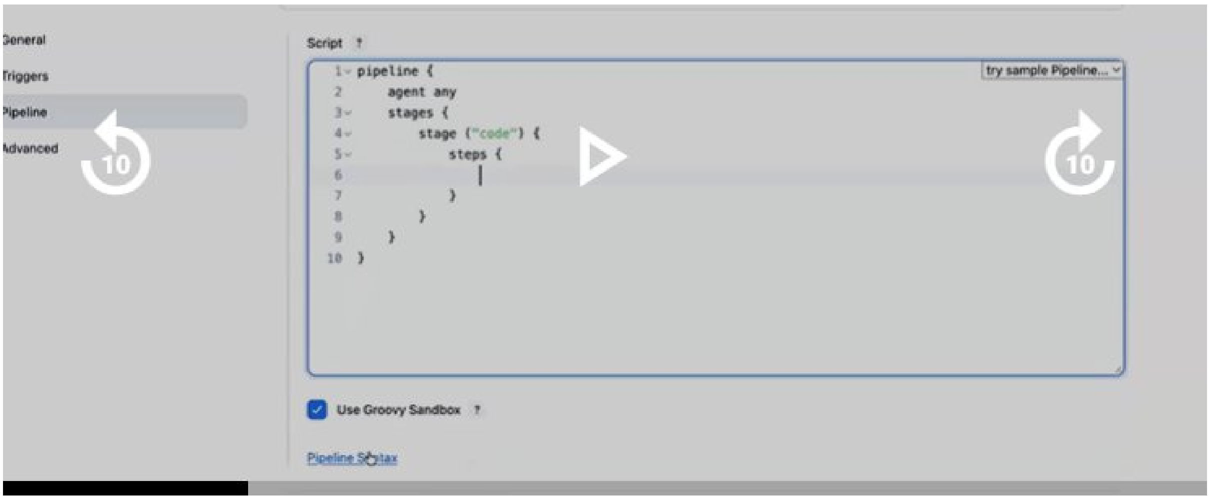
->to see pipeline o/p goto def path of Jenkins

```
[root@ip-172-31-93-197 ~]# cd /var/lib/jenkins/
[root@ip-172-31-93-197 jenkins]#
[root@ip-172-31-93-197 jenkins]#
[root@ip-172-31-93-197 jenkins]# ll
total 80
drwxr-xr-x.  3 jenkins jenkins    26 Apr 14 14:03 caches
-rw-r--r--.  1 jenkins jenkins  1660 Apr 14 13:43 config.xml
-rw-r--r--.  1 jenkins jenkins   156 Apr 14 13:42 hudson.model.UpdateCenter.xml
-rw-r--r--.  1 jenkins jenkins   370 Apr 14 13:43 hudson.plugins.git.GitTool.xml
-rw-------.  1 jenkins jenkins  1680 Apr 14 13:43 identity.key.enc
-rw-r--r--.  1 jenkins jenkins     7 Apr 14 13:44 jenkins.install.InstallUtil.lastExecVersion
-rw-r--r--.  1 jenkins jenkins     7 Apr 14 13:44 jenkins.install.UpgradeWizard.state
-rw-r--r--.  1 jenkins jenkins   184 Apr 14 13:44 jenkins.model.JenkinsLocationConfiguration.xml
-rw-r--r--.  1 jenkins jenkins   171 Apr 14 13:42 jenkins.telemetry.Correlator.xml
drwxr-xr-x.  4 jenkins jenkins    42 Apr 14 14:01 jobs
drwxr-xr-x.  3 jenkins jenkins    45 Apr 14 14:00 logs
-rw-r--r--.  1 jenkins jenkins  1037 Apr 14 13:42 nodeMonitors.xml
-rw-r--r--.  1 jenkins jenkins    46 Apr 14 14:03 org.jenkinsci.plugins.workflow.flow.FlowExecutionLi
st.xml
drwxr-xr-x. 97 jenkins jenkins 16384 Apr 14 13:59 plugins
-rw-r--r--.  1 jenkins jenkins   258 Apr 14 13:59 queue.xml
-rw-r--r--.  1 jenkins jenkins    64 Apr 14 13:42 secret.key
-rw-r--r--.  1 jenkins jenkins     0 Apr 14 13:42 secret.key.not-so-secret
drwx------.  2 jenkins jenkins 16384 Apr 14 14:00 secrets
drwxr-xr-x.  2 jenkins jenkins   149 Apr 14 13:43 updates
drwxr-xr-x.  2 jenkins jenkins    24 Apr 14 13:42 userContent
drwxr-xr-x.  3 jenkins jenkins    86 Apr 14 13:44 users
drwxr-xr-x.  4 jenkins jenkins    42 Apr 14 14:03 workspace
[root@ip-172-31-93-197 jenkins]#
[root@ip-172-31-93-197 jenkins]#
```

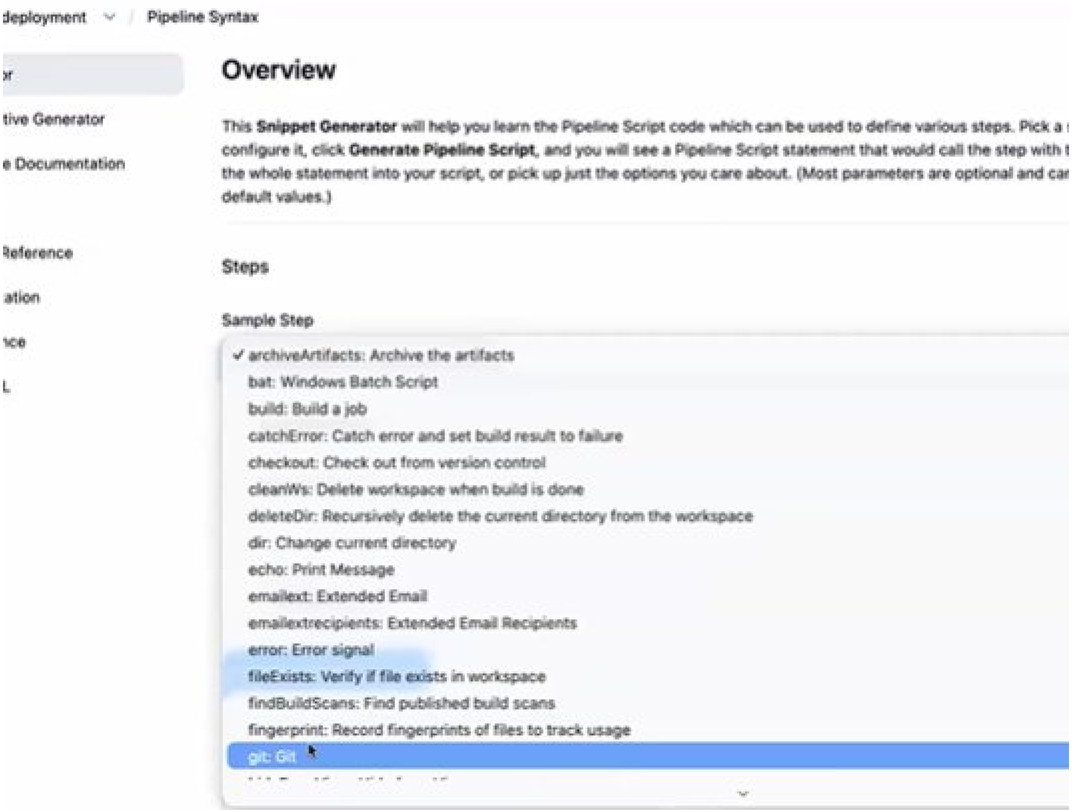


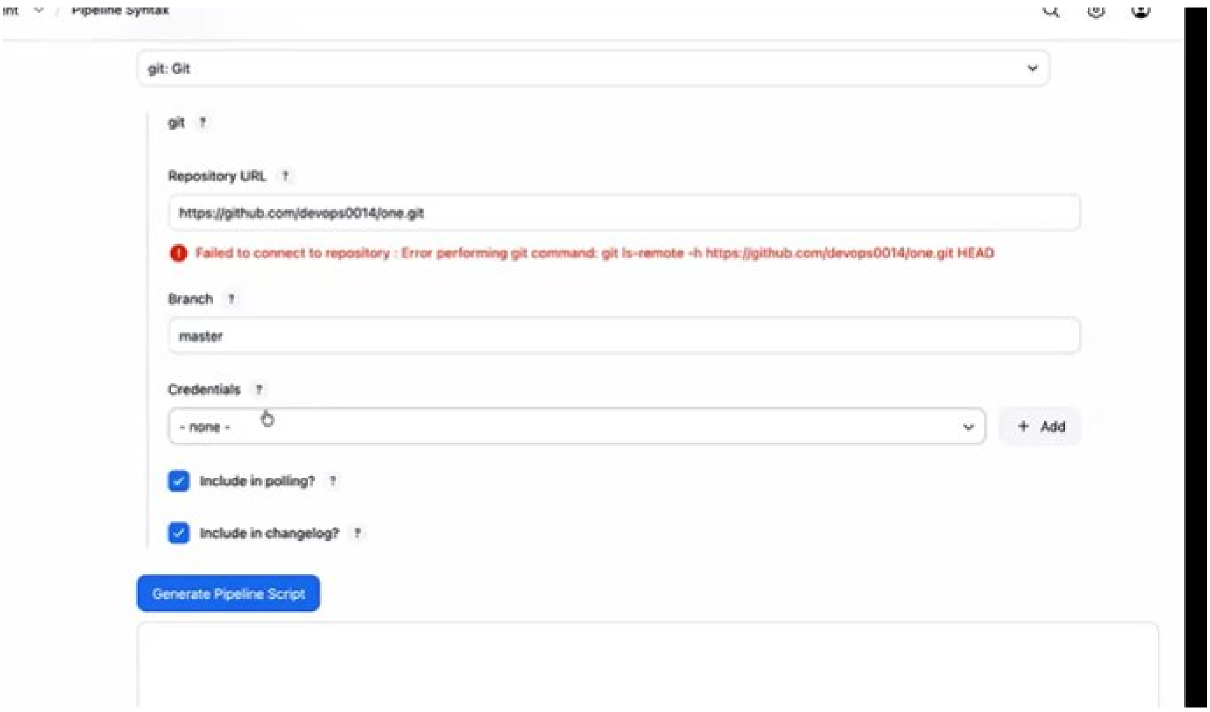
[1. CODE TO CI SERVER:]()

->getting code frm github using pipeline, use below pipeline syntax option



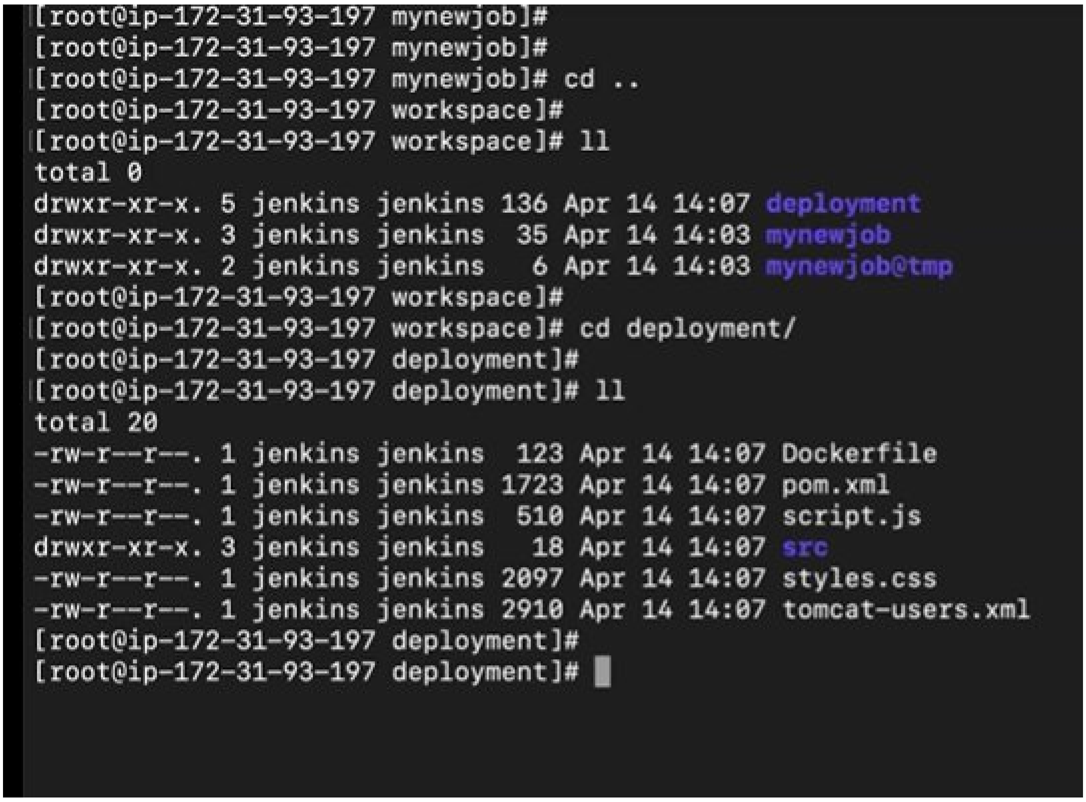
Click on pipeline syntax>sel git>give url>generate pipeline script





Now see pipeline created >save>build now

->in server chk whether code got into CI server or not



2. BUILD STAGE:

->fr build we need mvn, add it in tools

Goto pipeline>configure>in script after agent in tools give mvn>in new stage mention mvn clean package>

Script ?

```
pipeline {
    agent any
    tools {
        maven 'mymaven'
    }
    stages {
        stage ("code") {
            steps {
                git 'https://github.com/devops0014/one.git'
            }
        }
    }
}
```

Use Groovy Sandbox ?

Script ?

```
    tools {
        maven 'mymaven'
    }
    stages {
        stage ("code") {
            steps {
                git 'https://github.com/devops0014/one.git'
            }
        }
        stage ("Build") {
            steps {
                sh 'mvn clean package'
            }
        }
    }
```

Use Groovy Sandbox ?

Pipeline Syntax

->save it n build now>in server chk target file created or not

```
[root@ip-172-31-93-197 deployment]#
[root@ip-172-31-93-197 deployment]#
[root@ip-172-31-93-197 deployment]# ll
total 20
-rw-r--r--. 1 jenkins jenkins  123 Apr 14 14:07 Dockerfile
-rw-r--r--. 1 jenkins jenkins 1723 Apr 14 14:07 pom.xml
-rw-r--r--. 1 jenkins jenkins  510 Apr 14 14:07 script.js
drwxr-xr-x. 3 jenkins jenkins   18 Apr 14 14:07 src
-rw-r--r--. 1 jenkins jenkins 2097 Apr 14 14:07 styles.css
drwxr-xr-x. 4 jenkins jenkins   70 Apr 14 14:12 target
-rw-r--r--. 1 jenkins jenkins 2910 Apr 14 14:07 tomcat-users.xml
[root@ip-172-31-93-197 deployment]#
[root@ip-172-31-93-197 deployment]#
[root@ip-172-31-93-197 deployment]# ll target/
total 12
drwxr-xr-x. 2 jenkins jenkins    28 Apr 14 14:12 maven-archiver
drwxr-xr-x. 4 jenkins jenkins    89 Apr 14 14:12 myweb-8.7.1
-rw-r--r--. 1 jenkins jenkins 10894 Apr 14 14:12 myweb-8.7.1.war
[root@ip-172-31-93-197 deployment]#
[root@ip-172-31-93-197 deployment]#
[root@ip-172-31-93-197 deployment]#
```

3. DEPLOY STAGE

->deploy cheyyali antey tomcat server kavali, so tomcat server ni install chesi setup chesukovali

->deploy ki manam deploy to container plugin install chesamu

->click on pipeline syntax>sel deploy to war/ear container(okkokasari kanipinchadu restart jenkins) under sample step>under war/ear files lo give target/*.war>under context path give any name like myapp>under container sel tomcat 9>under credentials click on add>global>username with password>next>give uname n password of tomcat>give description>create>sel that created credentials>generate script>now build it

->now goto tomcat n search app got deployed or not

NEXUS INETGRATION

->create a instance fr nexus n install nexus(chk version in official web n install)

->goto settings>repositories>create a repo>maven2 hosted>name>under deployment policy give redeploy

->in manage Jenkins install nexus artifact uploader

->after build add stage fr artifact>pipeline syntax>nexus artifact uploader>nexus version3>hhtp>url>credentials>add>add uname with pword>sel that created credentials>group id frm pom.xml>version>repo>artifactid frm pom.xml>type war>in file target/myweb-8.7.1.war>generate pipeline script

->now paste that script in artifact stage and build it

->chk artifact in nexus

CLASS-11 PIPELINE TYPE JOB PART-2

->install java21 version in Jenkins server

->Jenkins.io>downloads>redhat>in cmds we can see which version is supporting

->pipeline syntax gurtunchukovalsina avasaram ledu>right side hello world click chesthey chalu

->by def ga master branch lodi vastadi

```
git "https://github.com/devops0014/one.git"
```

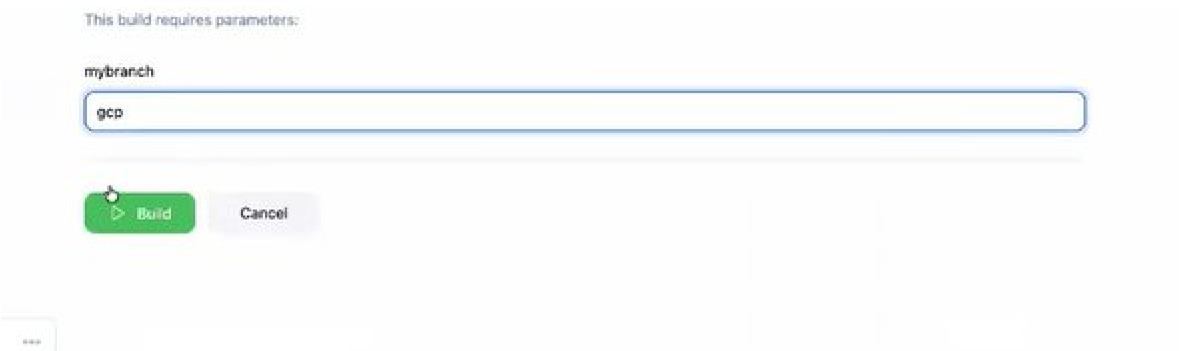
->oka particular branch dhi kavali antey pipeline syntax use chesukovadam ey

```
git branch: 'main', url: 'https://github.com/devops0014/one.git'
```

PARAMETERS

->job>configure>in general sel this project is paramterized>sel string param>give name>in pipeline give name>save it

->now build with param, give name of any branch>build



->chk whether code is into ci server or not

->u can use choice param as well

TASK – get code frm priv repo using pipeline

PIPELINE FOR SLAVE

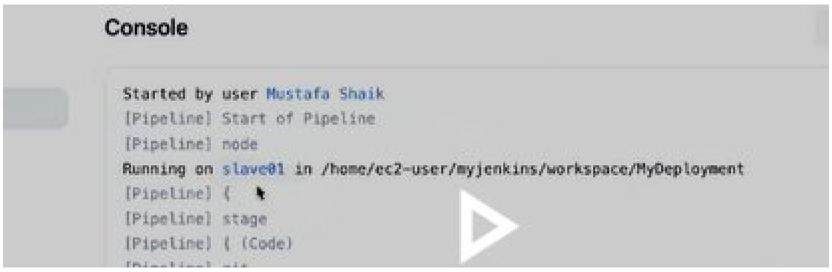
->Launch server fr slave using c7i flex large

->install java21 in slave

->Jenkins>manage Jenkins>nodes>add new node>permanent agent>create>no of executor lo 1>in remote root direc give /home/ec2-user/any_foldername>any label name>in usage sel only build jobs….>in launch method sel launch agents via ssh>in host give ip of slave>in cred click on add>global>ssh uname vth priv key>give any id n des>in unmae give ec2-user>in priv-key give u selected key-pair pem file>in host-key verific sel non-verifying verf strategy>save it

->now goto job>configure>in pipeline give agent n in node label name shd match vth while creating node

->in console o/p we can see that pipeline is running on slave only



NOTE- multiple slaves lo okey sari run cheyyacha

No

VARIABLES

1) global var's

->job>configure>pipeline>under agent>give var's

Script ?

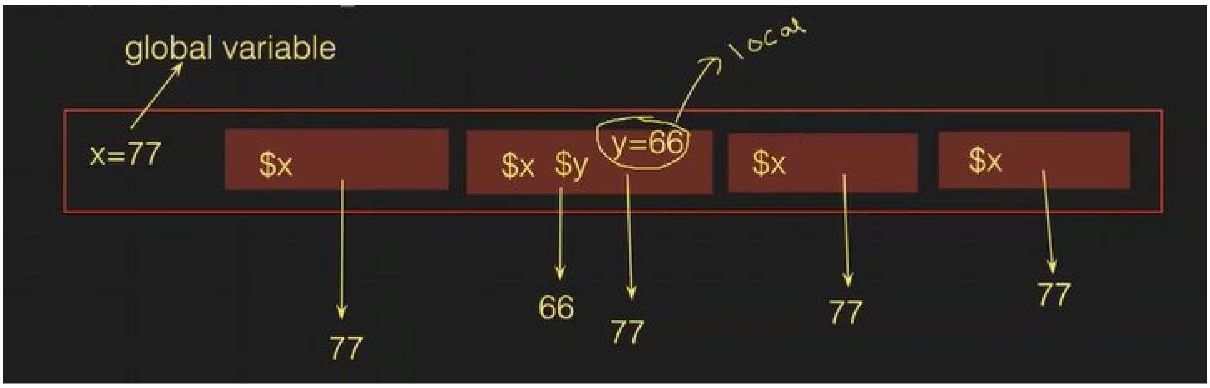
```
pipeline {
    agent {
        node {
            label 'dev'
        }
    }
    environment {
        tool = "docker"
        topic = "devops"
        day = "thursday"
    }

    stages {
        stage('Code') {
            steps {
```

```
    environment {
        tool = "docker"
        topic = "devops"
        day = "thursday"
    }

    stages {
        stage('Code') {
            steps {
                echo "Welcome to $topic, we are learning $tool on $day"
            }
        }
    }
}

```

Docker,k8s lo ee var’s use chestamu

->var’s of 2 types

1)local 2)global



Script ?

```
        }
    }
    environment {
        x = 77
    }

    stages {
        stage('Code') {
            steps {
                echo "the value of x is $x"
            }
        }
    }
}

```

Use Groovy Sandbox ?

```
        }
        stage('Build') {
            steps {
                echo "the value of x is $x"
            }
        }
        stage('Test') {
            steps {
                echo "the value of x is $x"
            }
        }
        stage('Deploy') {
            steps {
                echo "the value of x is $x"
```

Use Groovy Sandbox ?

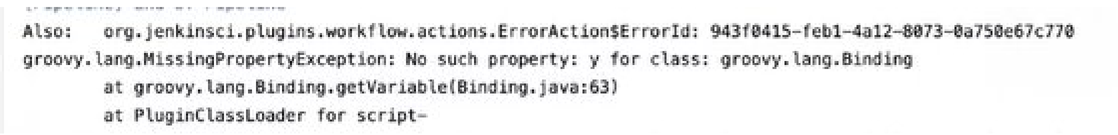
NOTE – oka stage ki ichina name inko stage ki ivvakodadhu

2) local var’s

Script ?

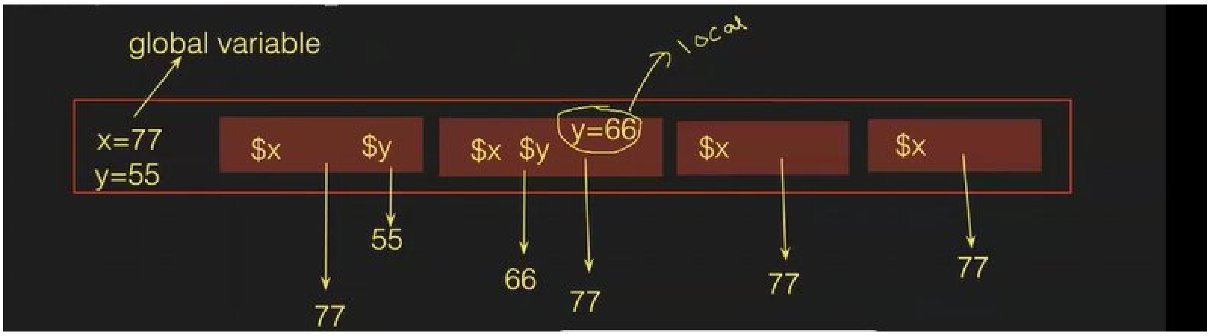
```
        }
        stage('Build') {
            environment {
                y = 66
            }
            steps {
                echo "the value of x is $x and y value is $y"
            }
        }
        stage('Test') {
            steps {
                echo "the value of x is $x"
            }
        }
```

NOTE- oka stage lo create chesina var inko stage lo use cheyyakodadhu err vastadi

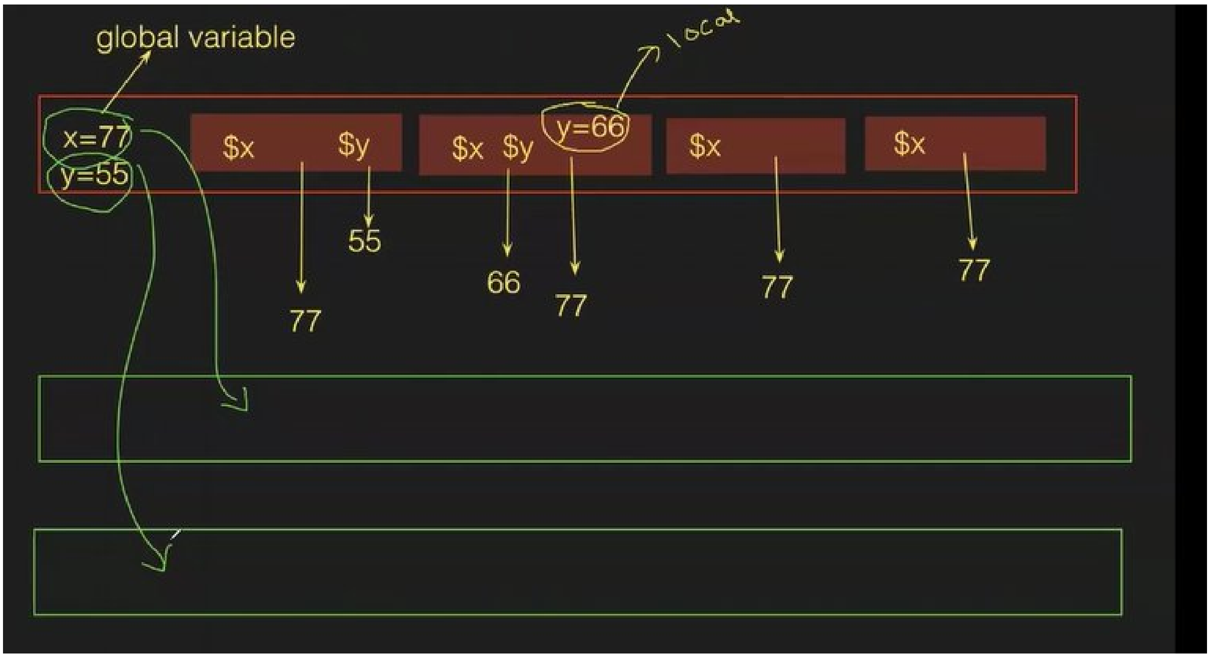


```
Also:   org.jenkinsci.plugins.workflow.actions.ErrorAction$ErrorId: 943f0415-feb1-4a12-8073-0a750e67c770
groovy.lang.MissingPropertyException: No such property: y for class: groovy.lang.Binding
        at groovy.lang.Binding.getVariable(Binding.java:63)
        at PluginClassLoader for script-
```

->local var ki preference istadi

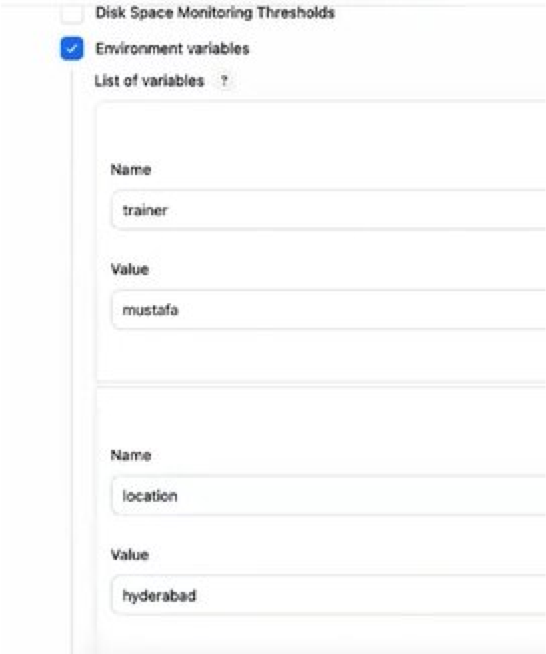


->oka var ni oka pipeline lo global ga declare chesamu, adi diff pipelines lo use chesukovacha –NO but Jenkins level lo create chesthey use chesukovachu

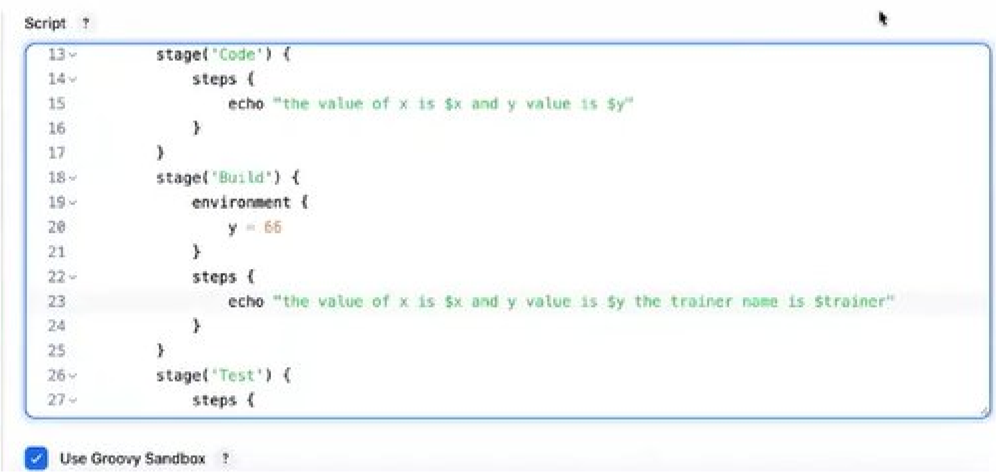


3) Jenkins global var's

->manage jenkins>sys>in global properties sel env var's>add>give any name



->now we are using that var in pipeline



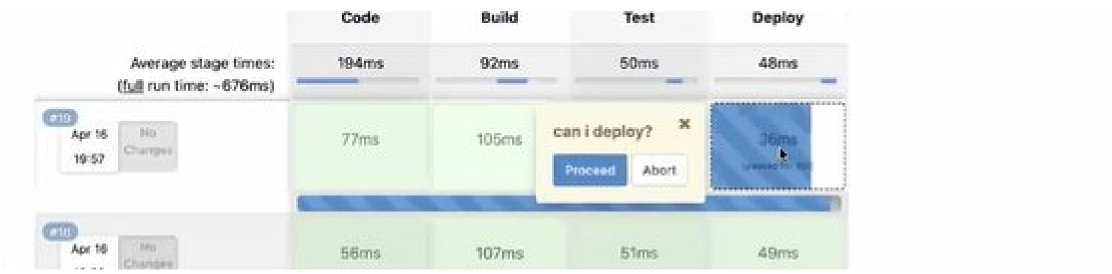
->now create new job vth pipeline>in pipeline use that var

->after building chk whether that var is able to use in any pipeline

->Jenkins lo create chesy var's koda global var ey

DEPLOY APPROVAL FOR PRODUCTION

```
            }
        }
        stage('Test') {
            steps {
                echo "the value of x is $x"
            }
        }
        stage('Deploy') {
            input {
                message 'can i deploy?'
            }
            steps {
                echo "the value of x is $x and our $trainer is good boy"
            }
        }
```

->proceed click chesthey deploy avutadi,abort chessthey cancel avutadi malli build chesthey proceed avutaru

MULTIPLE CMDS IN SCRIPT

->a file, folder server lo chusukovali

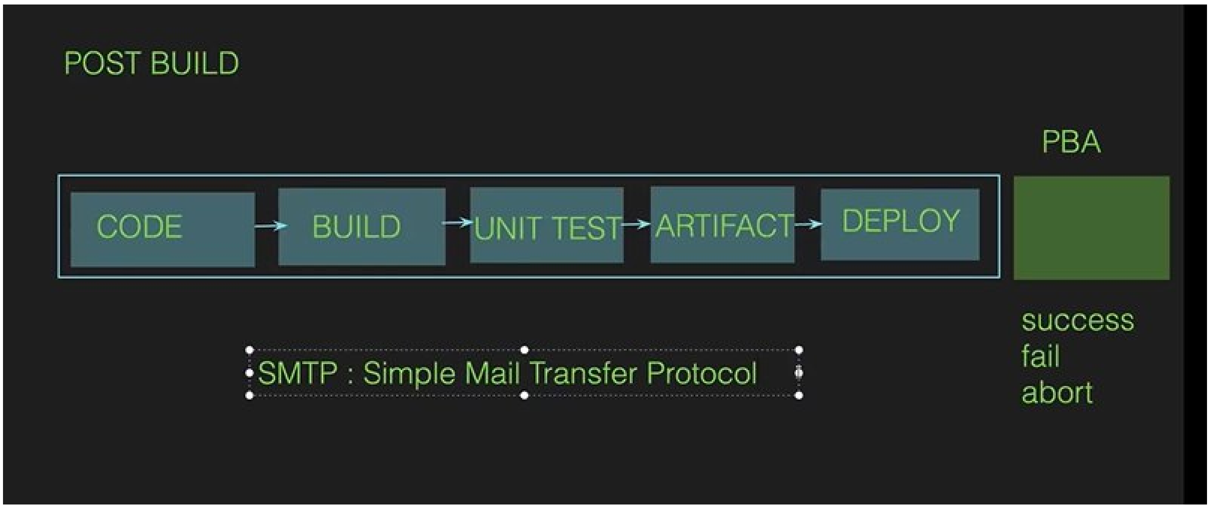
```
    stages {
        stage('stage-1') {
            steps {
                sh 'cal'
                sh 'timedatectl'
                sh 'mkdir mustafa'
                sh 'touch devops'
            }
        }
    }
}
```

Other way

CLASS-12 MAIL,SCRIPTED PIPELINE, CHANGING DEF PORT OF JENKINS n TOMCAT

POST BUILD ACTIONS

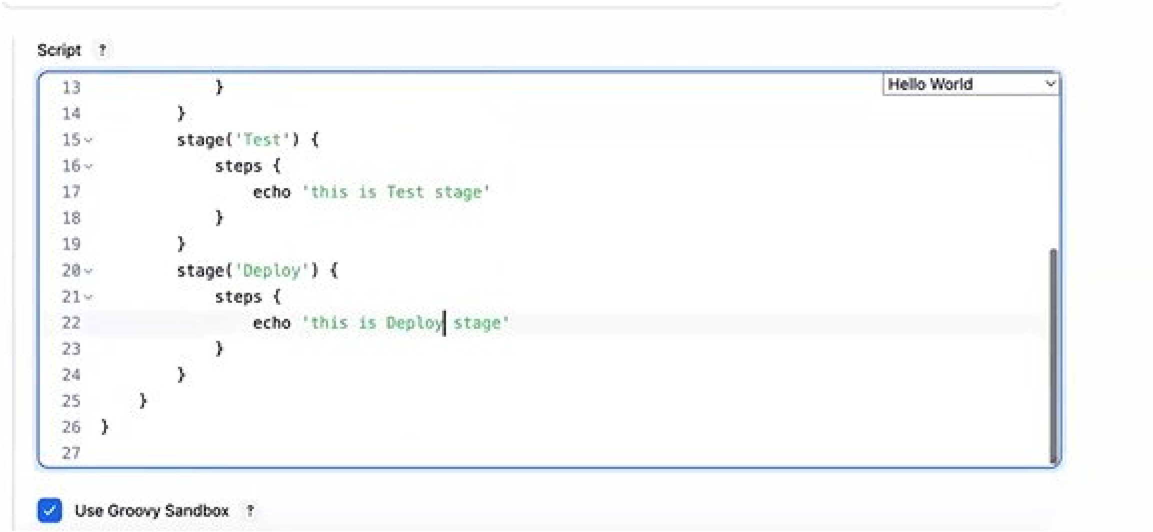
->manam deploy chesaka prathisari success/failure ani chudalemu and konni pipelines overnight run avutai so mana moka mail configure chesthey ade fail/success ani telustadi



CONFIGURING MAIL VTH JENKINS

->manage Jenkins>sys>in email notification in smtp server give def mail smtp.gmail.com>in advanced sel use smtp authentication n give ur personal mail in username> in pword shd give ur app pword(in browser goto to ur google acc>enable 2 step verification in security>search fr app passwords>give name>create>copy that n paste there)>select ssl>port 465>in reply to give ur's?chck test conf by giving ur mail>chk whether u got mail or not in ur mailbox

->now goto ur pipeline jst give stages and build



->do we receive any mail no, pipeline lo post-build actions lo mention cheyyaledu ga

->anni stages ayyaka ga build success/fail ani telisedi so stages taravta rayali

Create pipeline script>emailtext:extended email>in to give evariki mail ravali ani>give sub n body>generate script n paste in pipeline>save it

->now build n chk received mail or not

IQ- Jenkins wht are the def param's?

pipeline lo sh tho print env cmd ni run chesthey telustadi, stage-view plugin ni install chesi, buils chesaka left top corner lo chudachu print env lo em print ayyavi ami

```
}
post {
    success {
        mail to: 'awsanddevops18@gmail.com',
        subject: "Build Status: ${currentBuild.fullDisplayName}",
        body: "The build ${env.BUILD_NUMBER} finished with status: ${currentBuild.currentResult}. Vie
}
```

```
@gmail.com',
${currentBuild.fullDisplayName}",
UILD_NUMBER} finished with status: ${currentBuild.currentResult}. View logs at: ${env.BUILD_URL}"
```

oovy Sandbox ?

```
HUDSON_URL=http://13.218.196.189:8080/
NODE_NAME=built-in
USER=jenkins
HUDSON_SERVER_COOKIE=7591ea0d0763ce37
NOTIFY_SOCKET=/run/systemd/notify
SHLVL=2
BUILD_TAG=jenkins-Job-1-2
EXECUTOR_NUMBER=0
HUDSON_COOKIE=d68e94a1-4bc6-4534-a507-530e63143f42
JENKINS_HOME=/var/lib/jenkins
NODE_LABELS=built-in
STAGE_NAME=Hello
JOURNAL_STREAM=8:44238
WORKSPACE_TMP=/var/lib/jenkins/workspace/Job-1@tmp
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin
RUN_ARTIFACTS_DISPLAY_URL=http://13.218.196.189:8080/job/Job-1/2/display/redirect?page=artif
CI=true
BUILD_ID=2
```

Today

Permalinks

```
JENKINS_URL=http://13.218.196.189:8080/
JOB_BASE_NAME=Job-1
JOB_NAME=Job-1
INVOCATION_ID=ee2ab07058294bb1a1af78270f3b01cd
RUN_DISPLAY_URL=http://13.218.196.189:8080/job/Job-1/2/display/redirect
JOB_DISPLAY_URL=http://13.218.196.189:8080/job/Job-1/display/redirect
WORKSPACE=/var/lib/jenkins/workspace/Job-1
HUDSON_URL=http://13.218.196.189:8080/
NODE_NAME=built-in
USER=jenkins
HUDSON_SERVER_COOKIE=7591ea0d0763ce37
NOTIFY_SOCKET=/run/systemd/notify
SHLVL=2
BUILD_TAG=jenkins-Job-1-2
EXECUTOR_NUMBER=0
HUDSON_COOKIE=d68e94a1-4bc6-4534-a507-530e63143f42
JENKINS_HOME=/var/lib/jenkins
NODE_LABELS=built-in
STAGE_NAME=Hello
```

Build success aithey ney vastadi bcz post-build actions lo success ani pettamu

->failure case, pipeline lo edoka mistake cheyye

Script

```
        stage('Deploy') {
            steps {
                echo 'this is Deploy stage'
            }
        }
    }
    post {
        failure {
            mail to: 'awsanddevops18@gmail.com',
            subject: "Build Status: ${currentBuild.fullDisplayName}",
            body: "The build ${env.BUILD_NUMBER} finished with status: ${currentBuild.currentResult}
        }
    }
}
```

Use Groovy Sandbox

->abort case, deploy stage lo input ichi manam abort chedam mail radhu(pipeline lo post-build lo failure unna koda)

```
            steps {
                echo 'this is Test stage'
            }
        }
        stage('Deploy') {
            input {
                message "can i deploy?"
            }
            steps {
                echo 'this is Deploy stage'
            }
        }
    }
    post {
        failure {
```

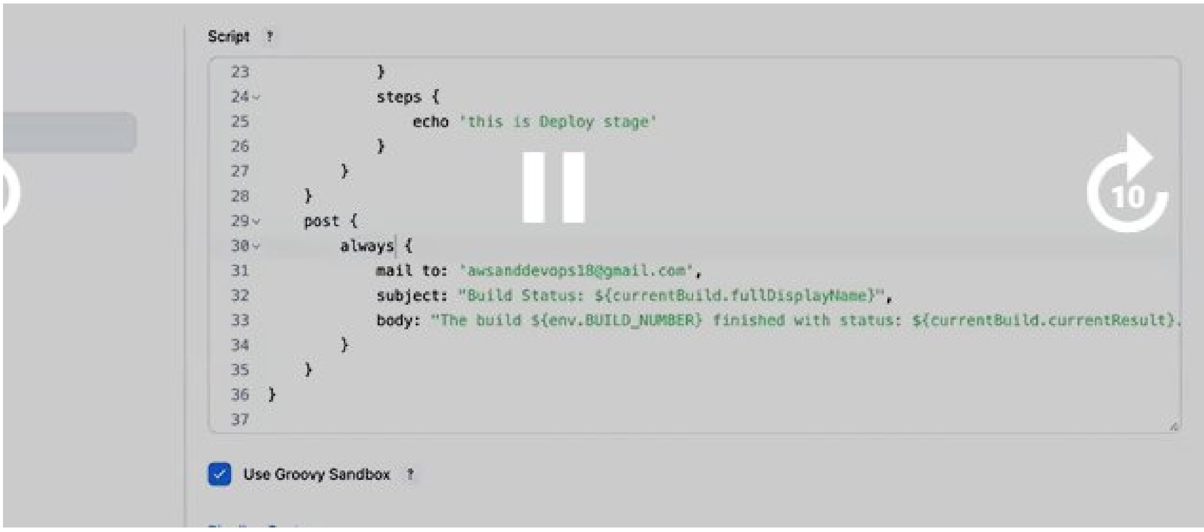
Use Groovy Sandbox

Fail case lo ravali mail but enduku raledu antey build inka complete avvaledu kabati

->ipuudu aborted pedadam

NOTE- irrespective of success/fail/aborted

Give always in post-build action so that ey case lo aina ravali antey



CHANGING DEF PORT OF A JENKINS

->stop Jenkins bu systemctl stop Jenkins

->goto cd /usr/lib/system/systems

->now serach for Jenkins file by find . -name jenkins.service

->vim jenkins.service

A port lo 8080 teesi ur own port

```
#Environment="JENKINS_LISTEN_ADDRESS="

# Port to listen on for HTTP requests. Set to -1 to disable.
# To be able to listen on privileged ports (port numbers less than 1024),
# add the CAP_NET_BIND_SERVICE capability to the AmbientCapabilities
# directive below.
Environment="JENKINS_PORT=

# IP address to listen on for HTTPS requests. Default is disabled.
#Environment="JENKINS_HTTPS_LISTEN_ADDRESS="
```

->now start Jenkins by systemctl start Jenkins

->in server in sec grp add that port no

->now access by ur port

CHANGING DEF PORT OF A TOMCAT

->goto apach… folder conf lo tomcat configurations untai

->bin loki velli ./shoutdown tho stop cheyyandi

->tomcat dhi apache>conf>vim server.xml

->connector port change cheyyandi

```
         APR (HTTP/AJP) Connector: /docs/apr.html
         Define a non-SSL/TLS HTTP/1.1 Connector on port 8080
    -->
    <Connector port="8080" protocol="HTTP/1.1"
               connectionTimeout="20000"
               redirectPort="8443"
               maxParameterCount="1000"
               />
    <!-- A "Connector" using the shared thread pool-->
    <!--
    <Connector executor="tomcatThreadPool"
               port="8080" protocol="HTTP/1.1"
               connectionTimeout="20000"
               redirectPort="8443"
               maxParameterCount="1000"
               />
:69
```

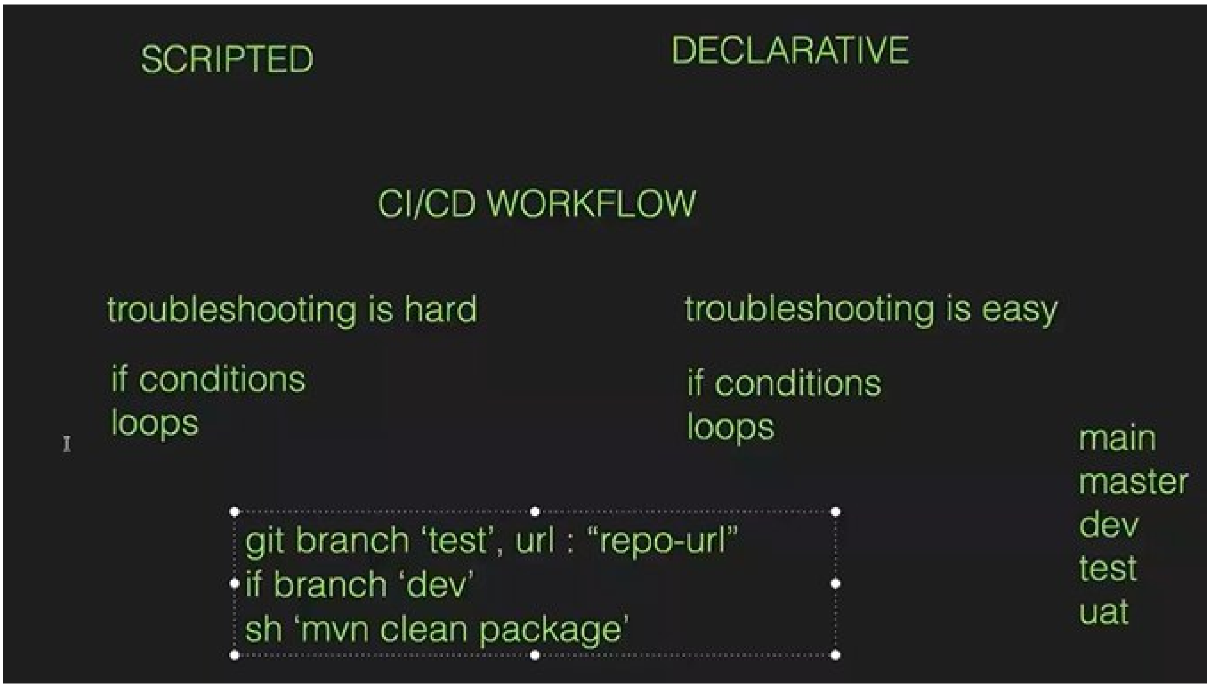
->start tomcat again

->in sec grp add that new port

->now try to access tomcat vth ur port

SCRIPTED PIPELINE

->scripted old n declarative new one



NOTE – realtime lo max declarative ey use chestaru

CLASS-13 git flow branching strategies

–>oka e-commerce app undi 1-dev frm bnglr,1-dev frm hyd n 1-dev frm Chennai vallu rasina code ni github ki push chestaru

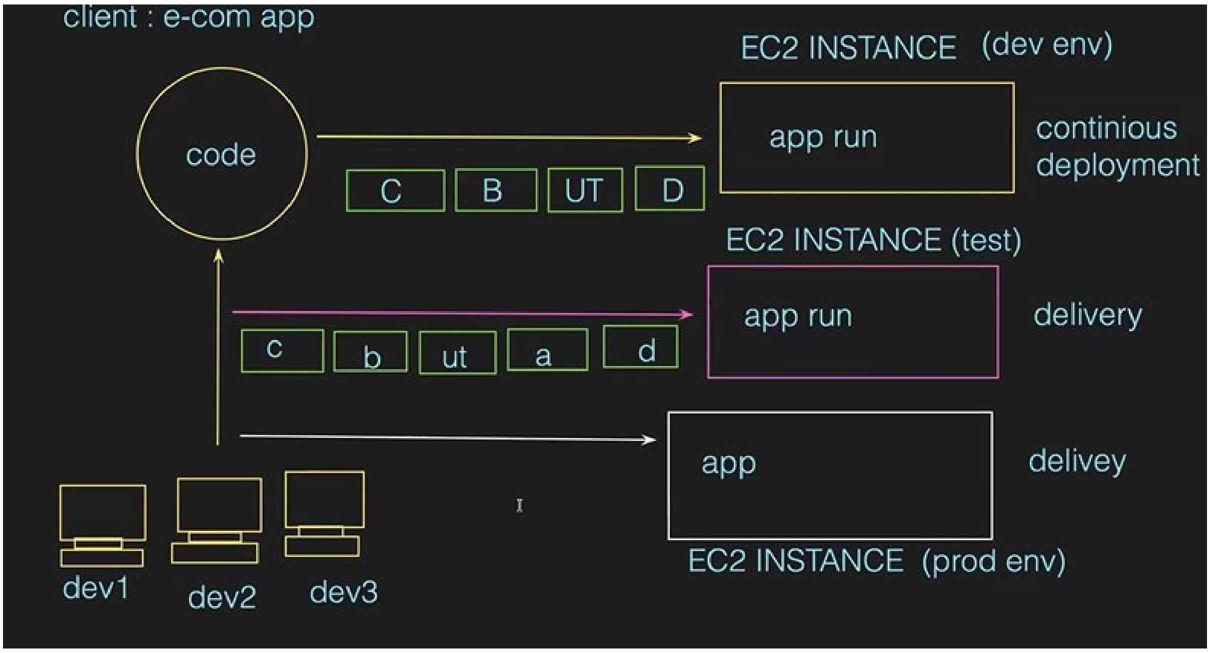
->realtime lo systems lo app's ni run cheyyaru, servers loney run chestaru

->a servers lo app's ni run cheyyali antey ee devops vallu ey chesedhi

->dev's push cheyyageny automatic ga code build+test aye deploy avvali

->paina process anta manam ey chesedi

->test env taravta UAT env untadi uat lo same test pipeline ey but label name okkati change chestamu

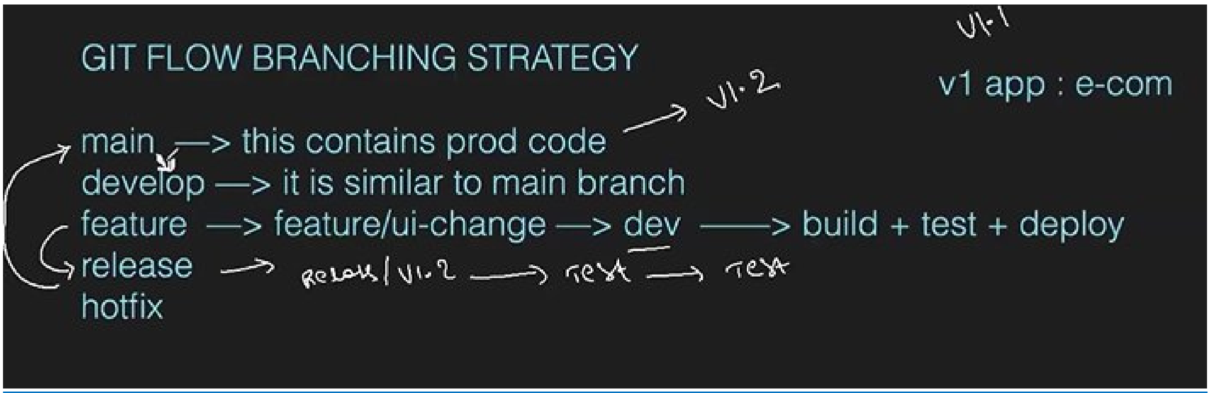


NOTE- 1.oka pedda dbt ey branch nunchi teesukovali ah

2. prathi stage ki okkoka branch untadi(anni stages ki same branch use cheyyamu)

3.realtime lo branches kinda image lo laga untai

REALTIME GIT BRANCHES



->main branch lo prod code untadi,deenlo ey modifications cheyyamu

->devlop branch lo same main code ey untadi

->client nunchi oka re qui change cheyyandi ani, so dev branch nunchi oka feature/ui-change aney branch create chesi, changes ayyaka pipeline create chesi deploy chestamu-anta working fine anipinchinapudu

->a feature/ui-change branch nunchi release aney branch create chestamu, pipeline create chesi deploy chesi testing team ki istamu testing ayyaka

->full testing ayyaka fixes em aina untey fix chesi, a fixes anni ayyaka final release1/release2.. a branch ni teesuku velli main branch ki merge chestamu-so prod lo run avutundi

->main lo unna changes develop-branch lo lev kabatti avi dev-branch ki merge chestamu

->ee hotfix aney branch enduku use chestamu antey pre-pod lo wrk aye, prod lo wrk avvadu-so ippudu immediate ga main branch nunchi hotfix branch create chesi changes fix chesi oka server lo deploy chestamu,adi wrk avutundhi antey ee hotfix branch ni main branch ki merge chestamu so prod lo koda wrk avutadi

NOTE – client req ni batti ee loop continue avutadi

JENKINS PIPELINE WITH BRANCHING STRATEGIES

->ec2 instance vth Jenkins-sg,t3 micr,8b volume

->install git n clone that one branch

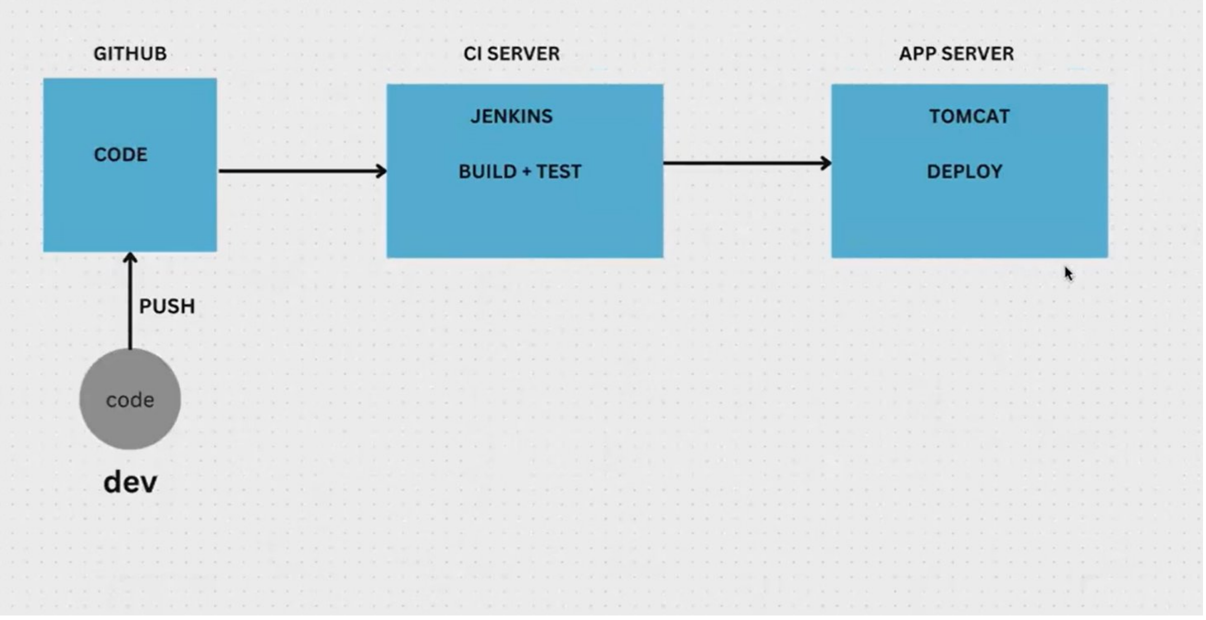
->goto one-repo>by def we'll be in main branch, but checkout to master

->master lo only src,pom.xml files chalu-so if u want u can remove those files

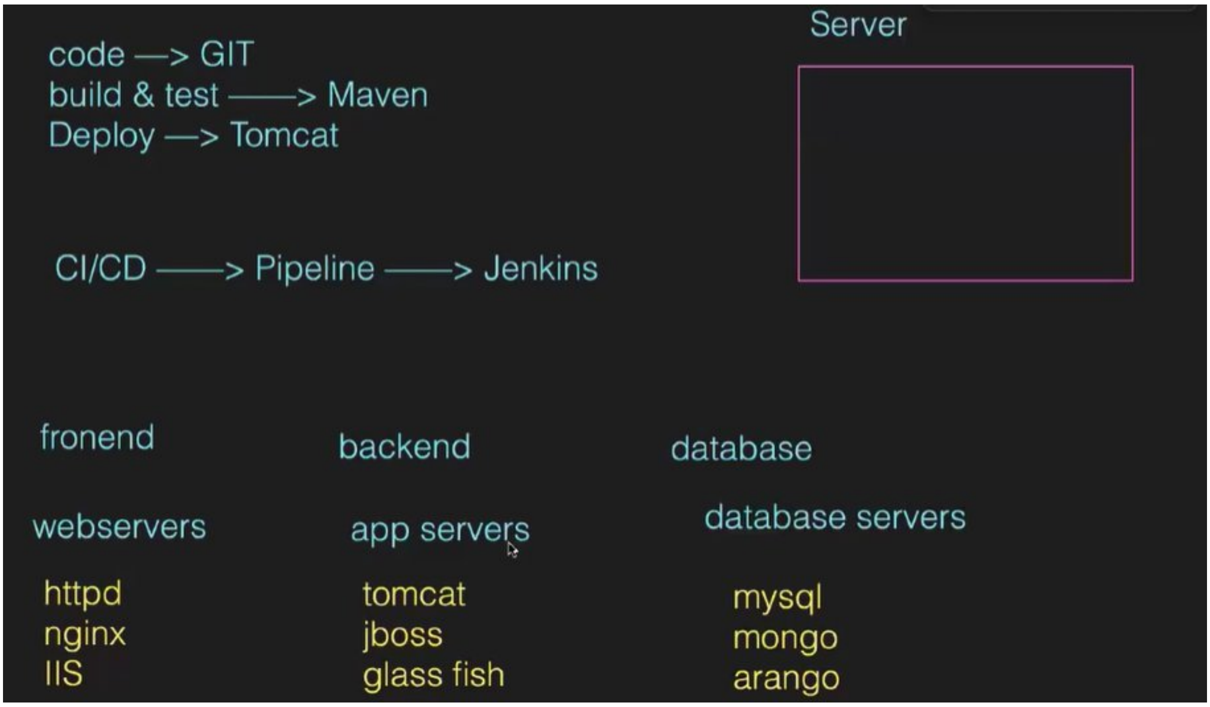
->now goto github n create a new-repo>clone that new-repo>copy all files frm one rep to that new-repo(copy -rm one/ new-repo/)

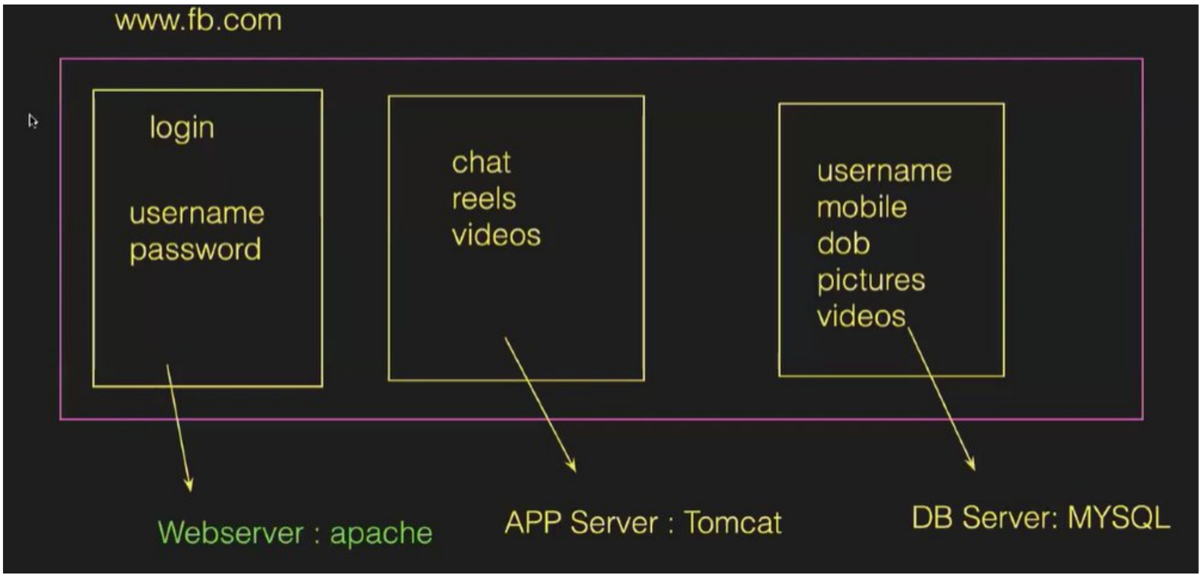
--resume at 34min ---
-

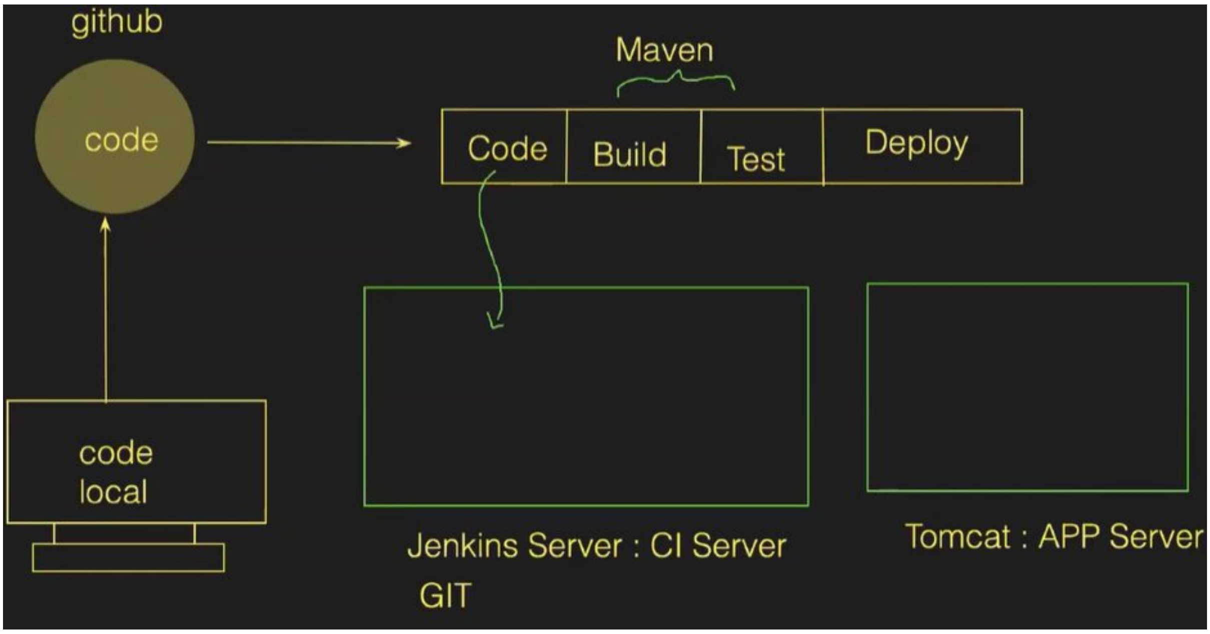
DAY – 4 DEPLOY JAVA APP USING CI/CD



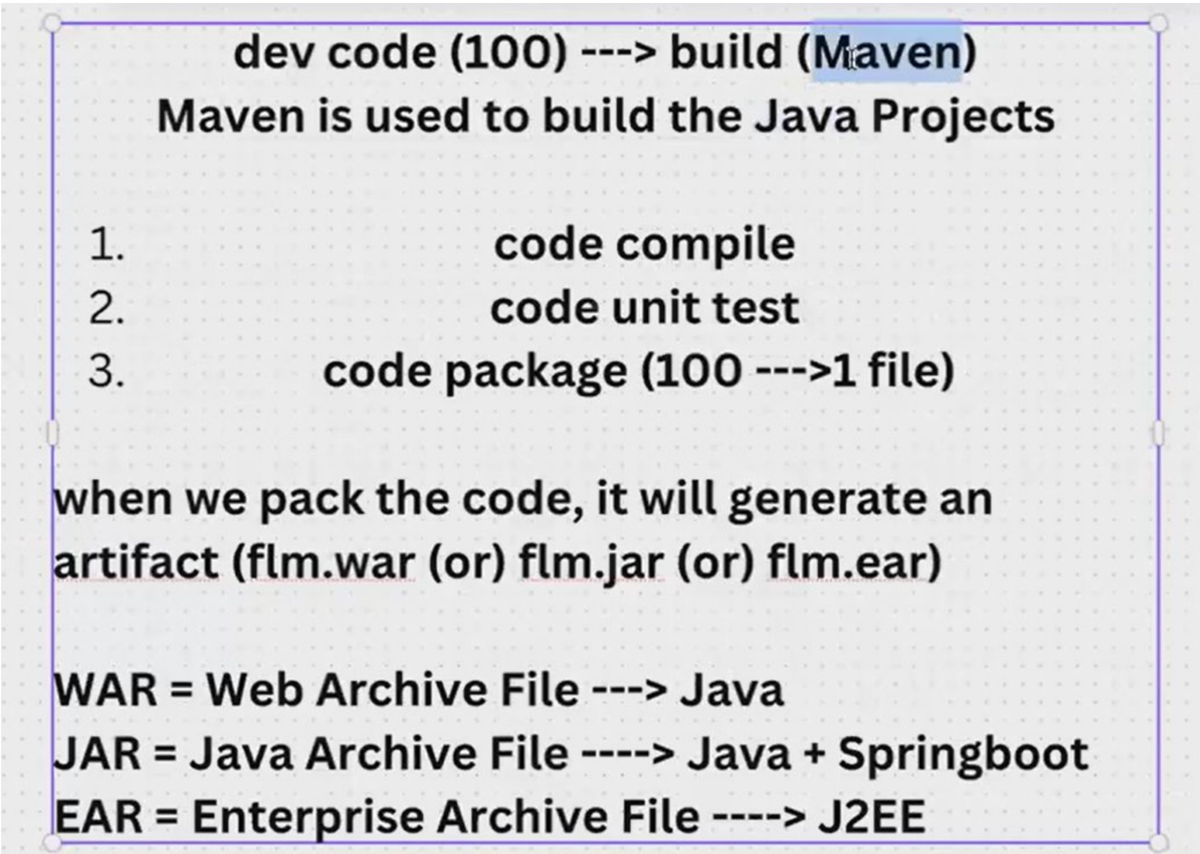
Unit test antey nothing code ni test like bugs em aina unnaya


code —> GIT
build & test ——> Maven
Deploy —> Tomcat
Server
CI/CD ——> Pipeline ——> Jenkins
fronend
backend
database
webservers
app servers
database servers
httpd
nginx
IIS
tomcat
jboss
glass fish
mysql
mongo
arango


www.fb.com
login
username
password
chat
reels
videos
username
mobile
dob
pictures
videos
Webserver : apache
APP Server : Tomcat
DB Server: MYSQL



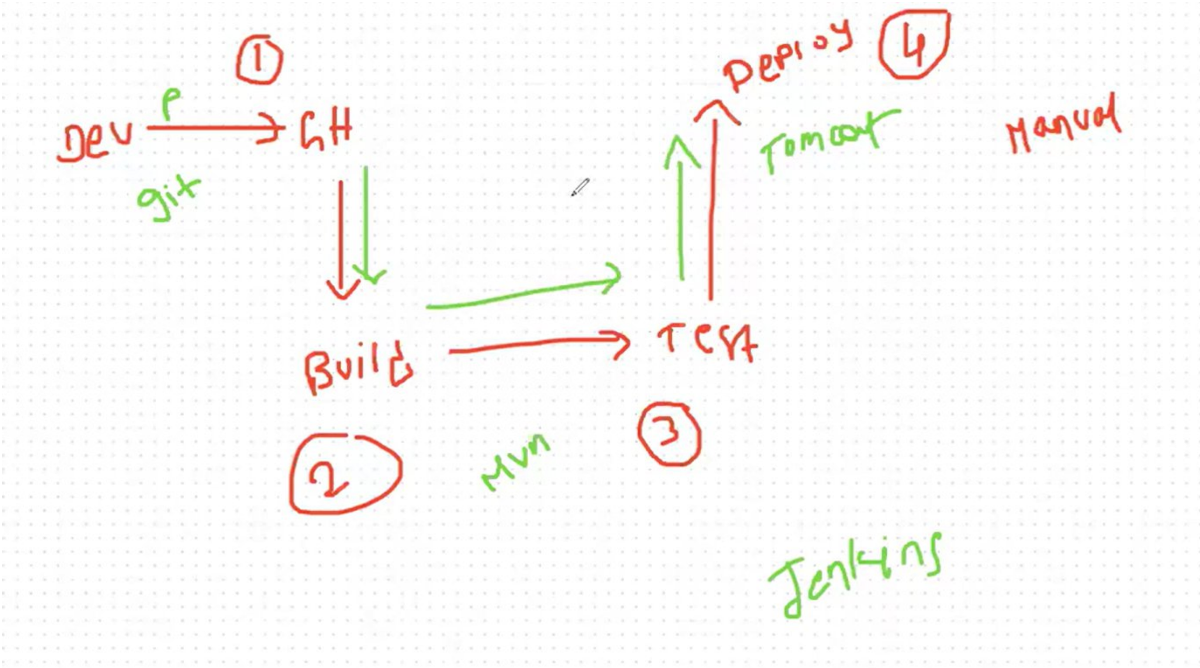
BUILD PROCESS



NOTE – 1 if source code is developed by python/node js/react no need of any build tools.

2 maven tho ney unit testing jarugutadi

3 maven tho compile,unit testing, packaging chestamu.ee process ni automate cheyyadaniki Jenkins use chestamu.



Dev’s code ni rasi, dani github loki push chestaru, github lo unna code teesukoni oka team build chestaru,build ayyakatesting team vallu test chestaru, testing ayyaka deploy team vallu prod lo deploy chestaru – MANUAL PROCESS

But once devlopers github lo code pettaganey, mvn ni use chesukoni build+testing chestamu, tomcat ni use chesi deploy chestamu---ee process ni automate cheyyadaniki manam Jenkins use chestamu

JAVA APP DEPLOYMENT

To get the source code into our github account

->way1 – fork source code frm the github

->way2 – srce code url ni browser lo paste chesi>repository vachaka>fork option vastadi

->way3 – goto Mustafa github(id-devops0014>search of all github>users>click on profile>goto repos>search fr one>click on it>click on fork(while forking disable the copy the main branch only))

STEP1 – since source code is in github, we need to get that code into CI server

->source code mana github acc loki vachaka, repo https url ni copy chesukondi

->in Jenkins create one job(java_app_deploy)>goto source code management, sel git,provide https github url,code master branch lo undi so give branch as master>save it

->now build it>once it built is successful, then src code is available in CI server.

STEP2 –source code is available, now we need to build+test

->in order to build+test we need maven tool, while setting up github we installed suggested plugins, in that maven plugin is installed.

->to integrate mvn plugin

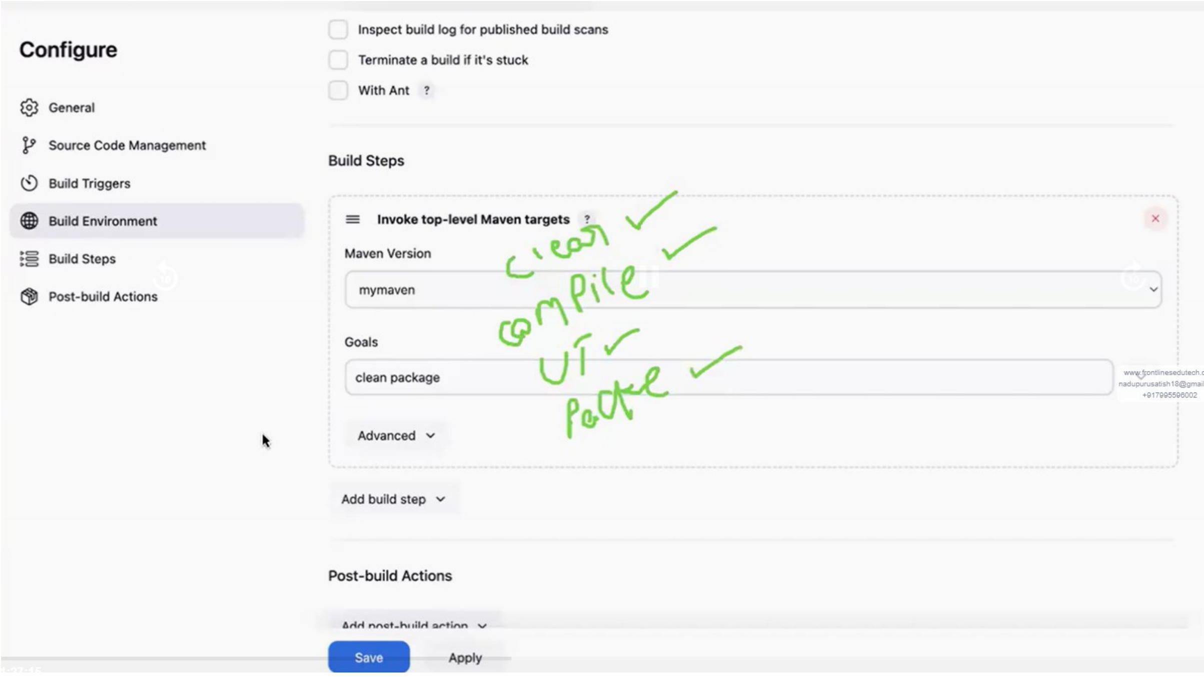
Goto dashboards>manage jenkins>tools>Under mvn installation>give name>save

->to build

Goto java_app_deploy job>build steps, under build steps option sel invoke-top-level mvn targets(since we are using mvn if nt use ant/gradle corresponding options)>in mvn version give which name u kept fr ur maven while integrating>under goals give clean package(which will clear the previous runs, again compile,unit test n package)>save

->now click on build now

NOTE – for building nt only mvn we can use ant/mvn/gradle



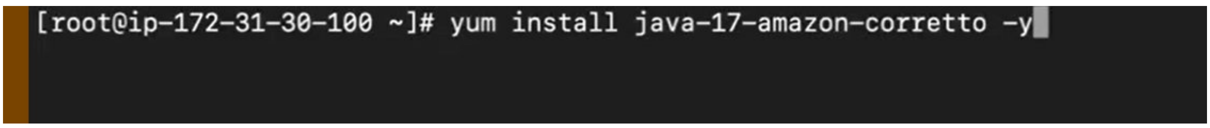
->once the build is success, we get the file(workspace>target>xxx.war file)

NOTE – a war file consist of code n dependencies req to run that code

<u>**STEP3–**</u> since build+test is successful, we need to deploy in app server(tomcat)

->ito install tomcat(2nd server)

Step1 – install java



Step2 – tomcat file download(dlcdn.apache.org>search for tomcat>click on tomcat>tomcat-9>v9.0.98>bin>sel 12MB file(if we click on it,downloads in local but to download in server copy link)

-to get download smtg frm browser to server use cmd **wget file_link**

```
[root@ip-172-31-30-100 ~]# wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.98/bin/apache-tomcat-9.0.98.tar.gz
--2024-12-24 14:24:15--  https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.98/bin/apache-tomcat-9.0.98.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12760610 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.98.tar.gz'

100%[==========================================================================>] 12,760,610  --.-K/s   in 0.1s

2024-12-24 14:24:15 (112 MB/s) - 'apache-tomcat-9.0.98.tar.gz' saved [12760610/12760610]

[root@ip-172-31-30-100 ~]#
```

->to unzip that file use cmd **tar -zxvf file_name**

```
[root@ip-172-31-30-100 ~]# ll
total 12464
-rw-r--r-- 1 root root 12760610 Dec  5 20:00 apache-tomcat-9.0.98.tar.gz
[root@ip-172-31-30-100 ~]#
[root@ip-172-31-30-100 ~]# tar -zxvf apache-tomcat-9.0.98.tar.gz
```

Goto that folder

```
[root@ip-172-31-30-100 ~]# ll
total 12464
drwxr-xr-x 9 root root      220 Dec 24 14:24 apache-tomcat-9.0.98
-rw-r--r-- 1 root root 12760610 Dec  5 20:00 apache-tomcat-9.0.98.tar.gz
[root@ip-172-31-30-100 ~]#
[root@ip-172-31-30-100 ~]# cd apache-tomcat-9.0.98/
[root@ip-172-31-30-100 apache-tomcat-9.0.98]#
[root@ip-172-31-30-100 apache-tomcat-9.0.98]#
```

->sigma rule to start/stop/restart tomcat goto bin folder

start service ——> bin

start
stop
restart

```
[root@ip-172-31-30-100 apache-tomcat-9.0.98]# ll
total 132
-rw-r----- 1 root root 20913 Dec  5 19:50 BUILDING.txt
-rw-r----- 1 root root  6166 Dec  5 19:50 CONTRIBUTING.md
-rw-r----- 1 root root 57092 Dec  5 19:50 LICENSE
-rw-r----- 1 root root  2333 Dec  5 19:50 NOTICE
-rw-r----- 1 root root  3283 Dec  5 19:50 README.md
-rw-r----- 1 root root  6901 Dec  5 19:50 RELEASE-NOTES
-rw-r----- 1 root root 16538 Dec  5 19:50 RUNNING.txt
drwxr-x--- 2 root root  4096 Dec 24 14:24 bin
drwx------ 2 root root   238 Dec  5 19:50 conf
drwxr-x--- 2 root root  4096 Dec 24 14:24 lib
drwxr-x--- 2 root root     6 Dec  5 19:50 logs
drwxr-x--- 2 root root    30 Dec 24 14:24 temp
drwxr-x--- 7 root root    81 Dec  5 19:50 webapps
drwxr-x--- 2 root root     6 Dec  5 19:50 work
[root@ip-172-31-30-100 apache-tomcat-9.0.98]#
[root@ip-172-31-30-100 apache-tomcat-9.0.98]#
```

->tomcat ni start cheyyali antey chala pedda process so run that startup.sh file

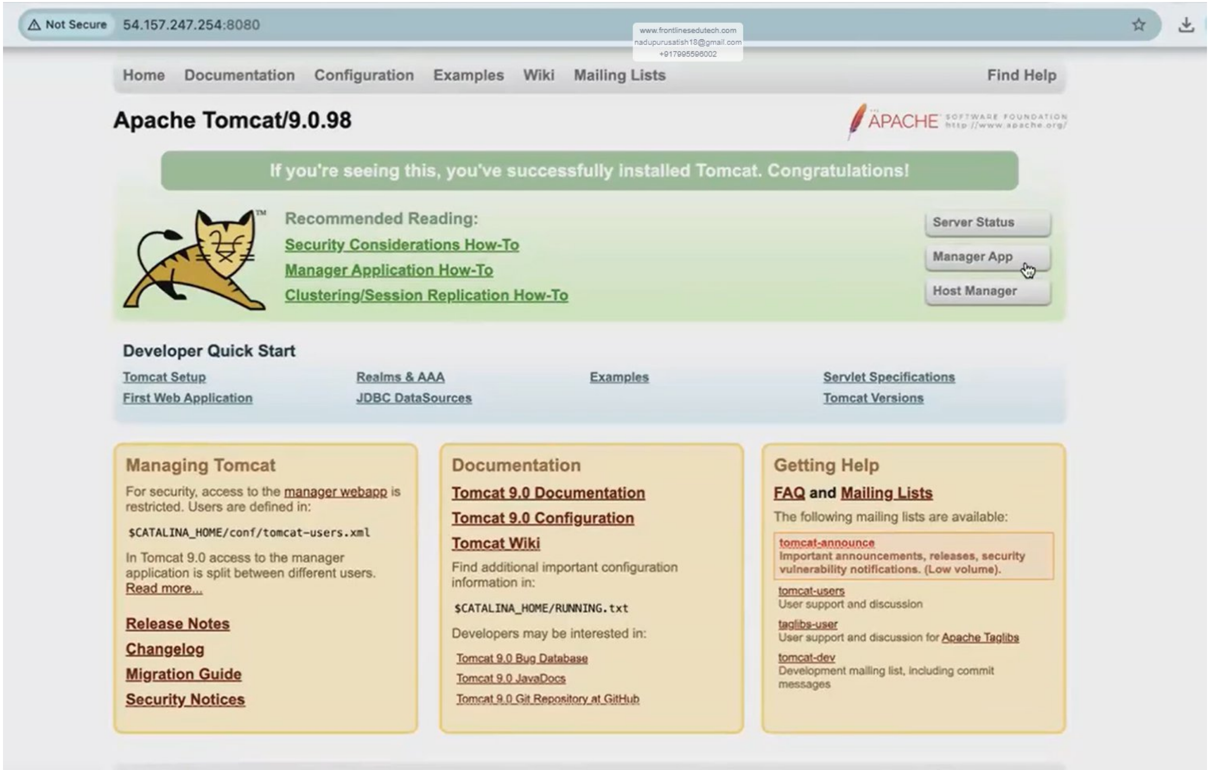
```
[root@ip-172-31-30-100 bin]# ll
total 816
-rw-r----- 1 root root  35459 Dec  5 19:50 bootstrap.jar
-rw-r----- 1 root root   1664 Dec  5 19:50 catalina-tasks.xml
-rw-r----- 1 root root  16856 Dec  5 19:50 catalina.bat
-rwxr-x--- 1 root root  25323 Dec  5 19:50 catalina.sh
-rw-r----- 1 root root   2123 Dec  5 19:50 ciphers.bat
-rwxr-x--- 1 root root   1997 Dec  5 19:50 ciphers.sh
-rw-r----- 1 root root 214459 Dec  5 19:50 commons-daemon-native.tar.gz
-rw-r----- 1 root root  25834 Dec  5 19:50 commons-daemon.jar
-rw-r----- 1 root root   2040 Dec  5 19:50 configtest.bat
-rwxr-x--- 1 root root   1922 Dec  5 19:50 configtest.sh
-rwxr-x--- 1 root root   9100 Dec  5 19:50 daemon.sh
-rw-r----- 1 root root   2091 Dec  5 19:50 digest.bat
-rwxr-x--- 1 root root   1965 Dec  5 19:50 digest.sh
-rw-r----- 1 root root   3606 Dec  5 19:50 makebase.bat
-rwxr-x--- 1 root root   3382 Dec  5 19:50 makebase.sh
-rw-r----- 1 root root   3814 Dec  5 19:50 setclasspath.bat
-rwxr-x--- 1 root root   4317 Dec  5 19:50 setclasspath.sh
-rw-r----- 1 root root   2020 Dec  5 19:50 shutdown.bat
-rwxr-x--- 1 root root   1902 Dec  5 19:50 shutdown.sh
-rw-r----- 1 root root   2022 Dec  5 19:50 startup.bat
-rwxr-x--- 1 root root   1904 Dec  5 19:50 startup.sh
-rw-r----- 1 root root  49607 Dec  5 19:50 tomcat-juli.jar
-rw-r----- 1 root root 346588 Dec  5 19:50 tomcat-native.tar.gz
-rw-r----- 1 root root   4576 Dec  5 19:50 tool-wrapper.bat
-rwxr-x--- 1 root root   5540 Dec  5 19:50 tool-wrapper.sh
-rw-r----- 1 root root   2026 Dec  5 19:50 version.bat
-rwxr-x--- 1 root root   1908 Dec  5 19:50 version.sh
[root@ip-172-31-30-100 bin]#
[root@ip-172-31-30-100 bin]#
```

To run/execute any file use cmd **./file_name**

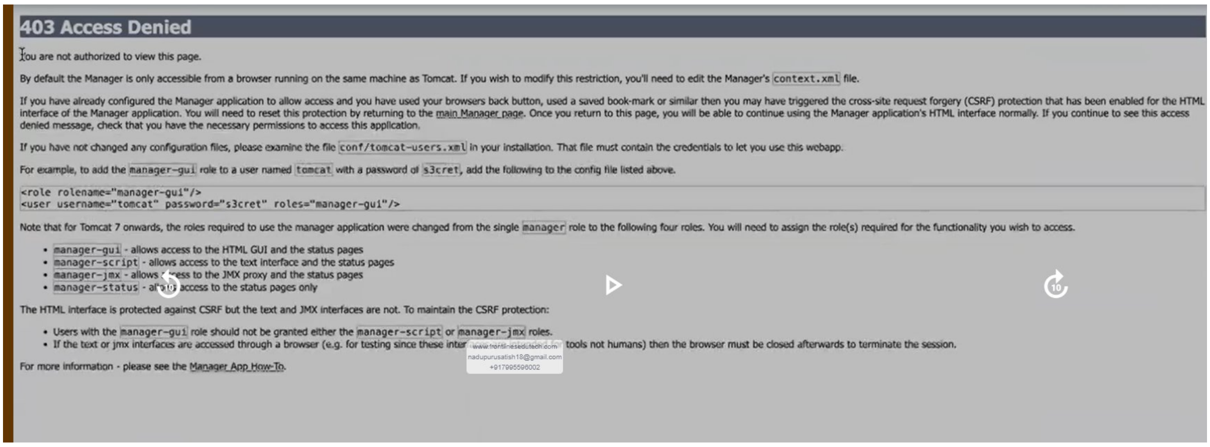
```
[root@ip-172-31-30-100 bin]# ./startup.sh
Using CATALINA_BASE:   /root/apache-tomcat-9.0.98
Using CATALINA_HOME:   /root/apache-tomcat-9.0.98
Using CATALINA_TMPDIR: /root/apache-tomcat-9.0.98/temp
Using JRE_HOME:        /
Using CLASSPATH:       /root/apache-tomcat-9.0.98/bin/bootstrap.jar:/root/apache-tomcat-9.0.98/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-30-100 bin]#
```

->to access tomcat dashboard use pub-ip of ur tomcat server vth 8080 port

Pub-ip:8080 in browser



We need to goto manager app, but we are getting err



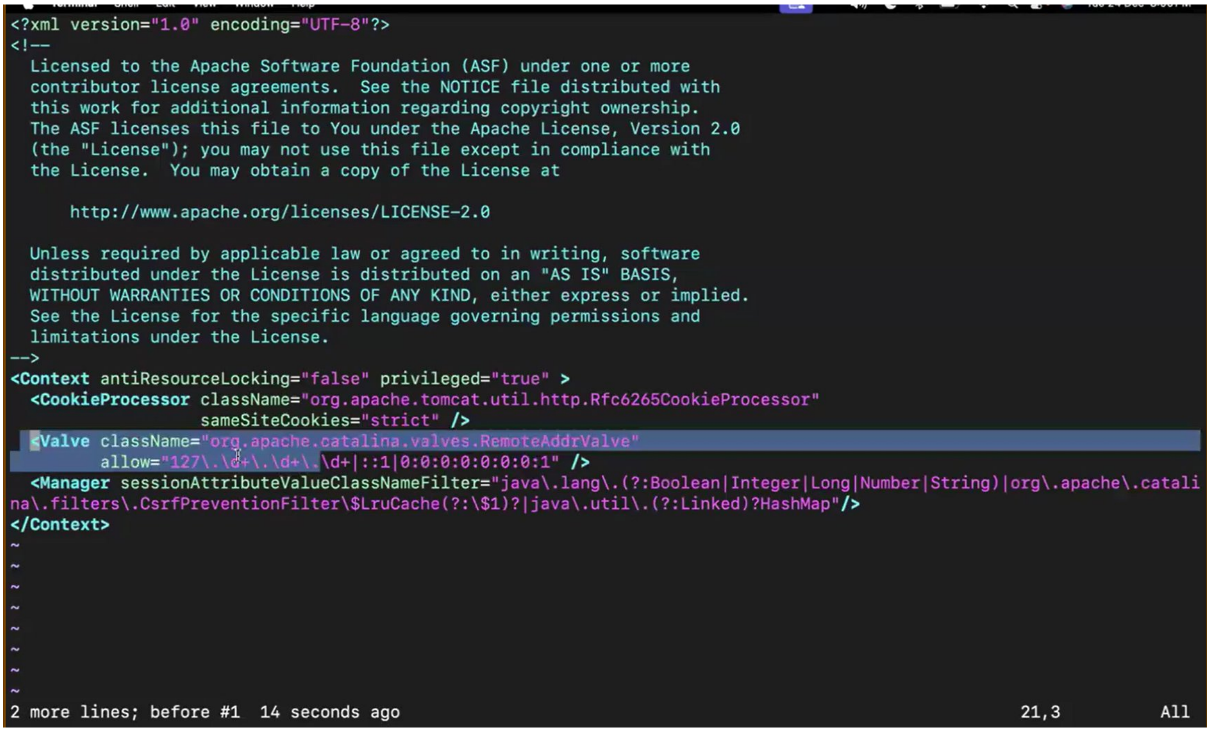
Ee restriction ni modify cheyyali antey context.xml file edit cheyyali

So goto main folder of tomcat n search fr context.xml file

```
[root@ip-172-31-30-100 bin]# cd ..
[root@ip-172-31-30-100 apache-tomcat-9.0.98]#
[root@ip-172-31-30-100 apache-tomcat-9.0.98]# ll
total 132
-rw-r----- 1 root root 20913 Dec  5 19:50 BUILDING.txt
-rw-r----- 1 root root  6166 Dec  5 19:50 CONTRIBUTING.md
-rw-r----- 1 root root 57092 Dec  5 19:50 LICENSE
-rw-r----- 1 root root  2333 Dec  5 19:50 NOTICE
-rw-r----- 1 root root  3283 Dec  5 19:50 README.md
-rw-r----- 1 root root  6901 Dec  5 19:50 RELEASE-NOTES
-rw-r----- 1 root root 16538 Dec  5 19:50 RUNNING.txt
drwxr-x--- 2 root root  4096 Dec 24 14:26 bin
drwx------ 3 root root   254 Dec 24 14:26 conf
drwxr-x--- 2 root root  4096 Dec 24 14:24 lib
drwxr-x--- 2 root root   197 Dec 24 14:26 logs
drwxr-x--- 2 root root    30 Dec 24 14:24 temp
drwxr-x--- 7 root root    81 Dec  5 19:50 webapps
drwxr-x--- 3 root root    22 Dec 24 14:26 work
[root@ip-172-31-30-100 apache-tomcat-9.0.98]#
```

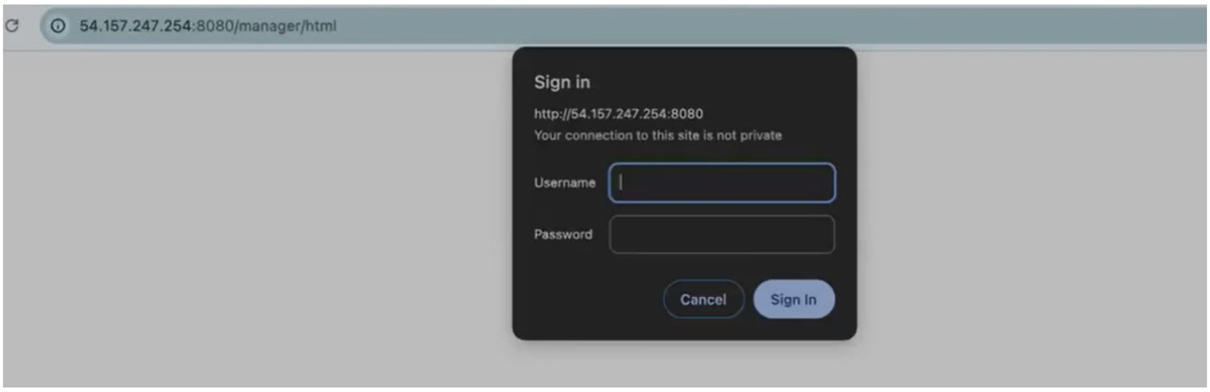
```
[root@ip-172-31-30-100 apache-tomcat-9.0.98]# ll
total 132
-rw-r----- 1 root root 20913 Dec  5 19:50 BUILDING.txt
-rw-r----- 1 root root  6166 Dec  5 19:50 CONTRIBUTING.md
-rw-r----- 1 root root 57092 Dec  5 19:50 LICENSE
-rw-r----- 1 root root  2333 Dec  5 19:50 NOTICE
-rw-r----- 1 root root  3283 Dec  5 19:50 README.md
-rw-r----- 1 root root  6901 Dec  5 19:50 RELEASE-NOTES
-rw-r----- 1 root root 16538 Dec  5 19:50 RUNNING.txt
drwxr-x--- 2 root root  4096 Dec 24 14:26 bin
drwx------ 3 root root   254 Dec 24 14:26 conf
drwxr-x--- 2 root root  4096 Dec 24 14:24 lib
drwxr-x--- 2 root root   197 Dec 24 14:26 logs
drwxr-x--- 2 root root    30 Dec 24 14:24 temp
drwxr-x--- 7 root root    81 Dec  5 19:50 webapps
drwxr-x--- 3 root root    22 Dec 24 14:26 work
[root@ip-172-31-30-100 apache-tomcat-9.0.98]#
[root@ip-172-31-30-100 apache-tomcat-9.0.98]# updatedb
[root@ip-172-31-30-100 apache-tomcat-9.0.98]#
[root@ip-172-31-30-100 apache-tomcat-9.0.98]# locate context.xml
/root/apache-tomcat-9.0.98/conf/context.xml
/root/apache-tomcat-9.0.98/webapps/docs/META-INF/context.xml
/root/apache-tomcat-9.0.98/webapps/examples/META-INF/context.xml
/root/apache-tomcat-9.0.98/webapps/host-manager/META-INF/context.xml
/root/apache-tomcat-9.0.98/webapps/manager/META-INF/context.xml
[root@ip-172-31-30-100 apache-tomcat-9.0.98]#
```

->a file ni vim file_name tho open chesi>:21 line ki velli>2dd(21,22 lines del chesatunnam)>save n continue(:wq)

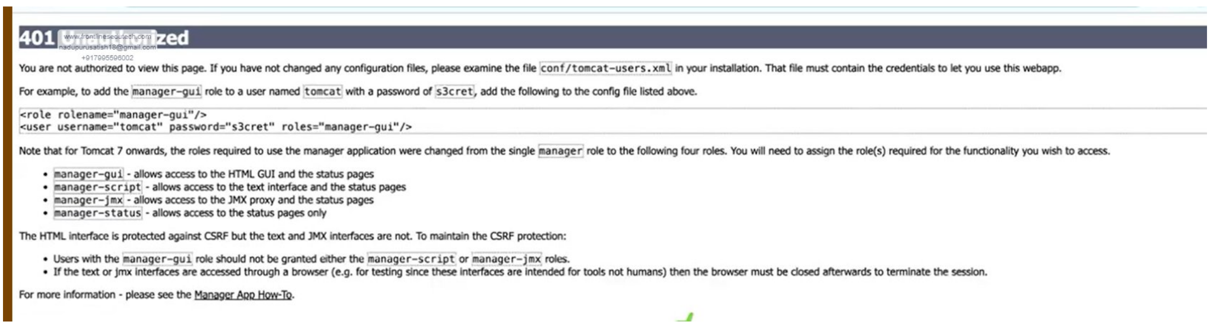


```
<?xml version="1.0" encoding="UTF-8"?>
<!--
  Licensed to the Apache Software Foundation (ASF) under one or more
  contributor license agreements.  See the NOTICE file distributed with
  this work for additional information regarding copyright ownership.
  The ASF licenses this file to You under the Apache License, Version 2.0
  (the "License"); you may not use this file except in compliance with
  the License.  You may obtain a copy of the License at

      http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
                   sameSiteCookies="strict" />
  <Valve className="org.apache.catalina.valves.RemoteAddrValve"
         allow="127\.\d+\.\d+\.\d+|::1|0:0:0:0:0:0:0:1" />
  <Manager sessionAttributeValueClassNameFilter="java\.lang\.(?:Boolean|Integer|Long|Number|String)|org\.apache\.catali
na\.filters\.CsrfPreventionFilter\$LruCache(?:\$1)?|java\.util\.(?:Linked)?HashMap"/>
</Context>
~
~
~
~
~
~
~
~
2 more lines; before #1  14 seconds ago                                   21,3          All
```

->ippudu tomcat authorization vachindi



->but credentials levu,so jst cancel it



401 zed

You are not authorized to view this page. If you have not changed any configuration files, please examine the file `conf/tomcat-users.xml` in your installation. That file must contain the credentials to let you use this webapp.

For example, to add the `manager-gui` role to a user named `tomcat` with a password of `s3cret`, add the following to the config file listed above.

```
<role rolename="manager-gui"/>
<user username="tomcat" password="s3cret" roles="manager-gui"/>
```

Note that for Tomcat 7 onwards, the roles required to use the manager application were changed from the single `manager` role to the following four roles. You will need to assign the role(s) required for the functionality you wish to access.

- `manager-gui` - allows access to the HTML GUI and the status pages
- `manager-script` - allows access to the text interface and the status pages
- `manager-jmx` - allows access to the JMX proxy and the status pages
- `manager-status` - allows access to the status pages only

The HTML interface is protected against CSRF but the text and JMX interfaces are not. To maintain the CSRF protection:

- Users with the `manager-gui` role should not be granted either the `manager-script` or `manager-jmx` roles.
- If the text or jmx interfaces are accessed through a browser (e.g. for testing since these interfaces are intended for tools not humans) then the browser must be closed afterwards to terminate the session.

For more information - please see the Manager App How-To.

->goto to conf folder

```
[root@ip-172-31-30-100 apache-tomcat-9.0.98]# ll
total 132
-rw-r----- 1 root root 20913 Dec  5 19:50 BUILDING.txt
-rw-r----- 1 root root  6166 Dec  5 19:50 CONTRIBUTING.md
-rw-r----- 1 root root 57092 Dec  5 19:50 LICENSE
-rw-r----- 1 root root  2333 Dec  5 19:50 NOTICE
-rw-r----- 1 root root  3283 Dec  5 19:50 README.md
-rw-r----- 1 root root  6901 Dec  5 19:50 RELEASE-NOTES
-rw-r----- 1 root root 16538 Dec  5 19:50 RUNNING.txt
drwxr-x--- 2 root root  4096 Dec 24 14:26 bin
drwx------ 3 root root   254 Dec 24 14:26 conf
drwxr-x--- 2 root root  4096 Dec 24 14:24 lib
drwxr-x--- 2 root root   197 Dec 24 14:26 logs
drwxr-x--- 2 root root    30 Dec 24 14:24 temp
drwxr-x--- 7 root root    81 Dec  5 19:50 webapps
drwxr-x--- 3 root root    22 Dec 24 14:26 work
[root@ip-172-31-30-100 apache-tomcat-9.0.98]#
[root@ip-172-31-30-100 apache-tomcat-9.0.98]# cd conf/
```

->vim tomcat-users.xml file open chesi>ee file lo last line ki vellali(G cmd)>line no 50 nunchi 3 lines copy chesukovali(3yy)>cursor ni 55 line lo petti paste cheyyandi(p)>a roles, pwords change cheyyali kinda chupinchinatlu

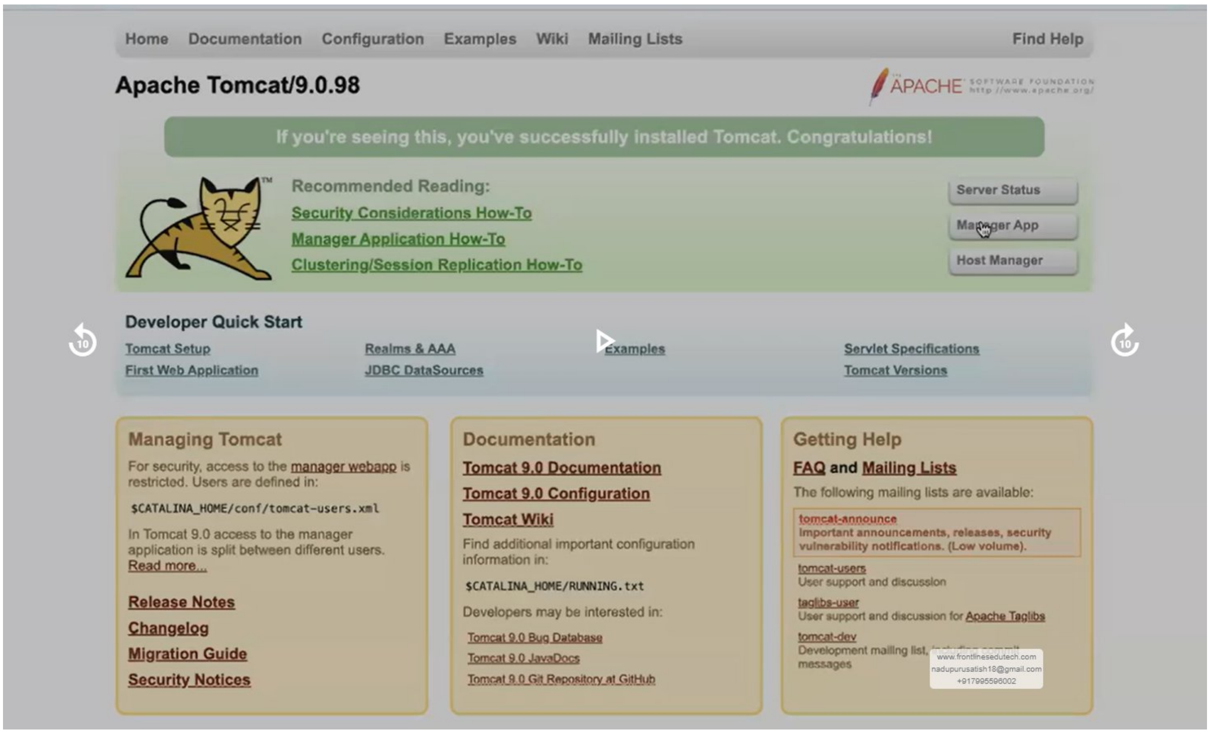
```
-->
<!--
  <role rolename="tomcat"/>
  <role rolename="role1"/>
  <user username="tomcat" password="<must-be-changed>" roles="tomcat"/>
  <user username="both" password="<must-be-changed>" roles="tomcat,role1"/>
  <user username="role1" password="<must-be-changed>" roles="role1"/>
-->
  <role rolename="manager-gui"/>
  <role rolename="manager-script"/>
  <user username="tomcat" password="admin@123" roles="manager-gui, manager-script"/>
</tomcat-users>
:wq
```

->whenever we change configurations we need to restart tomcat

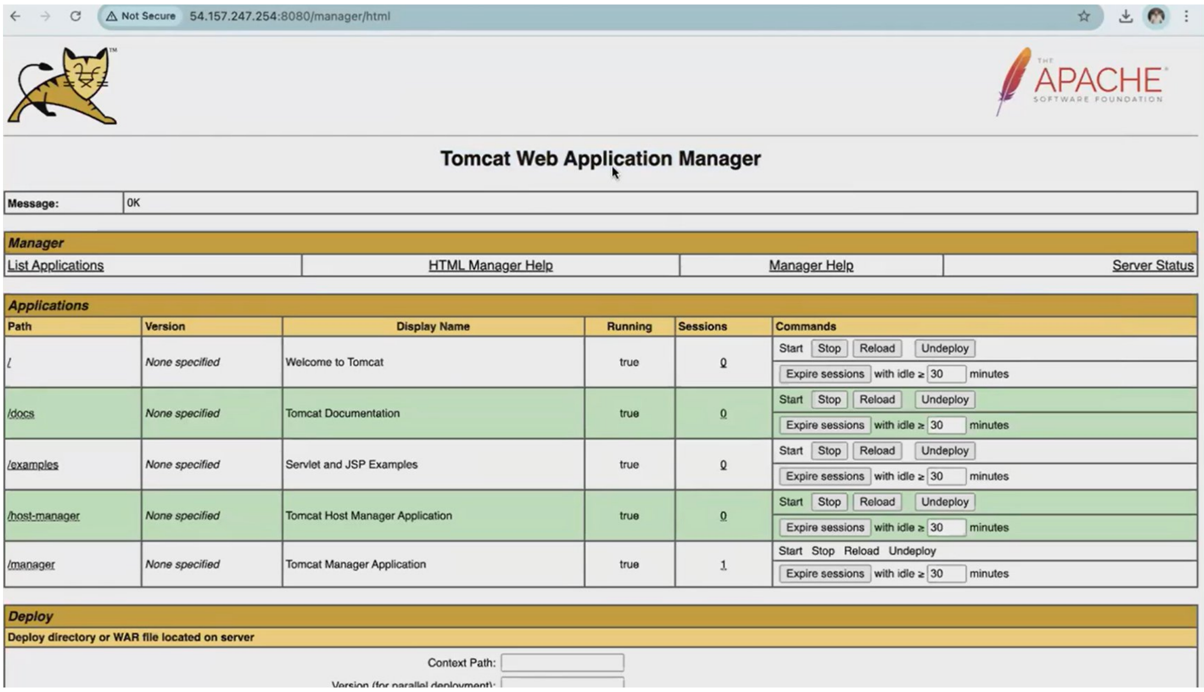
Restart = shutdown+start

```
[root@ip-172-31-30-100 bin]# ./shutdown.sh
Using CATALINA_BASE:   /root/apache-tomcat-9.0.98
Using CATALINA_HOME:   /root/apache-tomcat-9.0.98
Using CATALINA_TMPDIR: /root/apache-tomcat-9.0.98/temp
Using JRE_HOME:        /
Using CLASSPATH:       /root/apache-tomcat-9.0.98/bin/bootstrap.jar:/root/apache-tomcat-9.0.98/bin/tomcat-juli.jar
Using CATALINA_OPTS:
NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAME
D --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.r
mi/sun.rmi.transport=ALL-UNNAMED
[root@ip-172-31-30-100 bin]#
[root@ip-172-31-30-100 bin]# ./startup.sh
Using CATALINA_BASE:   /root/apache-tomcat-9.0.98
Using CATALINA_HOME:   /root/apache-tomcat-9.0.98
Using CATALINA_TMPDIR: /root/apache-tomcat-9.0.98/temp
Using JRE_HOME:        /
Using CLASSPATH:       /root/apache-tomcat-9.0.98/bin/bootstrap.jar:/root/apache-tomcat-9.0.98/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-30-100 bin]#
```

->now goto manager app n login



This is our dashboard



->deploy cheyyadaniki oka plugin untadi

->Jenkins>dashboards>manage Jenkins>plugins?available plugins>search fr deploy to container>install it

->Goto ur job>configuration>post-build actions>under post-build actions sel deploy war/ear to a container>under war/ear files give target/*.war>uder context path give ur app name>under containers sel tomcat9>give tomcat credentials n sel it>copy ur tomcat url n paste it in tomcat url(upto 8080)>save it.

->now build now

->if build is success, then app is deployed in app server

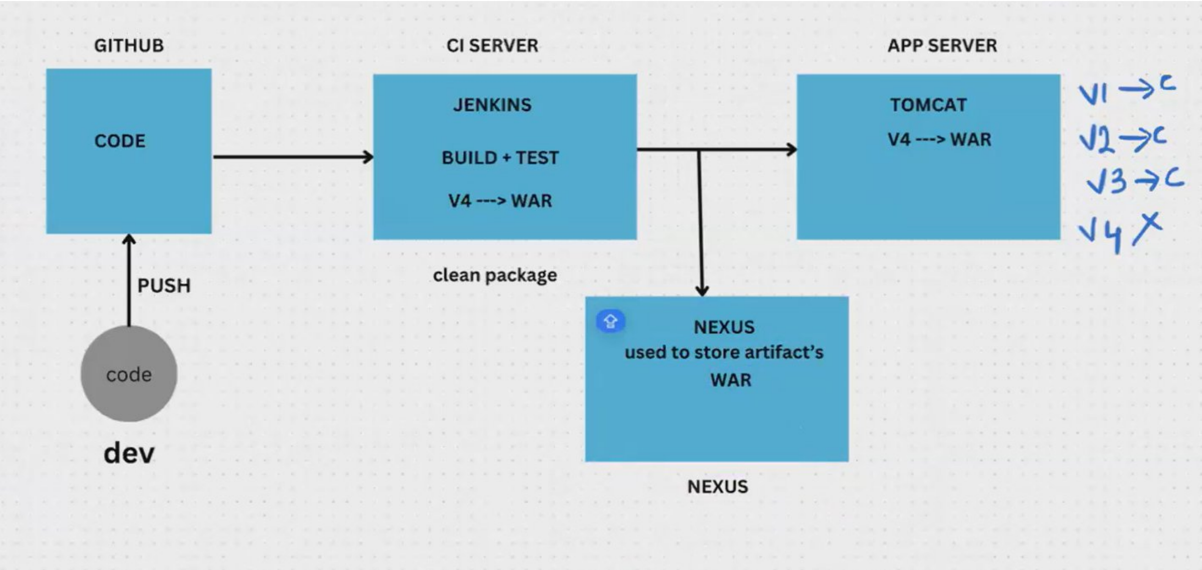
->now happily we can access the app

NOTE – developer edaina change chesaru oka file lo using Jenkins n committed, then automatically it'll reflect in app- NO. Bcz we need to build it again in Jenkins. In order to make it automated use webhooks.

TASK – deploy the app using webhooks

Q/A-

DAY – 5 NEXSUS INTEGRATION WITH JENKINS



->paina deployment chusthey, suppose v1,v2,v3 prod lo success ga deploy chesamu but v4 fail ayyindi anukundam. Since fail ayyindi kabatti manam prev's versions ki rollback avvali.manam clean package chestunnam deploy cheesy mundhu, so prev versions backup undavu.deeni overcome cheyyadaniki nexus ni use chestamu

JENKINS INSTALLATION

->Mustafa github ki velli all setups file ni mi local/github lo pettukondi

->manam manual ga Jenkins ni install cheyyali antey atleast it takes 15-20mins of time, so goto Jenkins.sh file n copy the lines in it.

->goto root>create a file vth .sh (vim Jenkins.sh)>goto insert mode n paste those lines>save n continue(:wq)

->run that file(sh Jenkins.sh)>it starts installing

->copy that pub-ip>paste in browser vth 8080 port(pub-ip:8080) to access Jenkins dashboard

TOMCAT INSTALLATION

->same as abv procedure but makesure the version is same in official web n script bcz the version gets changes every 1 to 2 weeks of duration.

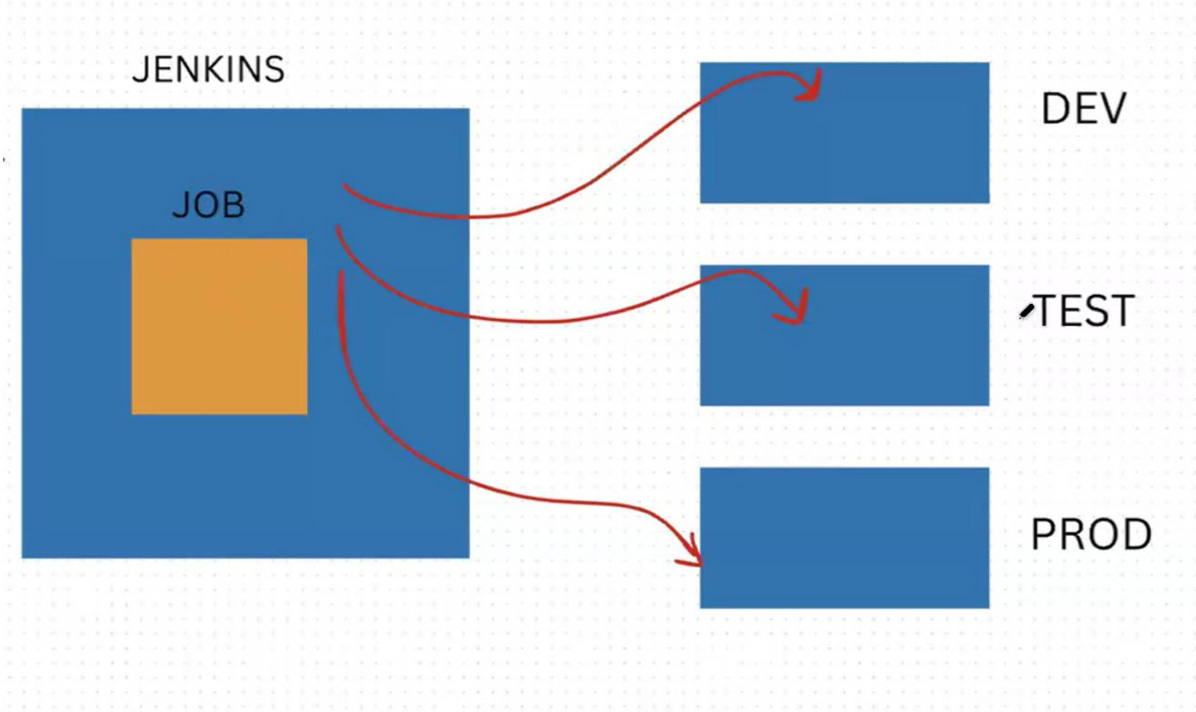
->credentials koda a file lo untadi

->it takes some 5mins to install tomcat.

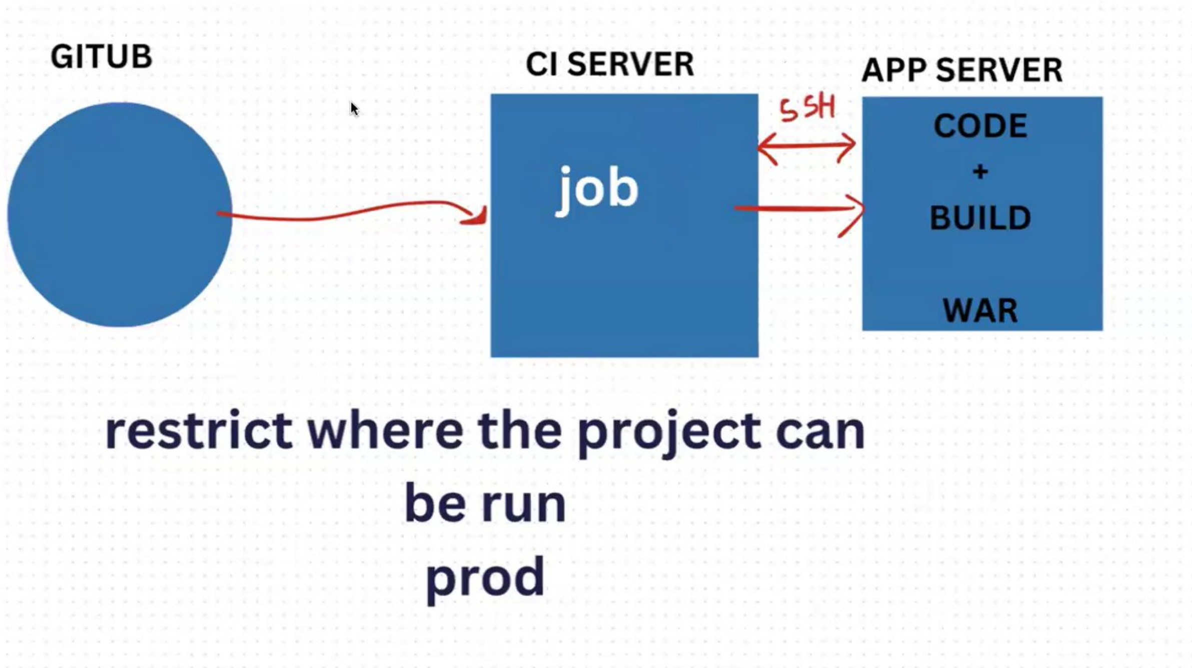
NOTE – makesure to have the maven n deploy to container plugins in ur Jenkins

NOTES PREPARE CHEYYALEDU, JST CHUSANU(33min varaku chusanu)

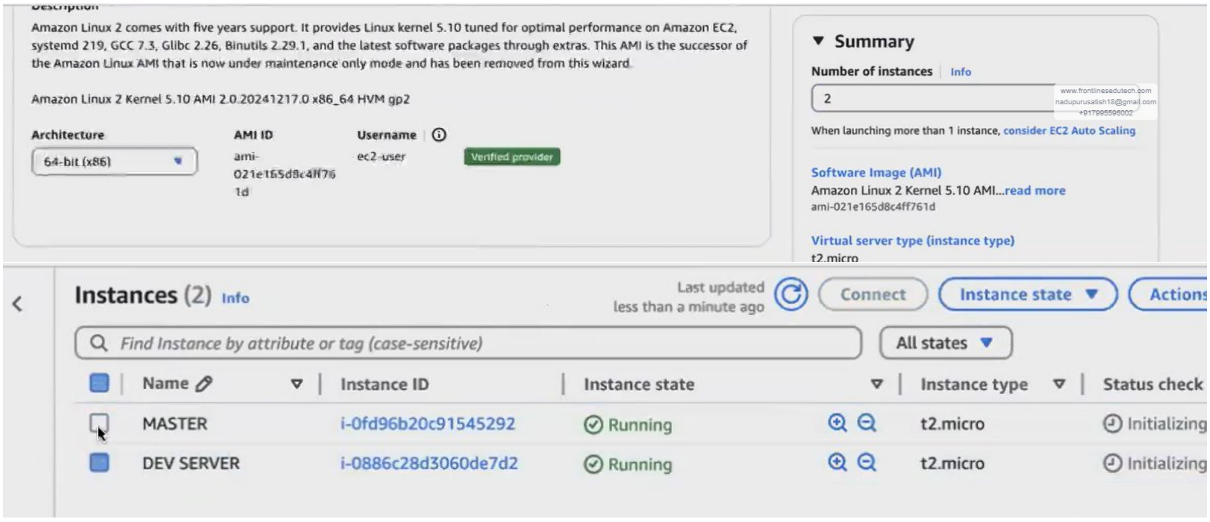
DAY – 6 MASTER-SLAVE & USER MANAGEMENT



->manam previous ga github lo unna code ni CI server loki vachaka (build+test avutadi)app server lo deploy cheesy vallam. But edi aithey build+test jarugutundho n manual ga deploy chestunamo,adi diect ga app server lo jaragali, jst manam Jenkins job trigger chesthey.



While creating instances give 2 as shown below.



->in master server install Jenkins using script

->in dev server install tomcat using script

->now in Jenkins create a job vth ggithub repo.

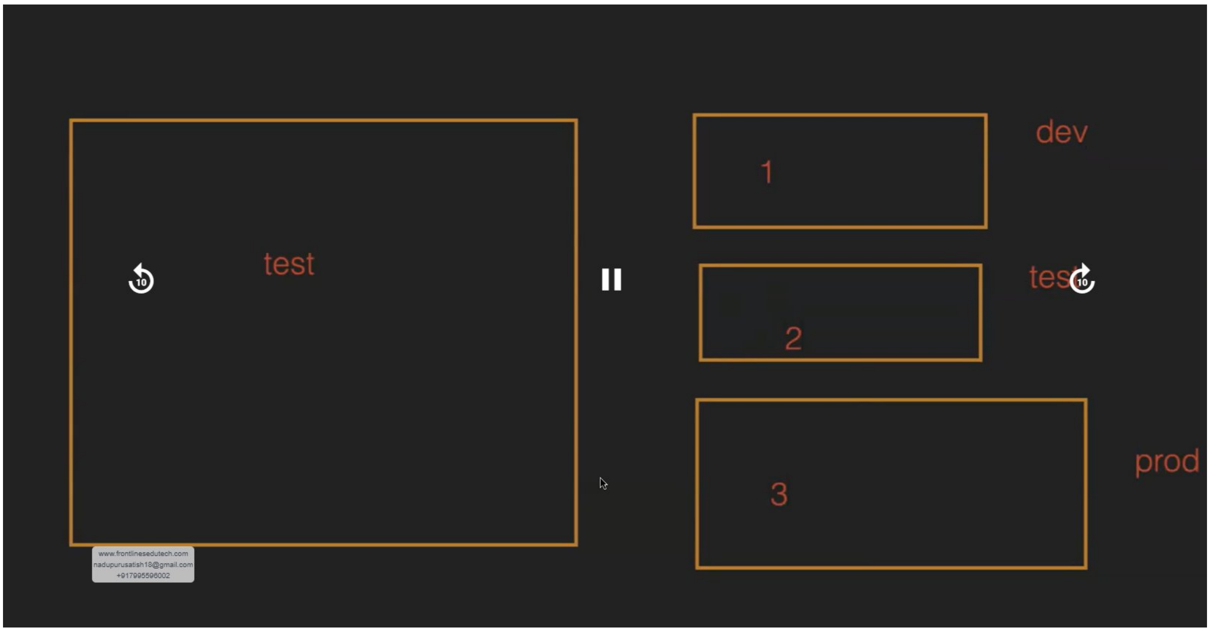
->if we build the job, then the code will goto CI server only. So, CI server ki vellakunda direct ga dev server ki vellali antey master-slave concept setup cheyyali

a)slave ni master tho attach cheyyali

b)slave server (app server) lo java17 ni install cheyyali

c)Jenkins dashboard>manage Jenkins>node(slave/server)>add new node(slave/app)>sel permanent agent>create>under no of executors>we can give any no of executors(based on req, ma team 3devs unnaru,so 3)>under remote root dir give app_server path(/home/ec2-user/Jenkins/=(/home/ec2-user/ path def ga untadi, last lo /Jenkins anedi automatic ga create avutadi)>label lo manaki ey server lo run avvali>under usage sel only build jobs vth label expressions matching this node(label exp match aithey ney build cheyye ani meaning)>under launch method sel launch agents via ssh>under host give priv-ip of app_server(bcz server stop chesina priv-ip constant ga untadi)>under credentials sel Jenkins, under kind sel ssh username vth pri-key(def_user ec2-user/give pem file pword)>click on add n sel that>under host key verification strategy sel non verifying verification strategy>save

d) we can left side that it is launching that slave



Label lo manam mention cheesy server lo adi run avutadi

->now goto job n build it

Job>configure>we can see option restrict where this project can be run, enable n give dev>save>now click on build now

->build fail avutadi, bcz git dev server lo install cheyyaledu

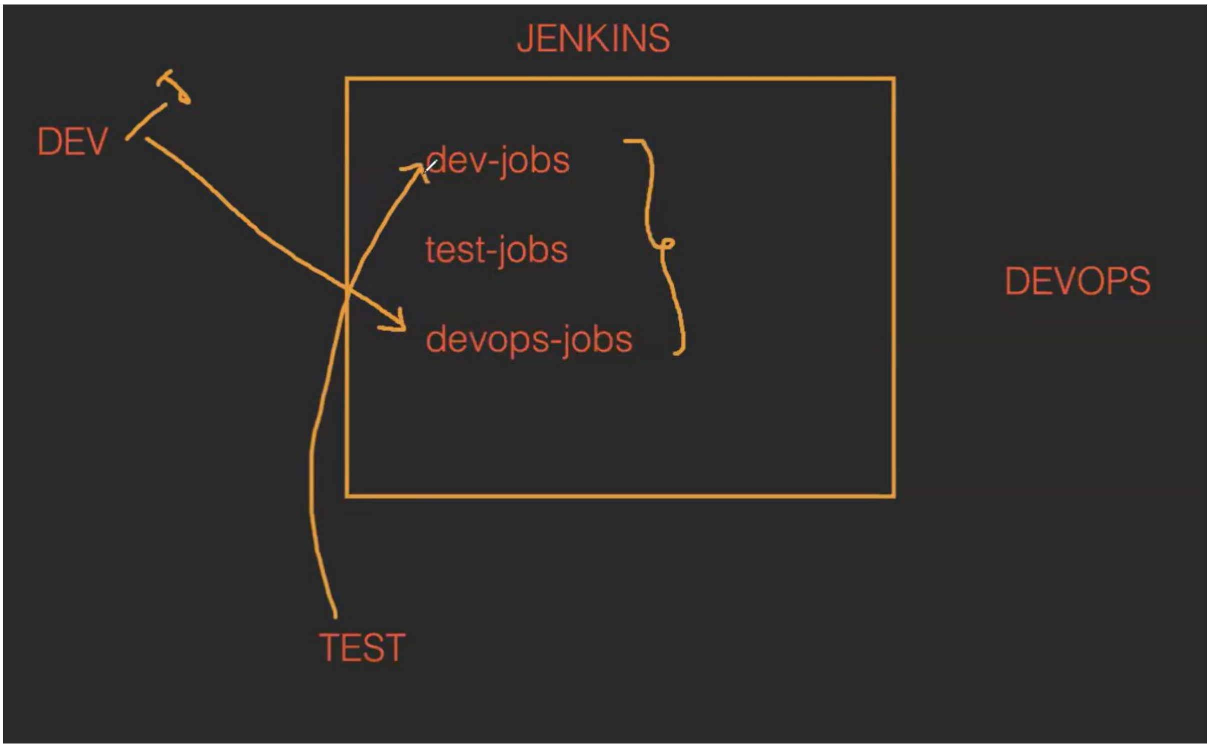
->install git n build it, success avutadi.

NOTE – we can't deploy the code at a tym in dev n test server

SECURITY OF A JENKINS

->realtime lo oka project lo diff teams(dev/test/devops) untai, mari veella andariki same Jenkins dashboard untadhi ahhhh???

Ans – yes, but Jenkins admins untaru veellu evariki kavalisina dashboards valla ki istaru(dev – dev dashboard, tester – test dashboards...etc) ala access istaru. Bcz oka dev team jr developer kotta ga vachadu anni dashboards ki access undi ga ani tanu prod jobs build chesthey em aina issues vastai. So, evariki kavalisina dashboards vallaki access istaru.



->malli job-create/build/delete/view policies edi kavali antey adi istaru

->eg – ma team lo oka ravi aney person ki job build cheyyadaniki permissions kavali ani adigithey as a Jenkins admin, nen ravi aney user ni create chesi- a credentials a person ki Istanu only vth build job permissions.

Real-time example

->1st oka plugin install cheyyali

Goto manage Jenkins>plugins>available plugins>search role-based authorization strategy>install

->creating user

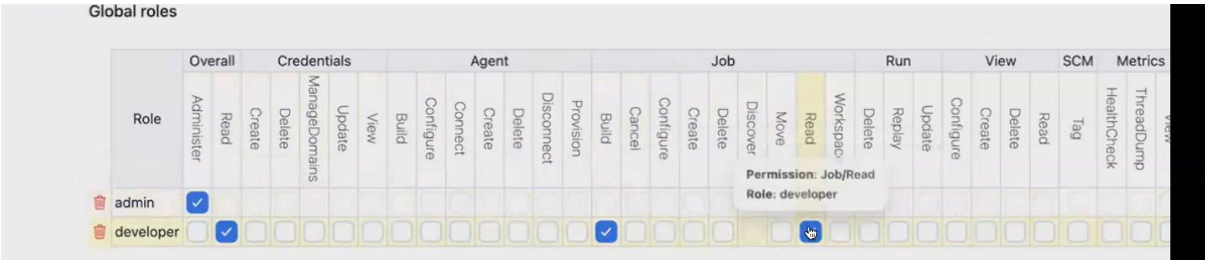
Manage Jenkins>users>create user>give credentials>create user

->after creating user, we need to assign roles

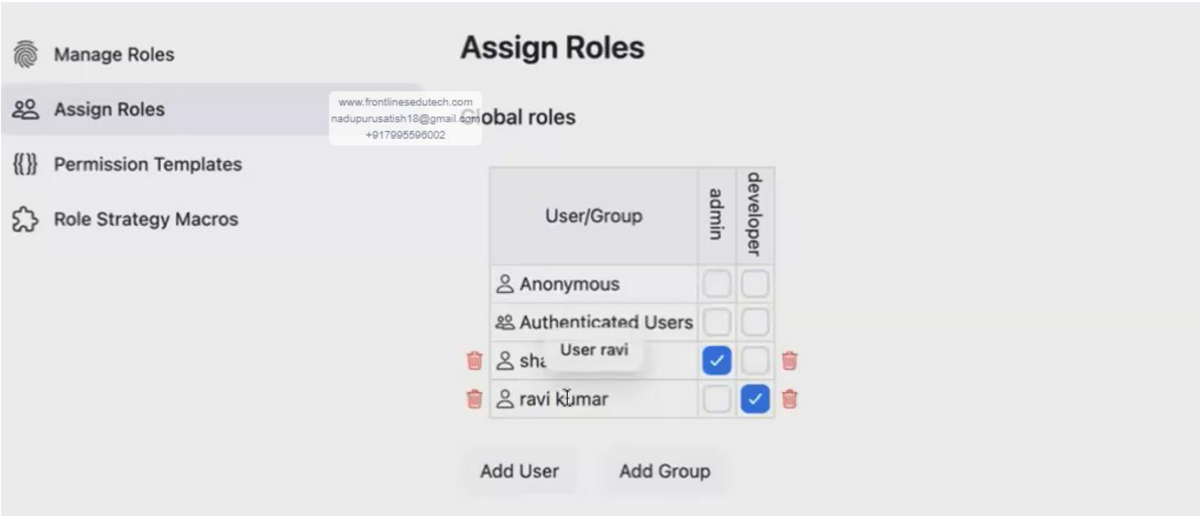
Goto manage Jenkins>security>under authorization sel role-based authorization strategy>save it

->now we can see option manage and assign roles in manage Jenkins

Goto manage and assign roles> under role to add give developer>assign wht roles required>click on add



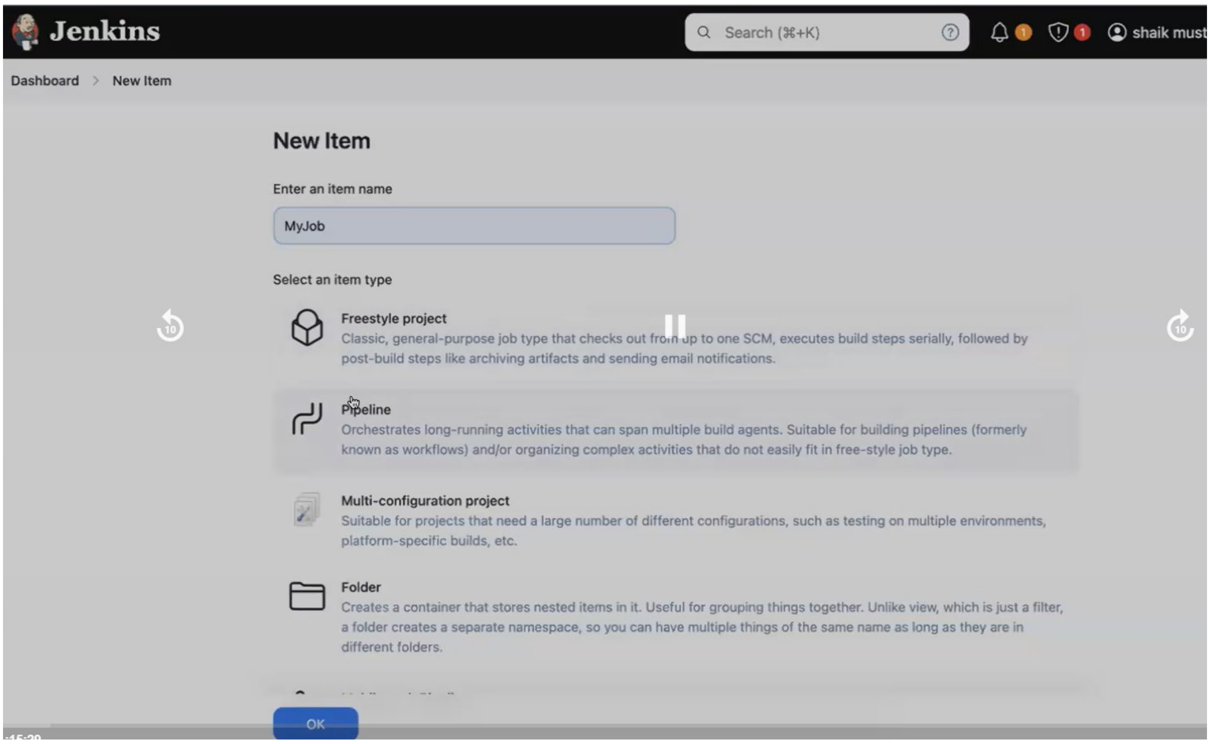
->goto assign roles, now we need to assign user to this role>click on save



NOTE – 1. we can't create a multiple users at a time.

2. we can reset the users pword.

DAY – 7 JENKINS PIPELINES

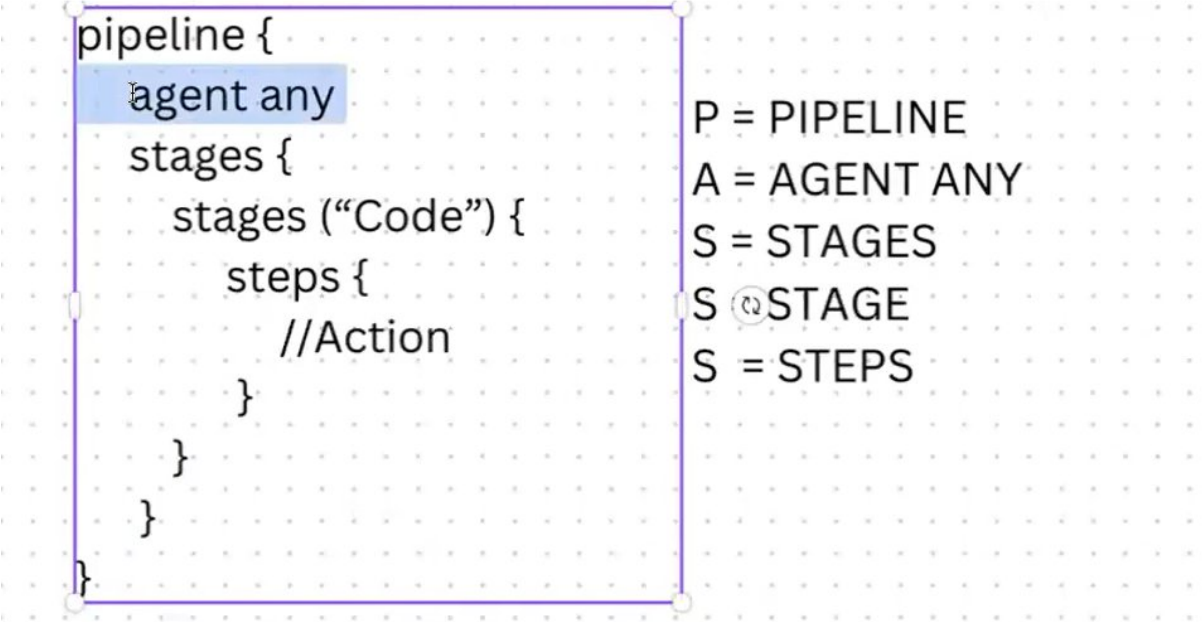


->realtime lo mostly pipelines ni use chesukoni jobs create chestamu, but konni freestyle jobs koda untai.

->in pipelines we use groovy scripts

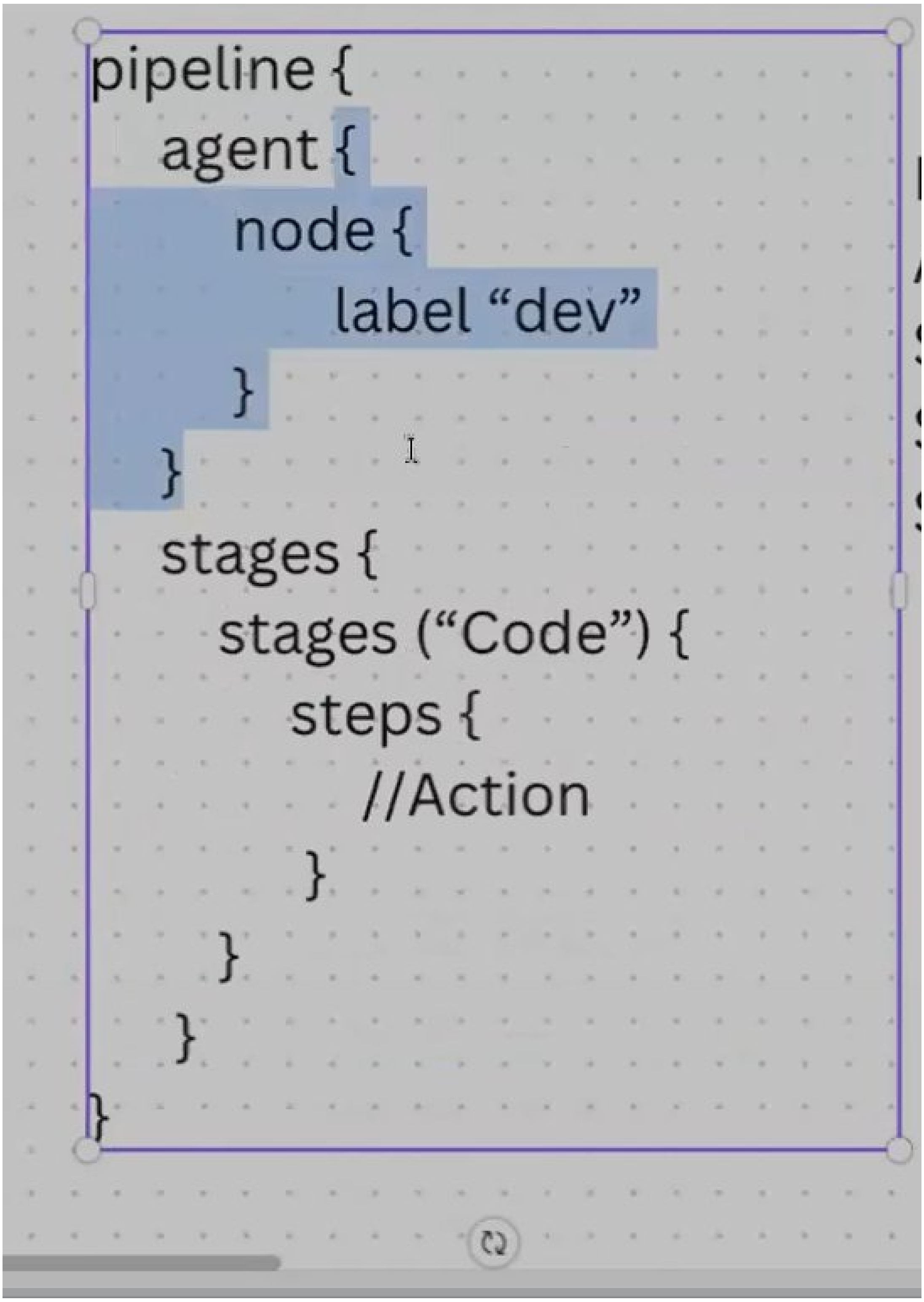
JOBS USING PIPELINES

1.groovy syntax for scripted pipeline



Agent any – ey dev/test/prod env lo run avvali ani, by default master lo execute.

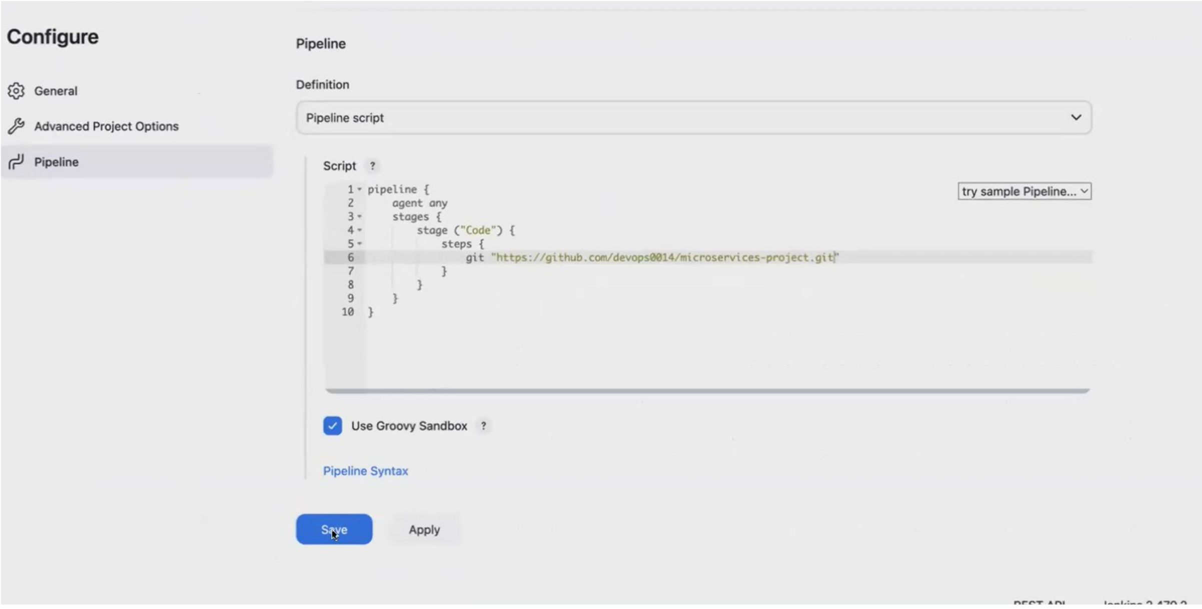
->particular ga dev lo run avvali antey


```
pipeline {
    agent {
        node {
            label “dev”
        }
    }
    stages {
        stages (“Code”) {
            steps {
                //Action
            }
        }
    }
}
```

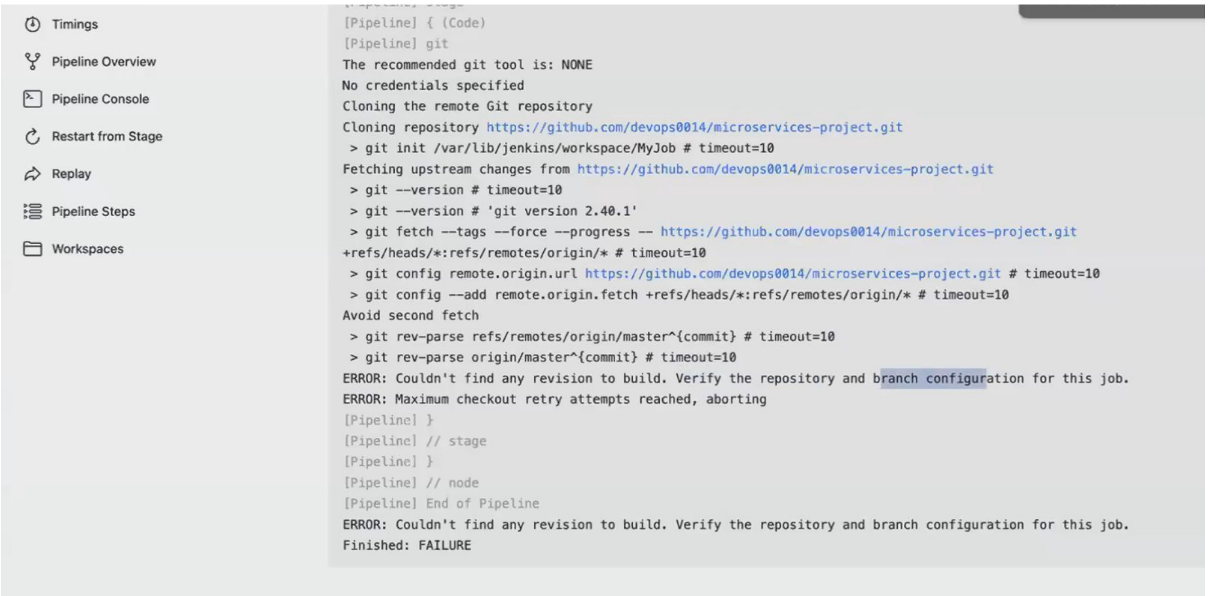
->create a job by selecting groovy>click on ok>goto pipe-line>under script write which task u r gng to perform(below we have wrote get the code frm github)> save it

->now click on build now,after success it gets code frm github to CICD server

->if it gets failes, 2 reasons – script is wrong/git is nt installed in server



->ee console o/p err bcz manam github nunchi techukomanamu but ey branch ani mention cheyyala ga



->to get particular branch

Goto job>configure>pipeline>pipieline syntax>under sample step sel git>in repository url give github url>under branch give main>generate pipeline script, copy that url n paste it in steps>save

->now click on build now

NOTE – inko branch ki job create cheyyali antey in script change the branch name alone n save it nd then build it.

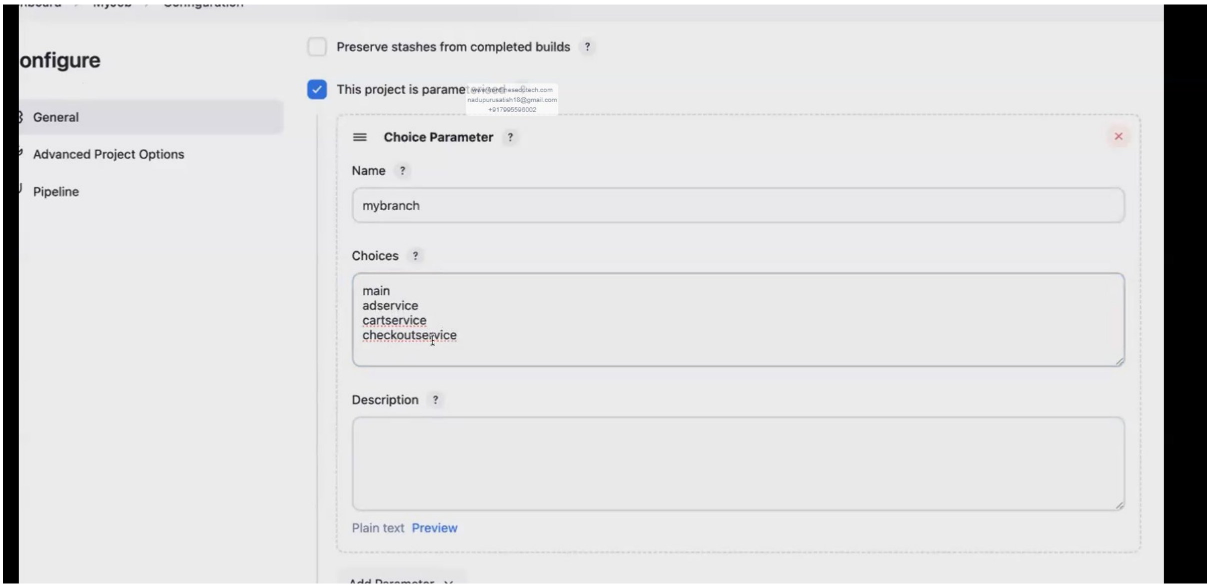
TO CHECK THE CODE REACHED TO CI SERVER OR NOT

->in dashboard we won't see any workspace option fr pipelines

->but goto console o/p copy that path n chk in server

TASK - pipeline job using private github repository

->mi manager prathi sari okkokasari okkoka branch ani cheptadi, so dyamic that branch we use choice paramterized option n in script give $parameter_name>save it



->now based on req sel the branch n build

->if we don't know the wht branches are available

Use string param>give name under name>in default don't provide anything>save it

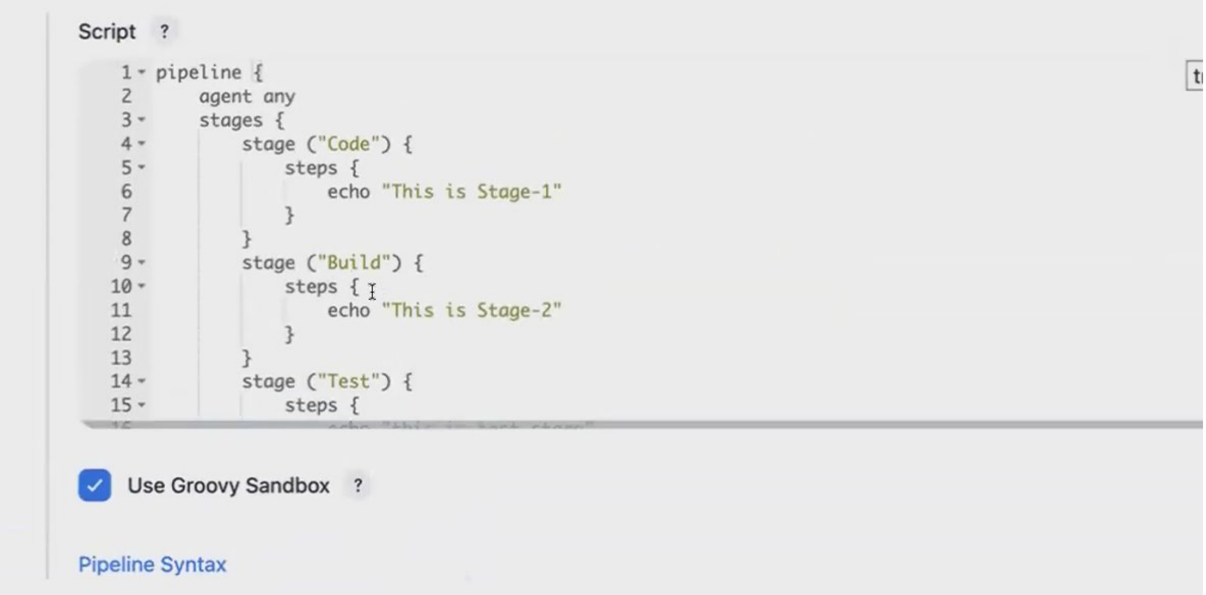
->while building vth build with parameters

Under my branch>give which branch u want to build.

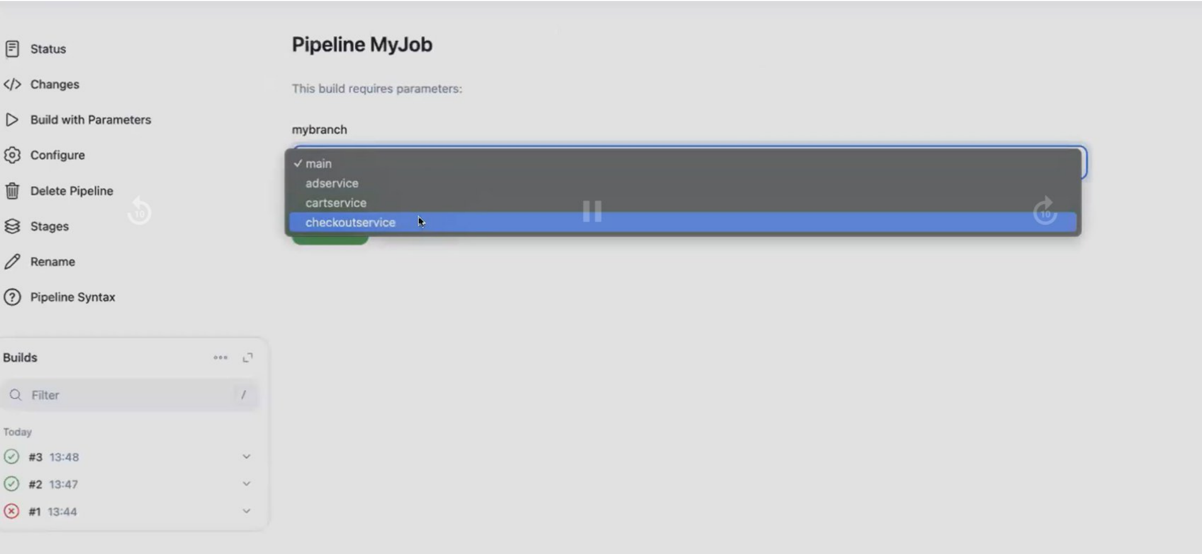
PIPELINE FUNDAS

1.multi-stage pipeline

->oka pipeline lo chala stages untai



NOTE – Each stage contains single step



->to know the stages goto stages option

MULTI-SATGE PIPELINE WITH COMMANDS

1. creating file

->edaina cmd mention cheyyali antey sh “cmd_to_perform”

->we can give single/double codes

->sh =shell/script

Script ?

```
pipeline {
    agent any
    stages {
        stage ("Code") {
            steps {
                sh "touch amazonwebservices"
            }
        }
    }
}
```


->after building we can see that file created or nt in server

2. creating folder

```
pipeline {
      agent any
      stages {
            stage ("Code") {
                  steps {
                        sh "mkdir azure"
                  }
            }
      }
}
```

2. we can give n no of cmds in steps

Script ?

```
pipeline {
      agent any
      stages {
            stage ("Code") {
                  steps {
                        sh "mkdir roopesh"
                        sh "touch krishna"
                        sh "timedatectl"
                        sh 'cal'
                  }
            }
      }
}
```

Use Groovy Sandbox ?

->we can use like this also, ila isthey diff shell o/p's ravu, only oka o/p lo vasthai

Script ?

```
pipeline {
    agent any
    stages {
        stage ("Code") {
            steps {
                sh '''
                touch likith
                mkdir anil
                cal
                getent passwd
                '''
            }
        }
    }
}
```

3.variables

->variables – local,global variables

->agent any taravata var’s declare cheyyali

Script ?

```
pipeline {
    agent any
    environment {
        abc = "Roopesh"
        place = "Hyd"
    }
    stages {
        stage ("code") {
            steps {
                echo "My name is $abc and i am from $place"
            }
        }
    }
}
```

Use Groovy Sandbox ?

->var’s ni anni stages lo use chesukovachu

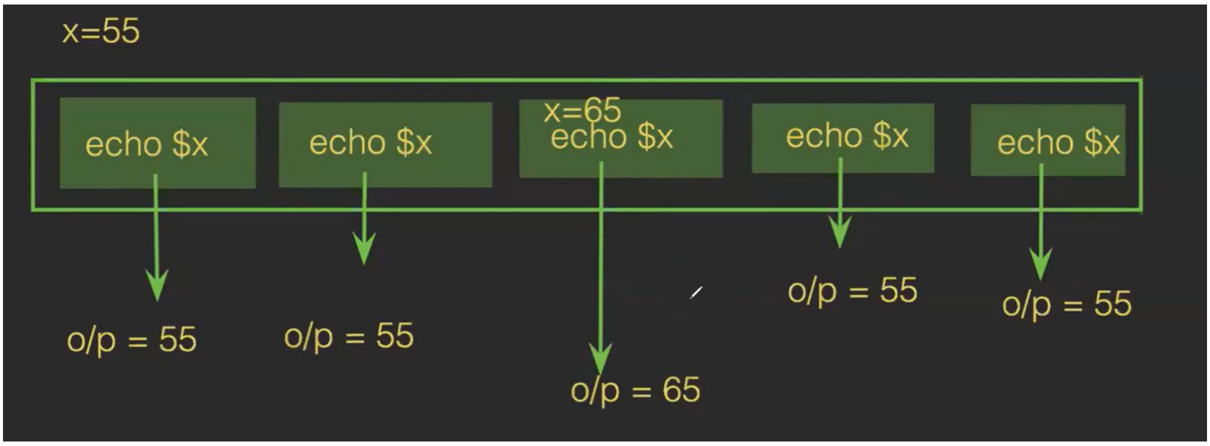
Script ?

```
    environment {
        abc = "Roopesh"
        place = "Hyd"
        company = "TCS"
    }
    stages {
        stage ("code") {
            steps {
                echo "My name is $abc and i am from $place"
            }
        }
        stage ("Build") {
            steps {
                echo "I am working for $company in $place"
            }
        }
```

->var vth numbers

Script ?

```
        place = "Hyd"
        company = "TCS"
        exp = 4
    }
    stages {
        stage ("code") {
            steps {
                echo "My name is $abc and i am from $place"
            }
        }
        stage ("Build") {
            steps {
                echo "I am working for $company in $place from last $exp years"
            }
        }
```



1.global var

Script ?

```
pipeline {
    agent any
    environment {
        x = 55
    }
    stages {
        stage ("Code") {
            steps {
                echo "The value is $x"
            }
        }
        stage ("Build") {
            steps {
                echo "The value is $x"
            }
```

2.local var

cript ?

```
        }
        stage ("Build") {
            environment {
                x = 65
            }
            steps {
                echo "The value is $x"
            }
        }
        stage ("Test") {
            steps {
                echo "The value is $x"
            }
        }
    }
}
```

NOTE – 1. Every stage name should be diff

2. local var has high priority

->if we want to print default values(like environment var's)

```
pipeline {
    agent any
    stages {
        stage ("Code") {
            steps {
                script {
                    sh 'printenv'
                }
            }
        }
    }
}
```

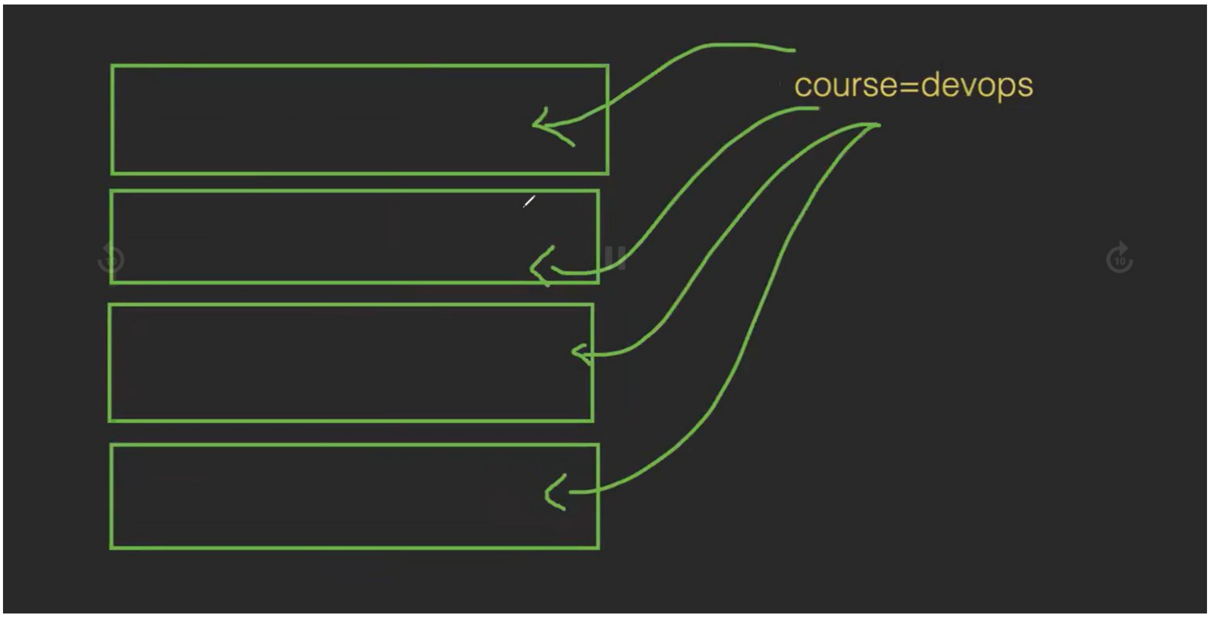
We need to see the o/p in console

```
[Pipeline] sh
+ printenv
JENKINS_NODE_COOKIE=bef4dbe3-edd6-4ca0-9fec-d43ed7a8d9db
BUILD_URL=http://3.95.198.94:8080/job/MyJob/20/
SHELL=/bin/false
HUDSON_SERVER_COOKIE=13f54ee5ba736329
STAGE_NAME=Code
BUILD_TAG=jenkins-MyJob-20
JOB_URL=http://3.95.198.94:8080/job/MyJob/
WORKSPACE=/var/lib/jenkins/workspace/MyJob
RUN_CHANGES_DISPLAY_URL=http://3.95.198.94:8080/job/MyJob/20/display/redirect?page=changes
USER=jenkins
RUN_ARTIFACTS_DISPLAY_URL=http://3.95.198.94:8080/job/MyJob/20/display/redirect?page=artifacts
JENKINS_HOME=/var/lib/jenkins
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin
RUN_DISPLAY_URL=http://3.95.198.94:8080/job/MyJob/20/display/redirect
_=/usr/bin/printenv
PWD=/var/lib/jenkins/workspace/MyJob
HUDSON_URL=http://3.95.198.94:8080/
JOB_NAME=MyJob
BUILD_DISPLAY_NAME=#20
BUILD_ID=20
JENKINS_URL=http://3.95.198.94:8080/
NOTIFY_SOCKET=/run/systemd/notify
JOB_BASE_NAME=MyJob
RUN_TESTS_DISPLAY_URL=http://3.95.198.94:8080/job/MyJob/20/display/redirect?page=tests
```

NOTE – 1. Oka pipeline lo 5 stages unnai, only 2nd stage lo access chesukovali antey declare it as local

2. 5 stages lo access chesukovali antey declare it as global

3. chala pipelines lo oka var ni use chesukovali antey declare it as global



Goto manage Jenkins>sys>under global properties sel env var's>add name n val>save it

->we can use this var fr in any pipeline


Pipeline script

Script ?

```
pipeline {
    agent any
    stages {
        stage ("Code") {
            steps {
                echo "I am learning $course"
            }
        }
    }
}
```

Q/A-

DAY – 8 JENKINS PIPELINE FOR DEPLOYMENT

MANUALLY ENABLING THE PARAMETERS

->jst created script vth a var(abc) using a job vth pipeline

->enable this project is parameterized>sel string param>under default give same name as abc>under val don't give any val while building we can give>save it

->now click on build with param>give val under abc>build

->ippudu console o/p chusthey, build chesetapudu ey val isthey adi print avutadi.

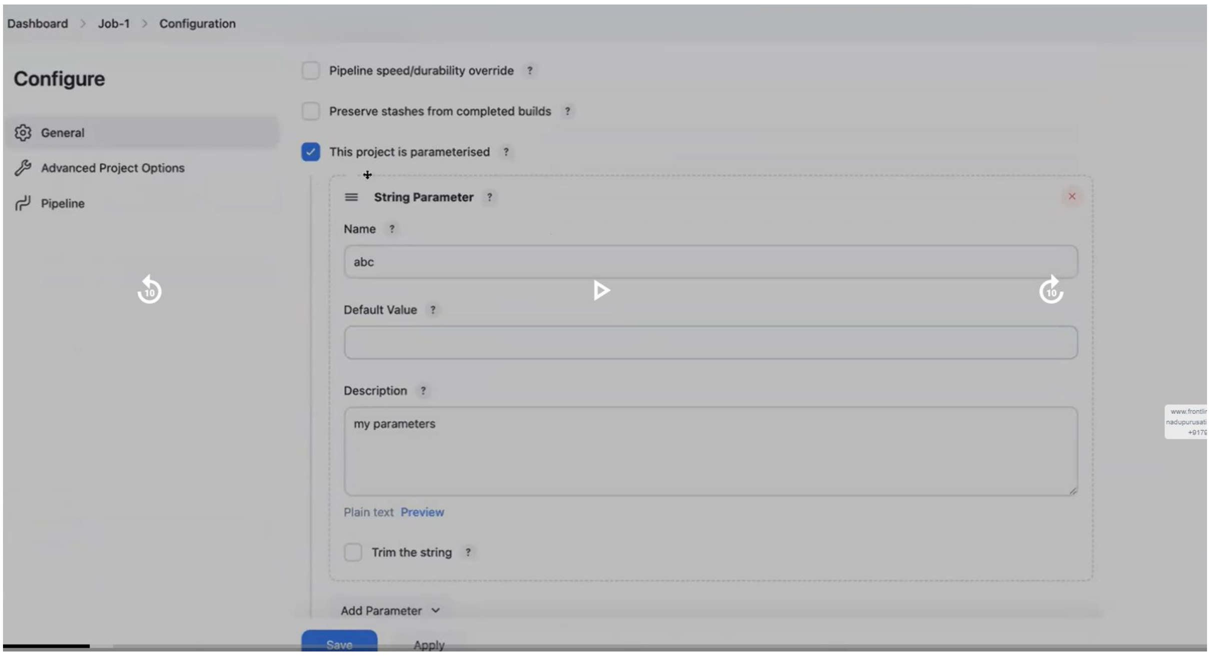
->idi manual process, we can automate it

AUTOMATICALLY ENABLING THE STRING PARAMETERS

->to enable the parameters automatically, use below script in pipeline

```
pipeline {
    agent any
    parameters {
        string(name: "abc", defaultValue: "", description: "my parameters")
    }
    stages {
        stage('Hello') {
            steps {
                sh 'echo "My name is $abc"'
            }
        }
    }
}
```

->goto configuration n chk whether it is enabled, after enabling save it n we'll get the build with param's option.



->click on build vth param's, pass the val. In console o/p it will print that val

AUTOMATICALLY ENABLING THE STRING PARAMETERS

->in the script give choice in param's n corresponding fields

Pipeline script

Script ?

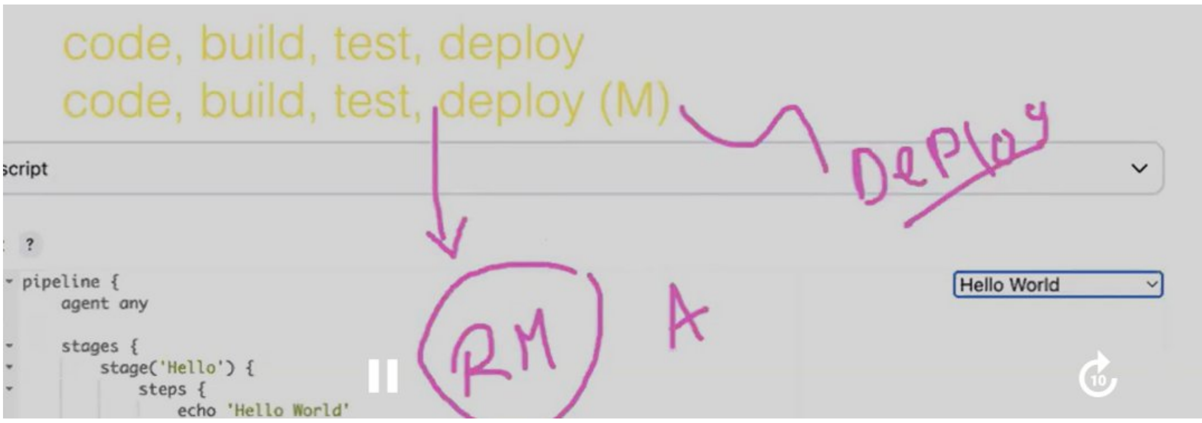
```
pipeline {
    agent any
    parameters {
        choice(name: "abc", choices: ['master', 'main', 'version1', 'version2'], description: "my parameters")
    }
    stages {
        stage('Hello') {
            steps {
                sh 'echo "My name is $abc"'
            }
        }
    }
}
```

Use Groovy Sandbox

Pipeline Syntax

->save it n build now

->once it is built, we’ll see the build vth param’s option



Continuoud deploy – code,build,test,deploy(everything is automation)

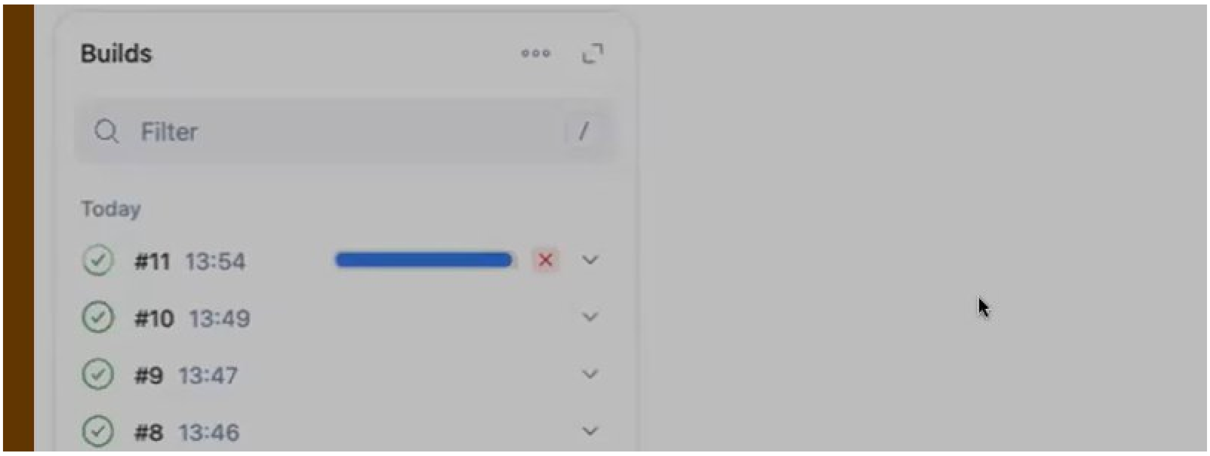
Continuoud delivery – code,build,test-- (everything is automation) only deploy manual(manager approval ayyaka ey deliver chestamu)

->continuous deploy automatic ga jarugutadi, but delivery lo adi build cheska ey stage lo approval kavalo adugutadi

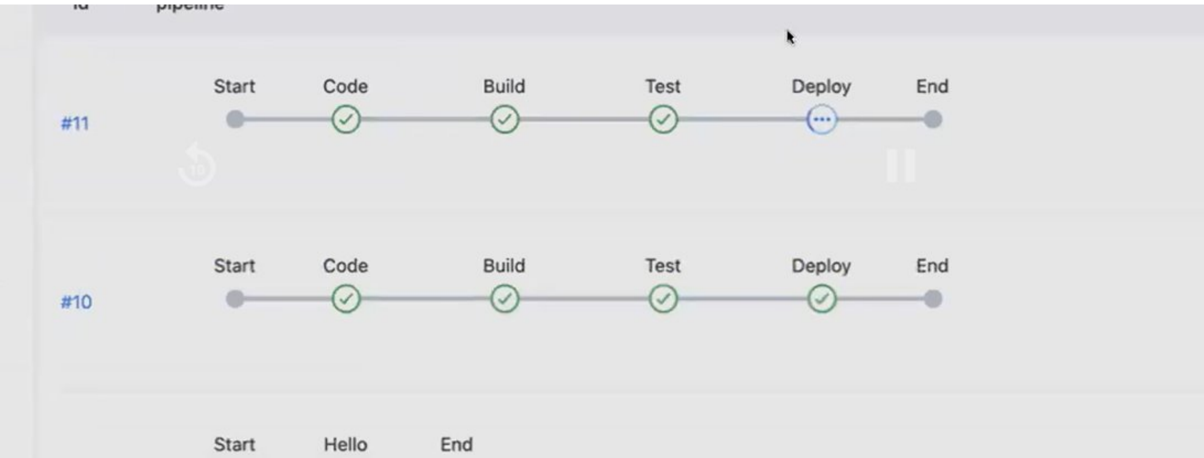
```
            steps {
                echo 'This is Test stage'
            }
        }
        stage('Deploy') {
            input {
                message "Can i deploy?"
            }
            steps {
                echo 'This is Deploy stage'
            }
        }
    }
}
```

A msg lo mention chesindhi adugutadi

->build now click chesaka, a build status alaney untadi kani complete avvadu, until unless it is approved



->goto stages to approve it

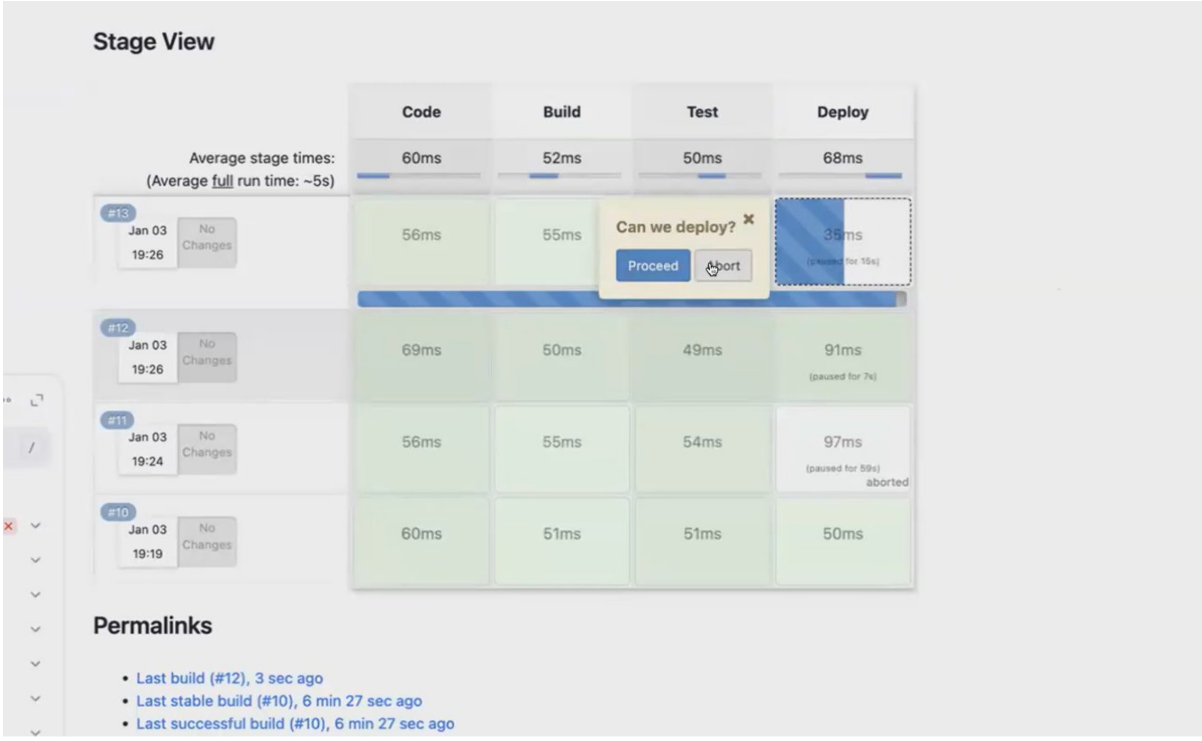


See it stoped in deploy stage,click on that

In wait for interactive input> expand it we can see 2 options>proceed it, then application will deploy

Ila proceed chestuntey avvadam ledu

Install that plugin for stages

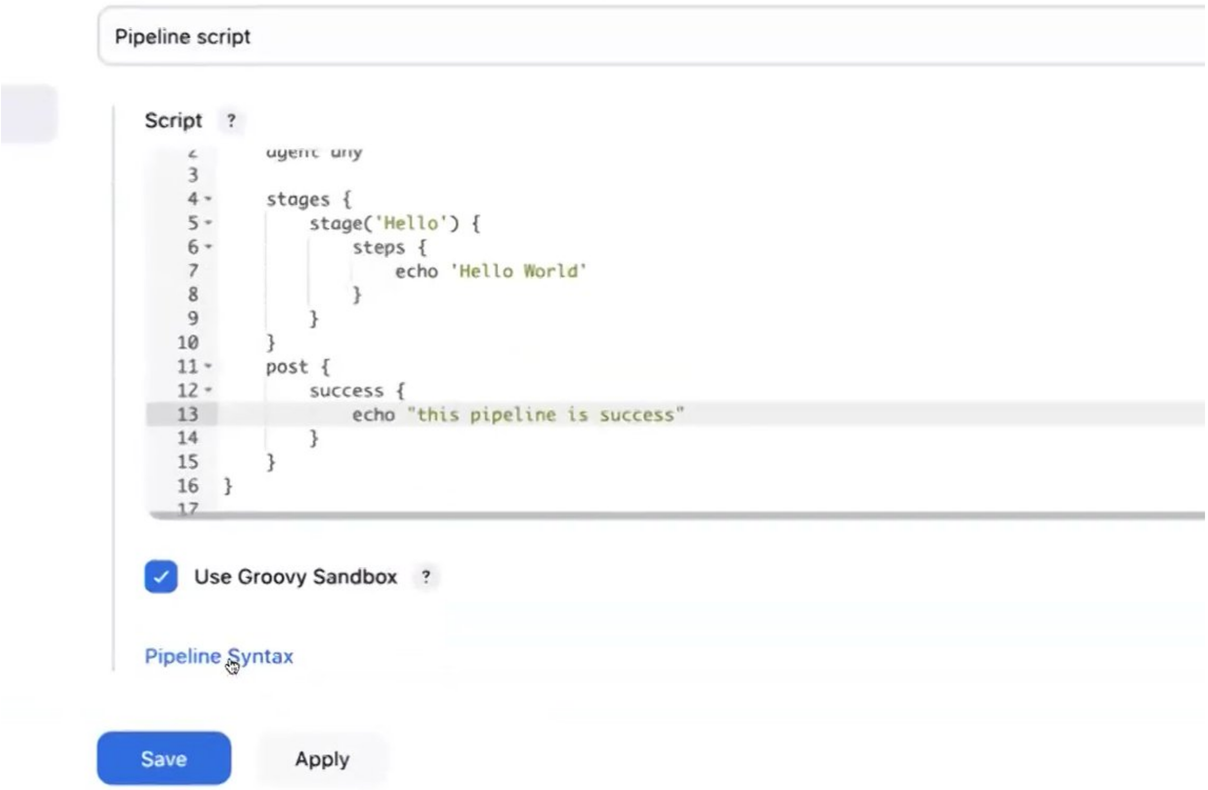


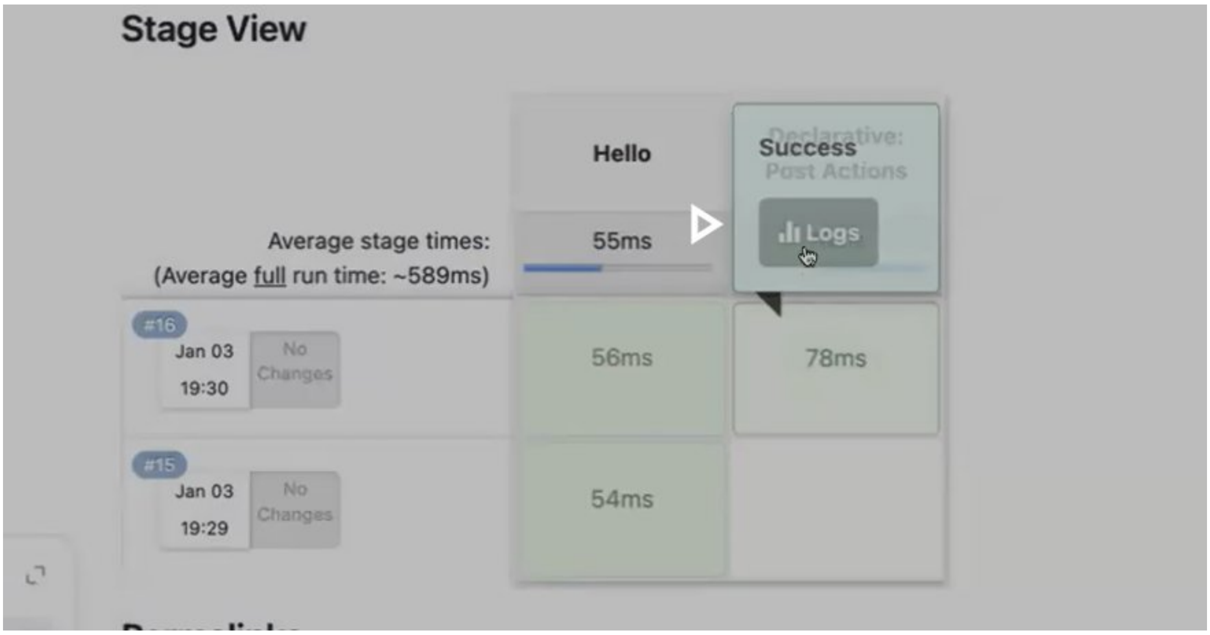
Ee plugin use chesthey, manaki clear ga ey stage lo stop ayyindi ani clear ga telustadi n proceed it

POST BUILD ACTIONS

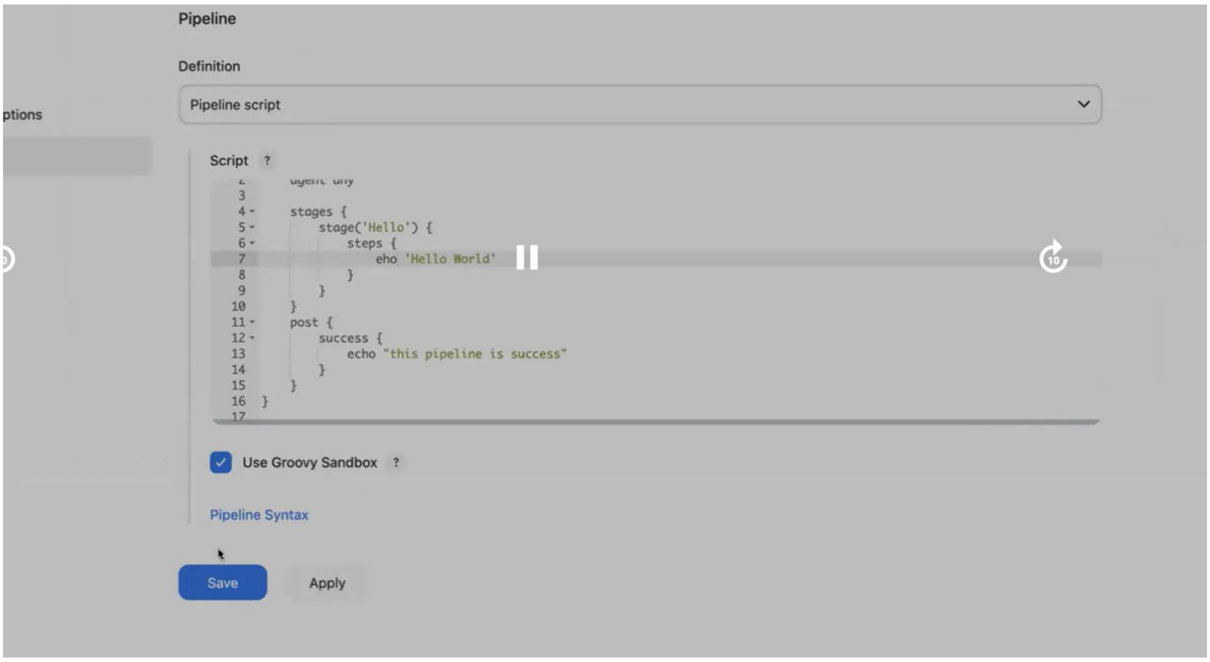
1.success msg after build success

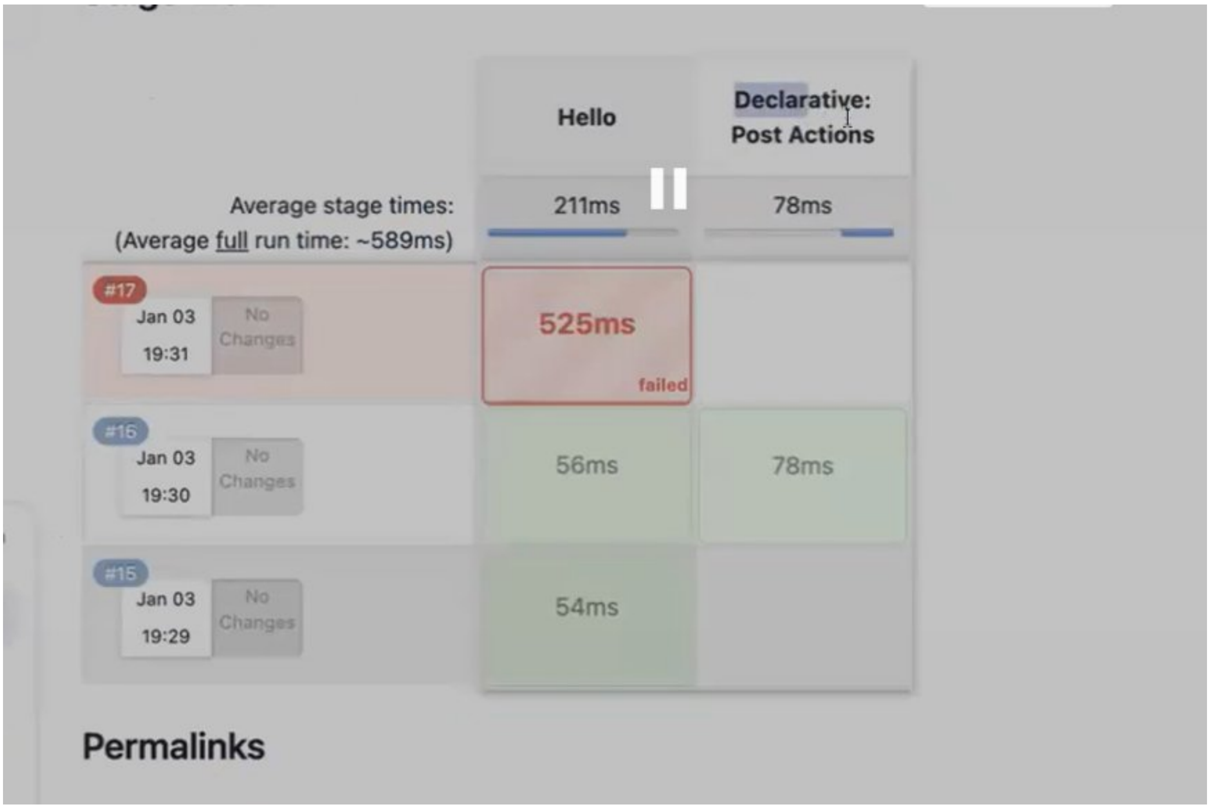
->realtime lo success msg pettaru, mail/teams/notification vachela set chesukuntaru





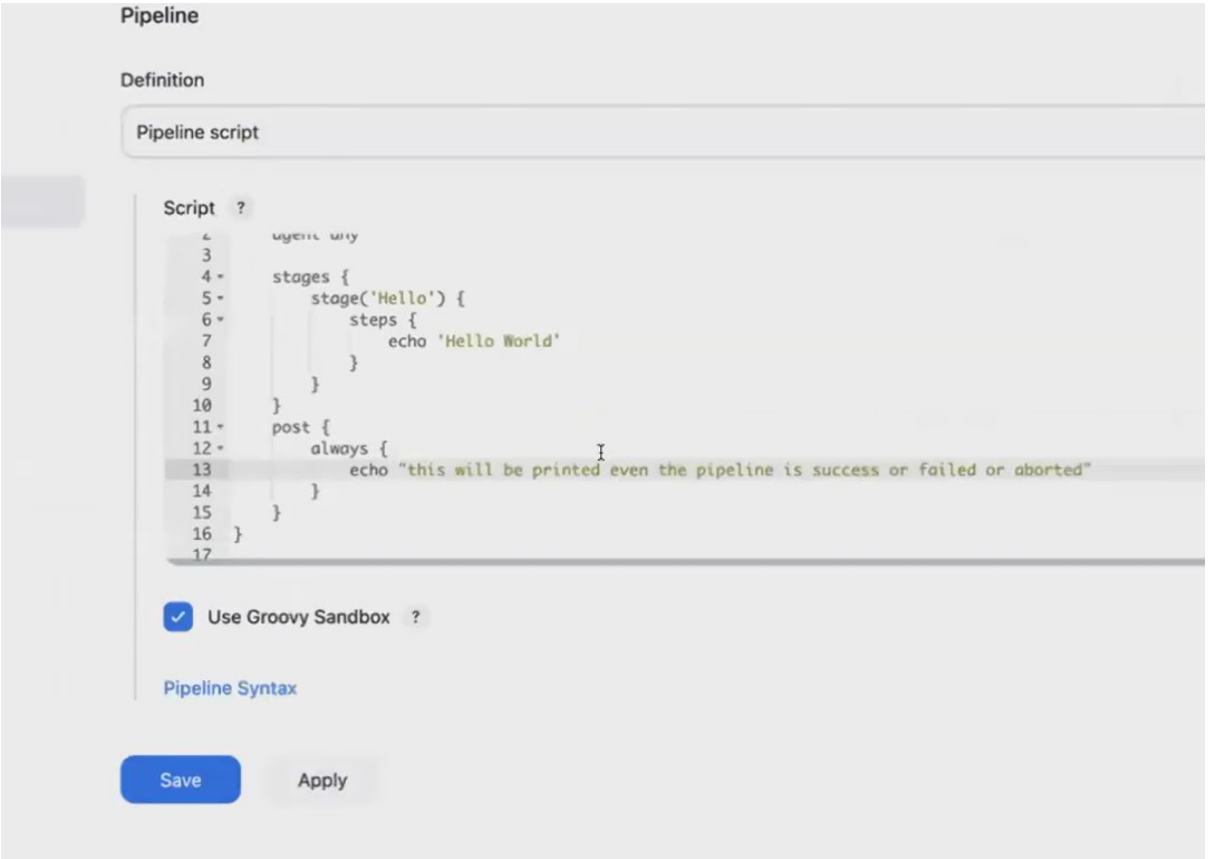
->echo spelling wrong ga ichanu, msg print avvadu bcz success aithey print cheyyamanamu

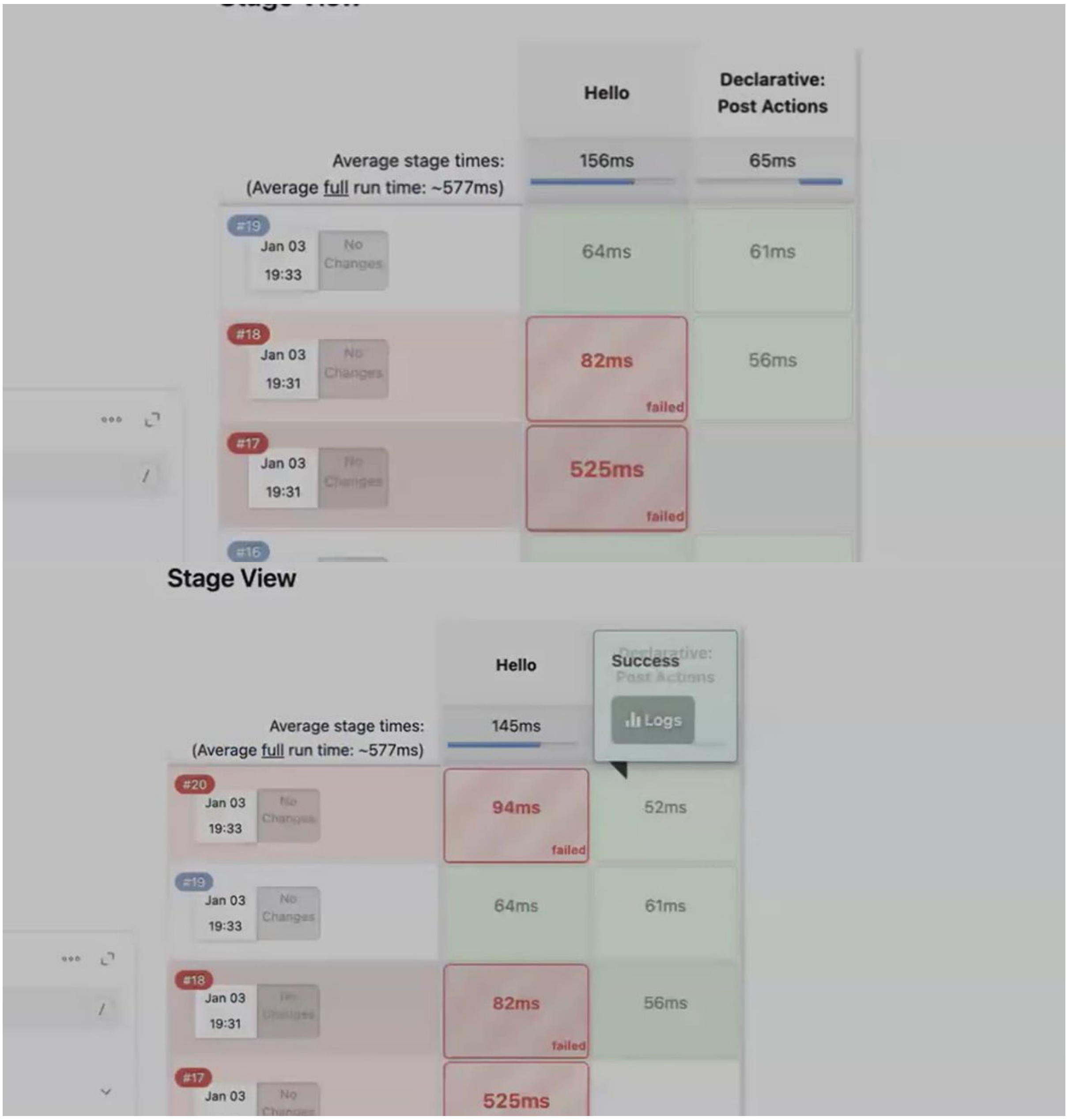




Em msg print avvaledu post buld actions lo bcz success aithey ney print cheyyamani mention chesamu

->success/failure it shd print the msg





For success/fail/abort it printing the post msg

DEPLOY AN APPLICATION USING PIPELINE

1.getting code frm github to CI server

-> create a Jenkins job using pipeline, using pipeline syntax get the github

Clickon pipeline syntax>git>repo url>master>generate pipeline script>copy that url>paste it in steps

Script ?

```
pipeline {
    agent any

    stages {
        stage('Code') {
            steps {
                git 'https://github.com/devops0014/one.git'
            }
        }
    }
}
```


->now build now n chk source code got it frm github or not in server

2. now code is in CI server, we need to build it

->to build we shd have Jenkins vth maven configuration

->to build by using pipeline, use the below stage

Script ?

```
pipeline {
    agent any

    stages {
        stage('Code') {
            steps {
                git 'https://github.com/devops0014/one.git'
            }
        }
        stage ("Build") {
            steps {
                sh 'mvn clean package'
            }
        }
    }
```

Mana server lo maven install cheyyaledu, but Jenkins lo mvn ni configure chesamu. So ni pipeline ki cheppali nen Jenkins lo configure chesina global mvn tool ni use chesukoni build chestanu ani

Script ?

```
pipeline {
    agent any
    tools {
        maven "mymaven"
    }

    stages {
        stage('Code') {
            steps {
                git 'https://github.com/devops0014/one.git'
            }
        }
        stage ("Build") {
            steps {
                sh 'mvn clean package'
```

3.build chesamu, ippudu app server lo deploy cheyyali

->deploy cheyyali antey app server kavali ga, so oka server ni create cheyye

->we shd have tomcat in our server

->install the tomcat using that githib tomcat script

->tomcat lo app ni deploy cheyyali antey Jenkins lo deploy to container plugin undali

->once plugin install chesaka, got our job>configure>pipeline>pipeline syntax> sel deploy:deploy war/ear to container, incase ee option kanipinchakapothey restart the Jenkins using cmd(systemctl restart Jenkins).>under war/ear files give targrt/*.war>under context path give myapp>under containers sel tomcat 9 n give credentials,url>click on generate pipeline script> copy that url n paste it in one stage in script

->build the job n chk whether app is deployed in tomcat or not

4.integrating artifact nexsus

->in free trail we integrated artifact nexsus, we can do it using pipeline also

->we'll take one server for nexsus

->server lo nexsus install cheyyali

->install chesaka, nexsus lo war file ni store cheyyali antey on repo ni create cheyyali

->Jenkins lo nexsus artifact uploader plugin install cheyyali

->goto job>configure>pipeline>pipeline syntax>sel nexsus artifact uploader>configure the req fields>generate pipeline script>paste that one stage(make sure that nexsus shd be before the deploy stage)

->build it, change the versions in pom.xml n pipeline lo koda change chesi build cheyyandi

DAY – 9 JENKINS PIPELINES(PART-2)

PARALLEL EXECUTION

->paina chesina pipelines anni declarative, 85% pipelines are declarative

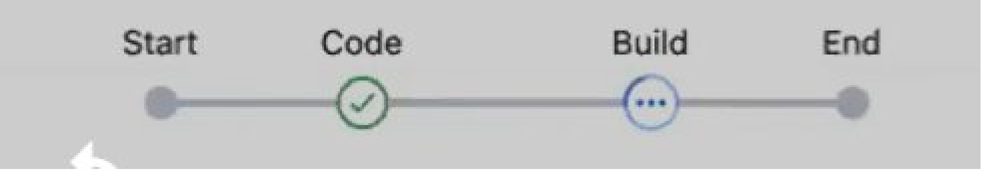
->pipeline lo 2 stages unnai, oka dani taravta okati execute avutadi

```
Script ?

pipeline {
    agent any
    stages {
        stage ("Code") {
            steps {
                echo "This is stage-1"
            }
        }
        stage ("Build") {
            steps {
                echo "This is stage-2"
            }
        }
    }
}
```

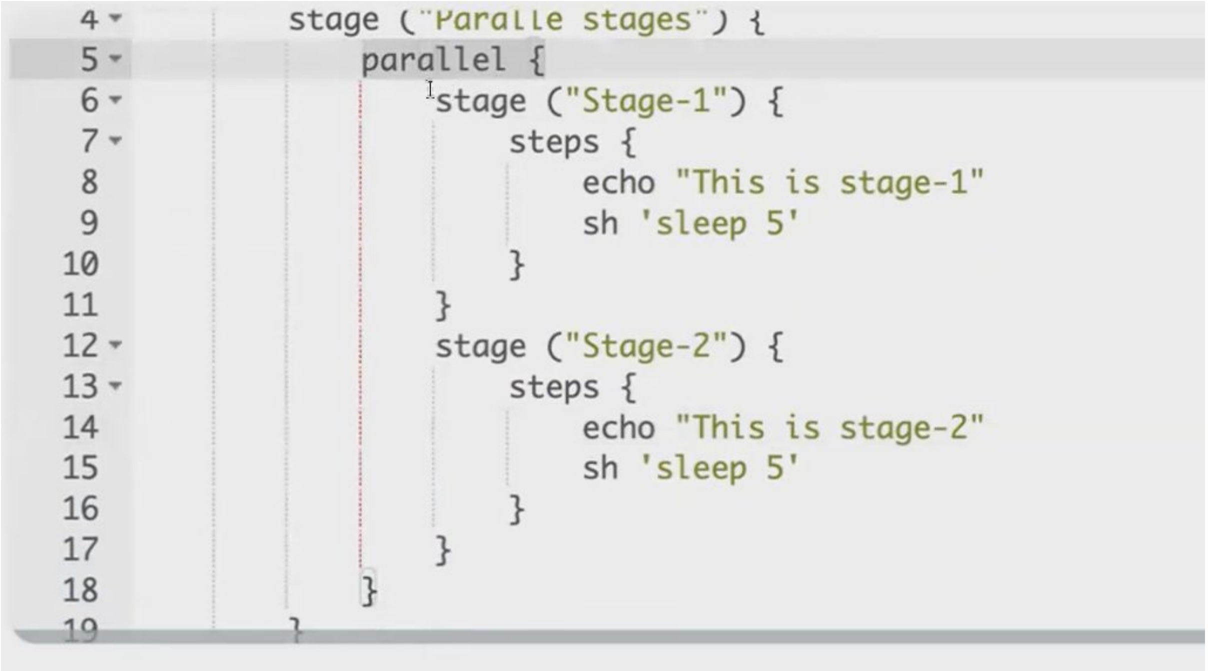
One by one execution chudataniki sleep pettanu

```
Script ?

pipeline {
    agent any
    stages {
        stage ("Code") {
            steps {
                echo "This is stage-1"
                sh 'sleep 5'
            }
        }
        stage ("Build") {
            steps {
                echo "This is stage-2"

            }
        }
```

See 1st stage execute ayyaka 2nd stage ki 5secs time padutundi



->suppose makai stages parallel ga execute avvali antey parallel use chesukovadam ey



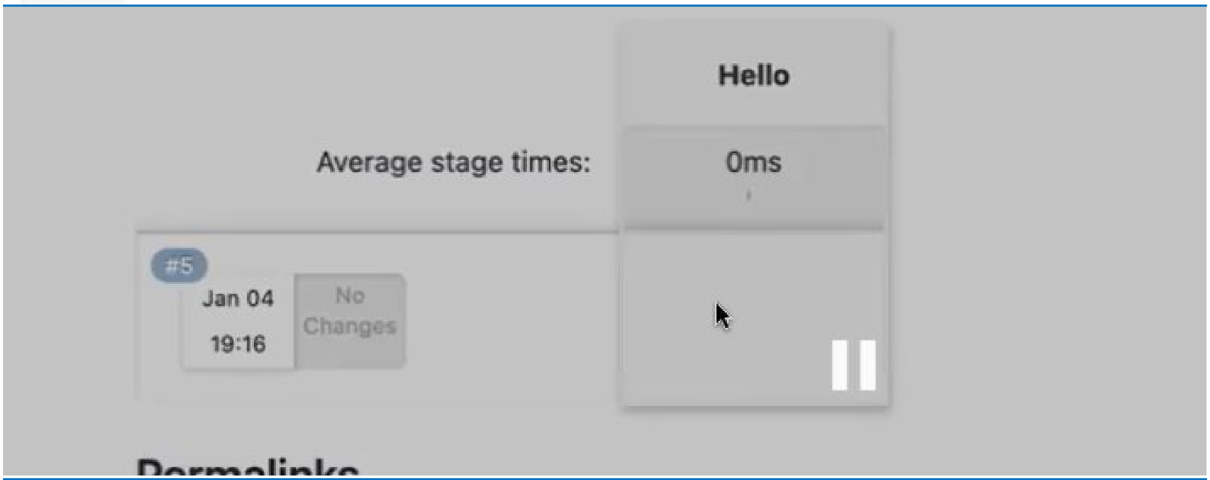
Parallel fun lo stages ni pettamu kabatti parallel ga execute avutai

->when we wanna perform task parallely(realtime – docker lo app container,db containers untai, so ee 2 at a time avvali antey use parallel tage execution)

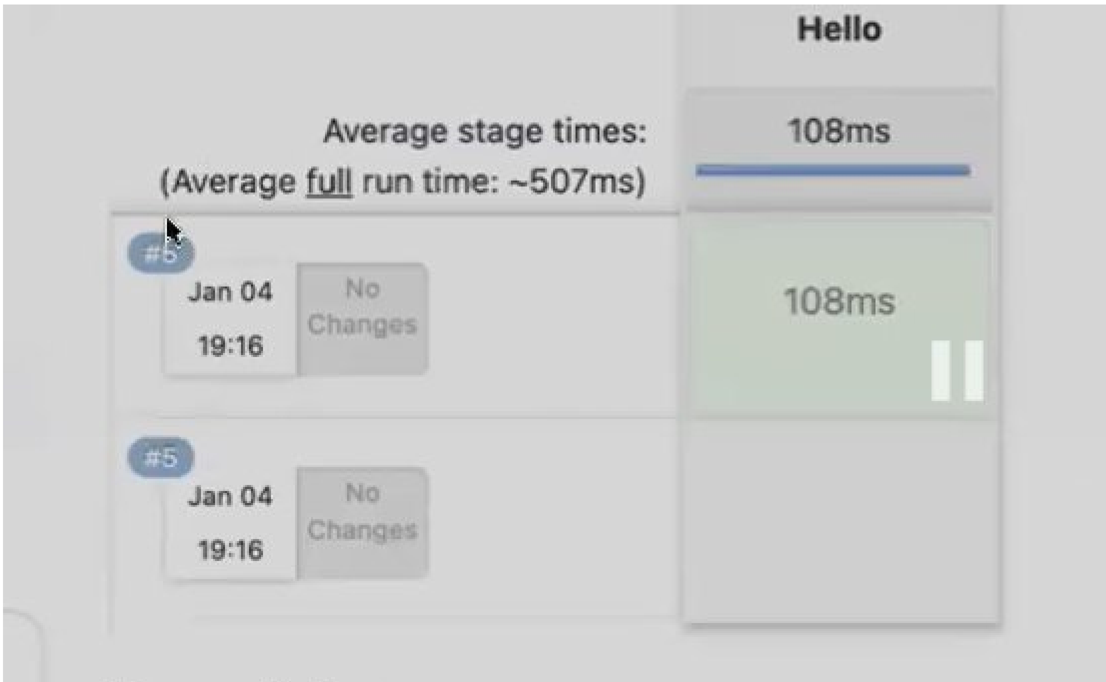
CONDITIONS IN A PIPELINE

->build no 6 aithey a stage execute avutadi

```
pipeline {
    agent any

    stages {
        stage('Hello') {
            when {
                equals(actual: currentBuild.number, expected: 6)
            }
            steps {
                echo "This is my stage"
            }
        }
    }
}
```

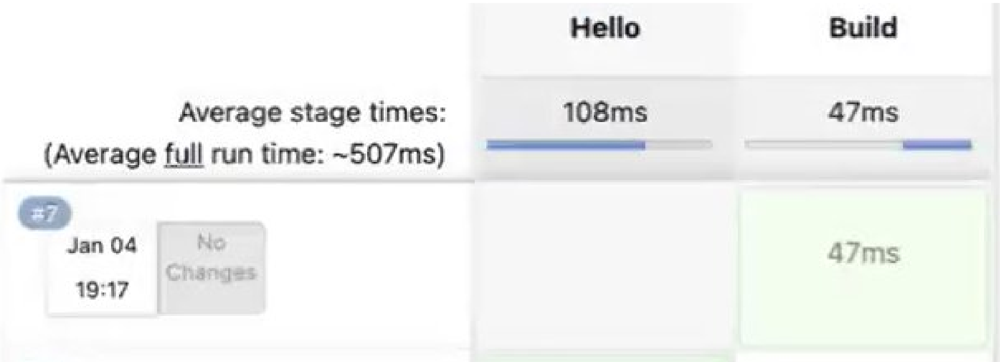


Condition meet ayyindi kabatti stage execute ayyindi

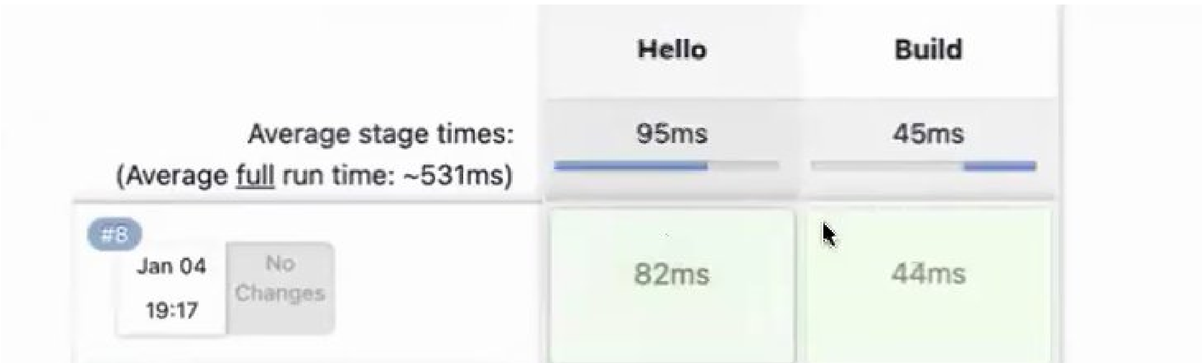


Condition meet avvaledu kabatti a stage exec avvaledu

```
    stages {
        stage('Hello') {
            when {
                equals(actual: currentBuild.number, expected: 8)
            }
            steps {
                echo "This is my stage"
            }
        }
        stage ("Build") {
            steps {
                echo "This is second stage"
            }
        }
    }
```

Condition meet ayyindi kabatti a stage exe ayyindi



REALTIME USAGE OF WHEN -1. github lo master branch aithey ney satge exe avvali

2. particular id aithey ney stage execute avvali – ee case lo when use chestamu

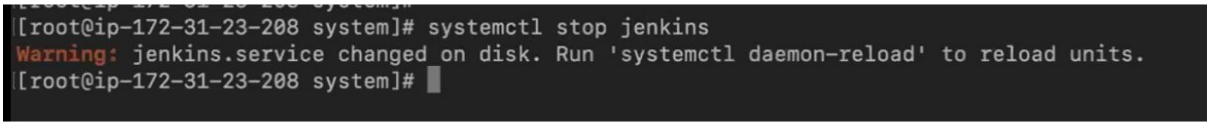
HOW TO CHANGE JENKINS DEFALUT PORT NUM

->goto path

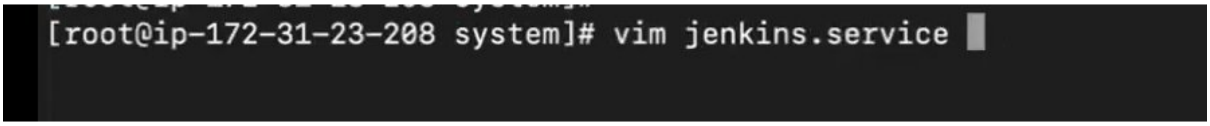
```
[root@ip-172-31-23-208 ~]# cd /usr/lib/systemd/system
[root@ip-172-31-23-208 system]#
```

->goto file Jenkins.service

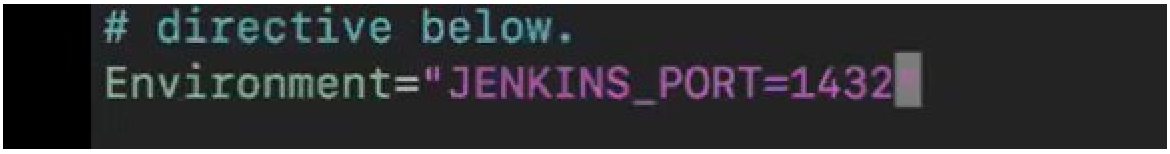
->before changing port we need to stop the Jenkins



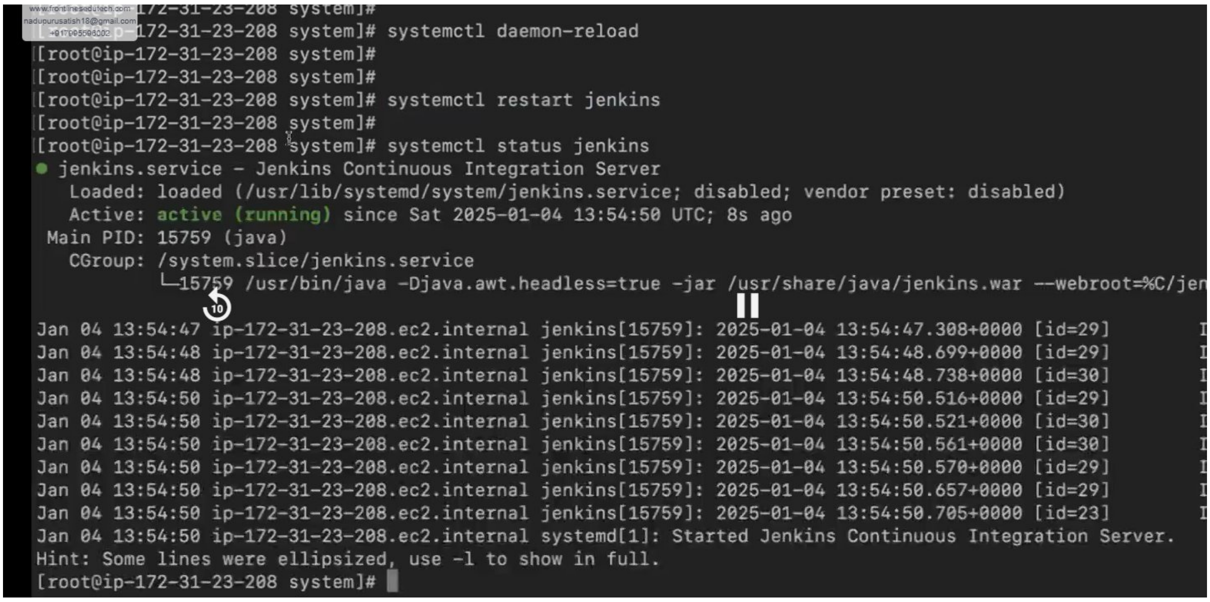
->open that file



->goto line 72 n change port no as u like nd save and continue

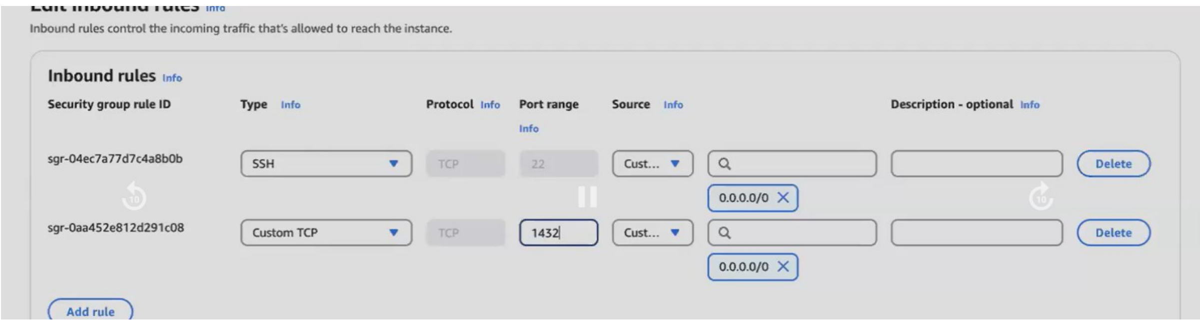


->restart the daemon n Jenkins



Nd chk the status

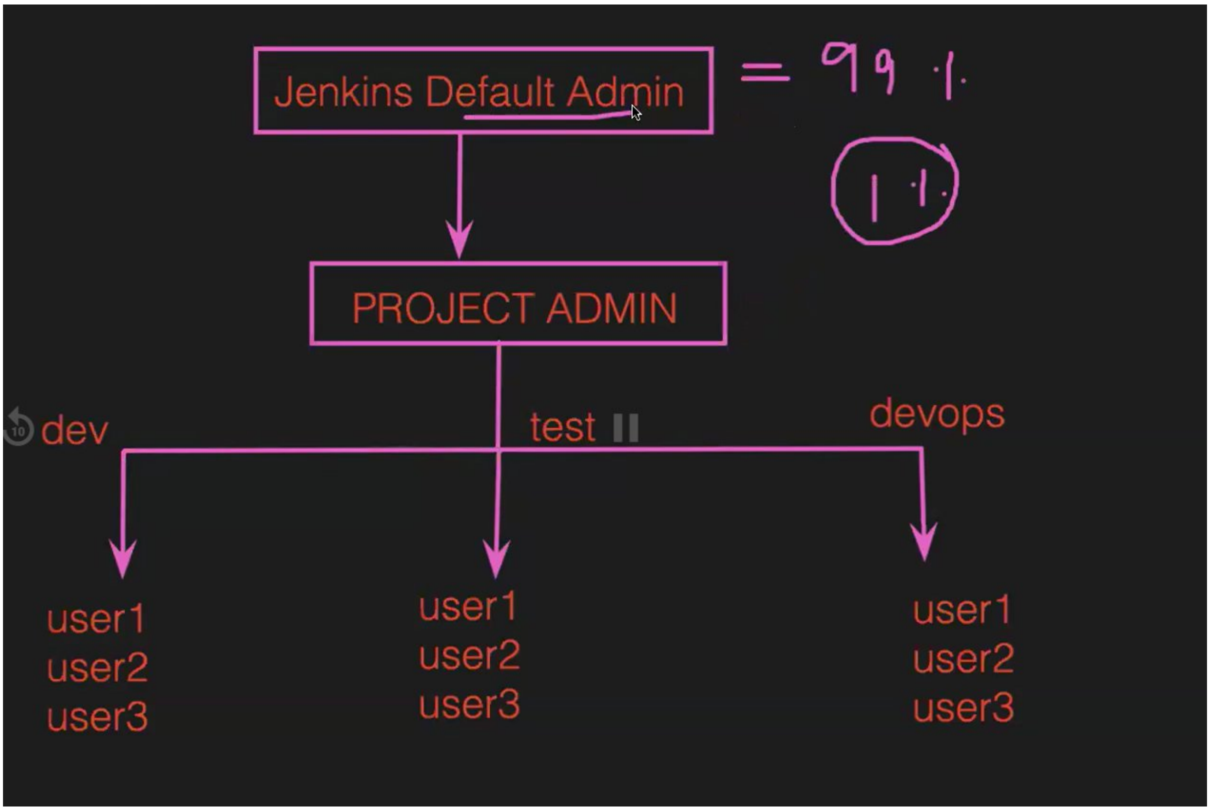
->ippudu Jenkins ni 1432 port tho cnt chesina connect avvadu bcz security grp lo 8080 untadi, so adi 1432 change chesthey cnt avutadi



TO RESET THE JENKINS PASSWORD

->Jenkins admin ey oka project admin user ni create chestadu, a proj admin users ni create chestaru

->incase a users pword marchipothey, ticket raise chesthey proj admin ki velthey reset chestaru,wht if proj admin ey pword marchipothey – Jenkins def admin ki velli reset chesukuntaru



->wht if Jenkins def admin pword marchipothey,

Goto def Jenkins path

```
[root@ip-172-31-23-208 ~]# cd /var/lib/jenkins/
```

Goto config.xml file

```
[root@ip-172-31-23-208 jenkins]# ll
total 60
drwxr-xr-x  3 jenkins jenkins   21 Jan  4 13:33 %C
drwxr-xr-x  3 jenkins jenkins   26 Jan  4 13:38 caches
-rw-r--r--  1 jenkins jenkins 1660 Jan  4 13:54 config.xml
```

Use security false petti, save n continue

```
Terminal  Shell  Edit  View  Window  Help
<?xml version='1.1' encoding='UTF-8'?>
<hudson>
  <disabledAdministrativeMonitors/>
  <version>2.479.2</version>
  <numExecutors>2</numExecutors>
  <mode>NORMAL</mode>
  <useSecurity>false</useSecurity>
  <authorizationStrategy class="hudson.security.FullControlOnceLoggedInAuthorizationStra
    <denyAnonymousReadAccess>true</denyAnonymousReadAccess>
  </authorizationStrategy>
  <securityRealm class="hudson.security.HudsonPrivateSecurityRealm">
    <disableSignup>true</disableSignup>
    <enableCaptcha>false</enableCaptcha>
  </securityRealm>
  <disableRememberMe>false</disableRememberMe>
  <projectNamingStrategy class="jenkins.model.ProjectNamingStrategy$DefaultProjectNaming$
  <workspaceDir>${JENKINS_HOME}/workspace/${ITEM_FULL_NAME}</workspaceDir>
  <buildsDir>${ITEM_ROOTDIR}/builds</buildsDir>
  <jdks/>
  <viewsTabBar class="hudson.views.DefaultViewsTabBar"/>
  <myViewsTabBar class="hudson.views.DefaultMyViewsTabBar"/>
  <clouds/>
  <scmCheckoutRetryCount>0</scmCheckoutRetryCount>
  <views>
    <hudson.model.AllView>
      <owner class="hudson" reference="../../.."/>
      <name>all</name>
      <filterExecutors>false</filterExecutors>
      <filterQueue>false</filterQueue>
      <properties class="hudson.model.View$PropertyList"/>
    </hudson.model.AllView>
  </views>
  <primaryView>all</primaryView>
```

Jenkins configuration change chesam kabatti, restart cheyyali

```
[root@ip-172-31-23-208 jenkins]# vim config.xml
[root@ip-172-31-23-208 jenkins]# systemctl restart jenkins
```

False pettamu kabatti, credentials lekunda login avvachu(un-authorized user login)

->ey user login ayyaru ani chudachu

Goto manage Jenkins>security>under security realm sel Jenkins own user database>now we can see users section in manage Jenkins>click on that users>we can see the user which we created while configuring Jenkins>click on that user, we can see reset option.

->now it'll ask fr pword while logging on

->config file lo true ni false pettamu, adi by default ga Jenkins restart chesaka true avutadi

SCHEDULE JOBS IN PIPELINE

->every minute

->

->now we see the scripted pipe-line(realtime lo antha ga use cheyyamu)

--- min--

Jenkins.io -> Download -> Red Hat Enterprise Linux and derivatives -> Long Term Support release. Copy the commands:

sudo wget -O /etc/yum.repos.d/jenkins.repo \ https://pkg.jenkins.io/rpm-stable/jenkins.repo -> It will download the Jenkins repository.

sudo yum upgrade /etc/yum.repos.d/ -> To check the repository downloaded path.

sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key -> To login in Jenkins we required password for that it will generate a key.

yum install Jenkins -y

systemctl status Jenkins -> To check the status of Jenkins

systemctl start Jenkins -> To start the Jenkins ->

If git is not installed in Jenkins server then we cannot access the GitHub repo in Jenkins pipeline. yum install git -y

Credentials -> for public no credentials needed   if private we have to give credentials.   Username : Git username   password : Git token(Settings->Developer settings-> Classic token generate)   ID: Our wish ID   Description: some matter Branch to Build-> Branch Specific-> we need to give which branch we are getting the code(ex: main) -> To run the pipeline we have to click on "Build now" -> To check the code or files in dashboard we have to click "Workspace" -> To check the code in server then we have to open "Console output"(To check the logs) -> To edit the pipeline we have to click on "configure" -> To delete the pipeline we have to click on "delete project" -> #1 -> it means that i build first time. -> After Jenkins creation, if u add data in gitub and click build now then only it will reflect data in Jenkins. -> For every add and deleting build now is must and should. ---